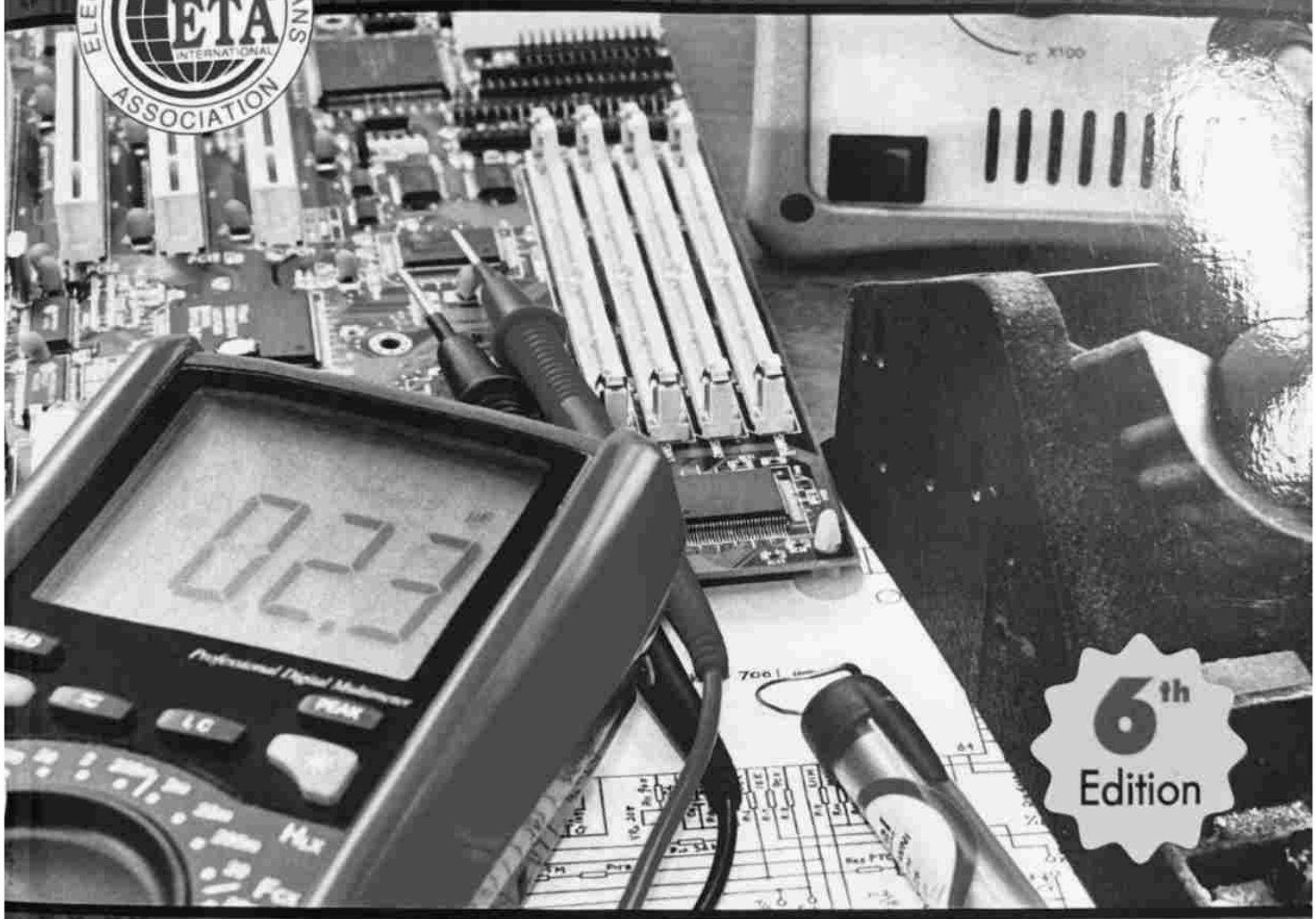
Associate CET

Study Guide



6th Edition

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The Associate CET Study Guide

6th Edition

Written by the ETA Associate Certification Committee and Test Administration Staff

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Preface

Sample wall certificate

This book is intended as an official study guide for students and electronics technicians who are seeking the status of CERTIFIED ELECTRONICS TECHNICIAN, ASSOCIATE LEVEL (CETa). It is also intended for technicians who are planning to become Journeyman (CET), Senior (CETsr) or Master (CETma) CETs. The latter three levels of certification require passage of both the Associate exam and one or more Specialty Option exams.

The book is not planned as an engineering or scientific text. While it contains many in-depth discussions of electronics technology basics, it is intended as a practical guide for working technicians without the mathematics, chemistry or physics requirements of rigorous academics.

That isn't to say ETA isn't interested in electronics technicians advancing to studies of galactic or quantum physics. Advanced study is encouraged. In fact, students who become electronics technicians, understanding the world of electricity, photonics, communications, electronic components and methods of measurement are already a step ahead of much of the population. Competent electronics technicians often find their interest attracted to hobbies or vocations which involve intellectual leading-edge technological comprehension of the Cosmos, as well as the atomic world we live in.

Possibly the most important technical worker today is the electronics technician. He or she may be involved in any one of hundreds of vitally important jobs. Today's technicians do not know it all. They must specialize in only a few segments of the electronics universe. That is because the technology has not only spread out vertically, but horizontally too. To become knowledgeable and experienced in every phase of electronics would require hundreds, if not thousands, of times the quantity of knowledge a technician needed half a century ago. The fact still remains that every technician is a better technician if he or she has a solid core, or basic, electronics knowledge. In a book that you can read in a day, ETA presents that core knowledge that can help you become a qualified electronics technician.

Sample wallet card

With an Associate CET status, you can qualify for hundreds of types of electronics jobs. You can stop there and become comfortable in a routine position, or you can use your knowledge to step up the ladder to Journeyman or higher recognition. With the basics firmly implanted in your mind, you will have a confidence in your work which can make your job gratifying, not drudgery. It can make many aspects of your life more fulfilling as you now have an understanding of how most things in life work.

Good luck in your career as an ELECTRONICS TECHNICIAN.

Richard K. Glass

Dick Glass, CETsr, EHF
CEO Emeritus

Acknowledgements

The 6th edition has been reorganized and edited from previous editions to provide more uniform content and better study tools for the Associate CET exam. Similar content has been grouped into units of study. Most chapters have review questions. There are unit review questions. Updated Associate CET practice exams are included. We hope you enjoy the new edition.

- Dick Glass, CETsr, Chief Editor

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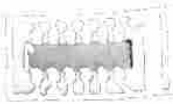

ETA® International Certifications

Basic Electronics:

Apprentice (APP)
Associate Electronics Technician (CETa)
Electronics Modules (EM1-5)
- DC (EM1)
- AC (EM2)
- Analog (EM3)
- Digital (EM4)
- Comprehensive (EM5)

Student Electronics Technician (SET)

Biomedical:

Biomedical Electronics Technician (BMD)
Biomedical Imaging Equipment Technician (BIET)

Communications:

Broadband Voice over Internet Protocol (B-VoIP)
Certified Satellite Installer (CSI)
Distributed Antenna Systems (DAS)
General Communications Technician-Level 1 (GCT1)
General Communications Technician-Level 2 (GCT2)
Line and Antenna Sweep (LAS)
Mobile Communications and Electronics Installer (MCEI)
Passive Intermodulation Testing (PIM)
RF Interference Mitigation (RFIM)
RADAR (RAD)
TRN Wireless Communications Technician (TRNI)
Telecommunications (TCM)
Wireless Communications (WCM)

Fiber Optics and Data Cabling:

ARINC Installer, Technician (AFI/AFT)
Data Cabling Installer (DCI)
Fiber Optics Designer (FOD)
Fiber Optics Installer (FOI)
Fiber Optics Technician (FOT)
Fiber Optics Technician-Outside Plant (FOT-OSP)
SAE Fabricator (SFF)
Termination and Testing Technician (TTT)

Photonics and Precision Optics:

Photonics Technician Operator (PTO)
Photonics Technician Specialist (PTS)
Specialist in Precision Optics (SPO)
Technician in Precision Optics (TPO)

Renewable Energy:

Electric Vehicle Technician (EVT)
Photovoltaic Installer-Level 1 (PVI-LVL1)
Small Wind Installer-Level 1 (SWI-LVL1)

Smart Home:

Certified Alarm Security Technician (CAST)
Residential Electronics Systems Integrator (RESI)
- Audio/Video (AV)
- Computer Networking (CN)
- Closed-Circuit TV (CCTV)
- Security/Surveillance (SS)
- Environmental Control (EC)

Master Residential Electronics Systems Integrator (RESIma)

Information Technology:

Computer Service Technician (CST)
Electronic Security Networking Technician (ESNT)
Network Computer Technician (NCT)
Network Systems Technician (NST)
Wireless Network Technician (WNT)

Additional Certifications:

Avionics (AVN)
Certified Service Manager (CSM)
Commercial Audio Technician (CAT)
Customer Service Specialist (CSS)
Gaming and Vending Technician (GVT)
Industrial Electronics (IND)
Radio Frequency Identification Technical Specialist (RFID)

ETA is also one of 13 COLEMs for FCC Licenses:

FCC

Marine Radio Operator Permit (MP)
General Radiotelephone Operator License (PG)
Global Maritime Distress and Safety System Operator (DO)
Global Maritime Distress and Safety System-Restricted (RG)
Global Maritime Distress and Safety System Maintainer (DM)
Radiotelegraph Operator (T)
Ship Radar Endorsement

ETA International certifications are accredited by:

ICAC
international certification accreditation council

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Addendum to This Book

Note: The Associate CET Study Guide was never intended to be an exhaustive textbook about e... erything involved in electronics technology. It was intended to be a practical book that is written in plain English. By involving multiple authors, some of the material is repeated occasionally. Some of the plain language can help those studying to become electronics technicians overcome difficult concepts and speed their understanding. Because everything in the practice exams has not been referenced, here are some additional truths that may help.

1. A video signal (such as TV or satellite) contains the video information plus the synchronization signals needed to properly display the video plus other information content or coding and messaging. It does not contain audio information.

2. Audio or an audio signal includes frequencies between 20 Hz and 15,000 Hz. It is common to see it defined as 20 to 20,000 Hz as the actual limits of the audio frequency band.

3. Troubleshooting: When faced with a product that is showing signs of a short, perhaps blowing fuses or circuit breakers, it may help that you can insert a low ohms, high wattage resistor in series with the AC power which will allow troubleshooting without as much danger of further damage or fuse blowing. A variac also can allow you to reduce the input AC power in a similar manner.

4. When identifying power supplies, if a filter type capacitor is found to be in the circuit between the power transformer secondary leads and the rectifiers, it indicates the circuit is likely a voltage-doubler power supply.

5. DMM's (digital multimeters) are invariably more accurate than analog multimeters.

6. In the Chapter 7 discussion of motors, know that motor speed controllers may use microprocessors or several other methods including op-amp comparator circuits.

7. A fuse should be replaced with the exact size and type. If a substitute must be made the amp rating should not be greater than the original. The voltage rating can be less but refers to the arc-over potential after the fuse blows. (e.g.: in a circuit with no more than 25 v, a blown 250 v fuse could be replaced with a 125 v).

8. A power source (battery or generator) for electronics power supplies might be compared with the water tower/tank in a municipal water system.

9. Transformers often use soft-metal iron core to increase lines of force concentration.

10. Because a small portion of the current in a transistor flows from emitter through the base, the most current flow in a transistor in the emitter, not the base or collector.

11. 'Q' is used in Chapter 3 to denote 'charge' in a capacitor (in Coulombs). 'Q' is also used to denote the 'selectivity' of a tuned circuit.

12. Electrolytic capacitors: Applying a 'backward' (reverse) voltage across this type will destroy it unless it is a special non-polar capacitor.

13. (See Ch 5) The 5th band on a resistor indicates tolerance (how far from its indicated value it may vary), or reliability (failure rate/1000 hrs.). Check the specs and measure to be sure.

14. A balun is an impedance matching transformer often used to connect coaxial cable to 300 Ohm transmission line. (as at the input to a TV receiver)

15. In performing an ohmmeter check of a transistor, checking the resistance between the E B and C legs should show at least a 3 to 1 forward-to-reverse reading and may be much higher.

Chapter 1

The CET Program Description

by Dick Glass, CETsr

Dick Glass is an author of previous ETA publications as well as more than a dozen books dealing with electronics technology and business management, published by Howard W. Sams Co, TAB Books, and McGraw-Hill. He has also written hundreds of articles in trade journals and association publications. Dick is a former Navy Avionics Technician, Industrial Electronics Computer Technician, Technical Writer, FCC Commercial License examination modernization committee Chairman in 1995 and 2005, inductee into the Electronics Hall of Fame and currently CEO Emeritus of ETA. He has a broad background in all phases of work performed by electronics technicians

In The Beginning

The CET program was initiated in 1965 by a national association incorporated under the name of NEA®, National Electronic Associations.

There were few credentialing organizations back in those days. Most certifications were aimed at personnel in the health care industries: RN, CNA, etc. There were licensing programs in some states (Indiana, Massachusetts, Louisiana, Florida) for TV technicians and the 'biggie' was the FCC Radiotelephone Operators License, administered by the U.S. Government.

NEA and the USDL

In the early 60's the Bureau of Apprenticeship and Training, U.S. Labor Dept., and NEA had initiated a retraining program which required on-the-job education, plus formal school classes. The course length was 8,000 hours, or four years. At the conclusion of the apprenticeship, if the student was successful in passing the final examination, he or she was given a certificate by the U.S. Government and NEA attesting to his/her schooling, abilities and knowledge. The title for the successful student now became "Journeyman Electronics Technician." It was a type of 'degree'—not an engineering degree—but more on the order of an Associate technical school degree.

Jerry Barkoukis, CET

As the program was implemented under the direction of NEA's contracted coordinator Jerry Barkoukis, CET, of Akron, Ohio, the association began to think about the many Radio-TV technicians already working all across the country. Many of these were well trained ex-military electronics technicians or those who had completed public school or commercial electronics training courses. DeVry, NRI, CIE, ITT, etc. had begun training technicians also. The question was: Should these well-trained technicians be able to prove their knowledge, abilities and training and receive a comparable 'degree' as those who completed the U.S. Labor Dept. 8000-hour program? The answer, of course, was "Yes."

So, under the direction of Dick Glass, CETsr, who was the Executive Vice President of NEA at the time, the CET program was initiated.

A Program for Radio and TV Techs

The first examination was aimed at Radio-TV technicians. Questions pertaining to antennas, transmission lines, basic components used in electronics, mathematics and formulas, test equipment, radio and TV and basic electronic circuits, electrical safety, audio and video, RF electromagnetic signal propagation, amplification, block diagrams and so forth were considered proper basic materials for the first certification exams.

There had been some earlier attempts to establish certification for professional technicians in the Seattle area and in California. Once a national association (NEA) took on the task, these programs joined in the national effort.

In the beginning there was only one examination. It included material related to radio and TV, simply because there were few other jobs in electronics. There were not many biomedical or industrial electronics products, satellites or fiber optics, and the avionics technician could still fall under the 'radio' portion of the certification. One exam for all.

Certification is Good

Certification proved to be a good program. It allowed proper recognition of those who had gained a level of education and had acquired sufficient on-the-job or experience credits. Technicians who were proud of their high tech job, and who recognized that the service they were performing was extremely important, could show their credentials to potential employers. TV-radio customers could affirm that the person working on their expensive new entertainment equipment was qualified, schools could determine just what needed to be taught to bring a student up to qualified worker status, and some states even used certification as the requirement for state licensing where laws were passed to regulate the repair and installation business.

Not many years after the CET program had begun certifying professional electronics technicians, most of whom were TV repair people, it was found that everyone wasn't involved in TV repair. Some technicians were exclusively wireless communications or radio workers. Many of these were amateur radio buffs and some were auto radio specialists. What to do?

Chapter 1

A Core Exam - The Associate

The association decided to recognize the various job skills which were becoming identifiable and important parts of the electronics industry. To do this, and to give a technician who was an expert in his area of electronics an affirmation that he could also serve in other allied fields, the association leaders determined that all CETs should be required to pass an examination based on the core elements of electronics of which all technicians should have knowledge. The idea was that a technician working with (for instance) computers (which by now had become important electronics products) should not be required to pass a test pertaining to the operation of a color TV. The specialized technician should know about soldering and basic electronics components and how to read block diagrams and schematics and to not get killed by electrocution. He need not know about the shut-down protection circuit in a color TV, even though this might be useful to him now or in the future. So, a core CET Exam was put in place and the Journeyman CET was expected to pass it, PLUS another written exam pertaining to the field of electronics work in which he or she specialized.

Over the years, the Specialty Exams were developed by technicians, instructors, employers and others, with each specialty CET program having its own committee chairman and committee members. These groups of subject matter experts averaged about a dozen members each.

Here are the Specialty areas as of 2016:

Avionics	(AVN)
Biomedical Electronics Technician	(BMD)
Biomedical Imaging Equipment Technician	(BIET)
Certified Alarm Security Technician	(CAST)
Certified Satellite Installer	(CSI)
Commercial Audio Technician	(CAT)
Computer Service Technician	(CST)
Distributed Antenna Systems	(DAS)
Electronic Security Networking Technician	(ESNT)
Fiber Optics Technician	(FOT)
Fiber Optics Technician - Outside Plant	(FOT-OSP)
General Communications Technician - Level 2	(GCT2)
Industrial Electronics	(IND)
Line and Antenna Sweep	(LAS)
Network Computer Technician	(NCT)
Network Systems Technician	(NST)
Radar	(RAD)
Radio Frequency Identification Technical Specialist	(RFID)
TRN Wireless Communications Technician	(TRN)
Telecommunications	(TCM)
Wireless Communications	(WCM)
Wireless Network Technician	(WNT)

By providing specialty areas [illegible] COPE on basics of electronics [illegible] broad knowledge all technicians [illegible] an experienced worker with specialized electronics job in which he was employed.

The Associate plus specialty [illegible] lem of deciding what to teach in [illegible] Industry would like each graduate to have universal knowledge of [illegible] ics technology today. To attempt [illegible] in even a two year course is an exercise technical education.

Today, many schools offer [illegible] may also offer separate courses [illegible] certifications which lead to CET [illegible] working and biomedical are examples logical way to manage the broad [illegible] ogy.

Associate Status

While the separating of the journeyman certification examinations into the [illegible] solved a giant problem and made [illegible] began to look at technical education [illegible] eye. It did not take four years [illegible] sharp teenager how to manipulate [illegible] ordinary computer programs. While [illegible] the computer keyboard technician [illegible] academic experience, he or she could [illegible] highly skilled and valuable worker [illegible] two years of formal computer training could become skilled workers with [illegible] training, not months or years of [illegible]

By the same token, a student could [illegible] formal basic electronics education [illegible] educational institution and be well [illegible] force, yet have no on-the-job experience [illegible] would love to hire a tech with a good [illegible] the inexperienced student or apprentice [illegible] than that of a common laborer.

The answer was to allow the [illegible] the-job trainee to attempt the Associate [illegible] exam. In ETA, we then call the [illegible] ciate CET. We even began allowing [illegible] graduate to attempt the specialty [illegible] be specifically schooled for. If he [illegible] the CET is also shown on his certificate [illegible] technical examination pertaining to [illegible] is important in getting a job with [illegible]

So we now have a beginners or apprentice-level certification program, as well as a program to certify the knowledge of one who is the mainstay of the work force. In older times that was called a journeyman or skilled craftsman. Today the term journeyman is less favored and rarely used except in formal labor union advancement programs.

Customer Service Specialist (CSS)

It was later decided that a certification devoted to human relations, people skills, problem resolution and job values was needed in the electronics industry. In fact, it was needed in all industries. It became apparent, however, that help-desk personnel, delivery and sales people, supervisors and non-electronics workers should not be tested for their basic or specialty electronics knowledge (even though some of this might help them better do their jobs). So, ETA, with the push of Don Howell, CETsr, of Lake Washington Technical College in Seattle, in 1990 designed an examination which could set a people skills level and examine workers to see if they know the practical thinking processes which can raise the standard for co-worker and customer skills. It was called the Customer Service Specialist or CSS certification program.

Because it does not require an experience prerequisite or technical skills, it is different than the Associate or the Journeyman levels of the CET program. It does, however, require the examinee to also sign and accept a code of conduct which commits the CSS to just plain, professional, decent behavior.

Have it Both Ways

Other certifications, technical in nature, also seemed to fit the stand-alone format. For instance, fiber optics installers would do well to have basic electronics knowledge. They would benefit from a thorough understanding of data communications. This isn't required in order to work with a crew of fiber installers. In the same manner, a keyboard technician (CST or A+) would be a more versatile worker if he or she knew basic electronics or cabling fundamentals, but can, in many jobs, serve well without those nice-but-not-required skills. So CST can also be a stand-alone certification by ETA. The successful CST cannot use the term CET after his name, as he isn't a Certified Electronics Technician. He can display CST after his name and can show his ETA certification proving that he has the required knowledge to perform as a computer keyboard worker. Thus, ETA has allowed the computer service technician to be certified and to also be a CET if he feels that certification will allow him to advance, or if a new job requires a CET.

So, now a worker can obtain an Associate, beginner or apprentice level, certification. He can become a journeyman or professional certified electronics technician, or he can obtain specialty certifications that fill special needs.

Senior CET

Today, ETA requires re-certification for most CETs and specialty certification holders. Prior to 1996 when re-certification was officially instituted, once you obtained an NEA, ISCET or ETA certification, it was for life. With the high-tech world demanding even more qualified workers, ETA instituted a four-year re-certification program. Before 1996 the Senior CET was offered to journeyman technicians who could show at least eight years of schooling/experience in electronics. The Senior also must retake the CET exam and this time, pass with a higher score: 85% rather than the ordinary 75%.

The Senior program allowed conscientious technicians to prove they were keeping up with advances in electronics. The motive was to not copy the policy of the Federal Communications Commission which allows a technician to be licensed once, then never have to prove he or she ever learned the new technologies or gained any further skills. So, ETA solved another problem by instituting re-certification and thus requiring all CETs to keep up with technology.

With re-certification the Senior certification was not dropped. In fact, it is still popular. The time requirement was reduced to six years, the passing score retained at 85%, and a beautiful wall plaque is offered. A majority of the electronics instructors who certify do attain the Senior status, as do many working technicians.

So now, a technician has several choices: He or she can be a beginner and attain the Associate CET status; within two years advance to the professional level of Journeyman (by far the most popular certification ETA issues), add a specialty certification such as the Customer Service Specialist, Fiber Optics Installer, etc., or he or she can attain the Senior CET which shows he/she has a very high level of knowledge, plus plenty of on-the-job experience. An employer can credit the applicant with the exact level of abilities for the job requirements.

Master CET

One almost revered level of certification is the Master CET. Over 350 technicians had attained this highest level of certification in ETA as of 2016.

Why would you want it? The Master CET must be able to prove himself or herself in not just one electronics category, but he or she must master the Associate CET exam, plus pass at least six of the Specialty exams at the 75% level. Because it is difficult to be proficient in several categories (i.e. Avionics and Biomedical, for instance), Master is not for everybody. It does qualify the professional technician as being proficient in several areas.

Chapter 1 The CET Program Description

For instance, a technician may have served in the military as a wireless communications tech. While in the service he or she may have become seriously involved in audio and video. Upon detaching from the service he or she may have gone to school to study computers, thus gaining the CST and CNCT abilities. While working on the job he or she also might have become interested in RADAR and decided to get an FCC RADAR endorsement for his FCC license. At this point he or she likely could pass five of the specialty exams. He or she only need choose one additional Specialty exam and he or she has enough exams passed to become qualified as a MASTER CET. This too is offered in a beautiful wall plaque, is lifetime (doesn't require re-certification), and is the highest level of certification. It is a unique honor and is recognized by many employers for the special broadly-trained and experienced electronics technician it attests to.

So, now you know about all five levels of the CET program:

- Student (high school students)
- Associate
- Journeyman
- Senior
- Master

(all categories are listed in the front of this book)

What are the costs to become certified by ETA? They are much less than most other similar type technical certifications.

Here are the established fees* (current as of 2016):

Journeyman Electronics Technician (CET). . . . .$75-$150
-Each additional Journeyman certification is $30, if taken with the Associate at the same time

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ARINC Installer, Technician (AFI/AFT) $175
Apprentice Electronics Technician (APP). $60
Associate Electronics Technician (CETa). $60
-Associate exam FREE if taken with a Journeyman exam at the same time
Avionics (AVN). $75
Biomedical Electronics Technician (BMD). $75
Biomedical Imaging Equipment Technician (BIET). $75
Broadband Voice over Internet Protocol (B-VoIP). . . $150
Certified Alarm Security Technician (CAST). $75
Commercial Audio Technician (CAT) $75
Certified Satellite Installer (CSI). $75
-Satellite Endorsement. $60
Certified Service Manager (CSM). $150
Computer Service Technician (CST). $75
Customer Service Specialist (CSS). $75
Data Cabling Installer (DCI). $150
Distributed Antenna Systems (DAS)
Electronics Modules. $75
Electronic Security Networking Technician (ESNT)
Electric Vehicle Technician (EVT). . . . . .
Fiber Optics Installer (FOI). . . . . .
Fiber Optics Technician (FOT). . . . . .
Fiber Optics Technician - Outside Plant (FOT-OSP)
Fiber Optics Designer (FOD). . . . . .
Gaming and Vending Technician (GVT). . . . . .
General Communications Technician - Level 1 (GCT1)
General Communications Technician - Level 2 (GCT2)
Industrial Electronics (IND) . . . . . .
Line and Antenna Sweep (LAS). . . . . .
Network Computer Technician (NCT). . . . . .
Network Systems Technician (NST). . . . . .
Master RESI (RESIma). . . . . .
Mobile Communications and Electronics Installer (MCEI)
Passive Intermodulation Testing (PIM). . . . . .
Photonics Technician Operator (PTO). . . . . .
Photonics Technician Specialist (PTS). . . . . .
Photovoltaic Installer - LVL1 (PVI-LVL1). . . . . .
RF Interference Mitigation (RFIM). . . . . .
Radar (RAD). . . . . .
Radio Frequency ID Technical Specialist (RFID). . .
Residential Electronic Systems Integrator (RESI). . .
-Basic RESI and one endorsement is $75, if taken same sitting
-RESI Endorsement. . . . . .
SAE Fabricator (SFF). . . . . .
Small Wind Energy Installer - LVL1 (SWI-LVL1). .
Specialist in Precision Optics (SPO). . . . . .
Student Electronics Technician (SET). . . . . .
Technician in Precision Optics (TPO). . . . . .
Telecommunications (TCM). . . . . .
Termination and Testing Technician. . . . . .
Wireless Communications (WCM). . . . . .
TRN Wireless Communications Technician (TRN)
Wireless Network Technician (WNT). . . . . .

Master Exam. . . . . .
Master Upgrade with plaque. . . . . .
Senior Exam with plaque. . . . . .
Senior Upgrade. . . . . .
-Senior upgrades must be done within the first certification

**Fees and certification availability, and procedur change. Please check the ETA website (www.eta-i most up-to-date information.*

Formal Schooling is Not Required

One might think a well-thought-out program like the ETA Certification would require mandatory classroom contact hours. That isn't the case. Since 1966, the association's attitude has been: If you know your stuff, can prove it by showing your work and/or training experience, and you can pass the CET exams, you then qualify.

Certification Maintenance

ETA initiated a certification maintenance program for the majority of its certification programs in 1996. CETs can retest or prove that they have attended 40 total contact hours of training during the previous four years. In 2003, the association implemented an even easier continuous certification maintenance process. This requires submitting upgrade education information, address changes, etc. each year, modeled upon the ISO 17024 standard.

The Certification Administrators

The success of the CET program in NEA, ISCET and ETA has been due to the individuals who promote and proctor the actual examinations. A typical CA is an electronics or computer instructor at vocational, technical, engineering public or commercial educational institutes. A list of the current test sites is included in the appendix of this publication and on our website (www.eta-i.org).

The CAs maintain security of the examinations and, with another school official, plan and proctor the examinations. Sometimes the exams are given at the conclusion of a technical course and are used as the final exam. Some schools will not graduate a student who cannot pass the appropriate exam.

CAs also are a major part of each of the certification examination development committees of ETA. Every year several CAs are elected to major ETA Board of Director and Officer positions.

Additionally, the CAs also keep a close watch on the content of each of the program examinations. The interaction of actual teachers between ETA and the school, with real knowledge of how well the student did during the course, helps both the school and ETA validate the certification. For instance, if a class of twelve students did very well in a tech school course, passing the textbook quizzes, the written and oral exams composed by the instructor and exhibiting a thorough understanding of the subject, but then failed the ETA exam, it would be a red flag to the instructor, the school and to ETA that something is not right, either with the training or the examination.

Free Retakes

Because everyone doesn't acquire ETA study guides, or may not be totally familiar with the content of the ETA exams, a free retake policy has always been in effect. One free retake is allowed after waiting at least thirty days, and before one year has elapsed. The association has always felt that the period between a failing of an exam and then retaking it and passing, is a time of maximum learning as the examinee goes back to the books, or engages in discussion of exam topics with other technicians or his instructor.

What's the Point?

The point of the entire CET program is to establish a level of technical ability and knowledge that assures the world that when it comes to electronics, cabling and computer installation, wireless communications, maintenance and service, that CETs are the best qualified workers. That the cost of gaining an outstanding technical education pays off in higher compensation and personal satisfaction in knowing your work is an important part of the human experience on this planet.

Each new technician who becomes certified raises the overall level of the entire profession. Each individual counts. CETs are proud of their profession and their part in it. They respect other CETs and the educators who help make it all happen. CETs belong to a great fraternity!

NOTES

Chapter 1 The CET Program Description

Chapter Quiz:

1. When did the CET certification program begin?

   a. 1965
   b. 1976
   c. 1986
   d. 1996

2. When did ETA initiate Certification Maintenance?

   a. 1966
   b. 1976
   c. 1986
   d. 1996

3. All ETA Certification categories require the passage of the Associate CET Examination.

   a. True
   b. False

4. To qualify for MASTER CET, the examinee must attain a score of 85% on each separate exam.

   a. True
   b. False

5. An Associate CET cannot attempt one of the specialty examinations.

   a. True
   b. False

6. Master CET status is lifetime and does not require certification maintenance.

   a. True
   b. False

7. The Journeyman level:

   a. denotes competency in one specialized area.
   b. requires 75% passing.
   c. requires 2 years of work or school.
   d. all of the above.

8. Certification maintenance requires payment of a fee and retesting or proving at least 40 hours of upgrade retraining within a four-year period.

   a. True
   b. False

9. A worker may become certified by ETA as a CST (Computer Service Technician), without also becoming certified as a CET (Certified Electronics Technician)

   a. True
   b. False

10. The Associate CET exam may be taken after the Journeyman exam.

    a. True
    b. False

ANSWERS on page 223

Chapter 2

Safety Precautions for Electronics

by Dick Glass, CETsr

Dick Glass is an author of previous ETA publications as well as more than a dozen books dealing with electronics technology and business management, published by Howard W. Sams Co., TAB Books, and McGraw-Hill. He has also written hundreds of articles in trade journals and association publications. Dick is a former Navy Avionics Technician, Industrial Electronics Computer Technician, Technical Writer, FCC Commercial License examination modernization committee Chairman in 1995 and 2005, inductee into the Electronics Hall of Fame and currently CEO Emeritus of ETA. He has a broad background in all phases of work performed by electronics technicians.

Shock

Technicians work with electricity. Therefore, they are in the vicinity of electric charges during much of their workday. Thus, they are potentially in a more hazardous position than the average worker. On the other hand, because electronics technicians understand the nature and the danger of electric shock, some think they are in less danger than those who may not always notice a lurking dangerous situation. Either way, it pays to know as much as possible about the dangers of working with electronics. This includes the danger to your person, as well as other dangers to components or others in the workplace, or the workplace itself.

Electrical shock is the number one concern we have when we speak of electrical safety. Everyone has experienced electrical shock, even if it is only the static discharge we get in dry weather when we touch a door knob after walking across a carpet.

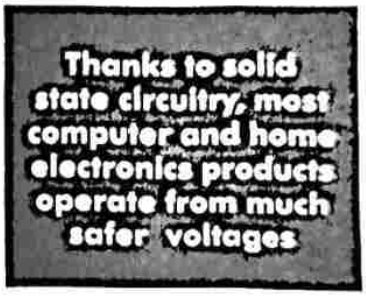
Change of Character

Dozens of years ago the U.S. Navy advised personnel who worked in and around electricity that even though contact with electricity might not be lethal, that any amount of electric current passing through the body could cause a change in character. For some of us this might be good! On the other hand, since the profound change it might make is unknown and unpredictable, we should be concerned. No one knows whether a slight change in DNA, or a shifting of brain cells takes place. We do know that even small doses of electric current tend to hurt. So, while you can't completely eliminate electric shock, you can respect electricity and keep shocks to a minimum.

A lethal dose of electricity is a good place to start in this discussion. Some say that the 115 V AC from the wall socket isn't lethal. Most of the time it isn't. But if you make a really good contact with it, say while you are standing in a pool of water in your bare feet, you may be killed.

B+ or supply voltages for the tubes or transistors in radios or other products used to be much higher, from 50 volts to 500 volts. Today, transistor products may utilize low voltages like +5 V and +12 V, which are very unlikely to be lethal. AC powered products, however, still have 115 V AC coming in on the line cord, so you should be concerned even with these much friendlier electronics products.

Reaction to Shock



While many of the products technicians work on are not dangerous, sometimes a shock from a lower voltage is such a surprise that the reaction to the shock causes some damage. If you react to the shock by dropping your screwdriver into the circuitry, you may destroy the product. If you jerk back from a small shock and fall off your chair or run into someone, you may feel pain, even if it is not caused by electrons flowing through your conductive body. So the reactions to the trauma that even small electric currents can cause are dangerous to life and limb.

Where Are The Volts?

Just where are dangerous voltages if we are mostly dealing with transistor circuitry, chips and such?

Keep in mind that there will still be legacy equipment to service such as TV and computer monitor kinescope displays. The high voltages which operate these high-vacuum picture tubes is as much as 26,000 volts. While the pigtail which connects this voltage to the button on the side of the picture tube is insulated with an insulating suction cup protector and the pigtail itself has an insulation value of well over 30 KV, it is possible to work your way into the circuitry. Today's plasma screen typically use 300-400 volts and LCDs use 1,500 volts, which you need to be aware of and cautious around.

Chapter 2

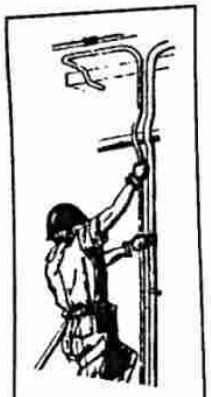
To check the high voltage levels you use a high-voltage probe which contains a small voltmeter and a large resistance, thus making it a safe way to check the presence and value of the second anode supply to the picture tube. It can become a shock hazard if you accidentally remove the ground wire and clip from the chassis, but even then, because of the large value resistance in the probe, the current is less than that which will burn your skin or produce trauma.

High Voltage Probe

The Electric Chair

How much voltage does it take to kill you? You might consider that the famous electric chairs in prisons use only 2,000 volts. Compare that with the 30 KV on the picture tube. As little as 10 milliamps of current can be felt, with 100 mA being lethal.

Wet skin can lower your normal skin resistance of about 100k ohms so that small currents can be felt and may be dangerous.



Lineman

Other common items which can zap you are stun guns and even bug zappers. But far more people are injured by power company service than these. Working with cable, you may find you have occasion to work on power poles. You might find in your career that you deal with antennas, towers and satellites, on roofs or in locations where you are in the vicinity of power lines. The insulated power service cables which deliver 230 volts to your electric meter base are not much of a hazard. The insulation on them protects against accidental touching by hand or with antenna elements, poles, etc.. Still, they are to be respected and not touched.

It is the bare primary power wires which most often kill. While they are usually high enough that a person really has to try to touch them, there are accidents which happen. You could be enjoying cutting a field with your tractor when the tractor tire catches the guy wire for the pole, breaking the pole and bringing the bare wires down on you. Or you could be working on an antenna tower or tall pole which breaks and falls across the power primary wires.

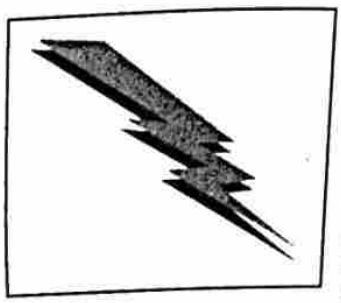
One day, on a street in Greencastle, during construction of a small bridge, one of the tri-axle dump trucks was spreading gravel, with the dump bed in the up position. It caught some drooping cable TV hard line which angled across the street near where the work was taking place. The cable wires stretched, then broke, then snapped back and coiled around the power primary wires. While no one got killed, many of the houses in the area had 4500 volts or more attached to their cable boxes, TV's, anything connected to them. No personal injuries occurred, but thousands of dollars in damage was done in the neighborhood. Anyone holding onto the wrong thing at that time could have been electrocuted as could workers who might have been in the way of the snapped off aluminum cables.

$$

Primary to cable accident was

Hot Chassis

Today's products often have only a small amount of metal their mounting chassis. Thus, even if the chassis is connected to the AC power ground and the house wiring to the AC outlet socket is reversed, there may be no observed hazard. TV sets and other electronic products are now made that any exposed metal, even the signal input "F" connector case, is isolated from the chassis circuitry. The technician will need to expose the chassis in order to service the unit, in which case a "hot chassis" may become a hazard.



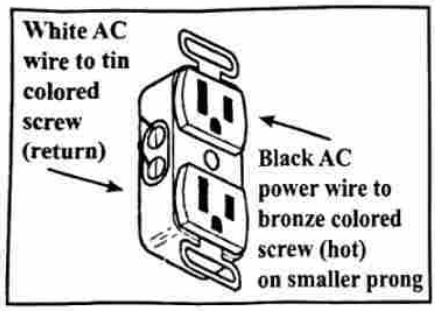
"Hot chassis" is a name given to equipment which is not isolated from the AC power source by a transformer. Instead, to reduce costs, the AC power hot side of the AC line is connected directly to the power supply diodes. The low side of the line is connected to the chassis. Under normal circumstances there is no hazard in servicing the unit. For instance, you could be holding the chassis with one hand and be touching the workbench ground with the other and no harm occurs.

What happens to make the "hot chassis" a hazard is when the AC power outlet is wired backwards (The AC power receptacle has the black to ground and the white wire to the hot side of the outlet, or the small slot), then the chassis is connected to the hot side of the AC power. Now if you touch the bench ground and the chassis at the same time...zappo! You don't have to touch the bench ground to accomplish this shock. You can merely touch another piece of equipment, plugged into a different AC outlet. Its ground may be correct. So your reversed AC outlet can still zap you.

One way to solve this problem at the shop is to always try to use an isolation transformer to operate your bench equipment.

Another cause of "hot chassis" type shocks is that lightning may travel down the power line into the house and arc across the 'hash filter' capacitors which are connected between each side of the AC power input to the product and the chassis. With these capacitors shorted, even a transformer type product becomes a possible shock hazard.

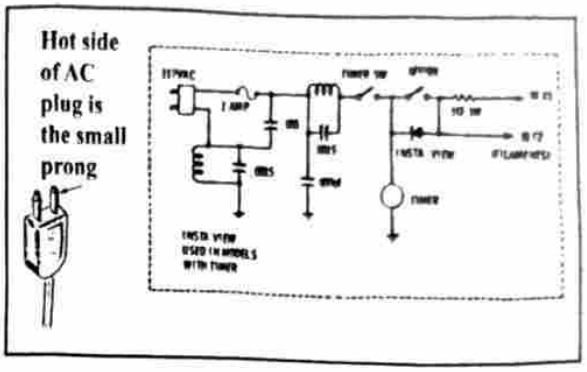
Most often you will notice the problem before a shock occurs. You will notice when you connect the cable TV coaxial cable to a DVD player or TV, that a small arc occurs. Static? Possibly, but probably not. When this occurs, you may even notice a tingle of a shock. It is a red flag, and should be checked out. The causes for the wrong voltage on the AC line are many. It isn't always easy even to check to see that they are wrong. You can usually use the AC power receptacle 3rd wire as a ground for your AC meter. You might use the cable TV or antenna coax shield as a ground. But you need to reference your voltage readings to ground. I once tracked such a problem back to a reverse wiring of an AC power outlet, right by the sink in the bathroom!



Check for proper wiring at the AC outlet

Most hot chassis problems go unnoticed as the product is in a house, isolated from good grounds. But when the TV or DVD player, etc. is now moved to the damp basement, it can become a lethal problem.

Technicians should always seek the cause of a wiring problem, even if this means warning the homeowner of the problem and suggesting an electrician be called to inspect the wiring. More often the problem is another piece of equipment connected on the same circuit. One that comes to mind is the case of the antenna amplifier that burned out frequently. The customer, of course, thought we were just installing weak or defective products. But after a while we decided to check the trailer home wiring, especially since our techs seemed to get a 'tingle' sometimes when dealing with the pre-amp or antenna coax cable. It turned out the skin of the trailer was 'hot' at times. These 'times' happened to be when the home heating system was in a particular mode. Which furnace wire was backwards, we didn't find out, but we did find out it was the furnace that was the cause of the hazard. It wasn't the homeowner who discovered the problem. It is most often the technician who recognizes something is wrong and who should advise the owner that there is a problem that could show up in other ways.

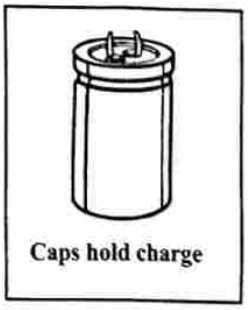


Product makers specify that technicians should check the power input cord for resistance every time the unit is serviced. Any leakage current above 500 microamps is too much. Search out the problem.

Capacitors

Large capacitors (electrically) are a lot like lead-acid storage batteries. If you short the terminals of a battery, you can do a nice welding job on your screwdriver. If you short the terminals of a large capacitor you can do a small welding job, or a large human shock job.

The battery, at least, does have an internal resistance which can limit the short-circuit current somewhat.

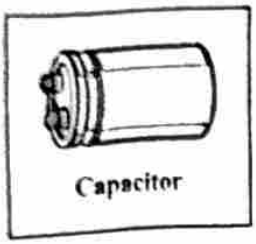


Caps hold charge

Capacitors have very little internal resistance when shorted and can supply all of their stored energy between your right hand and your left hand, right across your heart! Capacitors can be made to have virtually no leakage and thus can even be used in place of batteries in some types of equipment. Some are expected to retain a charge for 'years'! With that in mind, the technician must respect power supply and other filter capacitors. The voltage retained in a capacitor has been called residual voltage.

You assume there is no hazard, having unplugged the unit before attempting to service it, yet now, you get a severe shock after accidentally touching the capacitor terminals. It becomes second nature to technicians to discharge capacitors before using an ohm meter in the circuits or working with circuit components.

Arcing


Capacitor

Touching a wire or part which has a hazardous voltage potential on it is one thing to avoid. But higher voltages can jump and shock you the same as if you physically touched the metal part. Most common are the high voltages in TV and monitor displays. But RADAR and many other products operate with more than 1,000 volts. Inadvertently passing your hand or a meter lead by a high voltage potential can draw an arc. How far can voltage jump? Only three kilovolts can jump 1/4th inch. Could the 30,000 volts on a color picture tube jump over two inches? Think about it.

Water

We mentioned the isolation transformer above as being a safety device. It allows your work bench or equipment to be grounded while putting a barrier between your ground and a 'hot' ground some other piece of equipment may have in your shop or home. If someone defeats a polarized extension cord, grinds off the ground prong of an adapter, or the basement wiring is backwards, the isolation transformer can partially remove the hazard.

That is almost never the solution. The solution is to find the problem. But people are more likely to defeat polarized adapters and extension cords in garages and basements than in the safer parts of the home. Basements are notorious for being damp or having pools of water on the floor. AC power hazards are thus waiting to happen in these areas. Technicians can run across further potentials when they work in wet crawl spaces or find themselves doing wiring in monsoon type conditions outside a building.

Frayed cords should be replaced. Don't trust the insulation. Always try to keep AC potentials covered. If they must be left uncovered temporarily, leave a danger tag attached.

Modern Equipment

Computers are much safer to service than TV's. They operate the memory and program chips and transistors from 5 and 12 volt DC sources. The only danger, other than the monitor, is the power supply section where only the AC power plug, fuses, power transformer, filter capacitors and bridge rectifier are waiting to zap you. These are usually covered with an insulating box or metal cage.

Most electronic equipment is protected from internal shorts with one or more fuses. Some equipment, such as satellite receivers, has circuit breakers rather than fuses. TV sets and some other equipment have interlocks which disconnect the power if the cover or back is removed. Defeating is usually necessary in order to troubleshoot the feat them with care and do not leave them expos

One Hand

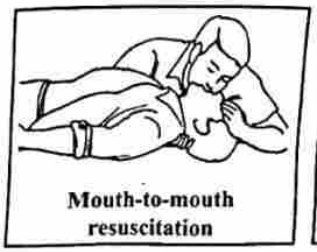
A rule followed for nearly a hundred years is for the technician to work with one hand where possible. Keep the other hand in your pocket when taking a voltage reading on powered equipment. Don't hold on to the chassis rail and make voltage even hold onto a grounded workbench or rest your cabinet of the test equipment.

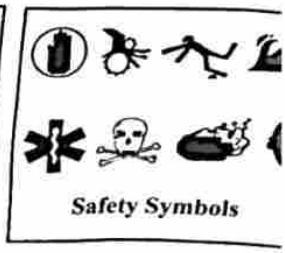
Second rule is use the buddy system. Don't work tronics products without having someone else near you do get shocked.

Shock Victims

When a person gets shocked badly, they pass out o Some shock victims drop onto the hot voltage source. is to remove the power and attempt to save that perso

There is a debate as to whether you should apply res tion procedures after removing the electrical hazard. If not trained in CPR, don't start learning now. If you are now is the time to do it. Mouth-to-mouth resuscitation i best method of getting a person breathing again.


Mouth-to-mouth resuscitation


Safety Symbols

Of course, attempting to get medical emergency help first is common sense. M-M resuscitation requires about 12-18 breaths per minute after you have checked to see that the throat is clear and have held the nostrils closed.

In addition to breathing problems, a shock victim may have severe burns. Treat these burns the same as other types of skin burns. Fortunately, most lesser electrical shocks cause the skin to burn and the charred burn contact point becomes an insulator of sorts, thus limiting the electron flow through your body.

Fire Hazards

The first rule about fire hazards is to always attempt to prevent them. Why use a grinder to sharpen your screwdriver, letting the sparks shower down onto your open gas can? Many chemicals can catch fire with a spark or from your hot soldering iron. Keep your work area neat, don't smoke, and be on the constant lookout for potential fire hazards. Keep fire extinguishers handy and checked for operation and let everyone know how to make them work. Powdered chemicals such as CO2 are used on electrical fires, rather than liquids. But whatever type of extinguisher you use, realize that the chemicals may put out the fire, but you should clear the area and ventilate it afterwards to protect your lungs.

When soldering, use a proper soldering stand. Don't breathe solder fumes - use ventilation. Beware of paper and cloth as well as chemicals which might start a fire when accidentally touched by the iron. Don't leave soldering equipment on after work is over.

Product makers have recognized that some parts, especially in TV receivers, are critical so far as their composition and current carrying capabilities. They can cause fire or other problems if operated wrongly or if replaced with non-exact parts. These are usually shaded on the schematic diagram. The rule is to always replace these parts with exact replacements.

Burns

Of course, you can get a burn by touching the muffler on a lawn mower, or on your truck when you are trying to get something that may have rolled under the vehicle. But technicians also get burned from soldering irons. They may find themselves working with heat shrink which covers cabling hardware and splices. Using a propane torch is a potential fire and burn hazard. The soldering iron may have a 600 degree temperature.

Burns of the first degree, those minor burns which redden the skin and hurt, but don't bubble up with blisters, usually require no treatment. Running cold water on the burn for a few minutes will relieve the pain. Then dry them off and protect them but don't apply a bandage.

Second-degree burns should not have water put on them or ice. Also don't break the blisters, if possible.

Third-degree burns are cases for the doctor. No water or ice should be applied. Only a thick clean bandage should be put on the area prior to getting to a doctor or hospital.

Chemicals

We have already discussed the fire safety rules when working with chemicals. There are other hazards:

Breathing carbon tetrachloride fumes is something that progressively damages your lungs. Before about 1980 it was commonly used as a solvent on technicians' workbenches. Don't use it. We all use tantalum capacitors. Don't play with them. Also the beryllium oxide transistor cases are hazardous to your lungs. Use ventilation and don't expose yourself to the chemicals or fumes.

Working with shop tools, technicians may use the bench grinder or power hand tools. Safety glasses are required. If you have never tried to polish or grind an odd shaped piece of metal and had it fly across the room like a bullet, you may not recognize the power of these tools used carelessly, or accidentally. Don't take a shortcut by doing a small job without glasses. An eye is hard to replace.

Your skin, your eyes and lungs, and those of your coworkers are dependent on your work safety. Obey the rules your company has set up and practice safety as an example to others around you.

Body Injuries

Glasses for eye protection; ear plugs for hearing protection; taking special care when cutting or working around glass, wood and abrasives; taking extreme care when handling kinescope tubes - these are common sense practices that will help you and your company avoid unnecessary and lost-time accidents. Remember that people don't plan most accidents. They are just that: accidents. Recognize the potential for them.

X-Rays and EMR

Electromagnetic radiation is a subject about which very little is known. The genetic damage caused by small or large doses of radiation is not fully understood. What is known is that a microwave oven does work. X-rays do cause burns and possibly initiate cancer and other ailments. No one knows if cell phones, operated next to your brain, cause brain damage, but there are a lot of weird people walking around these days!

Even low-powered satellite Internet transceivers are a concern for workers or users. Subjecting yourself to a direct beam of radiation is a potential hazard. Like smoking, everyone doesn't die of radiation sickness. But the overwhelming evidence is that it increases your susceptibility to life threatening sicknesses. That is why your x-ray technician always steps out of the room you are in and puts on a leaded apron before zapping you!

Even the old TV sets (tube types) were x ray hazards if technicians tried to operate them beyond their designed limits (or if parts started to fail causing abnormal current flow while using up to 30 KV potentials). Technicians must know the proper operation of the equipment they are working on and recognize that re-engineering the product can produce radiation or fire hazards.

NEC[e]

The National Electrical Code is a bible for electricians and everyone else installing, designing, modifying and working with electricity. The NEC rules have been designed to prevent electricity from harming humans and starting fires. The rules are modified every three years to bring about the latest changes needed to protect the public and workers.

All of the NEC rules need not be studied or memorized by technicians. Those rules that apply to your area of work should be studied and abided by. For instance, if you install fiber optics, you should know the TIA and EIA communications wiring standards, but you should also know the electrical code for wiring buildings. You might say: "But I only want to run a single fiber cable through the attic and down a wall, so why should I be concerned with NEC rules?" As an example, if you installed a cable which when burned releases toxic chemicals, that would be the reason to know about plenum cable NEC rules. There are dozens of NEC rules related to cabling and they must be followed. Technicians are in the installation, maintenance and repair business and thus in a position to either violate the rules, or conform to them.

OSHA - Occupational Safety and Health Agency of the federal government also has established rules for workplace safety. It is each technician's duty to follow these rules. For instance, technicians often climb poles, towers and buildings. OSHA requires that workers who might fall as much as six feet must have body harnesses which protect them should they fall. Violating the rules may save a moment's time, but the penalties for ignoring them can be severe in two ways. #1: You have a better chance of hurting yourself or a co-worker if you ignore them. #2: OSHA penalties are severe in monetary costs o your company.

NEMA, the National Electrical Mfrs. Assn., also publishes ules for installation and safe operations of electrical equipent. Depending on your job, you may need to become famil- r with NEMA rules also.

gnage

hat word may not be in the dictionary. It has come to mean signs that are posted to communicate to people. One need ommunicate is when there is a human hazard potential, and want to warn people of it to prevent an accident.

'Open Manhole' might be a good sign to keep people from falling in.

'Hard Hat Area' reminds people that the area may have falling hammers or debris that could kill them.

'Radiation Hazard' or 'Ear Protection Required' might be other valuable signs to post, temporarily or permanently

Electronics technicians need to post signs if they have exposed electric hazards to the public or co-workers. If you are working on an industrial motor and must have the electricity off while doing this, you will need a sign on the on-off switch or breaker to warn others not to turn the power on while you are working. Someone might do this thinking the power was turned off by mistake.

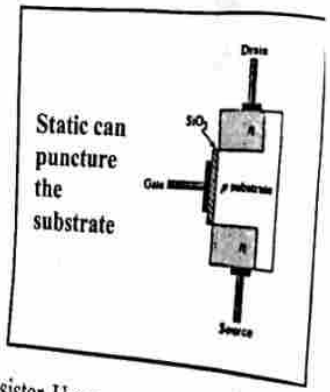
One of the most common warning devices is the street con used to advise traffic that work is going on near or on the roa and thus the drivers need to be cautious and on the lookout f the workers.

In addition to signs, there is a system of lockouts. These ar tags or actual padlocks and other devices which are placed o switches and equipment covers to prevent inadvertent energi ing of a circuit while work is going on.

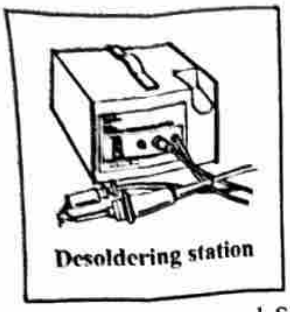
ESD - Static and Component Damage

ESD is known as Electro Static Discharge and does damage to small sensitive components. The technician is the person with the best opportunity to destroy circuit components via th static route or in other ways as he goes about his troubleshooting and repair tasks.

Today's electronics technicians are aware of static discharge problems with CMOS technology. The problem is the thin substrate which makes these transistors so sensitive and valuable. But a very small static charge can blow a hole in the substrate, destroying the component. The answer: discharge your body before and while working on or near computer type circuitry. While working, wear a wrist strap that is grounded through a 1 meg resistor. Use a composition work mat which drains off any charges that might accumulate on the bench.



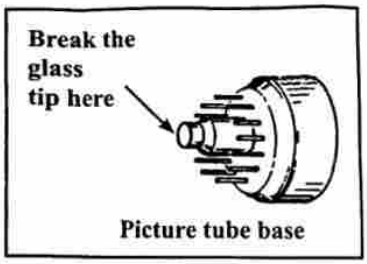
Soldering - Unsoldering


Desoldering station

You can do a lot of damage attempting to remove a transistor or IC chip. It takes a steady hand and knowledge of good soldering practices to do the job right. Even then, repeated solder-unsolder work will steadily deteriorate the PC boards. Troubleshooting with the idea that you must narrow the possible bad part down to only one has to be your creed. Simply replacing parts until you find the bad one is not the way to do it.

When you do remove a bad part, it is sometimes best to clip the legs and remove the part before attempting to unsolder the remaining legs and then solder in a new part. Too much heat can harm adjacent transistors or other components. Use heat shields (heat sinks) on the legs of pn junctions and use the least heat possible from the soldering iron.

While troubleshooting today's electronic circuits, it is increasingly difficult to measure voltages and signals without slipping off the lead you have aimed for and shorting between two of the legs. This often immediately destroys the transistor. Thus, taking your time and using thin test probes and taking every precaution is required.



Picture tube base

Legacy equipment such as kinescopes (picture tubes and vacuum display devices) can be ruined also. The base and pins are smaller and lighter than in the early years of TV. They can be broken off. The high voltage anode connection in some tubes is coaxial and also contains a connection for the focus element. A special tool is required to remove the HV pigtail. Without the special tool, you can damage the connection, destroying the expensive tube. Being an unwieldy, heavy and bulky part anyway, removing, storing, shipping and handling are hazardous undertakings. If the tube is removed and headed for the trash, most technicians remove the implosion hazard by breaking the neck of the kinescope. They protect those nearby by putting the tube in a corrugated box to accomplish this procedure.

Tools & Test Equipment

Digital multimeter

A dull screwdriver that falls off the slot most of the times you try to use it wastes time and does bad work. A DMM (digital multimeter) that needs new batteries and has an intermittent test lead may be worse than no meter at all. The best investment a shop-owner can make is taking the time to make your work bench as efficient as possible and to assure that you have all of the needed test equipment and tools.

Weller air filter

In checking hard-to-reach components, perhaps near dangerous exposed voltage sources, any meter will work, but a meter that includes a hold feature can allow you to concentrate on getting the leads on the right spot and not between two transistor legs or between the fuse and ground. An isolation transformer and a variable AC power source are both safer and very useful products to use daily on your bench. I have never seen a shop with too much test equipment. Some of the equipment will not accept unlimited voltages on the signal inputs. A good technician becomes familiar with the limits and specifications of the equipment he uses and thus avoids 'frying' his oscilloscope or signal generating equipment.

Clothing

What has your attire to do with safety? It depends on the type of electronics work you do. In all cases, however, a technician is at risk if he has dangling jewelry, trailing clothing, exposed skin and any of the varieties of skin piercings.

Some jobs require steel toe boots. In some you need climbing boots, even gaffs if you climb power poles. Many tech jobs require hard hats some or all of the time. Goggles are necessary in every shop on occasion. Handling poles and working with heavy construction equipment requires full time gloves as do many cabling and communications jobs.

Some jobs, for instance, working in auto-sound installation and repair, require the removal of all finger rings. If you have never gotten a ring between twelve volts and ground, you can't imagine the excitement it produces!

Long locks may be in style, but in work where moving equipment, fans, shock hazards and plenty of wires are involved, short hair wins every time. Loose sleeves look nice.

They also get caught on everything and limit your movements. You NEVER see the shuttle astronauts with loose clothing. It is the same for cool, about to fall off, holey, sloppy trousers. Don't wear them to work in a technician job. When you work with chemicals, use a breathing mask or get good ventilation. If you work with heavy objects, wear foot protection.

First Aid

As stated in this chapter, there is concern when you attempt to help a person who is an accident victim. I wouldn't advise you to try to administer CPR if you aren't trained and experienced at it. You can do more damage if you do it wrong. That's probably why mouth-to-mouth is today's favorite resuscitation method, but CPR may be the right thing to do. If you aren't trained at it, it is better to seek professional help even if you might dearly love to help the person.

First Aid kits are required in all work trucks and in all work places. Most aren't elaborate. They have some band aids and alcohol or hydrogen peroxide; perhaps some small bandages and tweezers and scissors. They probably have some aspirins. Know where the kit is and help keep it stocked to the level required at your workplace. If you work outdoors, on or under buildings or in trenches or in the suburban areas you should carry hornet spray. You may find dog repellants are necessary if you do cable hookups or home repairs and are frequently in unfamiliar neighborhoods.

I haven't mentioned truck safety before, but it is one place that has more than its share of accidents. Of course, prudent driving habits are required and technicians have a pretty good record as a group. Problems arise, however, when the truck is not regularly serviced and racks aren't made secure. Trucks carrying poles and other equipment that can become flying objects in the event of a crash must have fences behind the driver and passenger seats. Ladder racks are often added as an afterthought, which is better than trying to carry ladders around loose, but racks can be hard to either remove or replace the ladders on. The ladders also can be secured with bungee straps which can break and leave your ladder flying down the road at someone. Trying to get by on worn out tires and never giving the vehicles a close inspection for bad tie rods, loose wheel nuts, broken shocks, dangerous metal fender or door parts hanging out waiting to tear your flesh are all common sense things that need to be concerns. Always carry a fire extinguisher in the van. Don't use the two-way or cell phone while negotiating city streets or crowded roadways. Pull off.

Each individual job a technician performs will have additional safety rules and needs. Your job is to understand all of the common sense items described in this chapter, then find out all about the special requirements of your special job.

NOTES

Chapter Quiz

1. **B+, or supply voltages in electronics products, are considered to be:**

   a. +5 to +12 V DC.
   b. 0 to 30 V DC.
   c. +12 to 300 V DC.
   d. the anode or plate voltage supply.

2. **Leakage current in an electronic product, between either of the AC power cord conductors and the chassis or any exposed metal parts should not exceed _____.**

   a. 5 microamps
   b. 500 microamps
   c. 5 milliamps
   d. 50 milliamps

3. **Shaded components on electronics schematics mean:**

   a. only factory service personnel may change the part.
   b. only exact replacement parts may be used.
   c. the shading makes for better readability.
   d. high voltage is present on these parts.

4. **Ingested solder fumes or particles _____.**

   a. are not harmful to the human body
   b. are toxic
   c. from lead-tin solder aren't harmful
   d. from tin-alloys are most harmful

5. **Second degree skin burns should_____.**

   a. not be treated with ice and bandage pack
   b. be covered with thick bandages
   c. be treated with cold water or ice
   d. be treated by draining blisters

6. **Rules describing the fire or toxicity retardant types of cabling which may be installed in buildings is contained in _____.**

   a. NEC Article 770
   b. UL 568B
   c. TIA/EIA standard 578
   d. NEMA code 1316

7. **ESD and static discharge is most destructive to_____.**

   a. CMOS semiconductors
   b. TTL semiconductors
   c. Germanium PN junctions
   d. Silicon PN junctions

8. **If you have never received formal training and practiced CPR on a person or dummy, in the event of finding a fellow worker not breathing, you _____.**

   a. should use what knowledge you do have of the procedure and attempt it
   b. should not attempt it, but call for help
   c. should attempt it after calling for help
   d. should, if handy, read the instructions and attempt it in order to save the life

9. **When working at heights above 6 feet, a _____ is required by OSHA rules.**

   a. lineman's belt
   b. lineman's belt attached to a lanyard
   c. full body harness attached to a lanyard which prevents more than a 6' drop
   d. good insurance policy

10. **Residual voltage is associated with_____.**

    a. Northern Lights
    b. corona arcing from electrical generators
    c. charge retained by inductors
    d. charge retained by capacitors

ANSWERS on page 223

Chapter 3

by Fitz Herbert Reid, CET

Fitz Herbert Reid, CET, is a former Director of Technical/Vocational Education in his homeland of Trinidad and Tobago. He has 42 years in education at Primary and Tertiary and Administrative Levels. Fitz is the holder of BSc. (Electronics Technology) an special training in Testing and Evaluation. He has written three booklets for the Caribbean Examinations Council, where he se Examiner. These texts are Supplemental Study Guides for the Associate CET examination.

Powers of Ten

The powers of TEN are used extensively by engineers and technicians. Powers of Ten allow very large and very small numbers to be shown in a compact form. The name of the commonly used prefix is also listed in the table.

1,000,000,000	=	10^9	Giga
1,000,000	=	10^6	Mega
1000	=	10^3	kilo
1	=	10^0	Base Unit
.001	=	10^{-3}	milli
.000001	=	10^{-6}	micro
.000000001	=	10^{-9}	nano
.000000000001	=	10^{-12}	pico

The number of zeros determines the index or power to which 10 is raised. By shifting the decimal point to the right and left the following equations are true:

$$125{,}000 = 1.25 \times 10^5$$

$$.125 = 1.25 \times 10^{-1}$$

$$.07 = 7 \times 10^{-2}$$

$$.007 = 7 \times 10^{-3}$$

Move the decimal point to the right for large numbers and to the left for small numbers.

How do we relate the Power of Ten to the units used in electronics?

Refer to Table 1:

MULTIPLY ↓		
	Tera	10^{12}
	Giga	10^9
	Mega	10^6
	Kilo	10^3
	Hertz, Volts, etc.	
	Milli	10^{-3}
	Micro	10^{-6}
	Nano	10^{-9}
	Pico	10^{-12}

Table 1 - Units & Powers of 1

Example: Convert 1000pF to μF

a. 1 μF
b. 0.001 μF
c. 0.01 μF
d. 0.1 μF

SOLUTION:

We are moving "down" so we divide by 106 decimal point to the left (6 places).

$$1000\text{pF} = 0.001000\ \mu\text{F}$$
$$= 0.001\ \mu\text{F}$$

Example: Convert 700 KHz to Hertz (Hz)

a. 700
b. 70
c. 0.7
d. 700,000

SOLUTION:

The arrows on the Table tell us whether we should (shift the decimal point to the right) or divide (shift mal point to the left).

Converting from kiloHertz to Hertz, we must multiply by 1000 or shift the decimal point three places to the right.

Therefore: 700 KHz = 700,000 Hz

Choice (d)

Example: Convert 700 KHz to MegaHertz (MHz)

a. 700
b. 70
c. 0.7
d. 700,000

SOLUTION:

The arrows on the table tell us whether we should multiply (shift the decimal point to the right) or divide (shift the decimal point to the left).

Converting from kilohertz to MegaHertz, we must divide by 1000 or shift the decimal point three places to the left.

Therefore: 700 KHz = .7 MHz

Choice (c)

Example: Convert 2000pF to µF

a. 2 µF
b. 0.002 µF
c. 0.02 µF
d. 0.2 µF

SOLUTION:

We are moving down so we divide by 106 or shift the decimal point to the left (6 places).

2000pF = 0.002000 µF
= 0.002 µF

Choice (b)

Equations

An Equation is a mathematical statement. It tells us that two quantities or numbers are equal. By using letters to represent numerical quantities, any number can be used to form an equality.

Let a = 6, b = 3 and c = 2

If a = b x c 6 = 3 x 2

On rearranging the equation to maintain equality:

b = a / c, then 3 = 6 / 2 which also true

Or c = a / b, then 2 = 6 / 3
which also forms an equality

In the equation 4y = 2, "y" is termed an unknown quantity because we do not know the value of "y" However, we can solve for "y" thus

4y = 2; y = 2/4; y = 0.5

Ohm's Law

In an alternating current (AC) or a direct current (DC) circuit the relationship between voltage, resistance and current is called Ohm's Law.

This relationship is expressed mathematically as:

V = I x R where V is the applied voltage
I is the current
R is the resistance

I = V / R and **R = V / I**

With any two of the quantities of a circuit known, the third unknown quantity can be found.

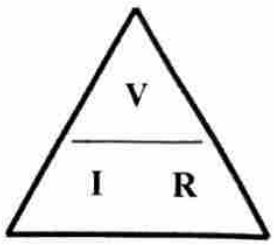


Fig. 1

(It's common for E (electromotive force) to be used instead of V in electronics math)

A triangle (Fig. 1) or circle is sometimes used to show the relationships between V, I and R. V is always on top or to the right in the Ohm's Law equations. Covering up the unknown with your finger in the triangle will help you remember the arrangement of the form of the equation to use.

To find I, cover up I and you see V over R
To find R, cover up R and you see V over I
To find V, cover up V and you see I times R

Watt's Law

Power is measured in units of Watts. The Power consumed in the circuit is calculated this way:

Power (P) = Current (I) x Voltage (V)

$P = I \times V$

rearrange to find $V = P/I$ and $I = P/V$

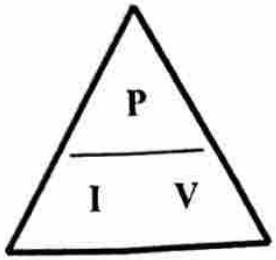


Fig. 2

As with Ohm's Law you can also use a triangle to help you find the unknown:

To find I, cover up I and you see P over V.
To find V, cover up V and you see P over I
To find P, cover up P and you see I times V.

With substitution from quantities in Ohm's law, the others can be found:

$P = I^2R$ or $P = V^2/R$

These are handy to find power when current and resistance are known or with voltage and resistance known:

R = square root of P / I and V = sq rt of P x R

With these combinations, 12 formulas are available for problem solving with Ohm's and Watt's Laws combined.

A graphic illustration in Figure 3 is provided in the shape of a wheel to help remember and use each.

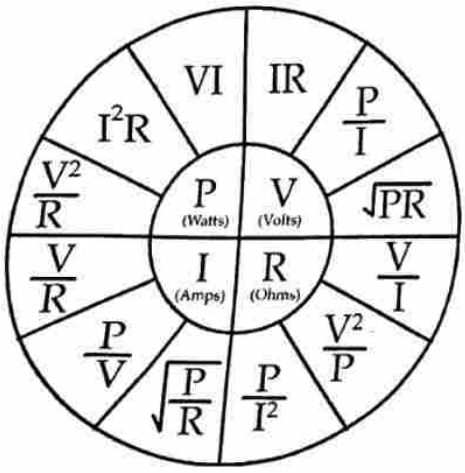


Fig. 3 Formula combinations (12) of Ohm's & Watt's laws

Internal Resistance

All electrical devices which produce electric energy are affected by their own internal resistance. These devices include generators, batteries, etc.

A battery is two or more cells connected in series, although a cell is sometimes referred to as a battery. Let us take a look at the inside of a battery.

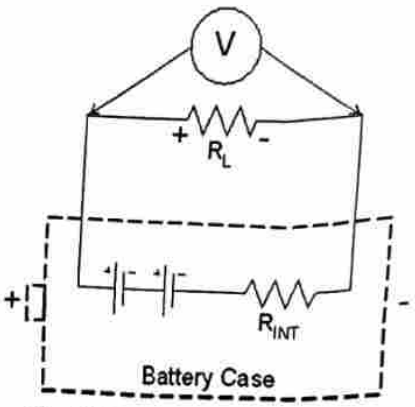


Fig. 4 Battery connected to a load

The total voltage developed by the chemical action is called the electromotive force (EMF). This EMF however, is not available at the terminals because a part of this voltage "sits" across the internal resistance whenever a current flows in the circuit. The size of this voltage drop is determined by the value of the current drawn by the load.

$$\text{Terminal voltage} = \text{EMF} - VR_{int}$$

Example: A cell with an EMF of 1.8 V delivers a current of 2.5A through an external resistance of 0.6Ω. Calculate the internal resistance.

a. 0.12Ω
b. 1.2Ω
c. 12Ω
d. 0.125Ω

SOLUTION:

$$V_{int} = EMF - V_{terminal}$$
$$I(R_{int}) = EMF - I(R_{ext})$$
$$2.5(R_{int}) = 1.8V - 2.5(0.6)$$
$$R_{int} = \frac{1.8 - 1.5}{2.5}$$
$$R_{int} = \frac{0.3V}{2.5A}$$
$$R_{int} = 0.12\Omega$$

Example: The EMF of a cell is 1.5V and its internal resistance is 0.25Ω. Calculate the current when the external resistance is 6Ω.

a. 24mA
b. 0.24mA
c. 2.4mA
d. 240mA

SOLUTION:

$$I(R_{ext} + r_{int}) = EMF$$

$$I = \frac{EMF}{(R_{ext} + r_{int})}$$

$$I = \frac{1.5V}{6.25\Omega}$$

$$I = 240mA$$

Choice (d)

Resistors in Series and Parallel

Resistors in series and parallel will be covered in Chapter 4 and 5 in detail, but the formulas are listed here for reference.

Resistors in Series $R_T = R_1 + R_2 + R_3 + ...R_n$

Resistors in Parallel

$$R_T = \frac{R}{n} \text{ or } R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ... \frac{1}{R_n}}$$

n = number of equal value resistors

With a calculator , the parallel formula can be entered:

$R_1 (1/x) + R_2 (1/x) + R_3 (1/x) + R_n (1/x) = (1/x)$

where (1/x) is the reciprocal key.

The Sine Wave

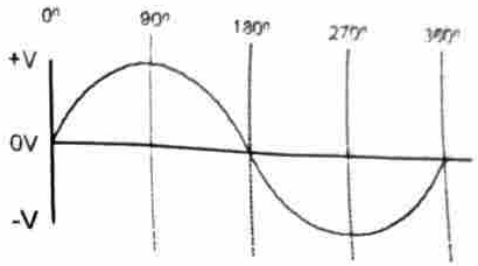


Fig. 5 AC Sine Wave

Fig. 5 represents one complete cycle of an AC sine wave which can represent voltage or current.

The following formulas are important in order to calculate various values pertaining to the AC sine wave:

Amplitude Conversion:

V Peak to Peak (Vp-p) = 2.828 x Vrms
V Peak (Vp) = 1.414 x Vrms
Root-mean-square (Vrms) = 0.707 Vpeak

Peak-to-peak = Peak value x 2
Peak = $\frac{\text{peak to peak}}{2}$

Time and Frequency Conversion:

Period (T) = 1 / F = seconds
Frequency (f) = 1 / T = Hertz (Hz)

Example: The RMS value of a 60Hz sine wave voltage is 120V. What is its peak-to-peak value?

a. 339.4 V
b. 42.4 V
c. 84.8 V
d. 169.7 V

SOLUTION:

$$\text{Peak value} = \frac{\text{RMS}}{0.707} = \frac{120}{.707} = 169.7 \text{ volts}$$

Peak-to-peak = Peak value x 2 = 339.4 V

Choice (a)

Example: A sine wave has a frequency of 60 Hz. Find the period, T:

a) 16.6 sec b) 16.6 ms
c) 016 ms d) .16 sec

SOLUTION:

Period T = 1 / F
= 1 / 60
= 16.6 ms

Choice (b)

Capacitance

A capacitor in an electrical circuit opposes a change of voltage. The basic unit of capacitance is the Farad, whereas the charge stored in a capacitor is measured in Coulombs.

$$Q = C \times V$$

where: Q is the charge
C is the value of the capacitor in Farads
V is the voltage in Volts

Example: A 12V DC supply is connected across a capacitor of 5μF. Calculate the charge:

a. 6 Coulombs
b. 0.6 Coulombs
c. 6 μCoulombs
d. 60 μCoulombs

SOLUTION:

Charge $Q = C \times V$
$= (5 \times 10^{-6}) \times 12$
$= 60 \times 10^{-6}$
= 60 μCoulombs

Choice (d)

The basic unit of energy is the Joule. The energy stored in the electric field between the plates of the capacitor is:

$$E = 0.5C \times V^2$$

Where: E is energy in Joules
C is the capacitance in Farads
V is the voltage in Volts

Example: Calculate the energy stored in the capa the previous example where V = 12 v and C = 5μF

a. 360 μjoules
b. 0.36 μjoules
c. 36 μjoules
d. 3.6 μjoules

SOLUTION:

Energy stored (E) = $0.5\ CE^2$
$= 0.5 \times (5 \times 10^{-6}) \times (12 \times 12)$ joule
$= (0.5 \times 5 \times 12 \times 12) \times 10^{-6}$ joules
$= 360 \times 10^{-6}$ joules
$= 36010^{-6}$ joules
= 360 μjoules

Choic

Capacitors in Series

Capacitors in series are like Resistors in Parallel. The capacitance can be found by the sum of the reciprocal v

$$C_T = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots \frac{1}{C_n}}$$

For 2 capacitors in series (a simpler formula):

$$C_T = \frac{C_1 \times C_2}{C_1 + C_2}$$

For equal value capacitors in series:

$C_T = C / n$ where n = number of capacitors

For capacitors in parallel:

$$C_{total} = C_1 + C_2 + C_3 + \ldots C_n$$

Example: If three equal value capacitors of 10μF each are connected in series, calculate total capacitance.

a. 33μF
b. 0.33μF
c. 330μF
d. 3.3μF

SOLUTION:

$C_T = C / n$
10 / 3 = 3.3 uF

Choic

Example: Calculate total capacitance of two 10 μF capacitors in parallel.

a. 20 μF
b. 10 μF
c. 5 μF
d. 0.5μF

SOLUTION:

$$C_T = C_1 + C_2 = 10 + 10 = 20\mu F$$

Choice (a)

Time Constants

A capacitor takes 5 time constants to be fully charged and one time constant equals the value of the resistance multiplied by the capacitance.

1 Time Constant (TC) in seconds = R x C

The charge takes place exponentially. The discharge action follows a reverse pattern (see Figure 6).

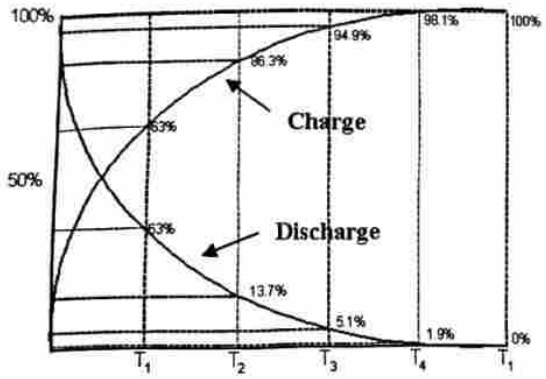


Fig. 6 Charge/Discharge Curve of Capacitor

Example: A series RC circuit is connected across a 100V DC supply. If C = 5μF and R = 10 kΩ, calculate one time constant.

a. 5 sec
b. 0.05 sec
c. 5 μsec
d. 0.05 μsec

SOLUTION:

One time constant = R x C
= 10^4 x 5 x 10^{-6}
= 0.05 sec

Choice (b)

Example: If 100 v is the applied voltage, after one time constant, what approximate voltage can be expected across the capacitor?

a. 0.63 V
b. 63 V
c. 6.3 V
d. 0.063 V

SOLUTION:

63.2% OF 100v = 63.2v

Choice (b)

Capacitive Reactance

Capacitive Reactance, XC, measures the opposition offered by a capacitor to the flow of an alternating current. XC is measured in Ohms and its formula is:

$$XC = \frac{1}{2\pi fC}$$

where: 2π is 6.28
f is the frequency in Hertz (Hz)
C is the capacitance (farads)

A memory aid that can help you remember that in a capacitive circuit current "I" leads the voltage "E" (E is voltage, not energy in this representation), hence the word "ICE".

Refer to Fig. 7 (A), circuit diagram; (B) waveform; and (C) vector diagram. In Fig. 7B, the peak voltage is represented by V, and not E. In Fig. 7C, the phasor pointing straight down is voltage (E) which is 90 degrees out of phase with the current (I), with its phasor pointing to the right.

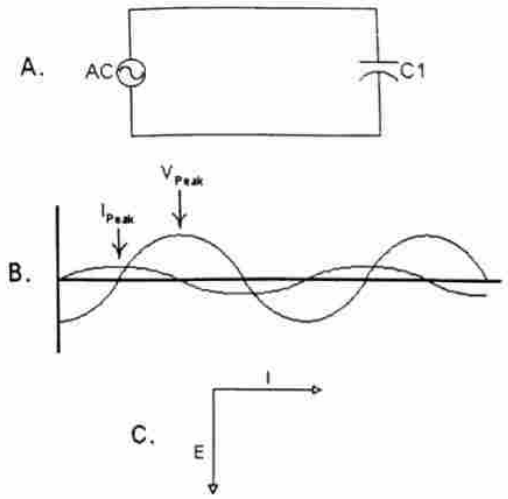


Fig. 7 Capacitive Reactance Phase Representations

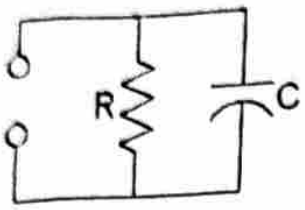


Fig. 8 RC circuit

Capacitors can be in series with resistance and also in parallel with resistance as shown in Fig. 8. When resistance is present along with capacitance, the phase angle between voltage and current in the circuit is less than 90 degrees, and can be found with vector or phasor representation. See chapter 6 for details.

Inductance

Inductance is that property of a coil or inductor which opposes a change of current in a circuit. Its unit is the henry (H).

Inductors in Series

Inductors in series are calculated like resistors in series, with mutual inductance added (if magnetic fields add) or subtracted (if the magnetic fields oppose) when the coils are close to each other.

$L_T = L_1 + L_2 + L_3 + \ldots L_n$ (+/- 2M for any mutual inductance, where 2M equals 2 x the Mutual Inductance)

Inductors in Parallel

Inductors are treated as resistors in parallel, with any mutual inductance either adding or subtracting from the total.

$$L_T = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \ldots + \frac{1}{L_n}}$$

Example: Three inductors, each 4mH, are connected in series. Calculate the total or effective inductance.

a. 12mH
b. .12mH
c. 1.2mH
d. .012mH

SOLUTION:

$L_T = L_1 + L_2 + L_3 = 4mH + 4mH + 4mH = 12mH$

Choice (a)

Example: Three inductors, each 4mH, are connected in parallel. Calculate the total inductance.

a. 13mH
b. 0.13mH
c. 0.013mH
d. 1.3mH

SOLUTION:

If equal in value, $L_T = L / n = 4 / 3 = 1.3mH$
Where n = number of equal value inductors

Ch

Inductive Reactance

Inductive Reactance, X_L, measures the opposition offe by an inductor to the flow of AC current. X_L is measured Ohms, and its formula is:

$X_L = 2 \pi fL.$

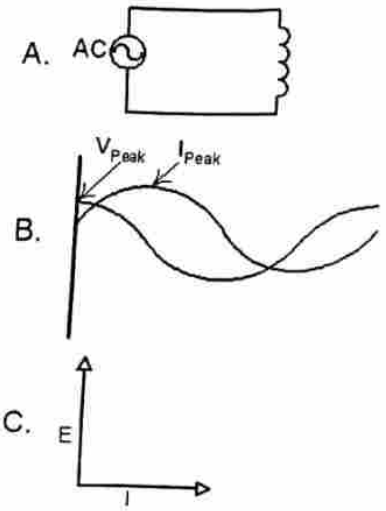


Fig. 9 Inductive Circuit Phase Representations

It is interesting to note that in an inductive circuit the vol age "e" leads the current "i," hence the word "ELI," where represents voltage and is shown as Vpeak on the graph. Ref to Fig. 9 (A), (B) and (C).

Additional references for inductive reactance can be foun in chapter 6.

Impedance

When both capacitors and inductors are present in a circuit in either series or parallel, along with resistance, special conditions take over and make the math calculations more difficult.

The total opposition to current flow in an RLC circuit is called **Impedance** and its formula symbol is Z. Z is made up of (R) resistance, (XC) capacitance reactance and (XL) inductance reactance values, which are inherently dependent on frequency of operation. The formula for impedance in a series RCL circuit is:

$$Z = \sqrt{(X_C - X_L)^2 + R^2}$$

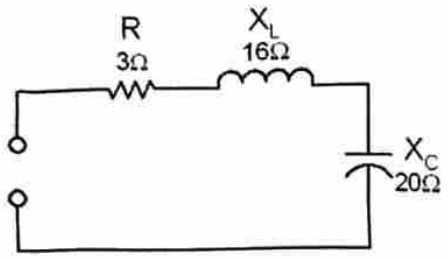


Fig. 11 Series RLC Circuit

Example: In the circuit in Fig. 11, calculate impedance, Z.

a) 39 Ω b) 5 Ω
c) 33 Ω d) 7 Ω

SOLUTION:

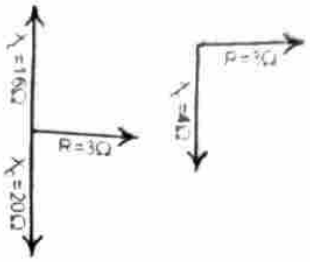
$$Z = \sqrt{(X_C - X_L)^2 + R^2}$$
$$Z = \sqrt{(20-16)^2 + 3^2}$$
$$Z = \sqrt{25}$$
$$Z = 5\Omega$$

Choice (b)

In examining the phase relationships between the voltage and current of the circuit in Fig. 11, it can be plotted on a graph to represent the 90 degree difference between resistance and the reactance. There is 180 phase difference between the values in the reactances of the capacitor and inductor, therefore they will subtract from each other.

Action through the resistance is in phase with the supply voltage, but there is "leading" and "lagging" with regards to the inductor and the capacitor. Because the Xc capacitive reactance dominates this circuit (it has the higher value), the circuit is said to be capacitive.

The circuit is more capacitive than inductive



(for R and X in parallel calculations, use the formula):

$$Z = \frac{RX}{\sqrt{R^2 + X^2}}$$

Resonance

Resonance is a special condition when the capacitive reactance and the inductive reactance are equal in a circuit (and exactly opposite in value, as noted on phasor representations). The circuit is said to be "Resonant."

$$X_C = X_L$$

Substituting the formula equations that contain the frequency and basic units of capacitance and inductance, we can find the resonant frequency, where this equality occurs:

Where fr is the resonant frequency and
L = units of inductance in henries of the coil
C = units of capacitance in Farads of the capacitor

$$2\pi fL = 1/(2\pi fC)$$
$$f_r = 1/(2\pi\sqrt{LC})$$

Also, the current in the capacitor and inductor are at maximum and the phase angle between the voltage and current is zero.

A circuit containing resistance and reactance is said to be in resonance when the supply voltage and the current are in phase. This makes the phase angle zero, and $X_L = X_C$.

Example: A 200μH coil whose resistance is 5Ω is connected in series with a 200pF capacitor. The resonant frequency of the combination is:

a. 795 KHz
b. 795 Hz
c. 7 MHz
d. 7.95 KHz

SOLUTION:

$$f_r = \frac{159}{\sqrt{(2 \times 10^{-4}) \times (2 \times 10^{-10})}}$$

$$f_r = \frac{159 \times 10^{-7}}{\sqrt{4 \times 10^{-14}}}$$

Resonant frequency = $$f_r = \frac{159 \times 10^{-7}}{2 \times 10^{-7}}$$

$f_r = 79.5 \times 10^{6}$

$f_r = 795$ kHz

Choice (a)

Units for Electronics Formulas:

Quantity	Formula Symbol	Unit	Unit Abbreviation
Voltage	V	Volt	V
Current	I	Amp	A
Resistance	R	Ohm	Ω
Power	P	Watt	W
Capacitance	C	Farad	F
Inductance	L	Henry	H
Reactance	X	Ohm	Ω
Impedance	Z	Ohm	Ω
Conductance	G	Siemens	S
Admittance	Y	Siemens	S
Susceptance	B	Siemens	S
Charge	Q	Coulomb	C
Energy	E	Joule	J
Time	t	second	s
quency	f	hertz	Hz

Greek Symbols

Many electronic abbreviations and formula symbols use Greek symbols for identification of units and values. The Greek name along with upper and lower case letters are summarized in the table.

Capital	Lower Case	Greek Name	English
A	α	Alpha	a
B	β	Beta	b
Γ	γ	Gamma	g
Δ	δ	Delta	d
E	ε	Epsilon	e
Z	ζ	Zeta	z
H	η	Eta	h
Θ	θ	Theta	th
I	ι	Iota	i
K	κ	Kappa	k
Λ	λ	Lambda	l
M	μ	Mu	m
N	ν	Nu	n
Ξ	ξ	Xi	x
O	o	Omicron	o
Π	π	Pi	p
P	ρ	Rho	r
Σ	σ	Sigma	s
T	τ	Tau	t
Y	υ	Upsilon	u
Φ	φ	Phi	ph
X	χ	Chi	ch
Ψ	ψ	Psi	ps
Ω	ω	Omega	o

Chapter 4

DC Circuits

by Rollin R. Okerberg, CETSr

Rollin Okerberg, CETSr, has been employed by the Department of Defense for 31 years. He started his career with the U.S. Air Force and received his technical training at Lowry AFB, Colorado. He also worked for the U.S Army at Sacramento Army Depot as an electronics technician, testing and testing test programs for circuit cards and line replaceable units. He is currently living in Syracuse, Utah with his wife and two daughters. He earned his Associate Degree at American River College, Sacramento, California. He possesses a General Radiotelephone Operator License with Radar Endorsement. He is currently an active member of ETA International and feels the Association has given him new and exciting opportunities for learning new ideas from fellow members.

Introduction

As with any subject of study, a student starts with a basic point of view to gain a basic understanding before pursuing more advanced topics. We will start with the atom. A technician needs to understand the structure of the atom to see how electrical circuits behave the way they do. Troubleshooting is a final part of this chapter.

The Atom

The hydrogen atom, the most basic element known in the universe, has one electron orbiting one proton. Its atomic number is one; this number tells us how many electrons and protons are in any particular element. They are the basic building block of all the known elements in the universe. Yes, of course, these particles can still be further broken down into more fundamental subparticles, but to do so is really beyond the scope of this chapter.

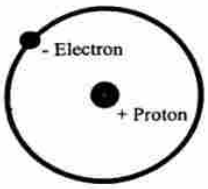


Fig. 1a Hydrogen Atom

Electrons and protons have equal but opposite charges. The electron has a negative charge, and we will see later on that it will play an important role in this chapter's discussion. The proton, of course, has a positive charge and is located in the nucleus of the atom.

Another part of the nucleus of the atom is called the neutron and, as the name implies, has a neutral charge. This particle also contributes to the atomic mass of an element. Having an equal number of electrons and protons, an element is electrically neutral. In a few words, some elements present themselves as good conductors while others are not conductors at all!

Atoms have up to 7 shells around the nucleus. These shells represent energy levels that electrons are allowed to occupy, and each one is designated with a different letter as shown in Figure *1b.*

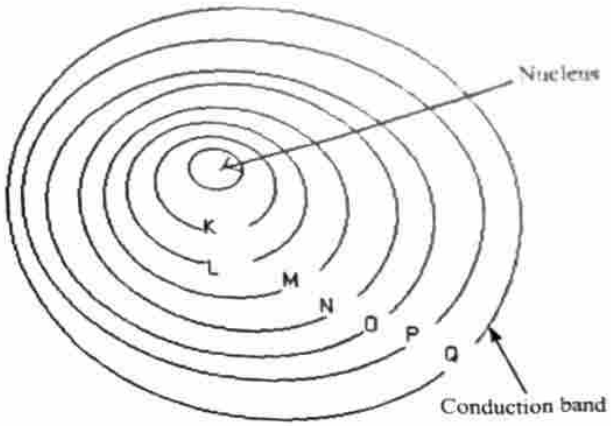


Fig. 1b

In electronics, the electrons we are concerned with are in the outermost shell; this is where conduction takes place when the electron is subjected to an external electrical source such as a DC power supply. In the conduction band, free electrons are allowed to move in a material that is subjected to a voltage source.

Three categories of materials in electronics work are:

1. Conductors
2. Semi-conductors
3. Insulators

If electrons move easily, the material is a good conductor. However, a material not having free electrons available for its conduction band is said to be an insulator. In an insulator, a large amount of energy is required to move electrons to this conduction band.

Examples of good conductors are: silver, gold, copper, aluminum, and iron. Examples of insulators are rubber, ceramics, glass, and wood. Some materials are also poor conductors, but not as poor as insulators. These are nichrome, carbon and silicon. Now that we have an understanding of conduction and the atom, let us go on to the next discussion.

Basic Units in Electronics

A table at the end of chapter 3 summarizes many of the commonly used electronic units and their assigned symbols and abbreviations. Formula symbols are used in math equations as unknown quantities. Abbreviations are used so that you don't have to write down the entire name of the unit when describing it.

The Coulomb is a unit of charge that is equal to 6.28 x 1018 electrons. The formula symbol for the unit of charge is Q.

The Ampere (I) is the unit of current flow in electronics. Current flow is measured in Amperes and is abbreviated A. More specifically, it is a rate of one Coulomb moving past a given point in 1 second as defined in:

I = Q (in Coulombs) / t (in seconds)

Voltage is defined as energy per unit of charge. The unit of voltage is the volt with a formula symbol of V, and the abbreviation is also V. This unit expresses the amount of force needed to move a given charge from one point in a circuit to another. The joule (E) is a unit of work or energy. The volt is a unit of electromotive force (EMF) that causes a current to flow in a circuit. Voltage is expressed as Joules per Coulomb in the formula:

V = E (in Joules) / C (in Coulombs)

It should be noted that in some circuits and references, "E" is used as the formula symbol for voltage. However, in this study guide, "V" will represent voltage, as it is in most new textbooks, so as not to confuse the student with the symbol "E" being taught in science classes as the symbol for energy. Energy efficiency is important today in product design and operation. In any circuit:

$$\% \text{ of Efficiency} = \frac{P_{out}}{P_{in}} \times 100$$

Power (P) is defined as an amount of work per unit of time. It is measured in Watts and is the amount of heat that is dissipated from a component such as a transistor or a resistor. Wasted energy ends up as heat and not as useful output.

P in - P out = P dissipated (heat)

The difference between applied power (input) and the consumed power (output) is the power dissipated as heat. Component sizes are rated with power ratings to express how much heat a device can dissipate. Technicians need to pay close attention when replacing components with an equivalent part, otherwise it may burn up if it has too small of a power dissipation.

Ohm's Law

Ohm's law is one of the most important tools for technicians and it is repeated several times in this book. You will use Ohm's *Law frequently.* Voltage is known as "potential difference" between two points. Voltage is also the product of current and resistance as shown in:

V = I x R

When resistance is held constant, as in the case of a fixed resistor, and current increases, the voltage increases. If, on the other hand, current through it is allowed to decrease, voltage across it will also decrease. So one can say that voltage is directly proportional to a change in current.

If we rearrange this basic formula to calculate current, we can see another relationship develop here. To calculate the amount of current flowing through a component, the current (I) is equal to voltage divided by resistance:

I = V / R

If voltage is held constant and resistance decreases, current increases. Or if resistance increases, current decreases. What this is really saying is that current is indirectly proportional to resistance

Resistance is defined as the opposition to current flow. Its unit is the Ohm (Ω), named in honor of George Simon Ohm. One Ohm lets one Ampere of current flow through a circuit having one volt applied. Rearranging terms in the Ohm's law equation above, resistance can be calculated from knowing the voltage and current. The relationship is shown below:

R = V / I

Again, notice that resistance is dependent on either voltage or current as discussed above. The three formulas above are important tools in the electronics trade. Please take the time to memorize them to be able to use them with ease.

Power (P) in Watts can also be shown as a product of current and voltage and is known as Watt's Law:

P = I x V

If either voltage and/or current increase/decrease, power will also increase or decrease in the same manner.

If resistance and current are known, square the current and multiply by the resistance to find power. The basic formula for this is:

$P = I^2 R$

After examining this formula, one can see a new relationship between resistance, current, and power.

When the resistance doubles in a circuit, the power also doubles. If the resistance is cut in half, the power is cut in half.

But what about current? If we double current, what happens to power? The answer to this question is that the power goes up by 4 times its previous value! Now, what happens when current is halved? The power also would be one-fourth of its previous value.

Another formula that is often used in the electronics world is the voltage squared divided by resistance:

$$P = V^2 / R$$

If an increase in resistance occurs in a circuit, the power decreases. To put this concept in another way, let's use an example. Suppose the resistance is cut in half. What happens to power? Power is doubled. Suppose resistance is doubled from a previous value? Power is now cut in half.

Understanding the relationships that go on in circuits makes circuit analysis quite simple. The whole idea is to simplify the complex circuitry to something more manageable to see what really is going on. The same relationship goes on with voltage, as does current in the previous equation. If voltage were to be doubled from a previous value, the power is increased 4 times its previous value.

Another important unit of measure that is not used quite as often is known as conductance. This unit is opposite of resistance and expresses least opposition to current flow, or how well devices conduct current. This unit has been known as the "mho" which is ohm spelled backwards. Conductance (G) is defined as the reciprocal of resistance as shown:

$$G = 1 / R$$

More recently, the conductance unit mho has been changed to Siemens. It is expressed in the same way and its symbol is the S.

Series Circuits

A series circuit is shown below.

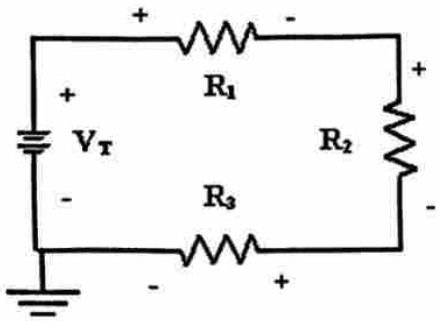


Fig. 2

As with any circuit, a complete path is necessary. Electrons need to have a destination to reach. Without this path, current flow would not be possible and electricity would be nonexistent. Electrons will flow from the negative terminal of the battery shown around the circuit and to the positive terminal. This is known as electron flow and will be discussed from here on. There is another explanation of current flow, and this is "conventional flow." This explanation says that current flows from positive to negative and Ben Franklin originated this idea before electron theory was understood.

As a practicing electronics technician, you interpret schematic diagrams. In examining the series circuit in the schematic diagram in Fig. 2, you note that there is only one path for current to travel. That current is the same at any point because there is only one path in the single loop. In this arrangement, only one current flows, therefore the current is said to be constant, and remains the same through any component in that path. In series circuits, it is easy to find unknown values using Ohm's and Kirchoff's laws.

The total resistance of a series circuit can be found by adding all of the resistance values, as shown in:

$$R_t = R_1 + R_2 + R_3 + ... + R_n$$

Another important fact about this circuit is that the total voltage drops around a series circuit can be also found in the same way. All of the voltage drops across each resistor can be found using Ohm's Law. The voltage drop across each resistor is added to arrive at a total. This total should be equal to the source voltage as shown in:

$$V_T = V_{R1} + V_{R2} + V_{R3}$$

With this, another law known as Kirchoff's voltage law develops. This is stated as:

$$0 = -V_T + V_{R1} + V_{R2} + V_{R3}$$

This says that the sum of all the voltages in a closed loop is zero.

You notice that a ground is used as a reference point for the sake of measuring, and in this case is a negative point with respect to any other point in the circuit. Notice the polarity of each voltage as we go around the circuit. If we add each voltage, while noting each voltage polarity, we will come to the same conclusion about this circuit as Kirchoff's law states:

The algebraic sum of all the voltage drops in a closed loop is equal to zero.

Voltages add up in a series circuit, and so does the total power. Each resistor, or load, is working to contribute to the total amount of work that a circuit is doing as a whole.

Everyone likes to be recognized for his/her work and to receive credit, and so does each load in a circuit. The power expended from each resistor can be added to arrive at a total power. This is expressed as follows:

$$P_{total} = P_1 + P_2 + P_3 + ... + P_n$$

If the total voltage is known, total power can also be found by using Ohm's Law. This is done by multiplying the current by the voltage applied to get the total power:

$$P_T = I_T \times V_T$$

Parallel Circuits

What is interesting about parallel circuits is that current flow is divided through several branches or paths of the circuit. Depending on an individual branch resistance, one branch can carry more current than another. This can be shown below where you can see that a 1000 ohm resistor will allow more current to flow through it than a 3000 ohm resistor, because the smaller resistance allows more current to flow. Ohm's Law shows that higher current takes the easiest path, which has less resistance.

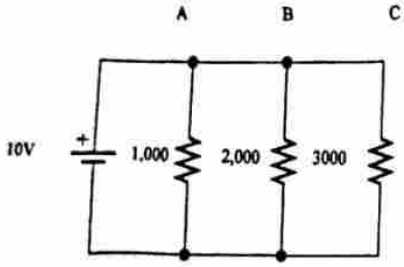


Fig. 3

Current through branch A = 10 / 1000 = 10 mA
Current through branch B = 10 / 2000 = 5 mA
Current through branch C = 10 / 3000 = 3.3 mA

As a result of this current division, the total current in a parallel circuit is the sum of each branch current. Another important law that has to be mentioned here is Kirchoff's Current Law. It states that the sum of current flowing into a junction (or node) is equal to the current flowing out of that same junction. This is shown in Fig. 4.

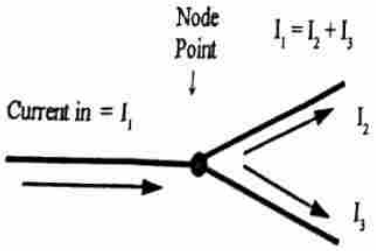


Fig. 4

mathematical form, $I_1 = I_2 + I_3$ is shown in Fig. 5, which ex- how current behaves in series and parallel circuits. Before the on was discovered, Gustav Kirchoff, a mid-nineteenth century n scientist, discovered these two important laws (voltage and) which make it possible to analyze different circuit configura- he total current in a parallel circuit is expressed as follows:

$$I_T = I_1 + I_2 + I_3 + \ldots + I_n$$

Since a parallel circuit consists of more than one branch, the total resistance of this circuit is less than the least resistance. How is this fact possible? Well, let us take a look at the above parallel circuit. The total resistance can be found with the following equation:

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

Notice that the total parallel resistance is the reciprocal of each branch resistance. The total resistance of a circuit can also be calculated as:

$$R_T = V_T / I_T$$

Vsource is the source voltage, and with this voltage, along with the total current, the total parallel resistance can be found. Notice that each current shown below (V/R) has the same source voltage (VT) applied to each resistance and is divided by that branch resistance.

$$V_T/R_T = V_T/R_1 + V_T/R_2 + V_T/R_3$$

This is why current divides: there is more than one path for current to flow. Each branch resistance determines how much current will flow through it. A little algebra is used to show that Vsource or VT is divided into both sides of the equation to get the resultant equation for total resistance:

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

If each branch resistance is equal to one another, then the total resistance is equal to the branch resistance divided by the number of branches. With this in mind, total resistance of parallel circuits with equal branch resistance values can be found quite easily:

$$R_T = R / n$$

Series-Parallel Circuits

Two simple forms of a series-parallel circuit can help you understand many of the possible combinations of circuits. Circuit can be in all different shapes and sizes. Doing a little circuit analysis is really all that is required to simplify a complicated circuit.

Do not let them confuse you, as they are not what they seem to be–a different guise to confuse the sharpest technician who needs to get a piece of equipment working. Shown in Figure 6 is one form of a series parallel circuit.

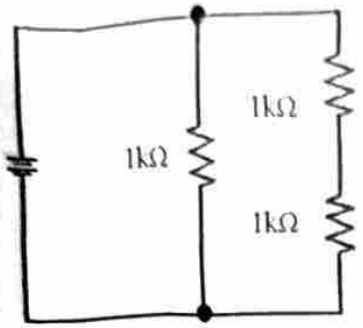


Fig. 5

To find the total resistance of Fig. 5, add the two resistors in ries on the right side. Then calculate the resultant resistance ith the single resistor in parallel on the left. If all three of these e 1kΩ, then the total will be:

1k + 1k = 2 kΩ equivalent for the right hand resistors.
Then the 2 kΩ equivalent value is in parallel with 1kΩ.
This can be solved with the combination formula:

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{1 \times 2}{1 + 2} = \frac{2}{3} = .66\ k\Omega$$

Sometimes this circuit in Fig. 5 is referred to as the "2/3rds" ircuit arrangement when equal value resistors are used, as the otal is 2/3rds of the value of an individual resistor.

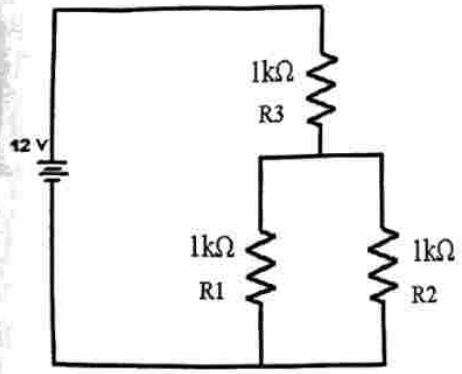


Fig. 6

The other simple form of a series-parallel arrangement is shown in Figure 6. First find the equivalent value of the two resistors in parallel on the lower part of the circuit. Use the combination formula again:

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{1 \times 1}{1 + 1} = \frac{1}{2} = .5\ k\Omega$$

Or $R_T = R / n$ where n is the number of equal value resistors in parallel.

$1k\Omega / 2 = .5\ k\Omega$

Then add .5 kΩ to the upper resistor of 1 k (R 3) because it is in series $R_T = R_1 + R_2 = .5\ k\Omega + 1k\Omega = 1.5\ k\Omega$

Figure 6 is sometimes referred to the "3 / 2" circuit because the total is one and a half times the value of one of the equal value resistors.

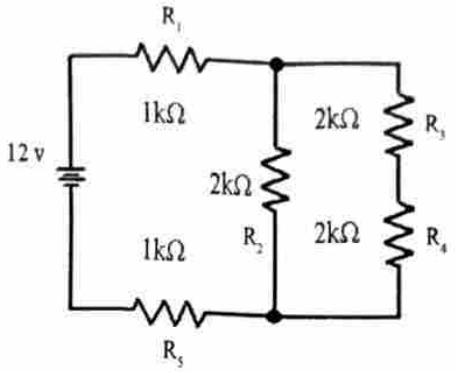


Fig. 7

A more complex circuit is shown in Figure 7.

To approach this circuit, simplify the most interior circuit combination first. It is the series circuit of the two 2K resistors, the right hand path of the parallel branch at the far right side made up of R_3 and R_4.

The resistor values of R_3 and R_4 are added to equal 4k or 4000 ohms. Now our circuit reduces to a circuit shown in Figure 8, a 4000 ohm resistor in parallel with R_2.

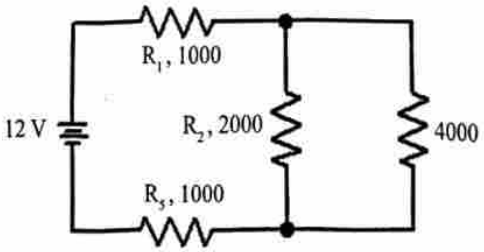


Fig. 8

A further reduction of the circuit with two resistors in parallel will equal the equivalent resistance that is in series with R_1 and R_5. To find this value, use the following:

$$R_{parallel} = R_1(R_2) / R_1 + R_2$$

Substituting each value in the above equation results in:

$$R_{parallel} = 2000(4000) / 2000 + 4000$$
$$R_{parallel} = 1333.3\ Ohms$$

The parallel combination now becomes a resistor in series with R_1 and R_5 as shown in Figure 9.

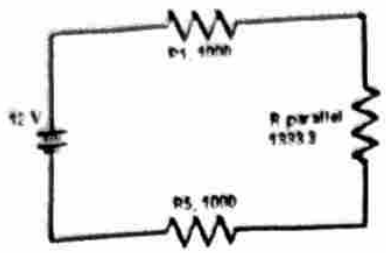


Fig 9

Next calculate the series combination to finally arrive at the final answer of a total resistance of 3333.3 ohms.

$$R_T = R_1 + R_{eq} + R_5 = 1k + 1.33k + 1k = 3.33\ k\Omega$$

As you can see from this discussion, a complex circuit can be reduced to something more manageable.

Now that we have found the total resistance of the series parallel circuit, suppose we need to find the voltage drops and currents through each branch of this circuit.

1. First find R_T the total resistance of the series parallel circuit. A practical technician may want to simply measure the complex circuit with a digital VOM (volt-ohm-meter) to find the total resistance.

2. Next to find the total current.

   This is found by dividing the applied voltage, 12 v, by the total resistance, 3.33kΩ.

   $I_T = V_T / R_T = 12v / 3.33k\Omega = 3.6\ mA$

   The result is 3.6 mA. With this current, we can now proceed to find each voltage drop around this circuit.

3. To find the voltage drops across each resistor, the current is multiplied by each resistance.

   $V_{R1} = I_T \times R_1 = 3.6\ mA \times 1k\Omega = 3.6\ volts$

The voltage across R1 is 3.6 volts. The voltage across the parallel (equivalent) portion of the circuit shown in Figure 9 is found by multiplying the total current times the total parallel resistance 1333.3 ohms.

$V_{Req} = I_T \times R_{eq} = 3.6\ mA \times 1.33k\Omega = 4.8\ volts$

The voltage across this portion of the circuit is 4.8 volts.

The voltage across R_5 is found in the same way as R_1 just mentioned. The voltage across this resistor is also 3.6 volts because the resistor is the same value as R_1 and the total current (3.6 mA) also flows thru it.

If Kirchoff's Voltage Law is applied to the series parallel network, the sum of the voltages around a closed loop is equal to zero. Adding all of the voltages around the loop as shown in Figure 9 will verify this is true. The voltage source is 12 volts.

The voltage drops across each resistor is as follows:

Voltage across R_1 = 3.6 volts
Voltage across $R_{parallel}$ = 4.8 volts
Voltage across R_5 = 3.6 volts

Adding all of the above voltage drops equals 12 volts. This is shown below.

3.6 volts + 4.8 volts + 3.6 volts = 12 volts

12 volts – 12 volts = 0 volts

Since we know the voltage drop across the parallel network, we can find the branch currents in the original series parallel circuit in Figure 7. The branch current for resistor R2 is found by dividing the 4.8 volts across this resistor by the resistance value of 2000 ohms. This equals 2.4 mA as shown:

$$I_{R2} = V_{eq} / R_2 = 4.8\ v / 2k\Omega = 2.4\ mA$$

To get the branch current for R3 and R4, we can simply subtract 2.4 mA from the total current 3.6 mA to get 1.2 mA.

$$I_{R3\text{-}R4\ branch} = I_T - I_{R2} = 3.6mA - 2.4\ mA = 1.2\ mA$$

Another way to do this is to divide the voltage drop across the parallel network by the total resistance (4kΩ) of R3 and R4:

$$V_{Rbranch} / R_{R3\text{-}R4\ branch} = 4.8\ v / 4k\Omega = 1.2\ mA$$

Then, to find the voltage drop across R3 and R4, take the current of 1.2 mA times the resistance of each:

$$V_{R3} = I_{R3\text{-}R4\ branch} \times R_3 = 1.2\ mA \times 2\ k\Omega = 2.4\ v$$

$$V_{R4} = I_{R3\text{-}R4\ branch} \times R_4 = 1.2\ mA \times 2\ k\Omega = 2.4\ v$$

Voltage Dividers

Now that we have discussed the series, parallel, and series parallel circuits in great length, our discussion takes us to an application of the above circuits. Voltage dividers are used extensively in power supplies to distribute voltages to various parts of a circuit for biasing amplifiers and such. An example of this is shown in Figure 10.

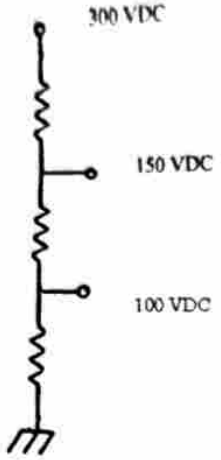


Fig. 10

In Figure 10, this voltage divider could be used for distributing the B+ plate voltage for tubes. You might think that tube circuits are a thing of the past, but do not be surprised if during your everyday encounters with circuits you find a few around.

Voltage dividers can also be used in setting up a reference voltage for one of the inputs to a comparator IC circuit such as the one shown below in Figure 11.

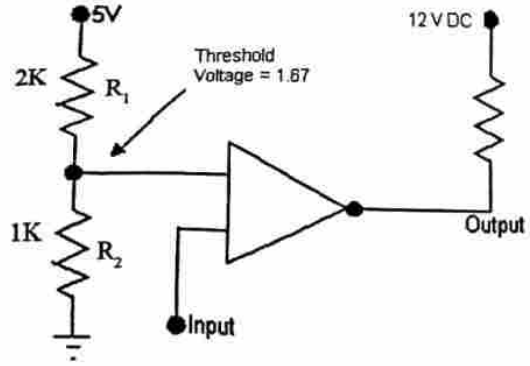


Fig. 11

In Figure 11, an LM139 voltage comparator is shown. The voltage divider sets the threshold voltage for the input to cause a change in the output when 1.67 v is reached. This voltage is also known as the trip point. The output will change from one maximum voltage to another, depending upon whether the (-) input exceeds the reference threshold voltage. This is a good example of where a voltage divider circuit is found in electronics.

Voltage dividers can be utilized in voltage comparator circuits to set the threshold voltage with respect to common ground. Use the following formula to determine the output across a lower resistor (R_2) in a voltage divider:

$$(R_2 / R_1 + R_2)\, V_{source} = \text{Voltage drop across } R_2$$

To find the threshold voltage first find the sum of R_1 and R_2 from the values shown in Figure 11

$$R_{(1+2)} = 2k + 1K = 3K \text{ or } 3000 \text{ ohms}$$

The next step is to divide R_2 (1K) by the sum of R_1 and R_2

$$(R_2 / R_1 + R_2) = 1k / 3K = .333$$

Multiply this ratio with the Vsource which is +5 vdc

$$.333 \times 5\text{ v} = 1.67 \text{ volts}$$

As you can see from this discussion, the voltage divider relationship is a powerful tool in finding a voltage in many electronic circuits.

The next part of this chapter takes us to another very important concept.

Maximum Power Transfer

A load is said to be matched to the source if the load impedance is equal to the source impedance.

With Resistance $R_i = R_L$ for DC circuits

With Impedance $Z_i = Z_L$ for AC circuits

Examples of impedance matching conditions when you will find maximum transfer of power are:

8 ohm amplifier connected to an 8 ohm speaker

50 ohm transmitter connected to a 50 ohm cable & antenna

75 ohm TV antenna connected to 75 ohm cable to TV input

600 ohm telephone connected to a 600 ohm line transformer

Common industry standard impedances are established to ensure maximum power transfer between devices from that same industry. For example, the RF industry uses 50 ohms, Video uses 75 ohms, Audio uses 8 ohms, and telecommunications uses 600 ohms devices.

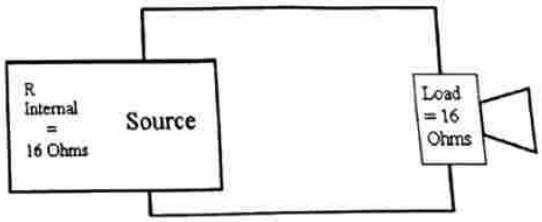


Fig. 12

An example of this is shown in Figure 12. Notice that the load is matched or equal to the source, an amplifier. The amplifier itself has an internal impedance of 16 ohm and the speaker is the same value. The same amount of power is dissipated in the internal impedance is also dissipated in the load.

An interesting point about the condition of maximum power transfer is that when the circuit is matched, with equal impedances, only 1/2 of the total power generated is delivered to the load. This is the best that can be realized with power transfer and that the efficiency of a matched condition is 50%.

There are, however, certain instances where matching impedance is not necessary simply because it is not very practical, but the load resistance has to be equal to or higher than the source internal impedance. Efficiency will go up with lighter loads that are higher in resistance. Take an example of a car battery and a small dome lamp in the vehicle. It is easy for the large capacity automotive battery with low internal resistance to power the dome light with a moderate resistance.

Another example, a battery having an internal resistance of, let's say, .1 ohms at 12 volts. What would be the maximum current capacity of this battery? The answer, using Ohm's Law, is 120 amps. From this information, we can understand that a load being connected to this battery (like a car starter) cannot be less than the internal resistance of this battery or power source.

Otherwise, more power is being expended internally to the power source than it can handle. This is why you would not place a load having a resistance lower than what a power supply can handle.

Troubleshooting

As a technician, you will find that troubleshooting involves doing a lot of circuit analysis as well as doing the divide and conquer method. You will also find that you will develop your own method of isolating a fault causing a problem in a circuit. There is no certain way to troubleshoot a faulty circuit. Just as a writer has his own writing style, a technician also has his own unique approach to troubleshooting a faulty circuit. You develop your way of doing this. Troubleshooting is a form of logic.

For starters, take a look at equipment service manuals for a piece of equipment. You will find a section that is devoted to trouble-shooting. In this section there should be a flow chart that will direct you by using listed possible symptoms.

In addition, these flow charts help the technician be in the right section in order to decrease the amount of man-hours required for troubleshooting. Use them if such charts are available. They are a good troubleshooting tool. They can get you in the right side of the ballpark.

Shown below is a simple series circuit to illustrate an interesting problem that is often overlooked.

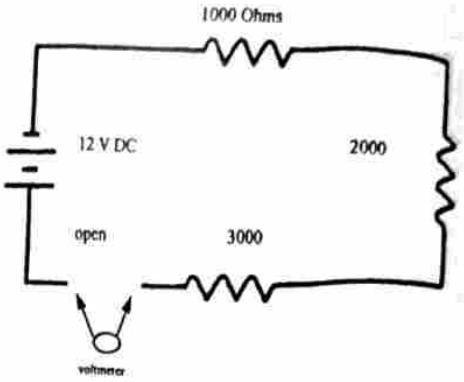


Fig. 13

After examining the above illustration, one can see there is an open that makes the series circuit incomplete, so electrons flowing from the power source have no way back home. You see, electrons seem to have a sixth sense about complete circuits. If one not complete, current cannot flow. What voltage reading will be indicated on the voltmeter if one were connected to this circuit?

The answer is that approximately 12 vdc will be indicated on the voltmeter, provided it is on a high enough scale. Why is this so? The voltmeter circuitry has high internal impedance, so most of the voltage applied to this circuit is felt across this measurement device. Not only is this true, but an open is essentially a high impedance anyway, so most of the voltage is felt across it. This is one symptom that should be considered carefully when encountered in complex circuit.

Let us take a look at another circuit that has a similar problem as discussed above.

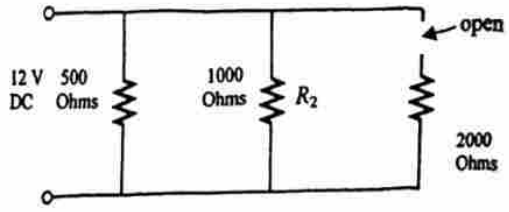


Fig. 14

A parallel circuit having resistance values as shown has an open in one of its branch circuits. What are the symptoms?

First of all, the total resistance of this circuit has increased from its 285.7 ohm original value.

The total current has decreased from 42 mA to now 36mA. The total voltage is really the same, even though an open exists in one of the branch circuits.

What about current through each branch circuit?

Will they change as a result of an open somewhere in the circuit?

Well, let us take a look. The branch with 500 ohms is still drawing the usual 24 mA, and the branch with 1000 ohms is still drawing its 12 mA current.

Now looking at the branch with 2000 ohms, there is an open. Yes, this does explain the total current decrease. No current is flowing through this portion of the circuit.

Keep in mind that these examples are simple in comparison to more complex circuitry, but with a good understanding of basic circuit theory, the most complex circuitry can be reduced to something more manageable. Another piece of good advice when approaching a circuit is to look for symptoms first before jumping to conclusions, as they will give you a clue as to where to look.

NOTES

Chapter Quiz

1. The atomic number of an element is the same as the element's atomic weight.

   a. True
   b. False

2. An increase in voltage with resistance held constant causes current to _____.

   a. decrease
   b. increase twice to value of voltage
   c. increase
   d. none of the above

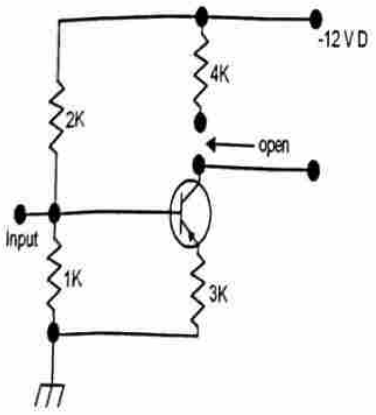
3. The ampere, expresses ____.

   a. volts/sec increase
   b. a unit of charge movement per second past a given point
   c. coulombs/charge
   d. an increase in resistance

4. The total resistance of a parallel circuit containing equal branch resistances is found by the following:

   a. by adding all of the branch resistance values.
   b. by taking the product of all the branch resistance values.
   c. by dividing the resistance value of a branch by the number of branches.
   d. the value cannot be found.

Circuit for Questions 5 thru 7.



5. An open is found in the series connected load of the transistor amplifier circuit as shown. If a voltmeter were connected from the open side of the resistor (the point of open as shown) to ground, what will be the reading?

   a. 0 volts
   b. -12 vdc
   c. Half of +Vcc
   d. None of the above

6. What is the voltage drop across the 2000 ohm resistor as shown in the amplifier?

   a. 3.3 vdc
   b. 8.0 vdc
   c. 9.0 vdc
   d. none of the above

7. If there is NO open in the circuit, what is the voltage across the 3000 ohm emitter resistor as shown above?

   a. 5.1 vdc
   b. 6.0 vdc
   c. 8.0 vdc
   d. None of the above

8. A 12 vdc power supply has a 15 ohm internal resistance. What load resistance can be connected to transfer maximum power?

   a. 15 ohms
   b. 20 ohms
   c. 10 ohms
   d. 5 ohms

9. Maximum power transfer occurs when the load impedance is equal to the source impedance.

   a. True
   b. False

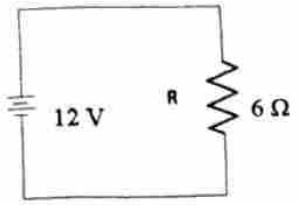
ANSWERS on page 223

Review Questions - DC Theory, Ohm's Law, Series/Parallel Circuits Chapters 3 and 4

Unit 2

1. **What is the current through the resistor in the schematic below?**

   a. .2 amps
   b. 2 amps
   c. 5 amps
   d. 50 amps



2. **In the circuit above, what is the power dissipated in the resistor?**

   a. 6 watts
   b. 12 watts
   c. 2 watts
   d. 24 watts

3. **What will happen to the current through a resistor if the resistor value increases?**

   a. Current increases
   b. Current remains unchanged
   c. Current decreases
   d. Current and the voltage across the resistor increases

4. **Which type of circuit has the same voltage drop across each component?**

   a. Parallel
   b. Series-parallel
   c. Series
   d. Combination

5. **Total power in a series circuit is determined by the:**

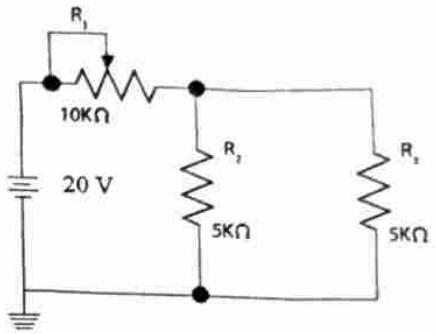
   a. source voltage times the current.
   b. total voltage applied to the circuit.
   c. current flowing through the switch.
   d. average wattage rating of each resistor.

6. **Which is the equivalent to 100 microamps?**

   a. .1 milliamps
   b. 1 milliamps
   c. 10 milliamps
   d. 10 nanoamps

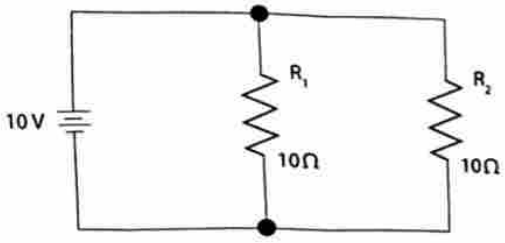
7. **If the current drain from the battery is 4 mA, what resistance is the rheostat in the drawing below?**

   a. 1.0k ohm
   b. 2.5k ohms
   c. 5.0k ohms
   d. 10k ohms



8. **In the circuit below how much current is flowing in R2?**

   a. 1 amp
   b. 2 amps
   c. 10 amps
   d. 100 amps



ANSWERS on page 223

Chapter 5 Capacitance, Inductance, Resonance and Filt

by Dr. Sohail Anwar, Ph.D, CETsr

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management. He is a member of ASEE, IEEE, Pennsylvania Academy of Science (PAS), and Electronics Technicians Association (ETA).

Review of Resistance

In chapter 4, resistance was covered with DC circuits and formulas discussed. The resistor symbol was used to represent loads for the circuits. Keep in mind that resistance and resistors have the same effect in both DC and AC circuits.

Resistance is defined as opposition to the flow of electrical current. It is designated as R and its schematic symbol is shown in Figure 1.

Fig. 1 Resistor schematic symbol

Resistance R is measured in ohms symbolized by the Greek letter omega (Ω). The resistance of a material is one ohm (1Ω) if a voltage of one volt (1V) applied across the material results in a flow of one ampere (1A) of current through the material. The common multiples of the unit ohm are kilo-ohm (KΩ) and Mega-ohm (MΩ). 1KΩ stands for 1,000Ω and 1MΩ means 1,000,000Ω or 106Ω.

A resistor is an electrical component which is specifically designed to have a certain amount of resistance. Two principal categories of resistors are: fixed resistors and variable resistors. A fixed resistor has a specific ohmic value which is set during manufacturing and cannot be altered easily. Fixed resistors are available in a large variety of resistance (ohmic) values. Various methods and materials are employed for their construction. A common type of fixed resistor is the carbon-composition type. Other types of fixed resistors include chip resistor, carbon film, metal oxide, metal film, metal glaze, and wirewound. (All of these are conductors, just not good ones.)

Fixed resistors that are in the 5%, 10% and 20% tolerance oups are color coded with four bands to indicate the resis- ce value and the tolerance. Figure 2 shows the color bands Figure 3 shows the color code.

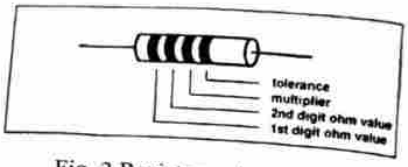


Fig. 2 Resistor color bands

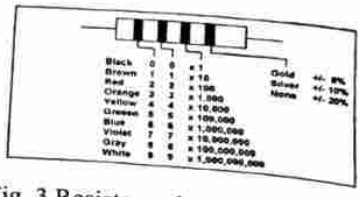


Fig. 3 Resistor color code meanings

The resistor color code is interpreted in the following way:

1. The first band is the first digit of the resistance value.
2. The second band is the second digit of the resistance value.
3. The third band is the multiplier (number of zeroes).
4. The fourth band gives the tolerance of the resistor and is usually gold or silver. For example, a 10% tolerance (silver band) indicates that the acceptable resistance value can be within 10% of the color-coded value. So a 200 ohm resistor with a tolerance of 10% can have acceptable resistance values in the range of 180 ohms to 220 ohms.

A few examples showing how to obtain resistance values us ing the resistor color code are presented below:

1. The color bands for a certain resistor are: first band is green, second band is violet, third band is yellow and the fourth band is silver. Using the resistor color code shown in Figure 3, the resistance value with tolerance is 570,000Ω ± 10% = 570KΩ ± 10%.

Therefore the acceptable resistance value for this resistor ranges from 513KΩ to 627KΩ in measured value.

2. The color bands for a certain resistor are: first band is orange, second band is brown, third band is red and the fourth band is gold. Using the color code shown in Figure 3, the resistance value with tolerance is 3100Ω ± 5% = 3.1KΩ ± 5%. Therefore the acceptable resistance values for this resistor range from 2945Ω (or 2.945KΩ) to 3255Ω (3.255KΩ).

3. The color bands for a certain resistor are: first band is blue, second band is red, third band is green, and the fourth band is silver. The resistance value with tolerance is 6,200,000Ω ± 10% = 6.2 MΩ ± 10%. Therefore the acceptable resistance values for this resistor range from 5.58MΩ to 6.82MΩ.

Some color-coded resistors have a fifth color band also. This color band is the indicator of the resistor's reliability in percent of failures per 1000 hours of use. The reliability color code is shown in Figure 4. As an example, a red fifth band indicates that if a group of similar resistors are operated under standard conditions for 1000 hours, 0.1% of the resistors in that group will fail.

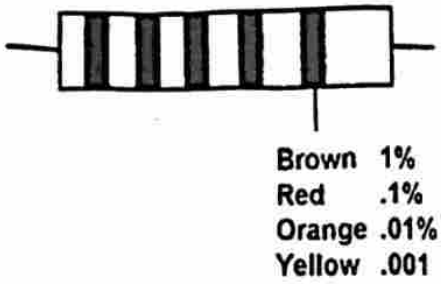


Fig. 4 Resistor Reliability Band

The second category of resistors comprises variable resistors that are designed so that their resistance values can be changed with a manual or an automatic adjustment. The potentiometer and the rheostat shown in Figure 5 constitute two basic types of manually adjusted resistors.

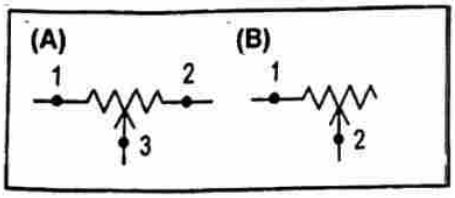


Fig. 5 Potentiometer (A) and Rheostat (B)

As illustrated in Figure 5(A), a potentiometer is a three-terminal electrical device. Terminals 1 and 2 have a fixed resistance between them which is the total resistance. Terminal 3 is connected to a moving contact. The electrical resistance between terminals 1 and 3 or between 2 and 3 can be varied by moving the contact up or down. A rheostat is a two-terminal device shown in Figure 5(B).

The power rating of a resistor is the maximum amount of power that a resistor can dissipate without being damaged by a buildup of an excessive amount of heat. Power rating is expressed in watts (w). The power rating of a resistor is not related to its resistance value but rather it is determined by the physical size and the shape of a resistor. The larger the surface area of a resistor, the greater amount of power it can dissipate. Carbon-composition resistors are usually available in standard power ratings ranging from 1/8 watt to 2 watts. The carbon-film and metal-film resistors have ratings up to 10 watts and wire wound resistors have power ratings up to 225 watts or greater.

Series Connected

When connected in series, resistors form a single path for the flow of electrical current. Figure 6 shows some examples of resistors connected in series.

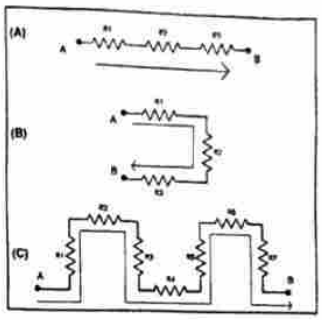


Fig. 6

In Figure 6(A) resistors R_1, R_2, and R_3 are connected in series forming a single path for the flow of electrical current from point A to point B.

In Figure 6(B) resistors R_1, R_2, R_3, and R_4 are connected in series forming a single path for the flow of electrical current from point A to point B.

In Figure 6(C) resistors R_1, R_2, R_3, R_4, R_5, R_6, and R_7 are connected in series forming a single path for the flow of electrical current from point A to point B. When the resistors are connected in series, the total resistance is the arithmetic sum of the ohmic values of the resistors. For example, the total resistance between the points A and B in Figure 6(A) is $R_T = R_1 + R_2 + R_3$. Similarly, the total resistance between the points A and B in Figure 6(C) is $R_T = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7$.

In general, for any number of individual resistors connected in series, the total resistance is the sum of the individual resis-

tance values. The total resistance is $R_T = R_1 + R_2 + R_3 + ...R_n$. An example of adding resistors in series is given In Figure 7.

Figure 7 shows resistors R_1, R_2, R_3, R_4, and R_5 in a series combination. The total resistance between the points A and B is $R_T = R_1 + R_2 + R_3 + R_4 + R_5 = 40\Omega + 50\Omega + 75\Omega + 90\Omega + 125\Omega = 380\Omega$

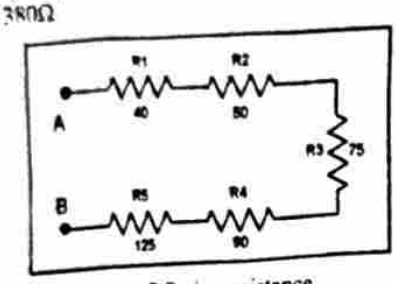


Fig. 7 Series resistance

Parallel Connected

When two or more resistors are individually connected between two points they are said to be in a parallel combination. A parallel circuit allows more than one path for the flow of electrical current. Figure 8 shows some examples of the parallel combinations of resistors.

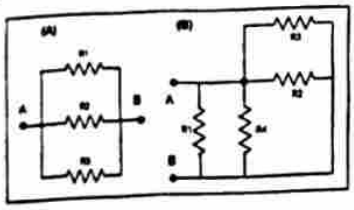


Fig. 8 Parallel resistance

Figure 8(A) shows resistors R_1, R_2, and R_3 connected in parallel between the points A and B. Figure 8(B) shows resistors R_1, R_2, R_3, and R_4 connected in parallel between the points A and B.

The equivalent (total) resistance R_{eq} for any number of resistors connected in parallel between two points is given by the formula:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ... \frac{1}{R_n}$$

where R_{eq} is the total resistance and R_1, R_2, R_3, ...R_n are resistors connected in parallel ("n" is an undetermined number). In order to illustrate the formula given above, an example involving parallel combinations of resistors is presented here:

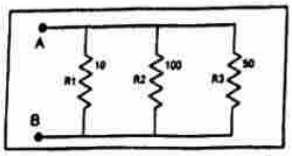


Fig. 9 Parallel resistance

In Figure 9 resistors R_1, R_2, and R_3 are connected in parallel between the points A and B. The equivalent resistance (or total resistance) between the points A and B is determined as follows:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{10\Omega} + \frac{1}{100\Omega} + \frac{1}{50\Omega} = \frac{13}{100\Omega}$$

So $R_{eq} = 100 \div 13$ or 7.69Ω

An important point should be noted here: the equivalent resistance of a parallel combination of resistors is always less than the smallest resistor connected in parallel. The example which was presented above attests to the accuracy of this statement.

When all the resistors connected in parallel between any two points are equal, the equivalent resistance or the total resistance of the parallel combination is simply the ohmic value of an individual resistor divided by the number of resistors connected in parallel. Thus if twenty 1.5kΩ resistors are connected in parallel, the total equivalent resistance of the parallel combination (R_T) is found by:

$$R_T = \frac{R}{n} = \frac{1.5k}{20} = 75\Omega$$

where n = number of resistors

Series-Parallel Resistors

Finally, an example is presented below to show how to determine total resistance in series-parallel resistive circuits.

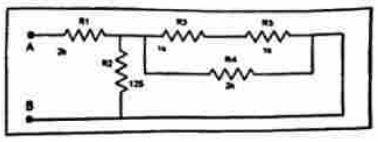


Fig. 10 Series-Parallel resistance

In order to determine the total resistance between the points A and B, we start from the extreme right of the given series-parallel circuit. It is noted that R_3 and R_5 are in series. So their series combination gives 2k Ω which is in parallel with R_4 and also in parallel with R_2.

This is shown in Figure 11.

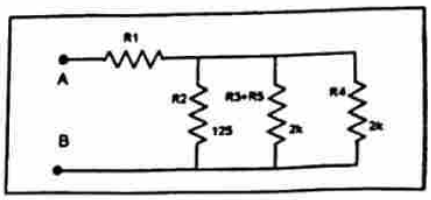


Fig. 11 Series Parallel resistance

The equivalent resistance R_{eq} of the parallel combination of R_1, R_2, and $R_3 + R_4$ is determined as follows:

$$\frac{1}{R_{eq}} = \frac{1}{125\Omega} + \frac{1}{2k\Omega} + \frac{1}{2k\Omega}$$

So $R_{eq} = 111.1\Omega$

Now R_{eq} is in series with R1 as shown in Figure 12.

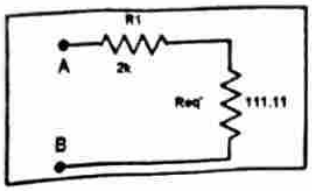


Fig. 12 Series resistance

Thus R_{eq} or the total resistance between the points A and B is $R_{eq} = R1 + R_{eq} = 2.111\ k\Omega$

Capacitance

A capacitor is an electric device that can store electrical charge. A capacitor consists of two parallel conductive plates separated by an insulating material called the dielectric.

Figure 13 shows the schematic symbol for a capacitor.

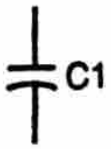


Fig.13 Capacitor schematic symbol

The ratio of electric charge Q of a capacitor to the voltage V across its plates is called capacitance C. Thus the capacitance C of a capacitor is:

$$C = \frac{Q}{V}$$

where Q is the electric charge stored by a capacitor and measured in coulombs. V is the voltage across the parallel plates of a capacitor and measured in volts.

The unit of capacitance is called a Farad (F). However, 1F is quite a large unit, so its fractional units, called microfarad (μF), nanofarad (nF), and picofarad (pF) are used instead. One μF equals 10^{-6}F, 1nF is equal to 10^{-9} F, and 1 pF equals 10^{-12}F. Several examples are presented below to show how various calculations related to capacitors are done:

1. A capacitor stores 100 microcoulombs (100 μC) with a voltage of 40 volts across its plates. Find its capacitance.

$$C = \frac{Q}{V} = \frac{100\mu C}{40V} = 2.5\mu F$$

2. Find the voltage across the plates of a capacitor having a capacitance value of 500pF and storing 40 μC of electric charge.

$$V = \frac{Q}{C} = \frac{40\mu C}{500pF} = \frac{40\times10^{-6}\ C}{500\times10^{-12}F} = 80{,}000V = 80kV$$

3. Convert 1000pF to μF.

$$1000 \times 10^{-12}F = \frac{1000 \times 10^{-12}\mu F}{10^{-6}} = 10^{-3}\mu F = .001\mu F$$

The amount of electrical energy that can be stored by a capacitor is $W=1/2CV^2$ where W is the electrical energy in joules (J), C the capacitance in farads, and V the voltage in volts. Thus the energy stored by a .01μF capacitor having 10 volts across its plates is $W=1/2(.01\mu F)(10\ volts)^2 = .5\mu J$.

The breakdown voltage or the working voltage, for a capacitor is the maximum DC voltage that can be applied to a capacitor without damage to the device. If this voltage rating is exceeded, permanent damage to the capacitor can result.

The permittivity of a medium is the measure of ease with which an electric field can be established in the material. Permittivity is represented by the Greek letter epsilon (ε). The permittivity of vacuum is (ε_0) and $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. The relative permittivity (ε_r) of a dielectric material is the ratio of the permittivity of the dielectric to the permittivity values for selected dielectric materials as shown in Fig. 14.

Dielectric Material	ε_r
Air (vacuum)	1
Paper	2.5
Mica	5.0
Ceramic	1200

Fig. 14

The capacitance C of a capacitor is given by:

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

where C is the capacitance measured in farads (F)

- ε_r is the relative permittivity of the dielectric material used in the capacitor
- $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m
- A is the area of cross section of the parallel plates of the capacitor in square meters (m^2)
- d is the separation between the plates measured in meters (m)

As the separation or distance between plates increases, capacitance decreases. As the area of the plates increases, the capacitance increases, and greatly affects the value. Both the distance between the plates and the area of the plates are used in small variable capacitors to change value.

Capacitors are generally categorized according to the type of dielectric material. The most commonly used dielectrics are mica, ceramic, plastic-film, and electrolytic. Ceramic capacitors have capacitance values in the range of 1pF to .01μF with voltage ratings up to 6000 Vdc. Ceramic capacitors are small, about the size of a dime, and are frequently marked with values in pF, and a 47pf may be marked as 47k. Plastic-film capacitors have several different types. Examples of some of the dielectrics used for these capacitors include mylar, tantalum, and polystyrene. Their capacitance values are up to 100μF. Electrolytic capacitors have high capacitance values up to and even over 200,000μF, however, they have relatively low breakdown voltages.

The capacitors described above are fixed capacitors. There is another class of capacitors known as variable capacitors. They are used in radio frequency circuits when changing frequency or tuning is required. As the plates separate or open, capacitance decreases. The schematic symbol for a variable capacitor is illustrated in Figure 15. The most common types of variable capacitors are air capacitors, trimmers, and varactor diodes, which are voltage controlled.

Fig. 15 Variable capacitor schematic symbol

When the capacitors are connected in series, the total capacitance C_t is determined by the formula:

$$\frac{1}{C_t} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots \frac{1}{C_n}$$

where $C_1, C_2, C_3 \ldots C_n$ are capacitance values of the individual capacitors in series.

When the capacitors are connected in parallel, the total capacitance value is the sum of all the individual capacitance values. So $C_t = C_1 + C_2 + C_3 + \ldots + C_n$ where C_t is the total capacitance and $C_1, C_2, C_3 \ldots C_n$ are the individual capacitance values.

Two examples illustrated below show how to perform calculations related to the series and the parallel combinations of capacitors.

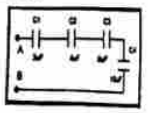


Fig. 16 Series capacitance

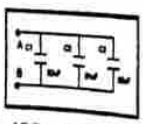


Fig. 17 Parallel capacitance

1. What is the total capacitance Ct of the series combination of capacitors shown in Figure 16?

$$\frac{1}{C_t} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{2\mu F} + \frac{1}{4\mu F} + \frac{1}{8\mu F} + \frac{1}{10\mu F} = \frac{1}{1.026\mu F}$$

So $C_t = 1.026\mu F$.

2. Find the total capacitance C_t of the parallel combination of capacitors shown in Figure 17.

$$C_t = C_1 + C_2 + C_3 = 0.02\mu F + 0.01\mu F + 0.05\mu F = 0.08\mu F$$

Inductance

Inductance is defined as the property of a coil of wire (called an inductor or a choke) that opposes a change in electric current. When there is a flow of electric current through an inductor, a magnetic field is established. When the current changes, the magnetic field also changes. An increase in electric current expands the magnetic field and a decrease in electric current reduces the magnetic field. Inductance or self-inductance is a measure of a coil's ability to set up an induced voltage across it as a result of a change in the electric current that flows through it. The schematic symbol for an inductor (or a coil) is shown in Figure 18.

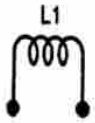


Fig. 18 Inductor schematic symbol

Inductance L is measured in a unit called Henry (H). Inductances are usually measured in a unit called milliHenry (mH). $1mH = 10^{-3}$ H.

An inductor stores energy in the magnetic field established by a flow of electric current through it. The inductor energy is given by the formula $W = \frac{1}{2}Li^2$ where W is the energy measured in Joules, L is the inductance measured in henrys, and i is the electric current.

The physical properties that determine the inductance of a single layer air-core coil are shown in the following formula:

$$L = \frac{r^2N^2}{9r + 10\ell} \text{ (inches)}$$

Where N = number of turns
r = diameter or radius of the coil
ℓ = spacing of turns or length of the coil

As the length or spacing between turns increases, the inductance of the coil decreases.

As the number of turns increases, the inductance increases.

The material around which a coil is formed is known as the core. A core made of ferromagnetic material permits the establishment of a stronger magnetic field. Examples of ferromagnetic materials include iron, nickel, steel, cobalt, or various alloys. Inductors are categorized according to the type of core material. Figure 19 shows three common types of inductors classified according to the core materials used.

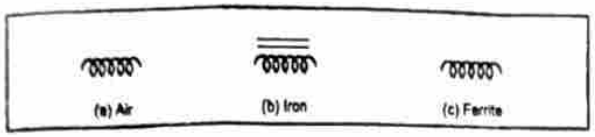


Fig. 19 Inductor schematic symbols

Iron and ferrite cores increase the inductance of a coil. Iron core types have large inductance measured in base units of Henries and are frequently called "chokes."

Ferrite cores are in shapes of round torroid coils or in radio frequency (RF) types that have adjustable moveable cores.

Some RF coils also have aluminum or brass cores, which decrease the inductance, rather than increase like iron cores. These are typically used where a very small amount of inductance is required in VHF and UHF circuitry.

The inductors described above are fixed inductors. Adjustable (variable) inductors have an adjustment that moves a sliding core in and out, causing a change in the value of inductance.

Figure 20 shows the schematic symbol for a variable inductor.

Fig. 20 Adjustable inductor schematic symbol

When inductors are connected in series, the total inductance of the series combination is given as the arithmetic sum of individual inductance values. Thus $L_t = L_1 + L_2 + L_3 + ...+L_n$ where L_t is the total inductance of the series combination and L_1, L_2, L_3....Ln are the inductances of the individual inductors connected in series.

When the inductors are connected in parallel, the total inductance value L of the inductors in parallel is given by the formula

$$\frac{1}{L_t} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots \frac{1}{L_n}$$

where L_1, L_2, L_n are the inductance values of individual inductors connected in parallel.

Transformers

A transformer is a device that transfers energy from one circuit to another by electromagnetic induction. The energy is transferred without a change in frequency. The basic construction of a transformer is shown in Figure 21.

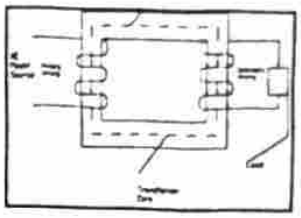


Fig. 21 Transformer

The transformer core is a closed loop of magnetically permeable material. Two coils of wire are wound around the core. The primary winding is connected to an AC power source and the secondary winding is connected to the load. The primary winding produces a changing magnetic flux through the secondary winding which induces an AC voltage in the secondary winding. If the number of turns of wire in the secondary winding are more than the number of turns in the primary winding, the voltage induced in the secondary winding will be more than the voltage in the primary winding. This is called a step-up transformer.

If the number of turns in the secondary winding is less than the number of turns in the primary winding, the voltage induced in the secondary winding will be less than the voltage in the primary winding. This is called the step-down transformer. This relationship is expressed mathematically as:

$$a = \frac{Vp}{Vs} = \frac{Np}{Ns}$$

where: N_p = number of turns in the primary winding
N_s = number of turns in the secondary winding
V_p = voltage in the primary winding
V_s = voltage in the secondary winding
a = turns ratio

The currents in the primary and secondary circuits are related as:

$$\frac{N_P}{N_S} = \frac{I_S}{I_P}$$

where: I_s = current in the secondary circuit
I_p = current in the primary circuit

The apparent power at the input to the primary circuit is equal to the apparent power rating at the secondary. This is expressed as:

$V_{1}I_{1} = V_{2}I_{2}$

The efficiency of a transformer is given as:

$$h = \frac{P_{out}}{P_{in}} \times 100$$

where: h = percent efficiency
P_{in} = input power
P_{out} = output power

A commonly used type of transformer is called autotransformer. Figure 22 shows two autotransformers.

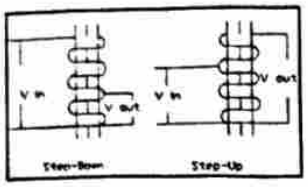


Fig. 22 Autotransformer

An autotransformer consists of a single winding that serves both as primary and secondary. Autotransformers can be used whenever both the input and output have a grounded line. Autotransformers are smaller, lighter in weight, and use less copper. The disadvantage of an autotransformer is that it does not provide the isolation of load from the power line.

Filters

Filters are frequency-selective electric circuits that permit signals of certain frequencies to pass from the input to the output while blocking all the other frequencies. A low-pass filter allows signals with lower frequencies to pass from input to output while rejecting higher frequencies.

The frequency response of a low-pass filter is shown in Figure 23.

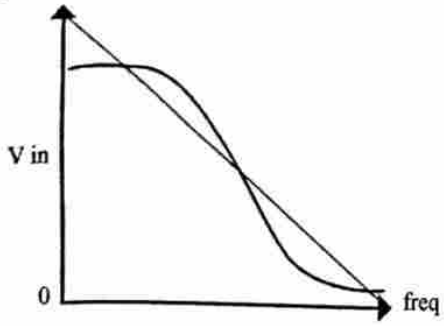


Fig. 23 Frequency response of a low-pass filter

A high-pass filter allows signals with higher frequencies to pass from input to output while rejecting lower frequencies. The frequency response of a high-pass filter is shown in Figure 24.

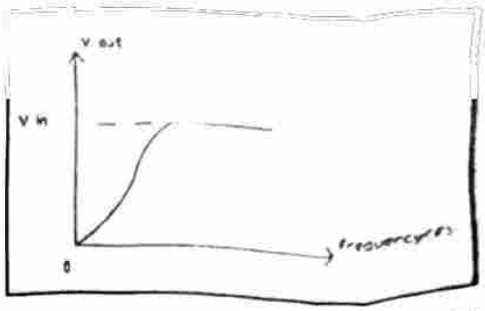


Fig. 24 Frequency response of a high-pass filter

A combination of low-pass and high-pass filters can be u to obtain a band-pass filter. A band-pass filter allows a certa band of frequencies to pass and rejects all frequencies belo and above the pass band.

The frequency response of a band-pass filter is shown in Figure 25.

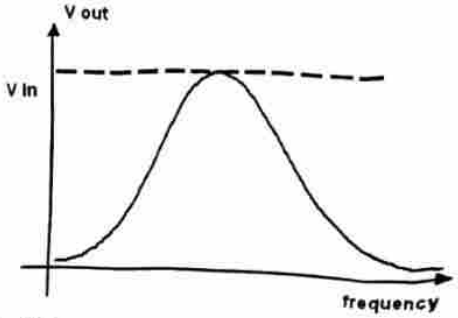


Fig. 25 Frequency response of a band-pass filter

Series RC (Resistance-Capacitance) networks exhibit a frequency-selective characteristic and therefore function as basic filters. A low-pass filter is implemented by taking the output across the capacitor shown in Figure 26. A high-pass filter is implemented by taking the output across the resistor shown in Figure 27. A band-pass filter is realized by taking the output across the resistor in a series resonant RLC circuit shown in Figure 28.

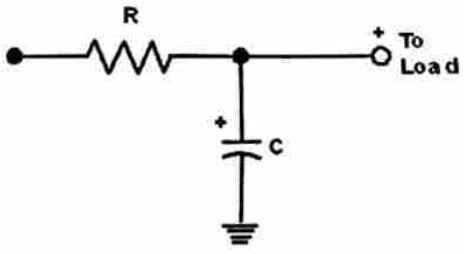


Fig. 26 A low-pass filter

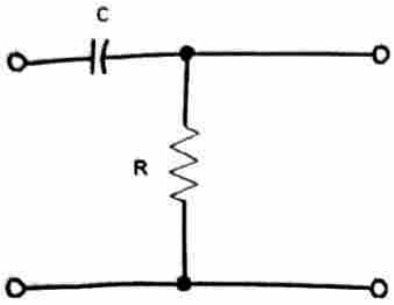


Fig. 27 A high-pass filter

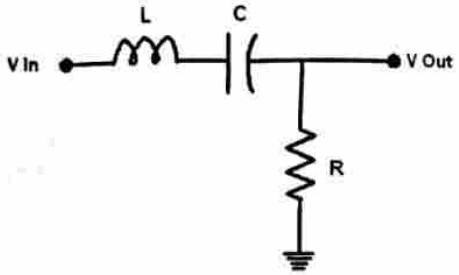


Fig. 28 A band-pass filter

The opposite of a band-pass filter is a band-reject filter, sometimes called a "trap." It is used to eliminate specific frequencies that may cause interference with desired signals.

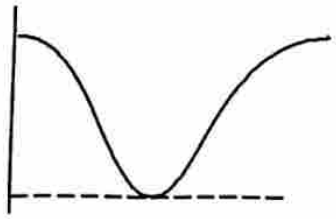


Fig. 29 Frequency response of band-reject filter

NOTES

Chapter Quiz

1. **A resistor _____.**

   a. stores an electric charge
   b. is measured in coulombs
   c. in a circuit causes a voltage drop from one end of the resistor to the other end
   d. is dependent on lines of force

2. **1 kΩ is the same as _____.**

   a. 1 thousand ohms
   b. 1 thousandth of an ohm
   c. 10^{-3} ohms
   d. 10000 ohms

3. **A resistor has color bands orange, red, brown. It is _____.**

   a. 232 ohms
   b. 3.2 megohms
   c. 2.3 kohms
   d. 320 ohms

4. **A circuit has one 500 ohm resistor in series with three 500 ohm resistors that are connected in parallel with each other. The total resistance would be _____.**

   a. 666 ohms
   b. 167.7 ohms
   c. 750 ohms
   d. 1000 ohms

5. **The voltage rating of a capacitor _____.**

   a. must be factored into the reactance formula
   b. relates to the maximum voltage that should be applied to it
   c. indicates the maximum current the capacitor can allow
   d. is the same as the power factor

6. **Calculating the total capacitance of several capacitors connected in parallel _____.**

   a. is not necessary as they are not used this way
   b. causes a need to consider inductance factors
   c. is done by adding each of the capacitance values
   d. is the same as the power factor

7. **Ferrite _____.**

   a. is a capacitor dielectric
   b. is a common inductor core material
   c. is the most common resistor coating
   d. causes faulty saturation of inductors

8. **The term Henry refers to _____.**

   a. the inductive property of a coil
   b. the inventor of inductors
   c. reactance of LR circuits
   d. resonant frequency of tank circuits

9. **A certain transformer has 250 turns in the primary winding and 1250 turns in the secondary winding. The turns ratio is _____.**

   a. 0.2
   b. 2.5
   c. 5
   d. 0.5

10. **When the turns ratio of a transformer is 10 and primary AC voltage is 8V, the secondary voltage _____.**

    a. 80 Volts
    b. 0.8 Volts
    c. 8 Volts
    d. 64 Volts

11. **Which of these color code patterns is found on a 270Ω 5% resistor?**

    a. Red, violet, brown, silver
    b. Red, violet, brown, gold
    c. Red, violet, black, silver
    d. Red, brown, black, silver

12. **One microfarad is equivalent to _____ picofa**

    a. 10
    b. 1000
    c. 100,000
    d. 1,000,000

13. **The capacitance of a capacitor with 10μC of electric charge and 100V across its plates is _____.**

    a. 0.1μF
    b. 10μF
    c. 1000μF
    d. 107μF

ANSWERS on page 223

Chapter 6

AC Circuit Operations

by John Ufer, CETsr

John Ufer, CT Tsr, has taught electronics as an adjunct instructor at Northwestern Michigan College for 8 years, as well as being contract course development for Sears, Syntec Inc., American Natural Gas, and Cone Drive Corporation. His background also includes publications of technical service manuals for the gaming industry. He is currently teaching a Business Computer Technology program at a prison facility located in Northern Michigan.

Series RLC Circuit

AC circuits can contain four different types of basic devices, a resistive component, electromagnetic component, electrostatic component, and a semi-conductive component. Within an AC circuit, any one or all of these elements can be found.

When troubleshooting a fault of one element of a circuit, it may require the understanding of how all the elements work together to accomplish a speedy and efficient repair.

In this chapter we will examine AC circuit devices and circuit actions from a technician's viewpoint.

Figure 1 shows a series AC circuit. In this circuit a common current is shared by all components, with a division of the source voltage by the various circuit devices. Individually, the voltage drops of each device may exceed the supply voltage but, added together, all voltage drops combined will equal the supply voltage. The total impedance Z_T is a result of the applied voltage divided by the circuit's total resultant current.

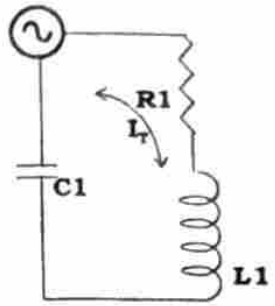
You can also find total impedance in a series circuit from the formula:

$$Z_T = \sqrt{R^2 + Xnet^2} \qquad Xnet = \text{total series reactance}$$

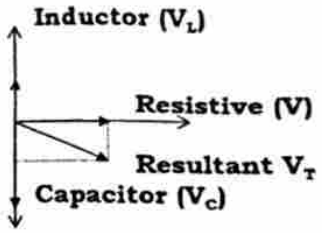
Characteristics of a RLC Series Circuit

* Division of Voltages
* One Common Current
* Sharing of Energy between the C and L components

In the series circuit shown in Figure 1, if in this circuit there is a single frequency applied, the resultant voltage phase angle and component voltage drops around the circuit would not change. In circuits that have various frequencies applied, the resultant addition of voltage drops and circuit current would change based on the applied frequency value. This is the main difference in looking at circuit actions with AC, rather than an applied DC voltage.



Voltage Addition within a Series RLC Circuit



If X_C is Larger than X_L then X_L Subtracts from X_C

Figure 1

Reactance will vary within the capacitor and inductor as the frequencies applied to the devices change. The change in reactance is directly proportional to frequency changes in the inductor and inversely proportional to changes in frequencies applied to the capacitor.

Inductive Reactance is X_L = 6.28 x Frequency x L

$$\textbf{Capacitive Reactance is } X_C = \frac{1}{\textbf{6.28 x Frequency x C}}$$

Frequency goes up, then X_L changes up. Frequency goes down, then X_C changes up.

The effects of changes in reactance of the two components would result in changes in total circuit current. The reactive series component voltage (V_L or V_C) will subtract from each other, the resistive components would add at a 90-degree phase shift with the net reactive component voltage. Inductive voltages add to other inductive voltages, capacitive voltages add to other capacitive voltages.

Note: The inductive and capacitive voltages subtract directly from each other. The net reactive and total resistive components add at a phase angle of 90 degrees from each other. Maximum circuit current would occur when X_L is equal to X_C. In a series circuit this is called Series Resonance.

Frequency of Resonance

$$F_R = \frac{.159}{\sqrt{LC}}$$

L = Henries of Inductance
C = Farads of Capacitance

In a series resonance circuit the frequency at which both reactances are equal is based primarily on the circuit component values of the inductors and capacitors within the circuit.

The bandwidth frequency response of the Figure 1 circuit on the previous page (below or above the resonance frequency) is a factor of the resistive components of the circuit and the Q factor of the inductor. The higher the non-reactive resistance is in the circuit, the broader the bandwidth response of the circuit. Changes in circuit frequency from the resonance frequency will make the circuit voltages more inductive as frequency changes above the frequency of resonance and the circuit will have higher capacitive voltages as the applied frequency drops below the frequency of resonance.

In a series AC circuit containing capacitors, inductors and some level of resistance, circuit current is based on the sharing of potentials developed across each device.

The capacitor electrostatic field contains a phase difference from the inductive component by 180 degrees. As the voltage across the capacitor goes positive, the voltage developed across the inductor will be negative. The two components can share energy as loading conditions reduce each field's energy level. Current demands on inductive and capacitive devices will reduce the back-EMF developed within each device, causing additional current to flow through the circuit. Most series circuits in electronic devices drive output devices. Output devices in series with reactive components increase peak current demands and may have short time periods in which to use the stored energy within each device. Power needs to be available to the output device sometimes within a fraction of the source wave time period.

Capacitors and inductors working together can supply energy during low voltage wave times of the source.

The current limiting factor at resonance is the circuit conductance in series with the filter components and the internal resistance of the inductive device.

From a troubleshooting standpoint, an open in the inductor or capacitor results in an inability for either device to share energy from the electrostatic field of the capacitor or the electromagnetic field of the inductor. A short in either device will result in additional circuit current and could result in low current to an output device if the output device uses the circuit energy in short duration peak demand cycles.

The source voltage should appear across the open with no circuit current in or through the device. Because the devices are using signal current (in many cases) to drive the capacitor or inductor, a lack of output signal may indicate an open in this circuit. If you measure across the open you may "see" the signal voltage. The best measurement is at the output to ground rather than across the device. Comparing the input signal to the output signal level from ground may show you where the open exists.

Parallel RLC Circuits

In the RLC parallel circuit, the common factor that is applied to each device in the circuit is the total applied circuit voltage. Each device, or series group of devices, has its own current path from and back to the source voltage. The reactance or resistance of each circuit component will limit the current through the device. The true total inductive or capacitive current is based on the subtraction or addition of each reactive device's current. Capacitive currents add to other capacitive currents, and inductive currents add to other inductive currents. Capacitive currents subtract from inductive currents. When you add or subtract currents in a parallel circuit or voltages in a series circuit, you must do so in a uniform way. Negative or positive results do matter, especially when dealing with more complex circuits. For the technician, use of schematic diagrams can be invaluable in "seeing" circuit actions and providing you with critical voltage measurement levels.

In a parallel circuit, current is shared between the capacitor and inductor. The phase difference between the inductor and capacitor are 180 degrees apart. As a result, a capacitor is developing a positive voltage across itself while the inductor develops a negative voltage. Current flows from a difference in potential. So currents will flow from the electrostatic field of the capacitor to the electromagnetic field of the inductor if a difference in potential exists. When the inductor and capacitor circuit devices have the same level of energy in their magnetic and electrostatic fields, the circuit is at parallel resonance. Total circuit current is at a minimum; circuit impedance is at a

maximum. If the fields have the same level of potential, there is no difference in potential, and current stops flowing between the devices or being supplied from the source to the devices. A source voltage is present, just no additional current flow.

$I_{net} = I_{CT} - I_{LT}$ $\quad I_T = \sqrt{I_R^2 + I_{net}^2}$

Inet represents only the out-of-phase currents

A Parallel RLC circuit has:

- Separate conductive paths for current flow through the circuit.
- Supply voltage potential that is supplied to all parallel circuit devices.
- Current that is shared by the inductive and capacitive components.
- The electrical potentials of the electromagnetic and electrostatic fields in the circuit that are 180 electrical degrees apart.
- Minimum source circuit current flows at the parallel resonance frequency.

Q and Resistance effects in Parallel RLC Circuits

In a parallel circuit, internal resistance in the coil can make a difference on the selectivity of the bandwidth response of the circuit. This is why it is critical when tuning a coil or troubleshooting defects in frequency responses that you do not think that, because an inductor is not open, it must be working correctly. As the reactance of the inductor or the reactance of the capacitor changes, the circuit allows current to flow out of the circuit and to the load as a result of these changes.

$$Q = \frac{X}{R} \qquad BW = \frac{Q}{Fr}$$

The level of this output current is based on the interactions at specific frequencies of both devices. Shorted turns on inductors or leakage through a capacitor's dielectric can cause major changes in the way these devices share energy, and how they react to energy demands at specific frequencies. For the technician, response of the circuit, or distortions in output signals, can be a result of a malfunction of a circuit component rated value. These types of problems are not always caused by just a short or open in a device.

Formulas for RLC Circuits

Vector Additions

In a series circuit the variable quantity is voltage. To find the unknowns of a series circuit you would first find the total impedance, then the circuit current, and from there, the voltage drops around the circuit. As a technician, this should help you have a better understanding of how a malfunction would cause changes in circuit readings.

Reactive net total = difference between the addition of all inductive reactance minus the addition of all capacitive reactances.

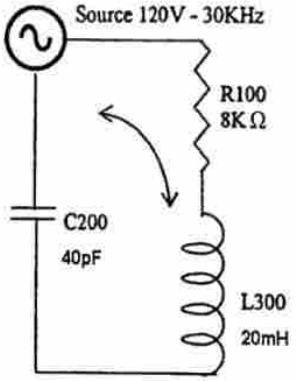
Remember - $X_L = 6.28 \times \text{Frequency} \times L$

$X_C = 1/6.28 \times \text{Frequency} \times C$

L is in Henries and C is in Farads

In a series circuit: (Current in a series circuit is the same through each component.)

R net = the addition of all series **resistive** components



$$XC = \frac{1}{6.28 * 30 \times 10^3 * 40 \times 10^{-12}} = 132.6K\Omega$$

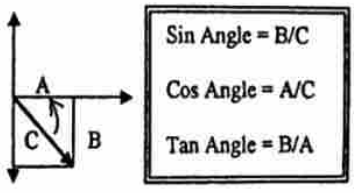
$XL = 6.28 * 30 \times 10^3 * 20 \times 10^{-3} = 3.7K\Omega$

$ZT = \sqrt{R^2 + \text{Reactance net}^2}$ $\quad I_T = V_T / Z_T$

$Z = 12.4K\Omega$ $\quad I = \frac{120V}{12.4K\Omega} = 9.7\ mA$

$\text{Tan}\,\theta = B/A = 9.5K\Omega / 8K\Omega = -55°$

CIRCUIT PHASE ANGLE



The analysis is based on a common frequency being applied to all components. If the frequency changes, the total circuit current and each reactance would change with it.

Voltage drops can be found by the use of Ohms law, once you have found the net resistance and circuit current flow.

$V_1 = I_T \times R_1$ $V_2 = X_L \times I_L$ $V_3 = X_C \times I_C$

In a Parallel Circuit

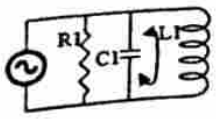
In a parallel AC circuit the source voltage is applied to each component. The phase components are current and impedance. Out-of-phase currents are shared (subtracted) between the inductive components and capacitor components. Inductor components add to other inductor currents, capacitor currents add to other capacitor currents. Resistive elements are in sync with the supply voltage and supply currents. All resistor branch currents add to other resistive branch currents.

Source Voltage = 120V
Source Frequency = 50 KHz
L1 = 10mH
C1 = 100pf
R1 = 300Ω

BASED ON THE VALUES GIVEN

$X_L = 6.28 * (50 \times 10^3) * (10 \times 10^{-3}) = 3.14K\Omega$

$X_C = 1 \div 6.28*(50 \times 10^3)*(100 \times 10^{-12}) = 31.8K\Omega$



$I_R = V_T \div R1 = 400mA$

$I_C = V_T/X_C = 3.77mA$

$I_L = V_T/X_L = 38.2mA$

$I_{net} = I_C - I_L$

I_{net} represents only the out-of-phase currents

$I_T = \sqrt{Ir^2 + I_{XNET}^2} = 401mA$

I_{XNET} = Difference between the Inductive and Capacitive Currents

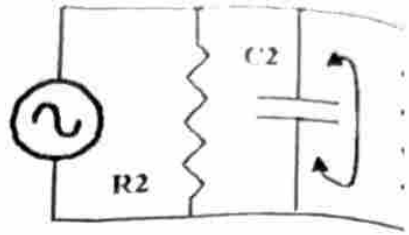
Impedance of an AC Circuit

$$Z_T = \frac{I^*_T}{I_T}$$

Impedance is the total opposition to current flow in an AC circuit; it has two components—magnitude and phase.

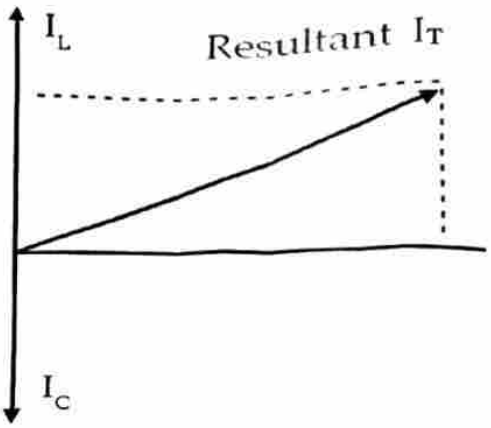
Parallel RLC Circuit

Figure 2



$V_T = V_{C2} = V_{L2} = V_{R2}$

Current Addition in a RLC Circu



In Figure 2 the current through the inductor the current through the capacitor, so the net cu circuit is a vector addition of the resistive curr resultant inductive current - (I_{L2} - I_{C2}).

If the current in the inductive branch was ec rent in the capacitive branch, the circuit woul parallel resonance and the circuit current wou current through the resistor branch section an leakage current through the reactive compone resonance, circuit current is at a minimum.

Another name for a parallel LC circuit is a tank energy from the capacitor electrostatic fi the energy from the inductor's electromagneti the circuit to share current between the induct The bandwidth response of the tank at resona the DC resistive characteristics of the inducto tive devices in the circuit.

Parallel Circuit Phase Angle F

$$\text{Sin } \theta = \frac{\text{Net Reactance Current}}{\text{Total Current in the circuit}}$$

$$\text{Cos } \theta = \frac{\text{Resistive Current}}{\text{Total Current in the Circuit}}$$

$$\text{Tan } \theta = \frac{\text{Net Reactive Current}}{\text{Resistive Current}}$$

Combinations of Series and Parallel Circuits

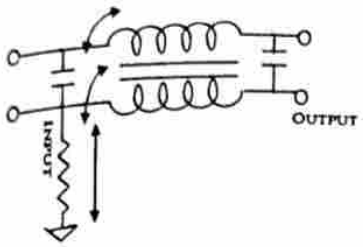


Figure 3

In Figure 3, a combination of a series and parallel configuration is shown. This is an example of the input choke (inductor) and capacitor network used to help reduce lightning and power surge damages to circuits. This configuration is very common in electronic devices. This circuit has the characteristics of both a series resonant and parallel resonant circuit.

In Figure 3, as input frequency increases (such as with a lightning strike or electrical noise), the inductor's reactance increases while the capacitor reactance decreases. Energy stored in the inductor will discharge through the capacitor with isolation provided and current limiting by the resistor to ground.

AC Passive Filters

Another common use for the series-parallel configuration is shown in Figure 4. The combination of a series and parallel resonant circuit is used to pass signals of specific frequencies while attenuation is provided for those frequencies outside of the resonant frequencies. L10 and C10 provide a series resonant bandpass filter while L20 and C20 provide a parallel resonance band stop circuit.

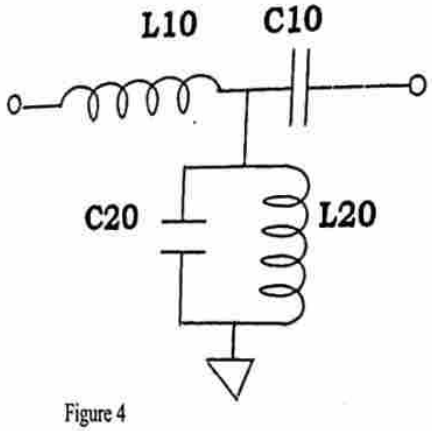


Figure 4

At specific frequency ranges the series circuit will pass the resonance frequency signals, while the parallel circuit will block the flow of these frequencies to ground. The C10 and L10 series components will attenuate those frequencies below or above the resonance frequency. L20 and C20 will aid in conduction of frequencies above and below the frequency of resonance to ground.

Each circuit component will have an effect on frequencies passed through the filter outside the bandwidth of the frequency of resonance. As frequency goes up, capacitive reactance will drop within C10 and C20. L10 and L20 inductive reactance will increase as frequencies rise above resonance frequency. The inductive reactance will increase, while the capacitive reactance decreases. At frequencies below the frequency of resonance, the capacitive reactance will increase in C10 and C20, while the inductive reactance of L10 and L20 will decrease.

At resonance the series components L10 and C10 will have their smallest impedance, and L20 and C20 will have their maximum impedance to the signal frequency. This type of combination allows set bandwidths of frequencies to pass or be conducted to ground. Signal processing is a very common use for these devices within highly integrated circuit applications. Use of a scope and voltmeter can easily locate faults in these filter circuits if you know what signal characteristics you need to find, and what levels of signal should be present.

Basics of AC Devices with R, L and C

Let's look at how AC circuits function from a technician's viewpoint. See Figure 5. When approaching an AC circuit troubleshooting problem you need to have a good understanding of how devices function under various frequency characteristics, as well as various circuit configurations. Some devices can defy logical thinking (such as capacitors). How does a capacitor really work?

One mystery to many is how can you couple (pass) an AC signal through a capacitor. The dielectric of the capacitor is an insulator that will block current flow, so how does the input signal reach the output side of the capacitor if its path in the circuit is blocked by the capacitor?

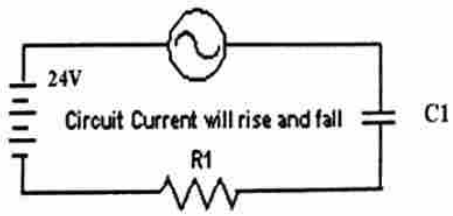


Figure 5

C1 will charge to 24 Volts - The AC source will increase and decrease the voltage drop across R1.

How can a capacitor stop the flow of DC current through the device, yet allow the flow of an AC current in the circuit? The answer is simply to understand that current flows from a difference in electrical potential. When the voltage of the signal is applied to the dielectric of the capacitor it creates a difference in potential in the dielectric material inside the capacitor. The stored charge induces a reflected voltage across the capacitor through its plates to the circuit it lives in.

The capacitor dielectric stores charge, which is why some capacitors can hold a charge even when power is removed. The capacitor dielectric's difference in potential causes the circuit to have a reflected difference in potential in the circuit. With the voltage developed within the capacitor, current changes occur in conductive paths that are felt by the circuits connected to the capacitor's leads.

So how can this help the technician?

Suppose a capacitor develops an open in the lead of the device. The capacitor will never develop a charge in the dielectric, so no charge - no potential difference in the conductive paths connected to the circuit. You would measure an input signal to the capacitor and may even see a small signal at the output of the capacitor. With a smaller potential difference into conductive paths from the capacitor, the circuit driven from the output side of the capacitor may reduce or stop its operation.

Power Use - True Power & Apparent Power

Capacitor devices and inductive devices have two distinct characteristics not shared by most purely resistive devices. Inductor components will develop opposition to the flow of current by the creation of a magnetic field that will oppose the source current. Capacitor devices develop electrostatic charge, which will oppose the source, as the voltages are developed directly opposing the source voltage. Resistive device applies a direct relational resistance to an applied source. Secondly, capacitors and inductors store energy and then return it to the circuit.

The electrostatic field held in the dielectric of a capacitor has a potential voltage in an opposite polarity to the inductive device magnetic field. When two sources of stored energy are connected in a conductive path, the voltages will either aid or reduce the effect of the energy stored in each device.

The inductor may be a motor, with big current demands when it first starts to turn. In the motor circuit could be a capacitor with stored energy in its electrostatic field to aid in applying energy to the motor. The source voltage has a lead of 90 degrees before it can supply a maximum potential into the motor. So when the motor inductor demands are at their peak, the motor's supply voltage is at 0V from the source. The capacitor voltage is available at this time to aid the induc its energy demands.

True power is the actual power used in the circuit to ge work out of the circuit devices. This comes from the sour and is always equal to current squared times the resistanc seen by the source. Apparent power is the supply voltage the supply current times the power factor. Transformers, r tors, and other inductive devices are rated by the Volt/Amp (VA rating) they use. They not only consume power but st power that is then used to create work by motors, lighting circuits, and other devices driven by inductive circuits.

Power factors can be adjusted by adding capacitor banks to inductive circuits to bring the power factor closer to unit Unity power factor of 1 allows maximum power to be used The higher the power factor, the more in-phase the voltage current are with respect to each other.

To efficiently use power you would want the source to be able to directly supply energy as needed to each device. Whe a perfect balance of sharing power takes place, the circuit has a power factor of 1.0. A power factor below 1 indicates an imbalance between the apparent and real power of the circuit. The technician must be able to look at circuit devices and "see" how the devices function together to produce the real power needed, not just the apparent power of each devic independently.

AC Circuit - Real Power = True Power

$$P_A = \text{Apparent power} = I^2R = (\text{or } V \times A)$$

$$P_T = \text{True Power} = I^2R = (VA) \times \text{Cos } \theta$$

$$PF = \text{Power factor} = \text{Cos } \theta = \frac{P_T \text{ (True Power)}}{P_A \text{ (Apparent Power)}}$$

$$PF \text{ also equals } PF = \frac{R \text{ (Resistance)}}{Z \text{ (Impedance)}}$$

In the Figure 5 circuit, the capacitor gives the motor additional starting current. This can be as high as four times its run current. Sensors detect a slow motor start and shut the motor off to protect it from overheating. You need to "see" how the capacitor helps the motor at starting times to repair this circuit. Capacitors can supply energy to an inductor field.

Analyzing an AC Circuit

If you find changes in voltages in series sections - remember

that a series capacitor stores signal voltages in an electrostatic field. If the capacitor has leakage within the device's dielectric, its ability to store signal energy is reduced. An open in the capacitor will stop, or reduce, signal energy from being shared with another component.

In parallel circuits, selectivity is the key factor. Distortion of the FM envelope, or passage of unwanted signals would be an indication of problems. This may result in large spikes that can destroy sensitive electronic devices, as occurs in TV horizontal output or high power amplifier circuits.

The Whole Picture

Lets start with an RF (Radio Frequency) Signal Circuit: A transmitter sends an AC signal out into the air. It travels across an antenna cut to a size to allow the wavelength of the signal to mechanically resonate across the antenna and induce signal energy which travels into the receiver. The signal contains a carrier signal and a modulated signal with the encoded transmitted information. The signal must pass through circuits and/or be blocked, or shunted to ground, by other AC circuits. Due to the extreme low level of voltage of the signal, the signal must also be amplified.

The signal will travel into a series AC circuit tuned to specific frequencies by the values of the inductor and capacitor. Resistors are used to set levels of current and biasing of other circuit devices such as transistors or IC chips. The series components allow specific bands and bandwidths of frequencies to pass through to the next stage. Connected across the signal path may be parallel tank circuits designed to take to ground signals outside of the selected bands and block the flow to ground of specific frequencies passing through the series components.

As the signals are amplified and selectively pass through the circuits, any unwanted signals are shunted to ground, and the final clean signal is sent to the final amplifiers, such as video or audio amplifiers.

When troubleshooting, you need to know what is passed in each section of the circuit and what level these passed signals should be. Substitute where needed to isolate faults, and watch closely for any changes in voltage measurements from those supplied on your schematics. For example, you measure a positive voltage where you should have a negative voltage. This may be an indication of an open or short in a capacitor in the circuit. Why a capacitor? They are the most likely culprits.

AC Circuit Configurations

AC circuits are used to amplify an AC waveform, or create a wave shape such as an oscillating waveform used in tuning or frequency phase adjustment circuits. They can be used as passive or active filters, or even to adjust DC voltage levels such as with switch mode power supplies. AC circuits are used for many additional circuit characteristics. From a technician's view, understand when working in an AC circuit it is critical to find the correct frequency, shape, magnitude and phase of an AC signal when you troubleshoot. Avoid the "Got Some" syndrome - understand that frequency is critical in order to pass energies through an AC circuit as well as magnitude. A low signal voltage may just be a signal being supplied around a defective circuit through other components and conductive paths.

NOTES

Chapter 6

Chapter Quiz

1. Total impedance in a series circuit is found by:

a. adding up all the series resistance and reactance values.
b. subtracting all the inductive reactances from all the capacitive reactances and adding the resistors' values.
c. finding the net reactance and the net resistance, square each separately, add them together and then find the square root of the sum.
d. divide the applied voltage by the addition of all the circuit components individual currents.

2. In a series AC circuit with a resistor, inductor and capacitor, each with the same resistance or reactance, a voltmeter measuring each individual voltage drop would find:

a. the voltages equal to each other.
b. the reactance voltage of the capacitor or the inductor would be greater than the resistance voltage measurement.
c. the resistance voltage would be higher than the voltage across the inductor or the capacitor.
d. the supply voltage would be measured across the resistor and the voltage across the capacitor and across the inductor would measure 0V.

3. Which of the following is not a characteristic of a series circuit?

a. Circuit components have a division of the supply voltage.
b. All components share one current.
c. Energy is shared by reactive devices.
d. There is a different level of current in the reactive components than in the resistive components.

4. In an AC circuit, as the applied voltage frequency changes upward, the inductive reactance in the circuit will:

a. decrease.
b. increase.
c. stay the same.
d. either increase or decrease.

5. In an AC circuit, as the applied voltage frequency increases the capacitive reactance in the circuit will:

a. decrease.
b. increase.
c. stay the same.
d. either increase or decrease.

6. A 50KHz AC voltage is applied to a 10mH inductor. Its reactance at this frequency is:

a. 3140 Ohms.
b. 300 Ohms.
c. 3100 Ohms.
d. 500 Ohms.

7. A .1 µF capacitor in a circuit with a frequency of 100 KHz applied has a reactance of:

a. 1.59 Ohms.
b. 15.9 Ohms.
c. 159 Ohms.
d. 15.9 kOhms.

8. In a series circuit when the inductive reactances equal the capacitor reactances the circuit has _____ impedance.

a. maximum
c. double the inductors
c. minimum
d. no

9. In a parallel circuit when the inductive reactances equal the capacitor reactances the circuit has _____ impedance.

a. maximum
b. double the capacitor's
c. minimum
d. no

10. Resonance is when _____.

a. all resistances in an AC circuit are equal to each other
b. all reactances are equal on each reactive component
c. the total of the inductive reactances in the circuit equal the total of all capacitive reactances
d. no current is flowing through an AC circuit when a voltage is applied

ANSWERS on page 223

Chapter 7

Generators, Alternators and Motors

by Chief Petty Officer James Bruening, CET

James Bruening currently serves in the US Navy as an Aviation Electronics Technician, Chief Petty Officer. In his 20 years of Naval service, Jim has fixed radar, radios, sonar, data link, military computers, and electronic navigation systems for Navy aircraft. In addition to certification as an Industrial Electronics C.E.T. he has a BS in Electronics Technology (Magna Cum Laude) from Regents College and an AS in Physics (Honors) from San Diego Mesa College.

Generators

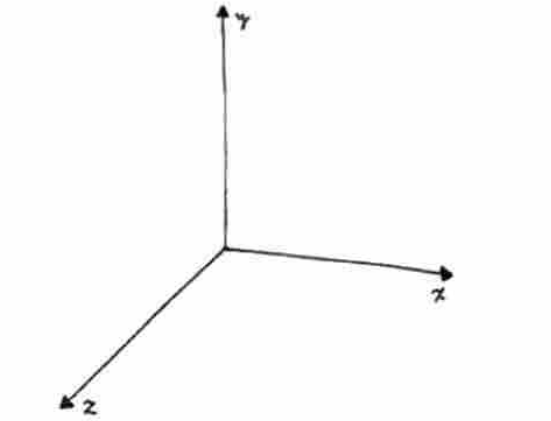
A generator is an electric machine that converts motion into electricity. All generators are driven by some external prime mover. Steam, wind, water, and internal combustion engines are examples of prime movers. The generator converts the mechanical energy of the prime mover into electrical energy according to Lenz's Law and the law of electromagnetic induction.

Lenz's Law and the Law of Electromagnetic Induction - Lenz's Law states that a changing magnetic field will induce a current in a conducting loop and that the induced current will seek to produce a magnetic field which opposes the changing magnetic field that produced it (an example of conservation of energy). Lenz's law explains the need for greater effort from the prime mover when a generator is heavily loaded.

Understanding Lenz's Law leads us to an understanding of the law of electromagnetic induction, which states that a voltage is produced in a conductor in motion across magnetic lines of flux.

The Left Hand Rule for Generators - The polarity of the voltage produced in the conductor depends on the direction of motion of the conductor and on the direction of the magnetic lines of flux through which the conductor is moved. The Left Hand Rule for generators is a useful tool for determining the direction of current flow and the polarity of the induced voltage. The thumb, forefinger and middle finger of the left hand are extended perpendicular to one another to determine the relationship between motion, flux, and current. The three dimensional axes and accompanying table of Figure 1 should assist in visualizing this concept.

Our interest in the Left Hand Rule is primarily to illustrate that current flow in a conducting loop that is rotated within a magnetic field will reverse direction as the loop is rotated on an axis. This continuously reversing current is known as Alternating Current (AC).



AXIS	"FINGER"	DIRECTION OF ...
y	Thumb	Conductor motion through magnetic flux lines
x	Fore	Magnetic flux lines ("flowing" from North pole to South pole)
z	Middle	Electrical current flow induced in conductor

Figure 1 - The "Left Hand Rule for Generators"

Alternators

AC Generators - An AC generator, also known as an alternator, consists of multiple conducting loops rotated within a magnetic field. The voltage produced from the conductor's motion through the field is the output of the generator. Figure 2 is a simplified single loop alternator illustrating the alternating output. As noted before, the reversing current flow induced in the rotating conductor yields our AC output.

The basic components of an alternator are the field, the armature, the slip rings, and brushes. Alternators typically use electromagnets to set up the magnetic field through which the conducting loops are rotated. The windings which produce this magnetic field are called the field coil(s), or simply, the field. The armature contains the windings from which the AC output is taken. In most alternators, the output voltage can be controlled by adjusting the field coils' excitation voltage.

The slip rings provide electrical contact from the rotating loops to an external circuit via the brushes. The slip rings may provide current to the rotating windings or take current from them. If the rotating windings are to act as the field, then current is supplied via the slip rings.

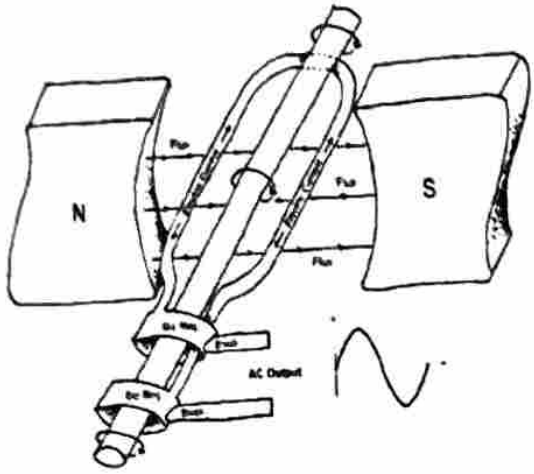


Figure 2 - Simplified AC Generator

Remember, the armature is the part of the alternator from which the output voltage is taken. The armature in an alternator may be the rotating windings or the stationary windings. The rotating windings, be they armature or field, are known as the rotor. The stationary windings are known as the stator.

Three-Phase Power - Older automobile generators (pre 1960) are single phase machines. The output taken from the armature was a single sine wave. However, the most common AC power is produced in three-phases.

Three-phase power consists of three distinct sine waves, each lagging the phase before it by 120 degrees when viewed on an o-scope.

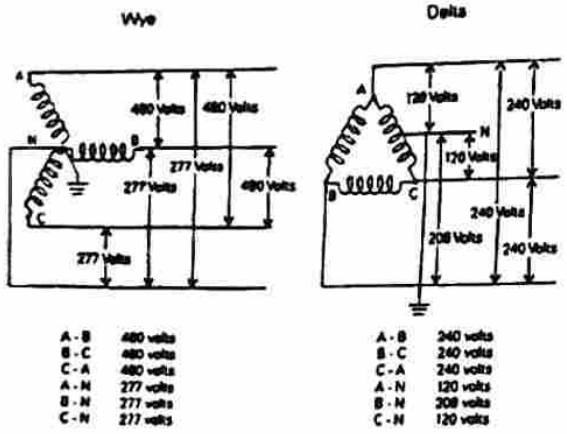


Figure 3 - Three-phase power connections

Three-phase power can be manipulated to obtain the desired voltage level and the desired number of phases by altering the electrical connection of the stator windings. For three-phase power, the stator is used as the armature. The two basic methods of connecting AC power are Wye or Delta connection of the stator. See Figure 3 for an illustration of three-phase connections.

DC Generators - Recall that an AC generator output is taken from slip rings. In a DC generator, the slip rings are replaced with a commutator. The commutator in a DC generator is a ring, cut into sections that are insulated from one another and that encircle the shaft of the rotor. Each section of the commutator is connected to one end of the rotating conductor loop such that a complete current path is provided from one section of the commutator to another, via the conducting loop.

The effect of commutation is that the alternating current in the rotating loop is picked off the commutator as voltage of a single polarity, i.e., Direct Current (DC). Figure 4 illustrates a basic commutator and its effect on generator output.

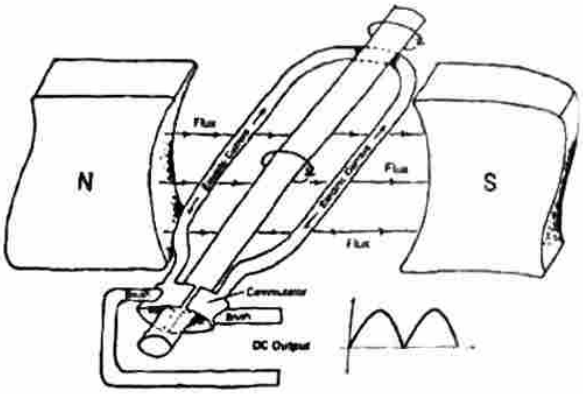


Figure 4 - Simplified DC Generator

DC Generator Output Ripple - The single rotor loop DC generator in Figure 4 is for illustration only. A single rotor loop would produce an output that looks like the top signal in Figure 5. In practice, the rotor will consist of many windings to produce a cleaner DC signal with greater voltage potential and less ripple. The bottom signal of Figure 5 illustrates the effect of adding armature coils.

Generator Self Excitation - Generators, both AC and DC, are often constructed such that the field excitation voltage is provided by the generator itself. This is accomplished by wiring the field coil in series with the armature, shunting the armature (in parallel), or wiring the field as a compound combination of both series and shunt.

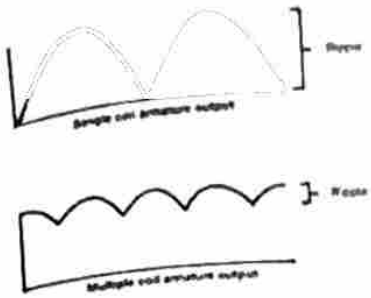


Figure 5 - The effect of multiple armature coils in a DC generator

Series wound generators are basically wired as illustrated in Figure 6a. Series wound field coil generators are not practical because the output voltage varies with the load current.

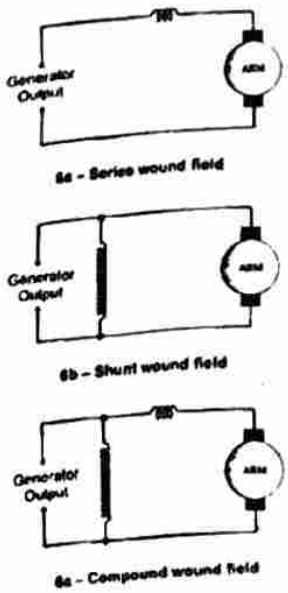


Figure 6 - Self-excited generator field configurations

Shunt wound generators basically shunt the armature with the field coil(s). Refer to figure 6b. Shunt wound field coils cause the output voltage of the generator to vary inversely with load current. Shunt wound generators are also not practical for this reason.

Compound wound generators are a practical method of generator self-excitation. The field coils are wired in a combination of series and shunt, causing the undesired output variations to cancel. See Figure 6c for an illustration of a compound wound generator.

Armature Losses - Most electric machines experience losses due to the unwanted conversion of usable energy to heat. These losses are known as copper loss, eddy current loss, and hysteresis.

Copper loss in the windings of a generator or a motor occurs because heat is produced whenever an electric current flows in a copper wire. Copper loss occurs when some of the energy of the current escapes in the form of heat.

Eddy current losses occur when the strong magnetic fields in a generator or a motor induce currents in the iron core(s) about which the windings are turned. The induced currents allow energy to escape from the machine, again in the form of heat. To avoid eddy current losses, the cores in an electric machine are made of laminated iron plates to reduce the amount of induced current flowing in the core.

Hysteresis losses occur as the molecules of the machine's core material attempt to align themselves with the magnetic fields of the machine. Since the fields are constantly in motion, molecular friction occurs as the molecules constantly attempt to realign themselves. The energy of the molecular friction is radiated as heat.

Motors

A motor is an electric machine that converts electrical energy to mechanical energy. Recall that magnetic poles of opposite polarity attract each other, while poles of the same polarity repel each other. Motors use magnetic attraction and repulsion to do work.

Motors consist of the same basic components as generators. In a motor, an electric current is supplied to the armature causing it to act as an electromagnet. When exposed to a magnetic field, the magnetized armature poles are attracted and repulsed by the poles of the field, causing the rotor to turn.

Lenz's Law and counter emf - Current is induced in the armature coils as the rotor of an electric motor rotates through the field coil's magnetic lines of flux. This is the same effect as found in generators to produce the generator output voltage, but in motors this current opposes motor operation. This current sets up a counter emf that opposes the applied voltage and limits the flow of drive current in the armature, limiting the speed of the motor. Lenz's Law is responsible for counter emf.

DC Motors - A DC motor is constructed exactly like a DC generator, except that current is supplied to the armature (rotor) instead of taken from it. Current direction is reversed in the armature windings with each revolution of the armature as the current is passed through the DC motor commutator. This reversing action of the current in the armature coils causes the magnetized armature to reverse polarity as it rotates through the field.

The commutator elements are placed such that the armature polarity attracts and repels the field polarity at the moment needed to promote rotation.

DC Motor Field Configurations - DC motors, like DC generators, can be connected with the field windings in series and/or in shunt with the armature. This allows the same current to magnetize the armature and to set up the field. Also, the field strength of a motor can be adjusted by adjusting the field voltage.

The series wound DC motor's speed varies greatly with the mechanical load placed on the motor. With no load applied, the series wound DC motor will speed dangerously out of control. For this reason, unloaded operation of a series wound DC motor is not recommended. Series wound DC motors have very good starting torque, making them ideal for use in small battery operated appliances and portable power tools.

The field strength of the shunt wound DC motor is not dependent upon the physical load on the motor as it is in the series wound motor. This is because the field current is in parallel with, and thus independent of, the armature current. Shunt wound motors have less starting torque than series wound motors, making them less attractive for applications where large torque from a dead stop is desired.

Compound wound DC motors enjoy the characteristics of both the series and shunt wound fields. A compound wound motor has strong starting torque (from its series wound field windings) and some speed regulation regardless of physical load (from its shunt wound field windings). See Figure 6 for a summary of series, shunt, and compound connections (note that instead of an output voltage, a motor will have an input voltage).

DC Motor Starting Current - The starting current of a DC motor can be as high as 600 percent of its rated current. For this reason, resistors are added to the armature circuit during start-up, limiting armature current to rated values. These resistors are removed from the armature circuit either manually or automatically as motor speed increases. Some external resistance may be left in the armature circuit to control final motor speed. Figure 7 illustrates motor current versus speed.

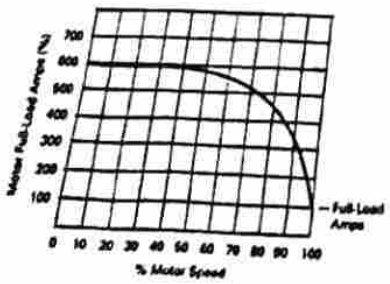


Figure 7 - Motor current vs speed

AC Motors - AC motor speed depends on the frequency of the voltage supplied to it. This makes AC motors well suited to constant speed applications.

In a single phase AC motor, the speed of the motor is directly dependent upon the frequency of the input voltage. The polarity shifts of the input voltage cause the armature polarity to alternate, promoting rotation as discussed in the section on DC motors.

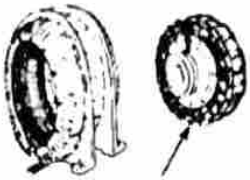
Multiphase AC motor operation requires an electrically rotating magnetic field to be set up by the stator. Remember that the stator windings are stationary and the rotor windings rotate. The primary power to the motor is a multiple phase AC signal that is applied to the stator and creates a field that rotates around the rotor. The rotor can be provided just enough external power to turn it into an electromagnet. The polarized rotor follows the rotating stator field similar to a compass following a magnet rotated around it.

The most common types of multiphase AC motors use three-phase power. Each phase of the supply voltage is applied to one of three sets of windings in the stator, producing the desired rotating field.

Types of AC Motors - Three types of AC motors are the series AC motor, the synchronous motor, and the induction motor. Series AC motors are wired similar to series wound DC motors. These motors are single-phase machines and their speed is directly dependent upon the frequency of the supply voltage. The series wound AC motor has strong starting torque just like the series DC motor. Having strong torque from a dead stop, series AC motors are ideal for fans, small appliances, and power tools.

Synchronous motors are most commonly three-phase machines. As such, they rely on a rotating magnetic field from the stator to operate. Synchronous motor speed is synchronized to the speed of the rotating magnetic field and is thus constant regardless of physical load.

Without assistance, a synchronous motor has no starting torque. This is overcome by wrapping the rotor windings in a conducting cage known as a squirrel cage. Enough current is induced in the squirrel cage to nudge the rotor up to a speed at which it can catch hold of the rotating magnetic field. See Figure 8 for an illustration of a rotor squirrel cage.



Squirrel-Cage Winding
Over Salient Pole Windings

Figure 8 - Self-starting synchronous AC motor

Induction motors are the most common type of AC motor. They are simple and rugged. The rotor windings of an induction motor are also wrapped in a squirrel cage. The current induced in the squirrel cage provides all of the rotor polarization required to run the motor. The induction motor rotor needs no external supply. The rotor in an induction motor, not being connected electrically to any external voltage, needs no slip rings or brushes.

When the induction motor is first powered up, the rotating magnetic field crosses the conducting loops of the squirrel cage in the rotor. This induces a current in the rotor that produces a magnetic field around the rotor. The now polarized rotor begins to rotate in the direction of the field rotation and eventually almost catches up to the rotating field. The rotor can never fully catch up to the rotating field because to do so would demagnetize the rotor (remember that a conductor must cut lines of flux for a current to be induced). The difference between the speed of the rotating magnetic field and the speed of the rotor in an induction motor is called slip. An induction motor must have slip in order to operate. The amount of slip is directly proportional to the physical load placed on the motor.

Another type of AC induction motor is a single phase machine that uses a capacitor and a starting coil in series to create starting torque. This starting circuit is wired in parallel with the rotor. The values of resistance and capacitive reactance in the starting circuit produce starting-winding currents 90 degrees out-of-phase with the current in the main field winding. This results in a two-phase rotating field which drives the rotor.

Summary and Additional Info

- Counter-voltage is a result of Lenz's Law.
- Lenz's law explains the need for greater effort from the prime mover when a generator is heavily loaded.
- The law of electromagnetic induction states that a conductor in motion in a magnetic field will produce a current in the conductor.
- The left hand rule applies to generators.
- An AC generator's output is coupled via slip rings.
- In most alternators, the output voltage can be controlled by adjusting the field winding voltage.
- A commutator will make a generator produce DC instead of AC.
- When the field current of a DC generator is provided by the generator, it is called a self excited generator.
- Eddy currents are controlled by laminating the iron core.
- In a motor, counter emf subtracts from supply voltage.
- The field strength of a motor can be adjusted by adjusting the field voltage.
- A commutator can be found in a DC motor but not in an AC motor.
- A series wound DC motor will self destruct without a mechanical load.
- A shunt wound motor has fairly constant speed with varying load.
- A drill or a mixer would be powered by a series wound motor.
- Manual starters for DC motors provide minimum armature current during startup.
- In a DC motor, resistors are placed in series with the armature to control startup current or to adjust the final speed of the motor.
- A rotating field in an AC motor is developed by using three phase power to three sets of stator windings.
- A squirrel cage is sometimes used in AC motor rotors.
- The rotor in an induction motor is not connected to any supply voltage.
- An induction motor must have slip in order to run.
- A capacitor start AC motor uses a two phase rotating magnetic field.

NOTES

Chapter Quiz

1. _____ cells are prone to battery memory.

   a. Zinc
   b. NiCad
   c. Alkaline
   d. Lithium

2. A conducting loop in motion through magnetic lines of flux has a current induced in it. The physical law that explains this is what?

   a. Lenz's Law
   b. The law of inertia
   c. The Compton shift equation
   d. Kirchoff's current law

3. An AC generator is also known as a _____.

   a. regulator
   b. alternator
   c. phasor
   d. slip ring

4. Three-phase power is obtained from an AC generator by wiring the armature windings in a _____ or _____ configuration.

   a. alpha, beta
   b. treble, tri-phase
   c. polymorph, one-to-three
   d. delta, wye

5. What element of a DC generator distinguishes it from an AC generator?

   a. Slip rings
   b. Commutator
   c. Series wound field coil
   d. Armature

6. What type of motor will speed dangerously out of control if not physically loaded?

   a. Induction motor
   b. Shunt wound DC motor
   c. Series wound DC motor
   d. Series wound AC motor

7. AC motor speed depends on _____.

   a. the frequency of the voltage applied to it
   b. the voltage level applied to it
   c. the phase of the voltage applied to it
   d. the power left in the dilithium crystals

8. Series wound AC motors have strong torque from a dead stop. This makes them ideally suited for use in small appliances.

   a. True
   b. False

9. What element of a synchronous AC motor provides startup torque for the motor?

   a. Slip rings
   b. Salient poles
   c. Squirrel cage
   d. Armature

10. An AC induction motor requires what to operate?

    a. Input power
    b. Slip rings
    c. Slip
    d. Rotor battery

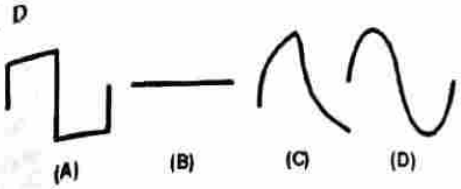
ANSWERS on page 223

1. **What happens to the value of a capacitor if the spacing between plates decreases?**

   a. Increases
   b. Remains the same
   c. Decreases
   d. Varies

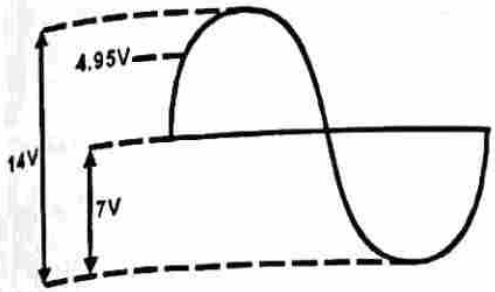
2. **Which waveform below is a sine wave?**

   a. A
   b. B
   c. C
   d. D



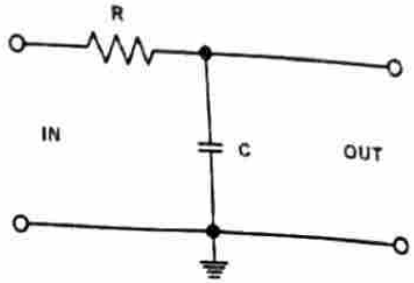
3. **The peak-to-peak voltage of the below waveform is:**

   a. 7 V.
   b. 14 V.
   c. 4.95 V.
   d. .707 V.



4. **The circuit below will require how many RC time constants to charge the capacitor to 63% of the applied voltage?**

   a. 1
   b. 2
   c. 63
   d. 100



5. **A variable capacitor's plates can open or close. As the plates open, capacitance:**

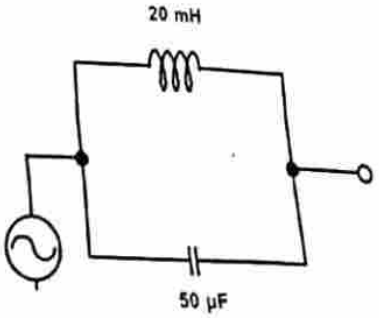
   a. remains the same.
   b. capacitors aren't made this way.
   c. decreases.
   d. increases.

6. **An inductor with the same length of wire but whose turns are spaced further apart will have:**

   a. less inductance.
   b. more inductance.
   c. equal inductance.
   d. more mutual inductance.

7. **What frequency will the parallel LC circuit below be resonant to?**

   a. 100 MHz
   b. 159 Hz
   c. 6.28 MHz
   d. 1590 KHz

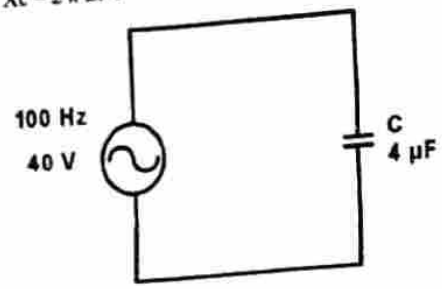


Unit 3

Review Questions - AC Electronics
Chapters 5, 6 and 7

8. **Which formula is used to calculate reactance in the following schematic?**

a. $Xc = \frac{1}{2\pi FC}$
b. $Xc = 2\pi FC$
c. $Xc = R^2 + L^2 + C^2$
d. $Xc = 2\pi LFC$



9. **An inductor in an electrical circuit:**

a. causes current to lead voltage.
b. opposes the flow of AC current.
c. appears capacitive to AC current.
d. appears as a short circuit to AC current.

10. **The term for the combination of resistance and reactance that restricts the flow of alternating current is:**

a. volt-amperes.
b. impedance.
c. susceptance.
d. power factor.

11. **Is the value of a 500 picofarad capacitor and a .5 nanofarad capacitor identical?**

a. Yes
b. No

12. **In a generator, a DC voltage applied to the field windings controls the output level.**

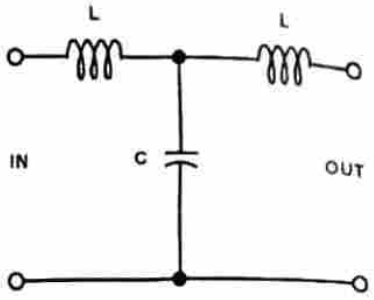
a. True
b. False

13. ***Which of these is within tolerance for a resistor color coded red, black, red, gold.***

a. 1850 ohms
b. 2150 ohms
c. 1950 ohms
d. 2250 ohms

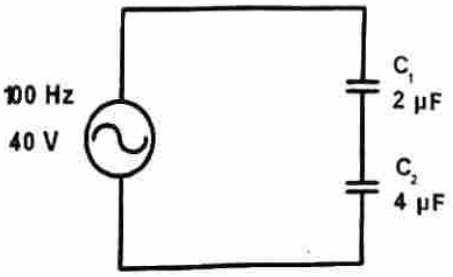
14. **Which one of the below is a common description of the filter shown below?**

a. Bandpass
b. Low pass L
c. Low pass T
d. High pass T



15. **What is the combined reactance of the two capacitors in series in the drawing below?**

a. Approximately 1,198 ohms
b. Approximately 265 ohms
c. Approximately 24 ohms
d. Approximately .00377 ohms



16. **Which one of the below will cause a generator to produce DC voltage rather than AC voltage?**

a. Brushes
b. Armature
c. Commutator
d. Field windings

17. **Total capacitance of capacitors connected in parallel is like:**

a. resistors connected in series.
b. capacitors connected in series.
c. resistors connected in parallel.
d. inductors connected in parallel.

18. Which factor determines the value of a capacitor?

a. Area of the plates
b. Temperature coefficient of the dielectric
c. Inductance of the dielectric
d. Insulation covering

19. A 3/8-inch diameter by 3/16-inch ceramic disk capacitor marked "47" is most likely what value?

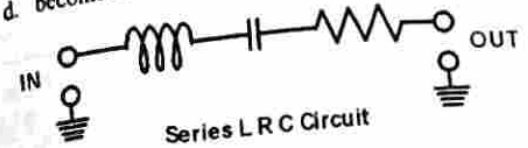
a. 47 nf
b. 47 uf
c. 47 pf
d. 47 mf

20. What happens to current flow in a capacitor when the DC voltage across it has increased to the applied voltage?

a. Current flow is maximum in capacitor.
b. Little current flows.
c. Polarity reverses.
d. Voltage drop is maximum.

21. Decreasing the inductance in a series tuned circuit (shown below) will cause the resonant frequency to:

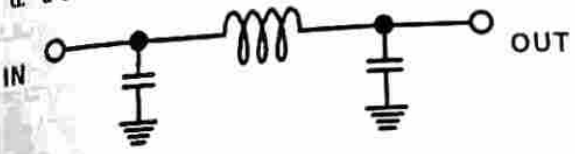
a. increase.
b. decrease.
c. there will be no change.
d. become unstable.



Series L R C Circuit

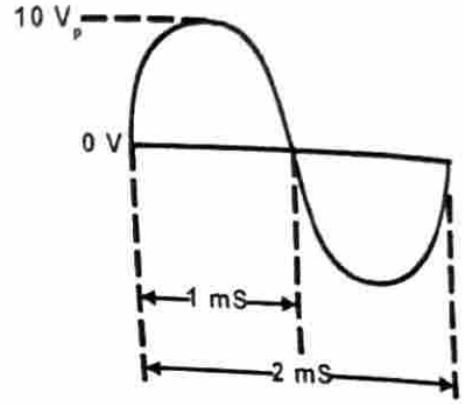
22. The filter configuration shown below is commonly called:

a. an "L" filter.
b. a "PI" filter.
c. an LCL filter.
d. a CLC filter.



23. The sine wave shown below is at what frequency?

a. 500 KHz
b. 50 KHz
c. 5 KHz
d. 500 Hz



ANSWERS on page 223

Chapter 8

Electronic Components

by Tom Glass, CET

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Introduction

In this chapter, we will discuss the components commonly used in electronic devices today.

Batteries *by James Bruening*

Types of Batteries

A battery is a series of cells, in which chemical energy is converted to electrical energy. Batteries deliver Direct Current (DC) to provide power to a device independent of an external supply. Primary cells cannot be recharged and are disposable where secondary cells are rechargeable. Types of primary batteries in use are carbon- zinc, alkaline, and lithium. Rechargeables are lead-acid, nickel-cadmium (NiCad), Nickel Metal Hydride (NiMH) and lithium ion.

Primary carbon-zinc cells have powered flashlights for years. A new carbon-zinc dry cell provides 1.5 vdc.

Alkaline batteries have greater capacity than carbon-zinc and will provide a 1.5 vdc output three to eight times longer.

Lithium batteries are the tiny button cells used in wrist watches and on PC mother-boards to retain BIOS settings. Lithium cells provide 1.5 v, and some are 3 v. A lithium cell will provide a minute amount of current for 5 to 10 years. And it is ideally suited to power portable micro-circuitry with low current demand such as volatile memory.

Rechargeable lead-acid batteries are used in automobiles, motorcycles, etc., to provide starting power and a reliable source of DC. Each cell of a lead-acid battery delivers 2.1 vdc when fully charged and exhibits very low internal resistance or very high current output demands.

A rechargeable type of alkaline cell will provide 1.5 vdc and withstand being recharged about twenty-five times before ceable performance degradation occurs.

ckel-cadmium (NiCad) cells deliver 1.2 vdc when fully ed. In practice, a NiCad cell can be recharged 1000 and will often develop a "memory effect" that degrades charging capacity. To avoid the "memory effect," a NiCad should be completely discharged before being recharged. Newer NiMH and Lithium Ion batteries do not have the memory effect and have greater energy-to-weight ratios.

Battery Ratings

The power delivering ability of a battery is given in "amp-hours (Ah)." A battery rated at 5-Ah will deliver the equivalent of 5 amps for one hour. To arrive at the practical power delivering ability of a battery from the rated Ah, divide the rated amps by the number of hours of desired current. For example, 5-Ah is just a convenient way of saying that a battery will deliver one tenth of 5 amps (500 milliamps) for ten hours.

Table 1

Primary Cells	Voltage	Ah	Cycle Life	Internal Resistance
Carbon-Zinc	1.5	1	-	Medium
Alkaline	1.5	4	-	Medium
Lithium	1.5 - 3.6	.5	-	High
Zinc-Air	1.5	1	-	Medium
Secondary Cells				
Lead Acid	2.1	100	High	Very Low
Nickel Cadmium	1.2	5	Medium	Low
Nickel Metal Hydride	1.2	10	High	Low
Lithium-Ion	1.2	4	High	Medium

Lithium-Ion rechargeable batteries are currently used in a majority of small consumer products.

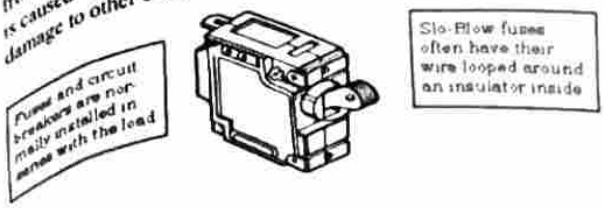
Other power producing components

Some devices like thermocouples produce a small amount of energy that is measured in the millivolt range. Other

devices such as photocells and solar cells will produce .5 volts to several volts. Piezo-electric crystals have a very low output and are found in RF applications for very stable frequencies

Fuses and Circuit Breakers

Circuit protection devices are designed to fail or interrupt when a certain current level is exceeded, protecting a circuit from excessive current flow. Excessive current flow usually is caused by a failing component or accident and can result in damage to other components and circuits.



Circuit Breaker

Fuses destruct when the rated current is exceeded. They are usually made of a soft metal or thin wire. When they fail, they must be replaced. Some, called "slo-blow," are made so that they fail only after a second or so, thus allowing short spikes of over-current to pass. Fuses come in a great number of physical shapes, sizes and mounting configurations. A fuse of 1A would be considered small, 30A would be large.

There also are thermal fuses that fail when their environmental temperature exceeds a design level. These protect a device when overheating occurs near the fuse.

Circuit Breakers protect circuits in the same manner as fuses do, but may be "reset" after failing. They usually are made of a coil that will open an internal switch when the design current level is exceeded. They are made in a wide variety of shapes, sizes and mounting configurations.

Insulated wires

Small gauge insulated wire is commonly found in electronic circuits. You may not think of wire as a component, but it is the connecting link between many devices. Two systems are used to identify wire size: the American Wire Gauge (AWG) and B&S. The larger the gauge number the smaller the conductor diameter. Electrical publications list wire tables that show diameters and current capacity of each size.

Resistors

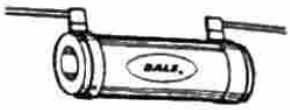
Resistors are used to control current flow or develop voltage levels in circuits. They are made of carbon, wire, resistive film or other resistive materials. They are designed for a value of measured in "ohms." With a given resistance, applying a voltage across a resistor will cause an amount of current (amperes) to flow through it, dissipating a certain amount of power (watts). If the power rating is exceeded, the resistor will fail (overheat or open).

Potentiometer

A resistor of .5 ohm would be small in value, whereas one of 500 meg-ohms would be a large value. They may be similar in physical size even thought the resistance value is much different. But with power rating, size varies, where 1/4 watt is small, while 20 watts is large in physical size.

Tolerance is how much the resistance value may be in error from the rated value. Actual resistance may vary from the rated value by as much as 20% and is indicated on most resistors. Common tolerance values are 5%, 10% and 20%, with newer precision and resistor packs as low as .1%.

Most resistors are fixed in resistance. But there are special types such as the rheostat, which is adjustable; the potentiometer, which has a moveable tap; the varistor, which will vary in resistance depending on the voltage applied.



Fixed Resistor

Other special components will also vary resistance. The thermistor will vary resistance depending on temperature. In a negative temperature coefficient (NTC) thermistor, when the temperature increases as with the addition of more heat, the resistance decreases.

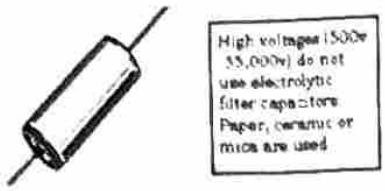
Resistor Pack

In the light dependent resistor (LDR), which is a resistor that varies depending on light being applied to it also decreases resistance with more light. Resistors have the same effect on AC and DC.

Capacitors

A capacitor, often called a "condenser," is made by placing two plates of metal in close proximity to each other, separated by a dielectric material. As a voltage is applied to it, a

is developed across it. They are used to couple signals between circuits or to filter signals in a circuit. They also are used in timing and wave-shaping circuits.



Capacitor

The size of a capacitor (capacitance) is defined in "farads" and it will have a maximum voltage limit. The size is determined by the area of the plates, the distance separating them and the dielectric constant of the dielectric material. The maximum voltage limit is determined by the dielectric material and thickness.

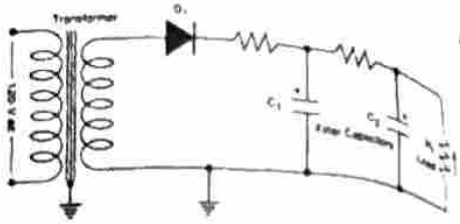
This material might be paper, mica, oil, air, ceramic, plastic, electrolytic or some other material that allows electromagnetic signals (AC) to pass through, but not electrical current (DC). Air has a dielectric constant of 1.00059 (a vacuum is 1).

A capacitor works by having a voltage applied across it. Initially, there is a high rate of current in the charge, but as it charges this rate decreases until the full charge is achieved. The larger the capacitance, the longer a full charge takes. Thus a smaller capacitor will react better to higher frequencies while larger ones react better to lower frequencies. A capacitor will charge to 63.2% of full charge in one RC time constant. In 5 RC times, it will be virtually fully charged.

Capacitors are used in circuits involving AC signals and pulses. A one farad capacitor will store one coulomb when one volt is applied to it. The discharging rate is the same as the charge rate.

Many capacitors are fixed and constructed so that the plates are separated by the dielectric, then that combination is encased in some solid material. Variable capacitors usually have a plastic or air dielectric and are made up of one or more pairs of plates that can be adjusted for more or less overlap (interleave). The more they overlap the larger the capacitance is. These are often found in radio tuning circuits and other circuits that must be tuned to certain frequencies.

A capacitor will be used in DC circuits to smooth out or hold constant the DC voltage. These are called "filter condensers." That capacitor will charge to the basic DC voltage of the circuit and then either charge more or discharge as variations (ripple) occur, resulting in a more pure DC voltage. Filter capacitors may be thought of as reservoirs of electrons.

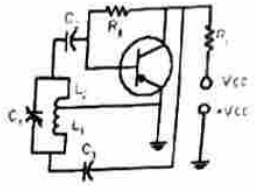


Filter capacitor circuit

Capacitors are used to couple signals from one circuit to another (coupling capacitor). These will be selected to have a reactance close to the frequency of the desired signals so that those may pass and unwanted signals do not.

Capacitors also are very often used in conjunction with resistors or coils to either block or pass signals of certain frequencies. These are called bandpass, bandstop, or band filter circuits. Some filters are designed to change the shape of signals to meet special needs.

An important variation of these filters is to be used as the timing circuit for many types of oscillators.



Oscillator with capacitor tuning

Connecting capacitors in parallel results in a capacitance equal to the sum of all of them. Connecting them in series results in a capacitance less than that of the smallest one. Actually, capacitors rated in "farads" would be very large and rarely used, so the most common references would be in "microfarad" (mf), "micro-microfarad" (mmf) or "picofarad" (pf). A 1 mf capacitor would be very large, while a 1 pf would be very small.

Inductor

Inductors, sometimes called "chokes," are coils of wire encircling a core that might be metal or air. While they do have a resistance in the wire itself, they are used for their inductive ability that creates a magnetic field. This inductive quality causes an opposition to a change in current. They are rated in this amount of inductance in "Henrys." A change in current of one amp will induce one volt across a one henry inductance.

Inductors

The inductance of a coil is determined by the size of wire used, the diameter of the coils, the number of coils and the type of core. As a voltage is applied across the inductor, current is opposed while lines of magnetism (force) are built up around the coils. Once the lines of magnetism are built with voltage remaining constant, current will flow normally through the coil. If the voltage remains constant, current will flow through. If the voltage drops, the field of force will begin to collapse, inducing voltage into the coils, which then results in sustaining current through the inductance.

An inductor works by having a voltage applied across it. Initially, there is a low rate of current through the coil, but as it charges, this rate increases until full charge is achieved and then the current remains at the maximum. The larger the inductance, the longer a full charge takes.

Thus a smaller inductor will react better to higher frequencies while larger ones react better to lower frequencies. An inductor will charge to 63.2% of full charge in one time constant. In five time constants, it will be fully charged. A one henry inductor will store one coulomb when one volt is applied to it. The discharge rate is the same as the charge rate.

Coils will usually be used in DC circuits where they will resist a change in current, thus "dampening" the line they are on. They also are used in conjunction with a resistor, capacitor or both as filters or "tank" circuits. These circuits are designed to pass or reject certain frequencies or bands of frequencies or to oscillate at a given frequency.

Like capacitors, due to size and usage, most inductors will be rated in the millihenry or microhenry range.

Transformers

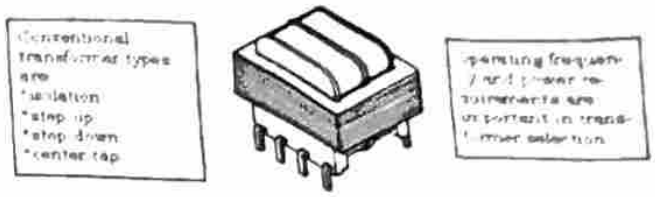
Transformers make use of the properties of coils (inductors) and come in a great variety. Usually, they will have two windings, called the primary and the secondary. Applying an AC voltage across the primary will result in an induced voltage across the secondary. If the two windings were exactly alike, this transformer would simply act as an isolation transformer, passing a signal unchanged between two circuits.

More often, there will be differences between the primary and the secondary. The construction of the transformer determines what it will do best. Listed here are the more common variables included in making a transformer:

Core type: Laminated iron and iron slugs are common, and have a significant effect on the coupling within the transformer. Toroid cores concentrate the lines of force.

Number of windings: One or more, depending on application. A single coil with unique winding and selective taps is called an "autotransformer." Other windings may be added depending on engineering needs.

Windings: Size and length of wire and number of windings in each coil affects stepping up or down of signals, frequency range of signals and maximum power rating of the transformer.



A transformer

The transformer is designed around what the input signal will be and what is needed on the output side. In the basic two winding transformer, if the voltage is stepped up the current will be reduced, and if the voltage is stepped down, the current will be increased in the secondary. The core will be designed to prevent "hysteresis loss" (hysteresis loss is caused by friction and a heating effect in transformer cores). A "buck-boost" winding might be added for certain power supply applications. Some will be built into an oil bath or have attachments for cooling purposes.

Some special application transformers are known as "IF," "Power," "Flyback," "Constant Voltage," "Isolation," "Power Output," "Balun," "Variac" or "RF input."

Vacuum Tubes

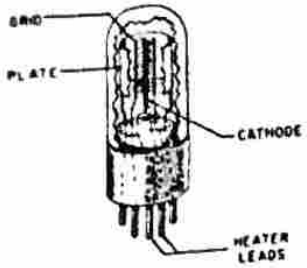
These are components that conduct current from a cathode to a plate and, under various designs, are able to perform a number of special jobs. In order to do this, the cathode must be made of a material that will emit electrons easily and will be heated to promote this process. The electrons given up by the cathode will be drawn to and collected by a positively charged plate. They require a vacuum to do this effectively. Beyond this, the makeup of the tube will vary in many ways to serve many purposes.

Many vacuum tubes are known by the purposes they serve. Some of these are: rectifier, diode, triode, tetrode, pentode, beam-power amplifier, gas diode, thyratron, klystron, magnetron and cathode ray.

In some diodes the heating filament and cathode are one and the same. Conduction in the tube depends on the plate being more positive than the cathode. This might be achieved by applying negative voltage to the cathode or positive voltage to the plate. Grids are added to vary what the tube can do.

The first grid is called the "control" grid and makes the tube a triode. Signals applied to the control grid cause variations in the current through the tube and thus an inverted signal at the plate. Often, but not always, this results in an "amplified" signal. The variations in how the signals on the control grid affect the tube current is referred to as "bias."

A second grid might be a "screen" grid, which too will be biased to perform special purposes like "reducing electrostatic influence between plate and control grid." This makes the tube a tetrode. There might be a "suppressor" grid near the plate to reduce the secondary emissions that can occur when the electrons strike the plate. A tube with three grids is called a pentode.



Vacuum Tube

Gas diodes effectively become a short when the ignition threshold of the gas (about 65v) occurs between the plate and cathode. They are diodes because the cathode can only emit and the plate can only receive electrons. The behavior of the gas is similar to neon and fluorescent tubes. Thyratrons share the gas property, but become a short when a grid raises to a potential that ignites the gas. It may be extinguished, or cut off, by breaking the plate or cathode circuit.

Klystrons are tubes that will oscillate by themselves for a short period when the cathode is pulsed. This occurs because of it having a set of "buncher" grids. The spacing of these two grids controls the frequency of oscillation, and this is adjustable from the outside of the tube. Magnetrons also oscillate when the cathode is pulsed, because of its containing resonant cavities inside. These cavities are designed to a certain frequency. The magnet acts as a guide for the electron stream, necessary in the operation of the tube. Magnetrons operate in a frequency range much higher than klystrons and are capable of much more power. RADAR and microwave ovens use magnetrons.

Cathode Ray tubes (CRT) use an "electron gun" to emit electrons in the direction of a phosphor coated surface such as a TV screen. The gun has elements that affect the speed and focus of the electrons. Turning the gun off and on modulates the electron stream producing light or dark from the phosphor coating. An "aquadag coating" on the tube's inner surface is made positive to draw the stream. Elements outside the tube will act to cause deflection from top to bottom and left to right. The phosphor coating might be a single color, or many colors. Color images are possible by three phosphor colors being arranged in very small "dots" and the use of three guns. An outside deflection coil (or internal deflection plates) will direct the stream of each gun to the color dots it is matched up with.

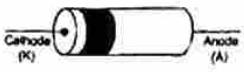
Vacuum tubes vary in size depending on their application, from very small glass tubes with fine wire for pins to giant power tubes used in radio, TV and radar transmitters. As the industry has developed, more and more vacuum tube applications have been taken over by semiconductor devices. Vacuum tubes have the inherent problems of power consumption, heat and noise and limited life due to filament failure, shorting, undesired internal gas buildup and cathode deterioration.

Semiconductors

There are many devices referred to as "semiconductors," and modern technology continually creates new ones. They share several fundamental properties. They usually are made of silicon or germanium. As long as they are not abused they can have a very long lifetime, maybe infinite (they don't wear or burn out). They are composed of at least two types of material, referred to as "P" or "N," which means it has impurities that make it relatively either positive or negative. They can be made very large or very small and, as a rule, are much less expensive than the components they replace.

Generally, these devices might be broken down into three groups: diodes, transistors and integrated circuits, (IC). Each may be modified to serve many different purposes.

Diodes: Composed of two materials, N as cathode and P as anode. Current will flow when positive voltage is applied to the anode or negative to the cathode. The materials, usually silicon or germanium, are impregnated with impurities to make them slightly negative or positive relative to the pure material. This allows current to flow from the N to the P and only in that direction. One of the characteristics of the diode would be the current handling, or power dissipation limit. Exceeding this, either in a forward or a reversed direction, will result in catastrophic failure of the diode.



Diode

Bipolar Transistors: are made up of three sections called the emitter, base and collector in either the NPN or PNP format. A positive voltage will be applied to the collector of an NPN while a negative voltage will be applied to the collector of a PNP. The base serves like the control grid does in a triode to vary transistor current. Conduction occurs when the base is properly biased in conjuction with the appropriate application of voltage to the emitter and collector.

Transistor theory is simplified by using the forward flow of electrons in the NPN, while in the PNP, holes travel from emitter to collector. The "holes" are the voids that occur backward as the electrons flow forward.

Transistor

Integrated Circuits (ICs): Beginning rather simply, modern technology has created myriads of integrated circuit types that perform innumerable complex functions. The inventory continues to grow at an exponential rate. Integrated circuits get more complicated, starting with a substrate, these components are "grown" by applying N, P, resistive materials and insulators in layers and connected to act as resistors, diodes and transistors according to what the IC is designed to do.

Other semiconductor types

Photodiode: When light strikes a PN junction, a voltage is developed. Diodes taking advantage of this property are used to develop current. Solar panels are a form of photodiode.

Light Emitting diode (LED): Illuminates when current flows. By using different frequencies and materials, these may be made to light in a number of different colors. These colors obey the rules of the light spectrum in that infra-red is near the low end of the visible spectrum while ultra-violet is near the high end. These rules are very useful in applications using fiber optic conductors and light sensitive devices. The light from an LED is very random and incoherent.

LED

Fiber optics cables utilize photons generated by LEDs to transmit intelligent signals

Laser diode: Emits a light when turned on. That light is a very small beam, is very coherent and may be modulated. They are an inexpensive substitute for a true (gas, solid-state or ruby) laser and are used in many printers.

Zener diode: A diode that is designed especially to conduct in a reverse direction when a certain voltage level is exceeded, effectively making it a voltage regulator or limiter. While usually used in DC circuits, it does have functions in AC.

Shottky diode: A metal/metal oxide diode. It performs very well at higher frequencies, conducts high current and has lower losses than other diode types. These are found in "switching" (DC-DC converter) power supplies.

Tunnel diode: Using special "doping" techniques (impurity application) these are very unique. They conduct at different rates at different bias levels. They are used in amplification, switching, oscillators and other special applications.

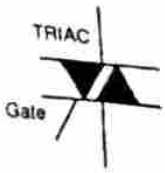
Metal Oxide Varistor (MOV): A special type of diode that is designed to conduct in either direction when a certain reverse bias level is exceeded, and is able to dissipate an extreme amount of energy. It is characteristic to totally destruct when their maximum energy level is reached, thus protecting circuits to the maximum. They are not polarity sensitive and are the basis of the best surge suppressors.

Varactor: A semiconductor with capacitive qualities. The capacitance can vary as the applied reverse bias voltage changes.

Silicon Controlled Rectifier (SCR): A diode with a gate between the anode and cathode. It will act as a shorted diode after the gate level reaches a designed level, and will continue to conduct until the anode or cathode circuit is broken.

Diac: Two zeners connected reverse in parallel. They are used in an AC environment to turn on, or avalanche, when the design maximum bias is exceeded.

Triac: Actually appears as two diodes of opposite polarity, parallel to each other and containing a gate like the SCR. The gate might be activated by a digital electronic signal or a mechanical switch to act as an ON/OFF switch. When the gate is biased properly, the device will be a short to AC and will then continue to conduct independently until it is turned off.



FET Transistor and other types: Some special transistor adaptations are the: Unijunction, Bipolar Junction, Darlingto Field Effect, Junction Field Effect, Metal Oxide Field Effect and Photo.

Operational Amplifier (Op-amp): A single circuit made of two transistor circuits that sample signals at two inputs.

Depending on the design, those two signals cause the Op-amp to output a third signal level.

Integrated Circuit (IC): Basically, a small 14-pin component that is designed for particular applications. They might contain some simple gates, amplifiers, oscillators or combinations of all. They have evolved from the original 14 pin dual in line

package devices to 8, 16, 24, 28, 40 and 88 pin LSIs (Large scale integration) and various other configurations. They also are packaged in 'flat packs' and cylindrical cases.

Read Only Memory (ROM): Comes in many sizes and configurations. The common characteristic is that they are manufactured so as to retain fixed information (data) when power is connected to them. This data may then be sampled by the device using it. These are used to provide the initial data a computer needs when it is powered on, as well as innumerable other purposes.

Random Access Memory (RAM): Integrated circuits that are designed especially as data memory. Usually made as 8K, 16K, 32K or 64K, they will store data in an 8-bit format. Each data location is addressable, and thus available at any time for writing or reading. Data is retained as long as there is power applied to the RAM. This is usually provided for by the power supply or an on-board battery.

Programmable Read Only Memory (PROM): A programmable ROM, but only with special equipment. Once programmed, it may not be altered.

Erasable Programmable Read Only Memory (EPROM): Like the PROM except that it has a window that allows it to be erased by use of ultraviolet energy after which it may be programmed again.

Microprocessor: An integrated circuit, probably rather large (32, 64 or more pins). It is designed and made to be the controlling component for an electric or electronic device. It actually contains all the circuits, memory and programming that will allow it to perform the job of controlling a device under all predictable conditions. They currently are used to control nearly all electronic devices including automobile ignition and electrical system, computers, heating systems, rockets and coffee makers.

Minicircuits: are very small components, maybe no larger than a dime or nickel, that might be permanently wired or mounted in sockets. They might be simply a semiconductor or IC, or they might also include such things as capacitors, coils, transformers, crystals and the like. They will be designed to perform special functions.

The list of integrated circuit types (solid state devices) includes many other types such as UART, MUX, DAC, A/D Encoders, Decoders, DRAM, SDRAM and promises to keep growing and packing more and more technology into smaller packages. A person must research each new component he or she encounters if he wants to stay up to-date on the components that are introduced into newer devices.

Thermocouple: A thermocouple is a junction between two different metals that produces a voltage related to a temperature difference. Thermocouples are a widely used type of temperature sensor for measurement and control and can also be used to convert heat into electric power. The output of thermocouples varies according to type, but is most often less than 100 microvolts per degree of temperature change.

Balun: This type of transformer couples an unbalanced transmission line (coax) to a balanced line (twin lead), port, etc. Thus the name for it: bal (balanced) un (unbalanced). Piezo-electric crystals create fairly high but very low-current when flexed or struck. They also work in reverse: when a voltage is applied across them, they flex. They are often used as receiver-speakers in cell phones. A crystal can be cut to physically resonate at a very specific frequency, making crystals the standard means of generating very precise frequency references.

NOTES

Chapter Quiz

1. **A high current MOV device would most likely be found:**

   a. in a surge suppressor
   b. as the radiator in a microwave oven
   c. in the DC power supply of a television
   d. as the oscillator in a superheterodyne receiver

2. **Which of these would most likely be used as a signal amplifier?**

   a. Triode
   b. Magnetron
   c. Triac
   d. DAC

3. **A capacitor and a resistor in series would be used in a signal line to:**

   a. cut off the peaks in AC signals at a set level.
   b. reduce or short signals above certain frequencies.
   c. pass the DC level, while blocking the AC signal.
   d. filter inductances from getting into an amplifier.

4. **The component used to control the functions of an electronic device most likely would be referred to as:**

   a. RAM.
   b. SCR.
   c. Microprocessor.
   d. Op-amp.

5. **A capacitor would most likely be rated:**

   a. 5 watts.
   b. 4 henrys.
   c. 7 ohms.
   d. .05 microfarad.

6. **Which of these is most true of a PNP transistor?**

   a. It will avalanche when the cutoff is reached.
   b. It conducts equally well forward or backward.
   c. A negative voltage will be applied to the collector.
   d. The cathode must be heated to operate.

7. **An advantage a transistor has over a vacuum tube triode is:**

   a. a triode cannot conduct from the cathode to the anode.
   b. a transistor does not require an anode load.
   c. a triode requires grid bias.
   d. a transistor normally has a longer life.

8. **The most likely use of a transformer would be:**

   a. to couple signals from one circuit to another
   b. to filter out unwanted signals
   c. as a regulator in power supplies.
   d. to protect a circuit from excessive currents

9. **Which of these would apply mostly to a "balun"?**

   a. Semiconductor
   b. Impedance matching device
   c. Integrated circuit
   d. Over voltage protection

10. **DIP refers to:**

    a. double integrated protection.
    b. dual ion positron
    c. discrete icon positioner.
    d. dual inline package.

ANSWERS on page 223

Chapter 9

by Dick Glass, CETsr

Dick Glass is an author of previous ETA publications as well as more than a dozen books dealing with electronics technology and business management, published by Howard W. Sams Co, TAB Books, and McGraw-Hill. He has also written hundreds of articles in trade journals and association publications. Dick is a former Navy Avionics Technician, Industrial Electronics Computer Technician, Technical Writer, FCC Commercial License examination modernization committee Chairman in 1995 and 2005, inductee into the Electronics Hall of Fame and currently CEO Emeritus of ETA. He has a broad background in all phases of work performed by electronics technicians.

Semiconductors

Just about everyone realizes that, without electronic semiconductors, our world would still appear to be in the dark ages. Imagine trying to carry your portable radio around with its five electron tubes and large batteries. Your watch would be spring driven. Only a few businesses would have computers. Typewriters would perform the letter writing chores. There would be no orbiting satellites or GPS or cell phones or CDs. Our world today is affected in nearly everything we do by semiconductors.

Early Semiconductors

Since the early days of the 20th century, electrical equipment has used semiconductors.

Here is a picture of one in Fig. 1:

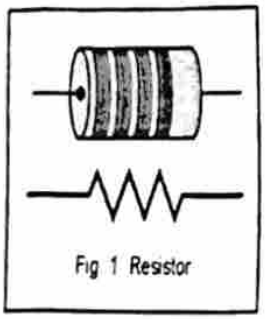


Fig 1 Resistor

The resistor, unlike a straight wire made of copper, aluminum or steel, resists the flow of electrons. It doesn't slow the electrons down, but it puts some logjams or roadblocks in their way. It sort of narrows down the pathways in the partially conductive materials so that fewer electrons can pass through.

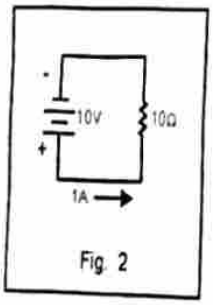


Fig. 2

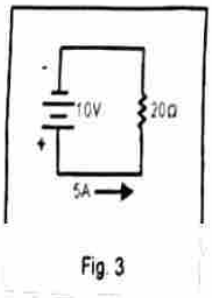


Fig. 3

In Fig. 2, the 10 ohm resistance allows the 10 volts of electrical 'pressure' to send 1 amp of current around the circuit. A resistor composed of a less-conducting material (such as carbon) measuring 20 ohms (Fig. 3) now allows only half as many electrons to move around the circuit. Electricity still moves at nearly the speed of light (186,000 MPS) but only a half-amp is flowing. (Fig. 2, 3, 4 and 5 show electron flow.)

We could control these electrons better if we also put a switch in the series circuit. (Fig. 4 and Fig. 5)

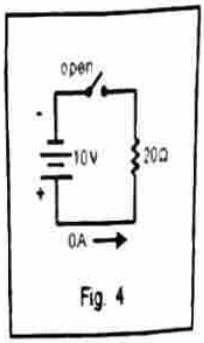


Fig. 4

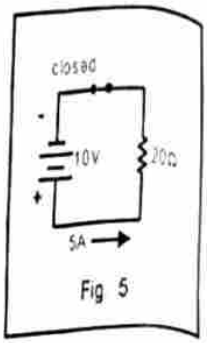


Fig. 5

So, now, instead of blowing up the battery by shorting the + and - terminals with a screwdriver, or by substituting a 10 ohm resistor for the screwdriver and thereby reducing the electron (current) flow to 1 amp, or reducing it even further with a 20 ohms resistor to where the current is only 1/2 amp (or 500 milliamps), we can turn the current on or off with the switch.

So what good is that? Let's see: we could put a buzzer in the circuit and turn the buzzer on or off. In fact, we could turn it on for a short time, or for a long time and communicate to anyone close enough to hear the buzzer by using Morse Code. One short buzz is a dot and one long buzz is a dash. Hey! That means the letter "A."

dot-dash or "A"

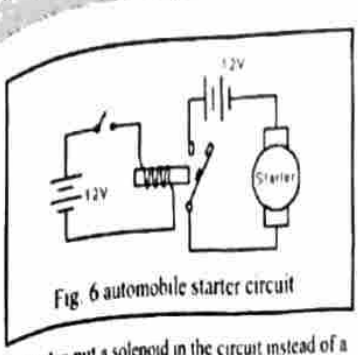


Fig. 6 automobile starter circuit

We could also put a solenoid in the circuit instead of a buzzer and cause an automobile starter to crank the engine.

Early Semiconductors

In the early days of TV, in the 50s and 60s (and even before that), semiconductors were used. It was discovered that if you coated a slab of metal with selenium, that for some reason or other, current would flow in only one direction through it. When the selenium-coated side of a metal-selenium plate was connected to a positive voltage and the metal side connected to a negative voltage , current flowed. Reverse the connections and no current flowed.

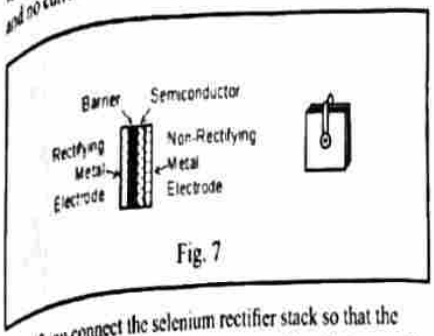


Fig. 7

So, if you connect the selenium rectifier stack so that the positive potential is on the selenium covered rectifying metal electrode and the negative potential on the non-rectifying electrode, you can cause current to flow. Connect it in an opposite manner and no current will flow.

Direction of Current Flow

Which way do the electrons flow in these metal-plate-and-selenium rectifiers? Current (electrons) flows against the arrow in the semiconductor symbol. Note the diode symbol in Fig. 8 and the reverse way we have biased it. No current flows. Look at Fig. 8, Fig. 9 and Fig. 10. Fig. 10 shows the most basic electronic product power supply and how the diode converts the transformer AC current into pulsating DC current.

You might think of the reverse biased drawing in Fig. 8 as a negative battery terminal creating a barrier between the arrow point of the diode and the perpendicular line part (the cathode). The higher the difference between the negative voltage and the positive potential, the greater the barrier becomes. If you think of this barrier as a thin slice between the semiconductor material and the metal plate, and this slice gets electrically wider with higher reverse voltage, you start to see how semiconductors work.

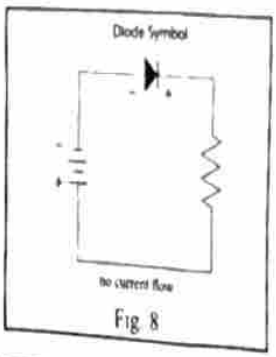


Fig. 8

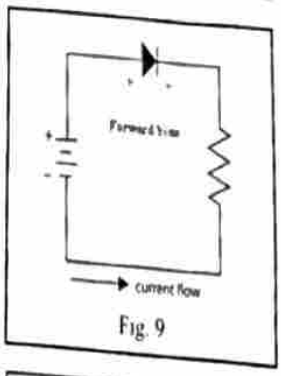


Fig. 9

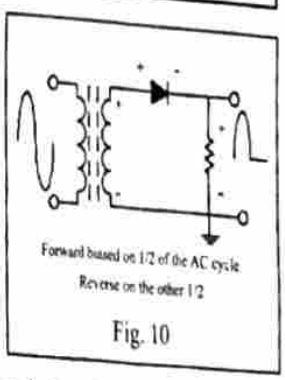


Fig. 10

With reverse bias, no current flows, but importantly (as you will see later on) this stress barrier becomes less or more intense depending on the reverse voltage levels applied. With a simple rectifier circuit as shown in Fig. 10, that stress barrier doesn't matter much, other than that it doesn't allow current to flow in the reverse direction.

In Fig. 9, we reverse the bias: Forward bias. Current now flows. The barrier portion electrons are enticed away from the barrier and the electrons can easily pass from the cathode to the anode. This causes a deficiency of electrons in the cathode circuit. A deficiency of electrons is a positive voltage. Just what you wanted!

Figure 10 shows the most common transformer half-wave power supply circuit. On positive half-cycles, current flows across the diode junction in a counterclockwise direction. Thus, across the resistor load there is a voltage drop from bottom to top of the resistor, leaving a positive voltage with reference to ground. That is exactly what you wanted to create in order to supply a voltage to properly operate the transistors in your little radio (or TV or computer).

On the negative half-cycle of the AC waveform which the transformer supplies to the diode, the top of the transformer secondary is now negative. So is the anode of the diode, thus current can't flow. So we just wasted half of our power from the transformer! There are ways to not waste the half-cycle of power from the transformer, but we need to do some other

things to change the current to a 'full-wave rectifier' or bridge rectifier circuit. If you understand how this simple single diode, half-wave rectifier circuit works, you will have little trouble understanding how a full-wave or bridge rectifier circuit works.

So that's how one kind of semiconductor is used. You take an alternating current (AC) from the wall socket - reduce the level of the voltage to a practical amount depending on the job that needs doing, rectify the AC to get rid of the negative half of the AC sine wave, then filter the diode's output pulses of DC to smooth it out to pure DC (like a battery provides) which can then power our TV, computer, DVD or other product's circuits.

Today, most rectifier diodes are made of silicon instead of selenium. Small signal-diodes are more often made of germanium.

PN junctions act like selenium diodes. They conduct in one direction only and have two elements, a cathode and anode. In Fig 11 is the Navy's NEETS manual sketch of the semiconductor PN junction diode.

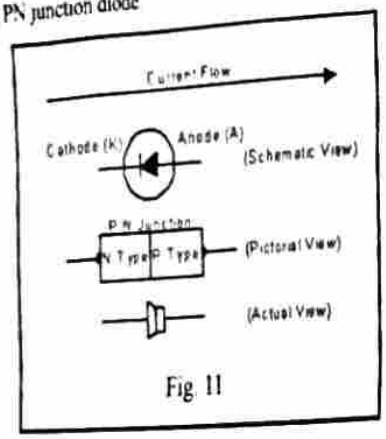


Fig. 11

HOLES! (search 'transistor depletion zone' online for more)

P type material (on the left side in Fig. 12) current flow is considered to be by holes rather than electrons. This is not easy to understand. In the N material current flow is normal, and negative electrons move through the crystal material. But in the P material it is the holes that move from the positive terminal of the P material to the negative terminal.

It becomes more confusing when you join P and N materials. The junction of the two is called the 'depletion' area, or barrier (as we discussed above). It is called a depletion area since there is a lack of free electrons, or holes that electrons could move into. (Fig. 12) One way to think about this mysterious barrier area is that the barrier slice between the P and N material has positive and negative ions which are attracted to the edges of the depletion area. The negative ions cozy up to the P material and the positive ions snuggle up alongside the N material. That leaves a no-man's land in the thin barrier slice between the P and N materials.

Applying positive (forward) bias voltage to the P material (and negative to the N material) reduces the electrostatic field that exists between the positive and negative ions in the barrier strip. By reducing the electrostatic field, electrons can flow from negative to positive in the junction (and holes can flow the opposite direction).

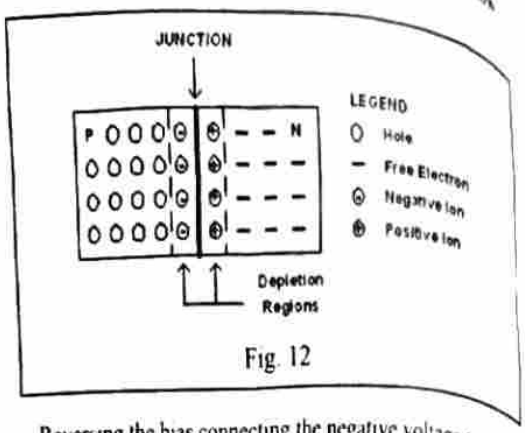


Fig. 12

Reversing the bias connecting the negative voltage to the P material and positive to the N material, the barrier is made even wider (electrically). The electrostatic field is stronger and thus it is much more difficult to move electrons or holes across the barrier.

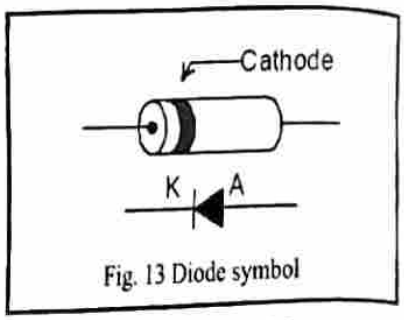


Fig. 13 Diode symbol

(K is used here as C is already used for capacitance.)

Forward Bias

Plus to P material. Minus to N material.

Silicon diodes require .7 volts forward bias to overcome the barrier potential and allow conduction to occur. Germanium diodes require about .3 volts. The natural barrier, or depletion slice, is narrower in germanium.

Once the barrier field has been overcome, an increase in bias voltage of only a few hundred millivolts—up to several volts—will cause an increase in forward current. There are limits specified for inverse voltage that can be tolerated before the junction breaks down and shorts out the diode for good.

There are specifications for forward current that can be safely handled without overheating the PN junction and causing it to fail. Technicians ordinarily replace defective diodes with exact replacements.

If you feel uncomfortable with the explanation of the PN junction and it makes little more sense than quantum mechanics, then just convince yourself of those truths which ARE believable:

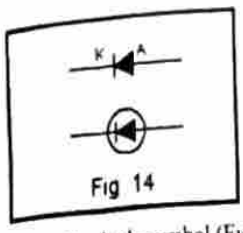


Fig 14

Recognize the diode symbol (Fig. 14)

1 Identify the cathode (K) and anode (A).
2 Believe that forward bias means putting the + voltage on the P material.
3 Believe that the identifying band on a diode is the cathode.
4 To conduct, the anode needs to be at a higher positive potential than the cathode.
5 The plus potential on the anode, perhaps the positive excursion of a 60 Hz sine wave, will suck electrons from the cathode, thus leaving the cathode in a power supply at a positive potential. So, in checking a diode to see if it is conducting, the plus lead of your voltmeter is placed on the cathode. In the symbol you can recognize the cathode as the place to check for positive voltage in a power supply. You notice the cathode symbol already nearly looks like a + sign. Generally diode ohm meter front-back ratio of 10 to 1 is OK.
6 Believe you will damage a diode (or any other PN junction) by applying too large reverse voltage or allowing too high current to flow, such as might be caused by a short circuit in the load (which is all of the circuitry the voltage is being supplied to).
7 Because diodes are most often located near the AC input circuit of an AC powered product, they are the most likely part to be damaged by lightning or power surges.

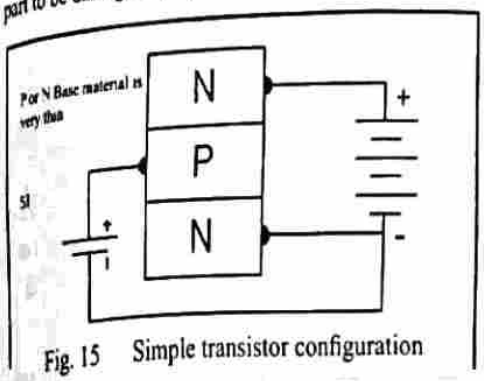


Fig. 15 Simple transistor configuration

Transistors

If you understand PN junctions then you will have little problem seeing how transistors work.

A difference between diodes and transistors is that the diode can't amplify a signal. It can only pass current up to the level of voltage applied to it and can pass only a specified amount of current. The transistor can produce an exact replica of an input signal (perhaps a sine wave) many times larger. That helps when you want to blast the cones out of your auto's speakers while listening to music.

NPN

We will talk mostly about NPN transistors. PNPs are virtually opposite in every respect, so if you understand NPNs you understand PNPs too.

The nice thing about the transistor is that the middle initial tells you what voltage to expect on the collector—P, or positive—for NPN types. And if you know that the transistor should operate with a positive DC collector, then you know that the base should also be a little bit positive compared with the emitter. That makes it easy.

There is a very small base current flowing in an NPN transistor if the base/emitter (B/E) junction is forward biased. That means when you expect a transistor to be doing its job in a circuit and you find zero volts difference between the emitter and base ...something is wrong. If you find .2 or .3 volts on the base of a germanium transistor and a more positive voltage on the collector, it should work unless one or both of the junctions have been destroyed.

How Do Transistors Work?

Maybe the NPN bias causes electrons to flow through the P material base, fooling the ions at the NP and PN junctions. Regardless of exactly how the holes and electrons move, the fact is that if you bias the base of an NPN on with .7 volts (for a silicon type), and you also have a higher positive potential on the collector, the transistor will conduct.

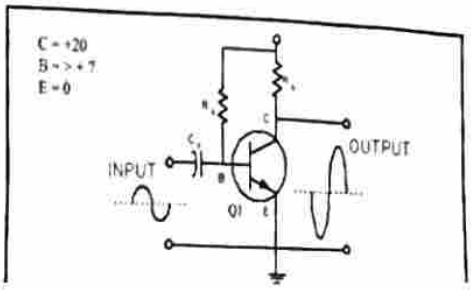


Fig. 16 Common emitter NPN

Chapter 9

Amplification

Diodes ordinarily are used in an on or off situation. They can conduct less than the full potential across their PN junction if they are barely biased on and if fully biased they will conduct to the potential input (less the junction bias required of .3 or .7 volts) but no more than the voltage across them.

The transistor has the ability to amplify an input to the base (or the emitter, but let's talk about the most common configuration for simplicity). The amplified signal appearing on the collector is not the signal appearing on the base, but you would think so since it is an exact replica of the base signal if everything is biased correctly. Check out Fig. 16.

Notice the smaller sine wave input to the NPN base through C appears much larger at the output on the collector. So the current flowing from the emitter of Q1 through the PN base-emitter junction is rather small, but the biased-on base allows larger current to flow from the emitter to the collector. The larger current produces a larger signal voltage across the load resistance RL. That is amplification.

Bipolar Transistor (NPN & PNP) Configurations

Transistors are very useful. They can be made to perform hundreds of electronic functions. You can study books which detail all of the ways they are used. You will find that IC (Integrated Circuit) chips may have many transistors, all doing similar jobs. Or you may find IC's with dozens of transistors in them, each one—or groups of them—performing different tasks. They may be used in an oscillator circuit, clock, regulator, gate, memory or power amplifier circuit for examples. Some are high power and some are used in small signal circuits. All will fall into one of three configurations:

1. **Common emitter** - like the circuit in Fig. 16

2. **Common base** - where the input signal is applied to the emitter rather than the base (highest voltage gain)

3. **Common collector** - has high input impedance (and low output impedance)

Common emitter is the most common hookup for a signal transistor. The emitter may be grounded or connected to a DC voltage supply. The emitter is common, or the reference for both the base and collector signals. As the base voltage goes higher, more current flows through the transistor to the collector. More current lowers the collector voltage and the collector voltage then is the opposite of that on the base, but it is still a replica.

The common base could work about the same as the common emitter except that the signal may be input to the emitter (assuming it isn't grounded). So, whether you input the signal to the base or the emitter, the difference between the base and emitter may be the same. Thus, the collector doesn't care which one you put the signal in on, and it goes ahead and produces the amplified signal on the collector. (Actually, the common base configuration usually does not produce as large an amplification, but it can produce as large a current gain.) One plus to putting the signal in on the emitter is that the collector signal is not inverted. That might be nice if you were only wanting to amplify a single pulse and you wanted it to remain a negative or a positive pulse after amplification.

The **common collector** might be useful in situations in which you wanted the collector to be grounded such as in your auto radio where the transistor is mounted externally and might get shorted out by a screwdriver. TV set horizontal output transistors and many other power transistors are mounted with the collector (which is connected to the shell) grounded. In this case, the output is off of the emitter with the input to the base.

Part number Identification

Why are transistors numbered like 2N123A? The 2 stands for the number of junctions. N means a semiconductor. The 123 is the manufacturer's ID number. An "A" or "B," etc. following the last number identifies an improved or later version of the 123. It is best to consult product catalogs or replacement guides to assure that a replacement transistor is the correct one.

Identifying Transistor Leads

A single semiconductor maker may use a standard layout or pin placement for the E - B - C legs. But don't expect any other maker to follow suit. You can check transistors with an ohm meter. Checking between the E - B - C legs, the ratio of forward to reverse resistance should be at least 30 to 1, and 100 to 1 is not uncommon.

PNP

We have previously discussed transistors above, using only the NPN type. The PNP is just as popular. Since the middle initial (N) shows the normal collector voltage potential, a quick check in-circuit of a PNP is to see if the collector truly is negative. The base also should be negative within a volt of the emitter and the emitter will be most positive. The polarity is reverse of the NPN type.

The biggest problem you have is in determining which leg is the collector, base or emitter. There is rarely any identification. Sometimes you can determine the C, E, or B by looking

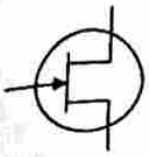
at the circuit, but in today's world of two-sided PC boards and miniature circuits, that too is something only to be attempted in an emergency. As with the NPN, you can short the emitter and base temporarily, shutting off the transistor. This can show you whether the collector is drawing current and thus if the transistor is conducting.

Field Effect Transistors - JFET's and MOSFETs

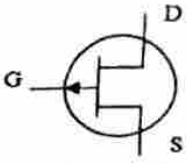
FETs can be categorized in 2 different high input impedance types of devices, Junction FETs and Metal Oxide Semiconductor FETs or MOSFETs. Within these types, there are N-Channel and P-Channel devices, which require opposite polarity of voltage to operate, just like NPN and PNP in bipolar devices.

NOTE: Remember that the ARROW on a semiconductor device always points to the N type material, whether it is a diode, transistor or FET.

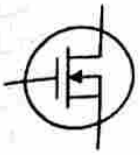
Both depletion mode and enhancement mode FET devices are available. Depletion mode means the device is normal conducting, and voltage applied to the control gate opens the conduction path between source and drain. Enhancement mode means that the device is already in an open state (dashed line as Channel—see below) and when voltage is applied to the gate, the device starts to conduct. The symbols for each type of FET are shown below.



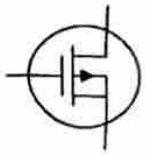
N-Channel JFET



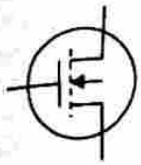
P-Channel JFET



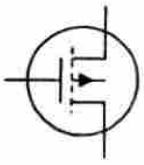
Depletion mode
N-Channel MOSFET



Depletion Mode
P-Channel MOSFET



Enhancement mode
N-Channel MOSFET



Enhancement Mode
P-Channel MOSFET

CMOS type semiconductors, like MOSFETs or IGFET (Insulated Gate FETs) are susceptible to damage from static electricity that everyone accumulates in cold weather in dry environments. When working with CMOS IC's and other components, the technician must wear a wrist ground strap and the bench surface must be grounded. If you are working in the field and are not certain of the potential, you should have static on your mind and continually ground yourself to the AC power outlet grounded cover. People get charged up to hundreds of volts. This discharge through a transistor can puncture the thin metal oxide substrate rendering the CMOS transistor useless.

IC's

Space-saving integrated circuits are composed of semiconductors. Some micro circuits may also contain built-in capacitors, resistors and even inductors. Some IC chips are standard, well- known combinations of components. Simple gate arrays, diode or resistor chips can be checked for opens or shorts in emergencies with an ohm meter. Computer gates can be tested with logic probes, pulsers and oscilloscopes. A multimeter can be used to ascertain that the leg voltages or resistances are proper.

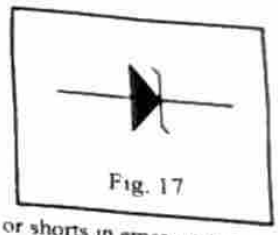


Fig. 17

Zeners

Zener diodes were named after Dr. Carl Zener, who discovered the effect in 1934. An ordinary diode, once the reverse voltage limit or "breakdown" voltage is exceeded, will short and is destroyed by the heat and high current. A zener diode is constructed of materials similar to ordinary diodes, but with carefully doped depletion areas which result in a strange phenomenon. Instead of burning up when the breakdown voltage is exceeded, an avalanche effect takes place with no harm to the PN junction. Electrons which have been released thermally in the depletion region or barrier zone between the PN materials gain energy and rupture the covalent bonds as they collide with lattice atoms. A virtual chain reaction results and reverse current then is limited only by the zener's current-limiting resistor.

Zeners may be specified from two volts to two hundred volts. They are widely used in regulated power supplies and overvoltage protection circuits. A zener diode acts like an ordinary diode when forward biased. Usually the zener is reverse biased.

Varactors

These diodes use the same symbol as an ordinary PN diode. But as shown they also have a tiny capacitor symbol beside the cathode.

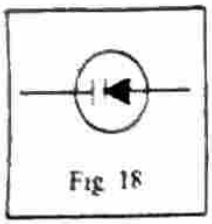


Fig. 18

When the capacitive effect of the PN junction was first understood, it created a revolution in the ways radios and TV's were tuned to the different stations. The first varactors were actually transistors. A technician would think he had discovered a transistor where someone had forgotten to solder a leg. Actually the product makers used the inexpensive transistor junction only as a diode–a varactor diode.

If you page back to Fig. 12, you will note the depletion zone of a PN junction. As different voltages are applied to the PN materials, this depletion zone electrically expands or contracts. The electrostatic field then is, in effect, changing. If you used a variable capacitor in place of the varactor, you could vary the electrostatic field by varying the separation of the plates of the capacitor. The varactor accomplishes the same thing. But the varactor is a very inexpensive semiconductor and takes up little space. It works great, so that is what you use for tuning and voltage control circuits today. You won't find any transistors being used in place of the varactors anymore. Increasing the voltage across the varactor PN junction decreases the effective capacitance.

SCRs (or Thyristors)

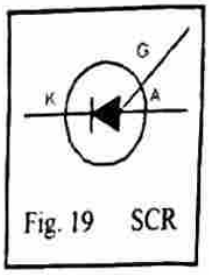


Fig. 19 SCR

Since most diodes and transistors are made from silicon you may wonder why one of them is named silicon controlled rectifier or SCR. It is also like a transistor and a little like a diode. Physically the SCR has four layers of N or P material. With no voltage on the gate or control element the SCR can't pass current. With the gate positive, the SCR is turned on and current flow takes place. The current is limited only by a limiting resistor externally. By removing the gate voltage, pulse, signal, etc., the SCR does not turn off. Once it is turned on it can only be turned off by removing the anode potential. SCRs are used as high speed switches. They can rectify controlled amounts of AC power. They can be turned on and off in microcircuits and do not have contact problems such as a solenoid or switch will develop.

One of the beauties of the SCR is that a very small signal voltage on the gate can turn on large current through the SCR. It is like the auto starter where you energize a solenoid with a tiny current to apply the full battery power to your automobile starter.

Triacs

Fig. 20 Triac

Triacs are three terminal semiconductor devices. Your TV remote control turns on the TV power by actuating a triac. A triac can be constructed by connecting two SCR's back-to-back with a common gate and common terminals.

Optical Semiconductors

Important devices in this class are those with optical capabilities. These are generally:

1. **Photodiode**
2. **Phototransistor**
3. **Photocell**
4. **LED and LCD**

Light Emitting Diodes (LEDs) need only about 1.6 volts forward bias to produce light. They draw around 10 milliamps of current and may last for over 100,000 hours. LEDs can be interconnected to form alpha-numeric characters. The seven segment LED display is a common example. The color of light the LED produces depends on the material used in manufacture. SUPER BRIGHT LEDs are very energy efficient and produce more light output (Lumens per Watt).

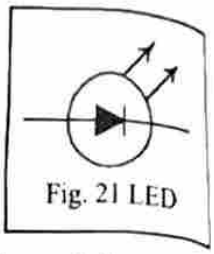


Fig. 21 LED

LCDs (Liquid Crystal Displays) are used in the same manner as LEDs but the liquid crystal display is energized to polarize the segments of the display to form alpha-numeric characters. LCDs use much less power than LEDs. The disadvantage of LCDs is that they do not operate well in very cold temperatures and need external illumination to be read in dim lighting.

Photodiode

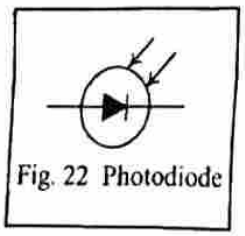


Fig. 22 Photodiode

The photodiode, phototransistor and photocells are semiconductors which are made to accept light through a protective lens which covers the photosensitive material. This PN junction has a certain resistance in darkness.

When exposed to light the resistance decreases, allowing current to flow in direct proportion to the amount of light beamed onto it. A characteristic of photodiodes is that they respond instantly to light, making them very useful in card-readers, light meters, optical scanning equipment and so forth.

NOTES

Phototransistor

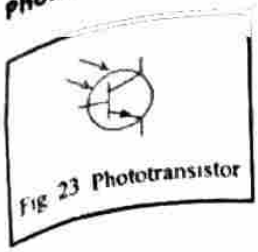


Fig. 23 Phototransistor

Much more sensitive to light than the photodiode is the photo transistor. These may be either of the PNP or NPN type. They may be used after a photodiode in their base circuit to utilize the speed of the diode and the larger current of the transistor.

Photocell

These are light-controlled variable resistors but with a wide resistance range. They are often used to control street lighting and timing circuits.

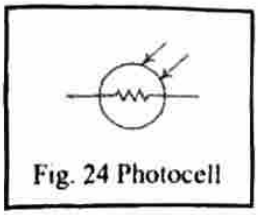


Fig. 24 Photocell

Photovoltaic Cell

Solar cells, as they are called, convert light energy into electrical energy. The space shuttle and space station use many solar cells as do satellite transponders. When light strikes the PN junctions of the photovoltaic cells, it produces movement of electrons. Each cell produces about a half-volt.

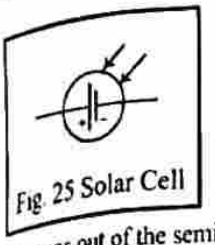


Fig. 25 Solar Cell

Light has been proven to be both a wave and particles (or photons). Photons impinge onto the light sensitive PN material and cause a change in the barrier or depletion region of the junction and current flows. A photon has an energy, so a photon knocks electrons out of the semiconductor material.

Optical Coupler

Technicians frequently encounter opto-isolators, or optical couplers. They can be used to couple AC signals or pulses while completely blocking any DC component. Capacitors can do this also, but may not entirely eliminate leakage current. The opto-isolator is a low-current device that can be used to trigger high current components such as SCRs and triacs.

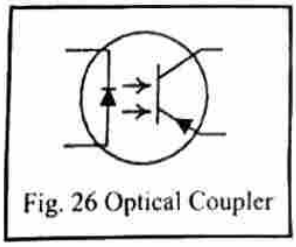


Fig. 26 Optical Coupler

Chapter 9

Chapter Quiz

1. **Diodes are marked with a band at the _____ end.**

   a. cathode
   b. anode
   c. center
   d. base element

2. **In testing a germanium diode, a forward-to-reverse ratio of 10-1 is indicative of a good diode.**

   a. True
   b. False

3. **To properly bias a NPN silicon signal amplifier transistor, which voltages appear correct for a signal amplifier (SI)?**

   a. E = 0; B = +.7; C = 10
   b. E = 1; B = 2; C = +20
   c. E = -1; B = -1.7; C = 9.3
   d. E = 0; B = 0; C = 10

4. **To properly bias a PNP transistor which voltages appear proper (SI)?**

   a. E = 0; B = +1; C = +10
   b. E = 1; B = 5; C = +20
   c. E = -1; B = -1.7; C = -10
   d. E = 0; B = 0; C = -10

5. **In a transistor, which section is made very thin compared with the other two?**

   a. Emitter
   b. Base
   c. Collector
   d. Each are exactly the same

6. **Which part of a transistor carries the most current?**

   a. Emitter
   b. Base
   c. Collector
   d. Base and emitter carry the same amount

7. **The primary difference between a NPN and a PNP transitor is:**

   a. physical shape.
   b. DC voltages are reversed.
   c. one is germanium and the other silicon.
   d. the polarity of the power supply voltage.

8. **Which transistor circuit normally produces the largest voltage gain?**

   a. Common emitter
   b. Common base
   c. Common collector
   d. Transistor circuits produce current gain, not voltage gain

9. **Name one safety precaution taken before replacing a transistor.**

   a. Remove power
   b. Remove transistor
   c. Short emitter to base to remove discharge voltage
   d. Put on rubber gloves for isolation

10. **Zener and varactor diodes are usually _____ biased.**

    a. reverse, meaning the positive voltage connects to the N material
    b. reverse, meaning the negative voltage connects to the N material
    c. forward, meaning the negative voltage connects to the N material
    d. forward, meaning the positive voltage connects to the N material

ANSWERS on page 223

Chapter 10

Discrete Solid State Circuits

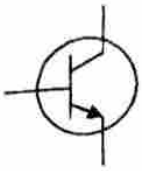
by John W. Baldwin, CETsr

John Baldwin has been involved in electronics for 45 years as a professional, 28 years teaching at the college level and 9 years in industry and the military. He taught electronics and wireless communications at South Central College in Mankato, MN as well as at Rasmussen College at Eagan and Brooklyn Park, MN. He was an instructor, designer for the Institute for Wireless Education, where he held seminars for working professionals. He holds a BA degree in Industrial Education from the University of Minnesota and teaching credentials in Minnesota and California. A charter member and Board member of the ETA, he is also an IEEE member, communications society member, holds an FCC GROL license, Amateur Radio License, and is a certification administrator for the ETA. John served in the USAF as a radio communications specialist and was formerly employed by FF Johnson Co. as a technical support supervisor and a technical instructor. He is the author of industry publications which highlight wireless and telecommunications. In 2006, John was the recipient of the GWEC Wireless Educator of the Year award for 2 year college programs.

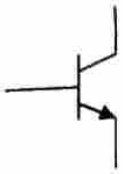
Introduction

Discrete solid state devices today are used in only a few areas where integrated circuits have not taken over the mass produced, high volume circuits.

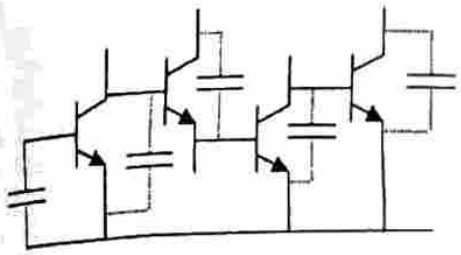
Some of the circuit areas where individual or "discrete" components are required are radio frequency (RF) circuits and those that require high frequencies to be amplified with a minimum number of components to reduce shunt capacitances on signal paths. Other areas could be high-speed circuit configurations that need minimum signal delay time from input to output.



Discrete solid state components, such as the NPN transistor above, have circles drawn around the device as shown in the schematic symbol. If this transistor were to be fabricated into an IC design, no circle is drawn around the device as shown below.



Because junction capacitance adds up, as in parallel capacitor circuits, where $C_T = C_1 + C_2 + C_3$, high density semiconductor integrated circuits using multiple transistor circuits cascaded together, have a greater signal loss at super high frequencies due to the low-pass filter effect of all of the shunt capacitance adding up.



Categories of Circuits

Solid state devices can be categorized many different ways - by the number of junctions, by the material or by their function. Diodes, transistors and integrated circuits make up the vast majority of discrete devices, with special input-output sensors, displays and readout devices making up most of the rest.

One way to classify circuits and electronic products is by their function.

Class 1 products typically are those items that will not cause a loss of time or money or are not used to support life functions. Such products are consumer electronic devices that we may use every day - a radio, television, satellite receiver or DVD player.

Class 2 products are in the business area and if they fail, will cause a loss of time or money. Any industrial process controller, cell phone, copy machine or computer system would be a class 2 product.

Class 3 products are usually thought of as in the medical or aviation field, and if they fail, they could cause loss of life.

Another way to classify solid state devices is by the number of semiconductor junctions or layers. Single junction semiconductors are a form of a diode. Rectifiers, signal detectors and light emitting diodes (LEDs) make up some of these. Two junction devices are sometimes called bipolar in that they can have current flow from two different current carriers - holes and electrons. Within the 2 junction devices are bipolar junction transistors (BJTs) and field effect transistors (FETs).

With more layers and junctions, some devices are variants of the basic types, and can control DC or AC circuits with gate inputs. Examples of these devices are silicon controlled rectifiers (SCRs) and triacs, which are called thyristors. Multi-layer semiconductor construction techniques are used extensively in integrated circuits, with low, medium and high numbers of transistors, or high gate density in digital applications.

Also, multilayer circuit boards, with several circuit layers embedded into the board, are very common today in high density applications such as PC computer boards, handheld devices and smart controllers. In these cases discrete components literally disappear from sight as higher and more complex circuit functions are manufactured into the product design. Super high density at super high frequencies in new smartphones is becoming the standard norm of expected performance in today's business world.

Types of Components

Types of components can be classified into 4 groups according to physical attributes. Throughhole are those designed to be inserted in a hole in a printed circuit board and soldered in place on the opposite side of the board. SMD, or surface mount devices, are mounted and soldered on the same side of a PC board. BGA or ball grid arrays are designed to be soldered in between the chip and the board, making it one of the more difficult to mount, as contacts and access pins are under the chip and making it almost impossible to troubleshoot with conventional means.

Four groups of component types are:

1. **THROUGHHOLE**
   includes DIP, PGA, TO-, DO-, Axial and Radial Lead Components
2. **SMD Chips and Discretes**
   includes SOT, SOD, Resistors, Capacitors, Inductors, and Crystals
3. **SMD ICs (lead frame types)**
   includes QFP, PLCC, TSOP, SOIC, SOJ, SSOP, TSSOP types
4. **BGA and FLIP CHIP**
   includes Ball Grid Arrays, Bumped Devices, CSP and Land Grid Arrays

Packages are the physical containers that house microelectronic devices. Devices can also be classified by package type within the above 4 types. Examples of these types of devices are shown below.

Axial lead component

Radial lead component

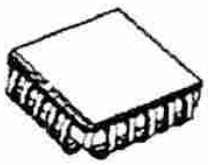
SIP - Single Inline Package

DIP - Dual Inline Package

Chip component, discrete capacitor or resistor

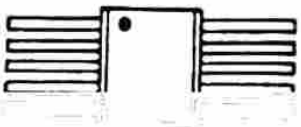
SMT - discrete Surface Mount Transistor

QFP - Quad Flat Pack



PLCC - Plastic (black) Leaded Chip Carrier
or CLCC - Ceramic (white) Leaded Chip Carrier

LLC – Leadless Ceramic Chip Carrier



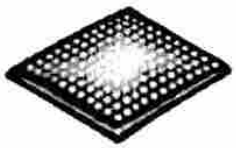
Flat Pack

TSOP - Thin Small Outline Package

MSOP - Miniature Small Outline Package

SSOP – Small Shrink Outline Package

TSSOP – Thin Small Shrink Outline Package



BGA - Ball Grid Array

From experience, typical large scale integrated circuit (LSI) chips used to be the most reliable. The Ball Grid Array (BGA) chip is commonly produced today, and is used extensively in most modern electronics products. It increases reliability tremendously.

Microprocessors and other chip sets rarely fail, except under extreme circuit conditions due to over voltage, excess heat or taking a soaking in water. In most cases today, troubleshooting high density circuits is reduced to only a few simple checks that can be done by a general service shop. Advanced troubleshooting requires a specialist with the same test equipment as the service shop as what is used in the manufacturing plant. Usually software controlled equipment is needed for automatic testing of complex circuit functions.

If a company today produces 1,000,000 widgets per month, that boils down to about one widget produced every 2.5 seconds !!! (2,592,000 seconds / month). There is little time for product testing without parallel production lines, or microcomputer assisted testing, where circuit tests are performed in microseconds.

So how can an electronics technician troubleshoot advanced circuits without the expensive automatic software controlled diagnostic equipment?

Some of the basic tests include looking for signs of misuse, bad connectors and incorrect settings of operator controls. Such is the case with consumer electronics, communications, aviation or medical equipment.

Simple Tests

One of the simple troubleshooting tests on discrete solid state circuits is to check for heat. Feeling the heat sink on a power amp or power converter will tell if the circuit is exceeding the designed temperature limits. Excessive current flow thru any circuit will generate higher than normal temperatures. Fuses and shutdown circuits protect most products today, so check glass fuses for any sign of excess current, soft blown or hard blown. A soft blown fuse results in a bent fuse link or an open link with no discoloration. A hard blown fuse is usually blown so fast due to a very high current overload that the fuse is discolored and the fuse element is completely gone.

Troubleshooting

Troubleshooting involves knowledge of the circuit function, expected performance compared to published specifications, and observed symptoms. Measurements of circuit parameters help to distinguish between the observations made and the area of the circuit or system that failed. The more precise the measurement made, the better the diagnosis will be. Simple steps in troubleshooting often help:

1. Observe symptoms
2. Compare to normal operation
3. Study flow charts and diagrams
4. Measure circuit parameters
5. Analyze results

The process may need to be repeated until the problem is found. Once a problem is found, proper documentation can help to resolve similar problems in the future.

Troubleshooting high density microelectronic boards adds another dimension of how to successfully remove the faulty component. Proper tools and specialized de-soldering equipment is required. An electro-static discharge (ESD) compliant workstation and soldering equipment is recommended for repairing and replacing all new types of components.

Also, with today's software control of hardware components, the appropriate test software is often required. Automatic diagnosis routines may be a tool that is available from the manufacturer of the product or diagnostic self-tests may be built into the equipment.

Chapter 10

Any way that it is performed, a computer is a tool needed for most software assisted troubleshooting today. That makes our life as an electronics technician just one notch higher on the skill-level ladder, but it also makes life easier to have computer assisted troubleshooting available.

NOTES

Chapter Quiz

1. **A BGA chip has _____.**

   a. thru hole pins
   b. leads on the sides
   c. solder balls underneath
   d. both gates anodized

2. **Troubleshooting discrete circuits involves a knowledge of _____.**

   a. circuit function
   b. where product was made
   c. how it failed
   d. a and c

3. **SMT stands for _____.**

   a. slow melting temperature
   b. surface mount technology
   c. sure mounts tough
   d. standard metal thickness

4. **When replacing a DIP chip, _____.**

   a. use an ungrounded soldering gun
   b. use an ESD static grounded soldering pencil or work station
   c. use the highest temperature possible to get the job done quickly
   d. use a solder sucker and 250 W gun

5. **One area where technicians will still find discrete electronic components is _____.**

   a. RF equipment
   b. mass produced video games
   c. high density circuit boards
   d. small portable handheld devices

6. **What basic circuit parameter makes it difficult to build integrated circuits with a high number of components to operate at super high frequencies?**

   a. Series resistance
   b. Parallel inductance
   c. High impedance
   d. Shunt capacitance

7. **When troubleshooting equipment that is software controlled, _____.**

   a. use of a computer may be required
   b. AC signals are not used
   c. DC voltages are always 5 volts
   d. diagnostics will be easy

8. **A category 2 type of electronics equipment could be a _____.**

   a. consumer home TV
   b. heart monitor
   c. fax machine
   d. FM walkman

9. **An example of a solid state device is _____.**

   a. Cermet resistor
   b. Tantalum capacitor
   c. silicon diode
   d. vacuum tube

10. **A four layer PN device is classified as a _____.**

    a. diode junction
    b. transistor
    c. thyristor
    d. integrated circuit

ANSWERS on page 223

Unit 4

Review Questions • Components, Semiconductors and Circuits
Chapters 8, 9 and 10

1. **How is a fuse installed in a circuit to insure proper operation?**

   a. Parallel to the load.
   b. In series with the load.
   c. In any way possible.
   d. At the ground point.

2. **A negative temperature coefficient thermistor will:**

   a. increase in resistance with temperature.
   b. decrease in resistance with temperature.
   c. have no change in resistance with heat or cold.
   d. increase in resistance with higher voltage.

3. **A NiMH battery cell, when charged, has a terminal voltage of:**

   a. 2.1 volts.
   b. 1.2 volts.
   c. 5.0 volts.
   d. 12 volts.

4. ***A thermocouple has an output in the range of:***

   a. millivolts.
   b. microvolts.
   c. volts.
   d. nanovolts.

nium diodes:

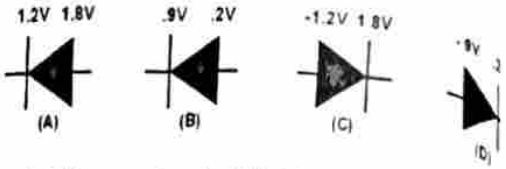
3V forward voltage drop.
often used in power supply circuits.
olerant of heat than silicon diodes.
rward voltage drop.

edance.
olar transistor.
nt changes.
nce.

es:

8. **Which of the silicon diodes below will conduct, given the voltages applied?**

   a. A
   b. B
   c. C
   d. D

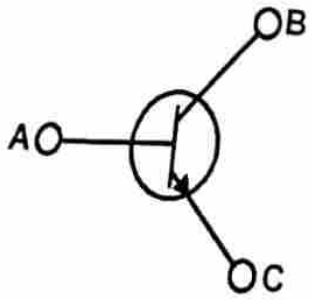


9. **If the gate voltage level, that turns an active SCR on, drops below the trigger point, while anode and cathode voltages are maintained, what will occur?**

   a. The SCR turns off.
   b. The SCR stays on.
   c. The SCR turns on.
   d. An output pulse is generated.

10. **The transistor configuration shown here is:**

    a. PNP.
    b. NPN.
    c. MOSFET.
    d. FET.



11. ***What is the effect on a varactor (varicap) when the reverse bias voltage is increased?***

    a. *Capacitance decreases and the tuned frequency increases.*
    b. *Capacitance increases and the tuned frequency decreases.*
    c. *The diode conducts more.*
    d. *The signal waveform becomes distorted.*

ANSWERS on page 223

Chapter 11

Electronic Power Supplies

by Ed Parady, CET

Ed Parady entered electronics as a Sonar Technician in the US Navy. In 1976 he joined the staff at Penn Technical Institute. In 2002 he accepted a position as the Data Cable Installer Instructor at the Pittsburgh Job Corps Center. Currently, Ed is employed as an Electronics Technician at Testex, Inc., a company that develops systems using Magnetism and Eddy Current technology to perform non-destructive testing of industrial plant equipment such as boilers, heat exchangers, piping, pressure vessels, condensers and storage tanks.

Introduction

Every technician needs to understand power supplies. In this chapter, you will find details on the:

- Purpose of a power supply.
- Sections of a simple DC (direct current) power supply.
- Function of each section in a DC power supply.
- Components that make up each section of a DC power supply.
- Symptoms that indicate a defective DC power supply.
- Types of power supplies.

An electronic power supply found in the typical piece of equipment is intended to change AC (alternating current) power line voltage into DC voltage levels required for the operation of electronic circuits.

The major sections of a simple DC power supply are the diode rectifier, capacitor filter and regulator circuit. Many types of power supplies are available today, including complex switching types that change the input AC to a higher frequency before changing the output to DC. This chapter will cover the fundamentals, so lets get started.

Purpose of a Power Supply

Power supplies are often classified in four categories:

- AC to AC frequency converters
- AC to DC rectifiers
- DC to AC inverters
- DC to DC converters

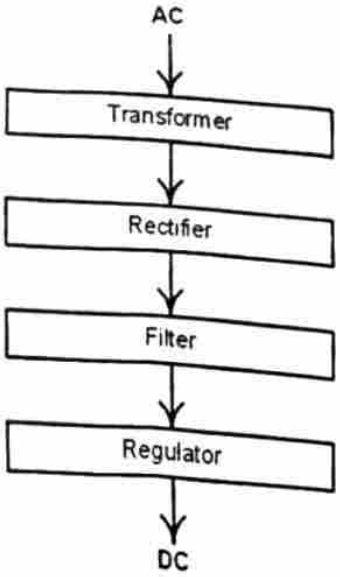
In the early 1900s George Westinghouse and Nicola Tesla, who were advocates of AC electrical power distribution, overtook the concepts of Thomas Edison using DC power as the accepted type of electric power distribution.

Subsequently, AC power became the standard utility. However, electronic systems are made up of amplifiers, oscillators, microprocessors and other circuits that require a steady, set voltage called DC. AC uses a changing voltage waveform and, therefore, must be changed to DC for systems to operate.

The most common power supply is one that converts AC to DC so we will focus on this type of power supply. The objective is to provide smooth, steady DC from the AC line source.

Sections of a DC power supply

The order of the sections within a power supply must be maintained, that is, you cannot filter before you rectify, etc. The sequence is diagrammed as:



(A regulator is not considered a basic section of a power supply but is most often included.)

Transformer

The transformer is used to step-up or step-down the voltage. 120 Volts AC can be changed to a smaller or larger amount by passing it through a transformer. Figure 1 is a picture of a small transformer and Figures 2 and 3 show the schematic symbols. The AC receptacle or outlet will connect to the Primary while the rectifier will connect to the Secondary.

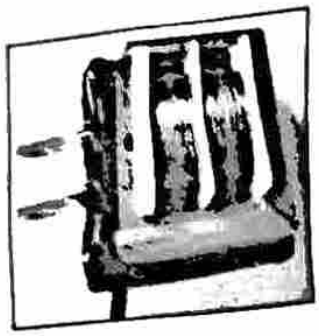


Fig. 1

Fig. 2

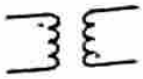


Fig. 3

If you examine Figure 1 (above), notice that it is nothing more than wire looped around a framework. Each loop is called a turn and you have primary turns and secondary turns.

The voltage at the secondary is determined by the ratio of the number of turns in the primary to the number of turns in secondary, and the voltage applied to the primary.

If you have 10 turns in the primary and 1 turn in the secondary a primary voltage of 120 volts would be stepped down to 12 volts on the secondary.

Rectifier Circuits

To review, AC is a waveform generated when current flows in one direction, stops and then flows in the opposite polarity. DC is a current flowing the same polarity or direction. A graph of AC voltage versus time is shown in Fig. 4.

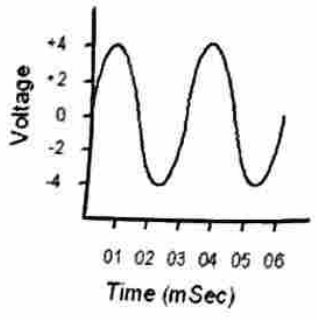


Fig. 4

A rectifier changes AC as shown in Fig. 4 to look like pulsating DC as in Fig. 5 shown as half wave, or Fig. 6 shown as full wave.

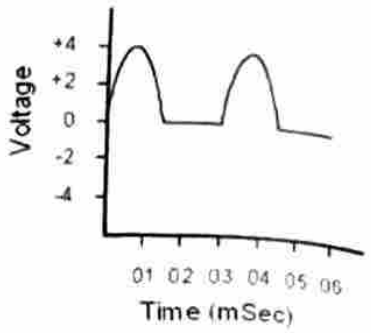


Fig. 5 Half Wave Pulses

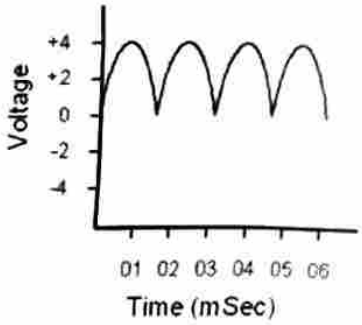


Fig. 6 Full Wave Pulses

The heart of the rectifier is the diode. The diode allows the electrons to move in only one direction, changing from a flow in two directions (AC) to a flow in one direction (DC).

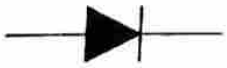


Fig. 7

The schematic symbol of a diode shows the anode (arrow) on the left and the cathode (bar) on the right. Electron flow is against the direction of the arrow much like water flow in a check valve.

The most common connection of diodes is shown below:

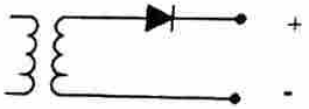


Fig. 8 Half Wave Rectifier

The circuit in Fig. 8 creates half wave (as shown in Fig. 5) and is called a half wave rectifier. It eliminates the

negative going part of the AC waveform by blocking the current flow when the anode (P-type semiconductor material) is negative, leaving only the positive going wave to be applied across the load.

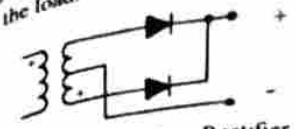


Fig. 9 Full-Wave Rectifier using a center-tapped (CT) transformer

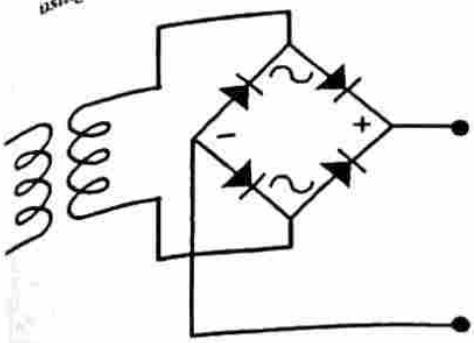


Fig. 10 Bridge Rectifier

The circuits in Fig. 9 and Fig. 10 both create full wave DC. Orientation of the diodes is important, as it will affect the polarity of the pulsating DC output. Note that Fig. 8 and Fig. 9 create a positive going waveform and that Fig. 10 will also produce a positive going waveform because of the direction of the diodes connected in the bridge circuit.

Without filtering these rectifier circuits produce pulsating DC which may not be very useful. Fig. 5 and Fig. 6 represent pulsating DC, where the rectifier changes AC to pulsating DC. A battery on the other hand outputs "pure DC" which has no variation in the waveform and is a steady level. A DC power supply tries to achieve "pure DC" by filtering and regulating the changing values, so as to be steady for the required load.

The half-wave rectifier Fig. 8 produces one pulse per cycle and the full-wave Fig. 9 and bridge rectifier Fig. 10 produce two pulses per input cycle of AC. As a result, the ripple frequency of a half-wave rectifier will be the same as the AC input frequency and that of the full-wave or bridge will be twice the AC input frequency.

Trouble Symptom:

If one diode in the full-wave or bridge opens, the rectifier will act as a half-wave and the ripple frequency will be the same as the AC input frequency.

Filters

Take another look at Fig. 5. Notice how the rectifier left us with high and low points. This is not the constant DC voltage we need. Suppose Fig. 5 represented water in a glass. If the high points represent a full glass, then the low points would be an empty glass. Suppose we refilled the glass after just a little water was removed. What would that do? Right! The glass would never be empty.

Fig. 11 and Fig. 12 demonstrate this.

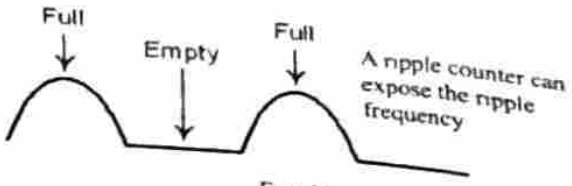


Fig. 11

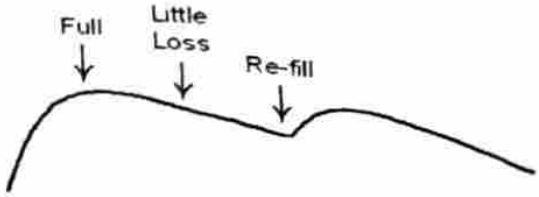


Fig. 12

This is what the filter does: In Fig. 11, pulsating half wave DC is shown with empty voids between the positive pulses. Fig. 12 shows the empty void filled in from the charge held in a capacitor that is connected in parallel with the rectifier circuit.

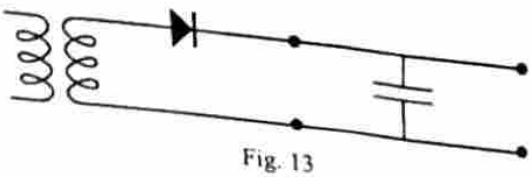


Fig. 13

The capacitor reduces the pulse amplitude and reduces the ripple or change in the DC output.

How does the parallel connected capacitor do that?

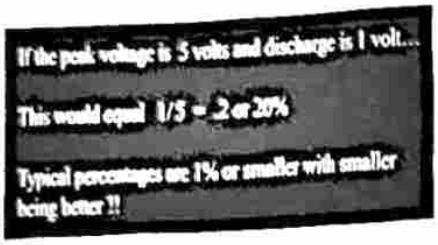
Like the glass holding the water, a capacitor holds electricity in the form of a charge. The capacitor is charged or filled and only loses a portion of the charge before it is filled up once again. The result is a fairly smooth DC voltage.

How smooth is it?

The peak voltage, when divided by the lowest discharge value, is a value that is approximately equal to the percentage of ripple.

$$\% \text{ of Ripple} = \frac{\text{AC ripple change}}{\text{DC peak out}} \times 100$$

The percent of ripple is difficult to calculate because of the wave shape of the discharging capacitor. That discharge is controlled by the value of the load connected to the DC power supply, and it is related to TC = R x C which is the time constant for the capacitor filter and load resistance.



The larger the capacitor, the less ripple in the DC output.

Notes:

- Filters consist of capacitors, resistors and inductors.
- More capacitance, that is, replacing a 10 mF with a 50 mF, would cause a lower ripple voltage and smoother DC output.
- BE CAREFUL! The capacitors in a power supply hold a charge even after the system is turned off. Before working on a power supply always:

Turn off the power
Discharge all capacitors

To calculate the DC voltage output of a power supply, and the voltage ratings of the capacitor required across the rectifier, the following formulas can be used, when the AC voltage is known at the transformer secondary.

Vp-p = 2.828 x Vrms
Vp = 1.414 x Vrms
Vrms = .707 x Vp
Vave = .636 x Vp

For example, if the secondary of the transformer is rated at 10 Vrms, then the peak DC output of a half wave rectifier is :

Vp = 1.414 x 10 vrms = 14.14 volts peak

This does not account for the voltage drop across the diode rectifier which may be .3 v for Germanium and Shottky diodes and .7 volts for Silicon diodes. The actual DC output will be reduced by this amount.

Trouble Symptom:

If the ripple voltage is higher than the specifications but the frequency is correct, the filter is the problem.

Here is an example of how ripple would be shown (in the shaded gray area below) in the specifications of a power supply.	
Operating Range	95 - 132 VAC 190 - 264 VAC
Frequency	47 - 63 Hz
Current	10.0 @ 115 V 5.0 @ 230 V
EMI	FCC B, CE
Efficiency	65%
Output	+5V @ 22A +12V @ 8A -5V @ 0.3A -12V @ 0.5A +3.3V @ 13A +5VSB @ 2A +5V & 3.3V<125W
Regulation	5% (+5, +12) 10% (-5, -12) 5% (+3.3)
Ripple	1% (p-p)
You can research more specifications by visiting www.pcpowercooling.com	

Regulators

The regulator is used to keep the output voltage of the supply constant in spite of changes in the power demands of the system load connected to it.

Did you know?

Your radio demands more power when you turn up the volume than it does at a low listening level. Think of this the next time your favorite song is played. Actually, changing the volume level would be just another event to a regulator that must do its job with every little change in the music.

How does it do that?

The secret is in a very specialized part called a Zener Diode.

This special diode will lock at a set voltage and maintain that level as long as the demands remain within a specified range. It is connected in the circuit so that it is reverse-biased, and conducts over its operating range, so that it regulates the voltage across it. Other components such as resistors and transistors act as supporting components and will also be found in the regulator section.

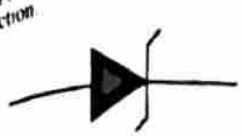


Fig 14
Zener Diode

Note: The Zener diode is vital to regulation but other components will also be found.

Trouble Symptoms:

When the Zener diode becomes defective, a higher than normal output voltage could result. This higher voltage is likely to cause other sections of the system to become defective or present errors. When these other sections are replaced, the unit may operate properly for a time but eventually the trouble will reappear. Before changing the disk drive, hard drive cables and motherboard, you may want to consider this fact.

Percent of Regulation

$$\%\,Reg = \frac{(V\ no\ load) - (V\ full\ load)}{(V\ full\ load)} \times 100$$

Just like percent of ripple, the lower the percentage of regulation, the better the power supply.

Chapter 11

Other types of power supplies

The power supply may take many forms which are often unique to the system being supplied.

Some lessons learned through experience include:

- Stereo Receivers often used a power supply that consisted of a half-wave, full-wave and bridge type rectifier. Multiple power supplies provide multiple output voltages for different circuits. Power amplifiers usually require higher voltage and demand larger current, so power supplies tend to be heavier to provide greater output.

- Battery power is as popular today as ever. Better batteries are being produced and found in computers, cell phones, power tools and automobiles.

- Occasions requiring a change from DC back to AC are becoming more popular today. Power inverters allow us to use 110 VAC appliances in our car. Some of the available power inverters may be explored by visiting web sites such as www.voltageconverters.com.

- One of the more popular DC power supplies is the one found in our PC. It is a switching type that changes the AC line frequency to a much higher frequency. This design creates a power supply that is more efficient and uses smaller components. For a detailed discussion on the PC power supply visit:
www.howstuffworks.com/power-supply.htm.

Summary

Take a look at the sections of the power supply in Figure 15. It consists of a transformer, half-wave rectifier, capacitor type filter and a Zener diode regulator, all of the basics covered in this chapter.

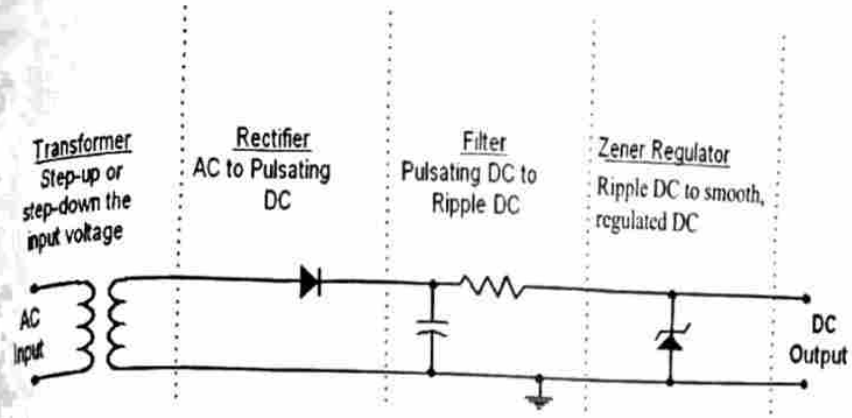


Fig 15

Chapter Quiz:

1. **An electronic power supply may be designed to _____.**

   a. change DC to DC
   b. change AC to AC
   c. change the frequency
   d. change AC to DC
   e. All the above answers are correct

2. **In the normal layout (block diagram) of a DC power supply the order of the sections would be:**

   a. Filter, Rectifier, Regulator.
   b. Rectifier, Filter, Regulator.
   c. Rectifier, Regulator, Filter.
   d. None of the above.

3. **This section of the DC power supply changes AC into pulsating DC.**

   a. Transformer
   b. Rectifier
   c. Filter
   d. Regulator

4. **This section of the DC power supply would step-up or step-down the voltage.**

   a. Transformer
   b. Rectifier
   c. Filter
   d. Regulator

5. **This section of the DC power supply would provide a steady DC voltage to a system.**

   a. Transformer
   b. Rectifier
   c. Filter
   d. Regulator

6. ***The zener diode is the most important component in this section.***

   a. Filter
   b. Regulator
   c. Both a & b

7. ***A ripple frequency of twice the AC frequency is normally associated with:***

   a. a half-wave rectifier.
   b. a full wave rectifier.

8. **Changing the filter capacitor from 10 mF to 50 mF would result in:**

   a. a lower ripple voltage.
   b. a lower ripple frequency.
   c. both a and b.

9. **The ripple frequency of the output of a full wave rectifier power supply measures the same as the AC input. The section causing this problem is the _____.**

   a. transformer
   b. rectifier
   c. filter
   d. regulator

10. **The ripple voltage is higher than the specifications but the frequency is correct. The section causing this problem is the _____.**

    a. transformer
    b. rectifier
    c. filter
    d. regulator

11. **A new computer gave a POST disk drive error the first time it was powered up. A new disk drive was installed and the same error was displayed. The hard disk cables were replaced and still the problem persisted. The motherboard was replaced and the system finally operated properly. After a day of operation the error returned. The most likely cause of the trouble is _____. (No reference in text, use logic.)**

    a. the power supply
    b. the disk drive
    c. the hard drive cables
    d. the motherboard

12. ***The switching power supply used in a PC takes the input from the AC line and changes it to _____.***

    a. a lower frequency
    b. a higher frequency

ANSWERS on page 223

Chapter 12

Amplifier Circuits, Types and dBs

by David E. Caldwell, CETsr

David E. Caldwell is a senior certified electronics technician in avionics and communications. He retired from the USAF after working on advanced fighter aircraft and state-of-the-art weapons systems for twenty years. He then taught avionics technicians at Pittsburgh Institute of Aeronautics, before developing and teaching his current two-year electronics technician course for the PA Department of Corrections. His course enables graduates to become certified as associate level electronics technicians. He has written several articles for FTA, chaired the FCC review of Element 3 of the General Radio Operators License (GROL), and is actively involved with his church and Habitat for Humanity.

Amplification

Amplification is the process of increasing the magnitude of a variable quantity (especially voltage, current or power), without altering any other quantity. In the study of electronics, to increase the magnitude of a signal's voltage, current or power, we use an amplifier. An amplifier consists of an active device, supporting circuitry for DC power and an input signal. The amplifier will provide a function called gain. Gain may be expressed as:

Voltage Gain:
A_V = (output signal voltage) / (input signal voltage)

Current Gain:
A_I = (output signal current) / (input signal current)

Power Gain:
A_P = (output signal power) / (input signal power)

The active devices used as discrete amplifiers are:

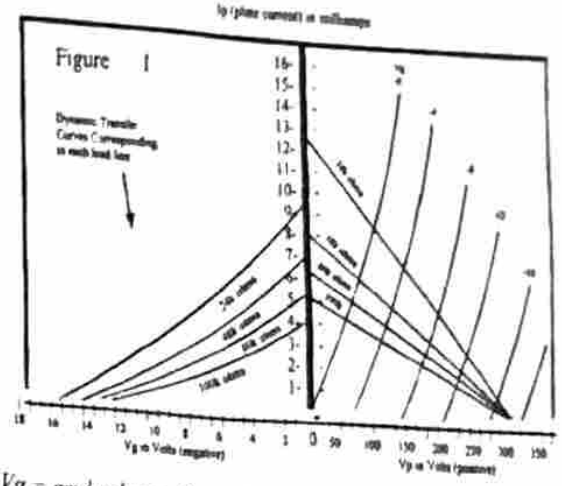
- Vacuum tubes (VT)
- Field effect transistors (FET)
- Bipolar junction transistors (BJT)

Integrated circuits (ICs) that incorporate some of these devices and their biasing circuits internally are in wide use today. One of the less complex ICs is known as an Operational Amplifier (Op Amp). We will briefly examine each of these devices' characteristic curves and bias them to operate within their specified parameters.

Vacuum Tube Amplifiers

Although vacuum tubes are not widely used these days, they still warrant study. Notice in Figure 1 a graph representing plate characteristic curves on the right and the corresponding dynamic transfer curves for each load line on the left. The load lines on the right are calculated by first assigning a maximum value for V_P.

Next, assume the tube is shorted and calculate Ip by using (VP) / (RL). For 24k, this equals (300) / (24k) → 12.5 mA. For 48k, we set (300) / (48k) → 6.25 mA.



Vg = grid voltage, Vp = plate voltage, Ip = plate current, and R_L = load resistance

With V_P and I_P calculated, simply draw a line connecting the two points on the plate curve side of the graph. To establish the dynamic curves on the left, assign various values of V_G and note the intersections of V_G and your load line. Plot these points on the left side of the graph and connect them. This will yield the dynamic transfer curves on the left side of the graph.

Next, let's assume we have a 10 V_{P-P} 50 Hz to 15k Hz input signal we wish to amplify. Examine the dynamic curve on the left corresponding to our load line and establish an amount of VG that will let the signal ride on the most linear portion of the load line. For our 48k load line, this would be -7 volts. Our signal will drive V_G down to -12V and up to -2V. With V_G = -7, we have I_P = 2.5ma and V_P = 175V. This will be our quiescent operating point (Q).

To determine R_K we need to have the cathode 7V more positive than the grid and we know we will have IP = to 2.5mA so: R_K = (Q_{VG}) / (Q_{IP}) → (7) / (2.5mA) → 2.8kΩ. Let X_C of C_K = (R_K) / (10) at the lowest frequency to be bypassed, so: C_K = (10) / (2π x f x R_K) → (10) / (6.28x50x2.8K) → 11.4μF. So we choose to use a standard 10μF capacitor.

For a grid resistor we will use a typical value for R_G of 470kΩ, this will allow any space charge that accumulates around the grid to bleed to ground.

When the negative alternation of the input signal is applied, V_g will drop to -12V causing I_p to decrease to approximately 1mA and V_p to decrease to 40V. During the other alternation of the input signal V_g will increase to -2V causing I_p to increase to approximately 4mA and V_p to increase to 192V.

One $10V_{p-p}$ input signal has caused our output signal V_{out} to vary by 143V resulting in a V gain of 141.10 = 14.3

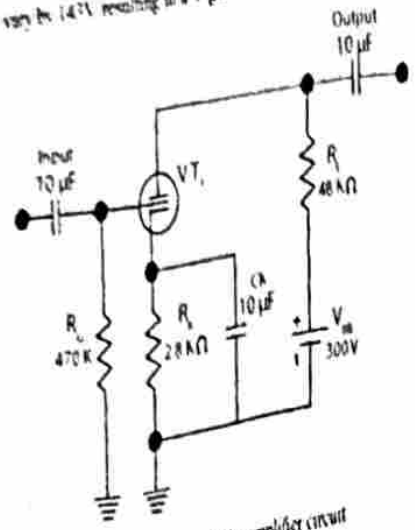


Figure 2 Vacuum Tube amplifier circuit

Let's compare this to the amplification factor of the tube. If we use the quiescent current of 2.5ma as a reference and draw a horizontal line across the plate curves, we can calculate the µ (mu) or (amplification factor) of the tube by using $\Delta V_p / \Delta V_g$ for our circuit V_p = 263V at V_g = -12, and V_p = 70V at V_g = -2. Our µ = (263-70) (12-2) → 19.3

The plate curves can also be used to calculate r_p (AC plate resistance) by following one of the curves and noting $\Delta V_p / \Delta I_p$. If we use Vg = -4, note that I_p = 6mA when V_p = 188 and when I_p = 1mA our V_p = 120 so we get

r_p = (68) ÷ (5m) → 13.6 kΩ

Voltage gain can then be calculated by the formula

(µ x R_L) ÷ (R_L + r_p) →

(19.3 x 48k) ÷ (48k + 13.6k) → 15.

Note how both methods used to establish Vgain resulted in nearly the same results, however, µ was not equal to either. This is because the circuit biasing components also affect gain.

Notice that we also kept the input signal on the linear portion of the load line and we kept the tube conducting for the entire cycle of the input signal.

Class of Operation

When output current flows for 360° of the inp
say the amplifier is **Class A** and also linear

If the DC bias or the amplitude of the input si
justed, the class can change

If output current flows for 280°, it is **Class A**
mately 200°, it is **Class B** and if less than 180°,
it is **Class C**

Class C amps are basically reserved for radio
plifiers, the tank circuit associated with these a
the rest of the signal. Class AB and B can be u
reproduction if connected as push-pull amps

FET Operation

For our next example, let's examine another
trolled device similar to the vacuum tube. Refe
and 4 and note the following:

V_{GS} is equivalent to V_g
I_D is equivalent to I_p
V_{GS} is equivalent V_g from our vacuum tube ex

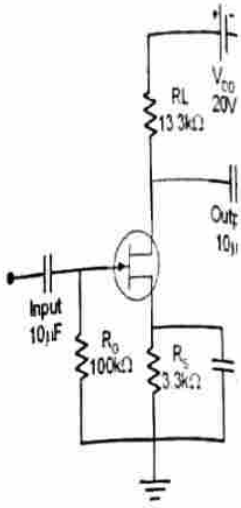


Figure 3 FET amplifier circu

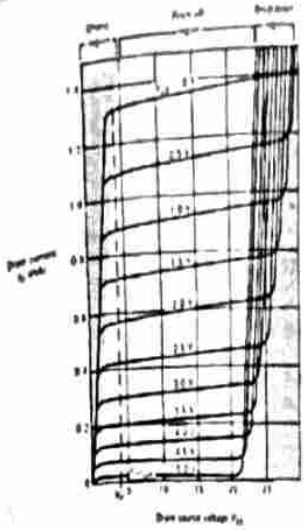


Figure 4

The JFET has an amplification factor:
$\mu = (\Delta V_{DS}) / (\Delta V_{GS}) \mid I_D$ constant,

and dynamic drain resistance:
$r_D = (\Delta V_{GS}) / (\Delta I_D) \mid V_{GS}$ constant.

The gain of the circuit may be determined by:
voltage gain = $(\mu \times R_L) / (R_L + r_D)$.

Let's first establish the two end points of our load line. If we have a 20V power supply to bias the circuit and minimal current is flowing the $V_{DS} = 20$. If the JFET is biased on all the way (saturated), it will appear as a short and I_D will be maximum at $(V_{DS}) / (R_D)$.

Let's establish I_D maximum to be 1.2mA. If we wish to amplify a $2V_{p-p}$ input signal in a linear fashion, we must establish a V_{GS} that will allow the V_{GS} to swing +1V and -1V from our bias point while remaining within the pinch off region for linear amplification.

Let's choose -2V as our V_{GS}. This establishes our quiescent V_{DS} to be ≅ 10V and I_D ≅ .6mA, giving us a R_D of 16.75kΩ. Let $R_S = (V_{GS}) / (I_D) \rightarrow (2) / (.6m) \rightarrow 3.3k\Omega$. Finally let $R_L = R_D - R_S \rightarrow 16.6k - 3.3k \rightarrow 13.3k\Omega$. X_C of C_S should be approximately 1/10 of this value.

When we apply our input signal, the gate voltage will decrease to –1V allowing I_D = .9mA. This results in V_{RL} = 12V. On the next alternation the V_G = -3V decreasing I_D to .3mA through R_L. This results in V_{RL} = 4V. We achieve a voltage gain $A_V = (8V) / (2V) \rightarrow 4$.

Let's compare $A_V = 4$ to the amplification factor of the JFET (μ). Let's use V_{DS} = 20V and 5V and draw a line horizontally at I_D = 6mA. At V_{DS} = 5V we have V_{GS} = 1.9375 and at V_{DS} = 20 we have V_{GS} = 2.0625

This gives us a ΔV_{GS} = .1250 and
$\mu = (\Delta V_{DS}) / (\Delta V_{GS}) \rightarrow (15) / (.1250) \rightarrow 120$

Second, calculate $r_D = (\Delta V_{DS}) / (\Delta I_D) / V_{GS}$ held constant by following the V_{GS} = -2.0 V curve and note I_D = 0.575mA at V_{DS} = 5V, and I_D = 6.250 mA at V_{DS} = 20V.

therefore, $r_D = (15) / (.05m) = 300\ k\Omega$

Computing voltage gain AV with these values:

$A_V = (\mu \times R_L) / (R_L + r_D) \rightarrow$
$(120 \times 13.3k) / (13.3k + 300k) \rightarrow 5$

Allowing for errors reading the graph, we get nearly the same results. This also further illustrates how much impact the biasing circuit has on gain.

Our μ (amplification factor) was approximately 120 but our gain was held to 4 or 5. Biasing the JFET into Ohmic region will result in a non-linear operation similar to the VT but necessary if you plan on operating class AB, B, or C.

Bipolar Transistor Operation

Our third device, the BJT (Bipolar Junction Transistor), is a current operated device.

Study Figures 5 and 6 and note that if the transistor is biased off, it will have V_{CE} = 12V, and if we bias it on to the point of saturation, we will have 8.2 mA of I_C.

Draw a load line between these two points.

Next, we establish the quiescent operating point (Q) for our amp. For Class A operation, choose a point midway along the load line.

We choose I_B = 40μA giving:
V_{CE} = 4.875V or I_C = 4.74 mA.

Note how I_B = 40μA controls I_C = 4.75 mA giving a β (I_C / I_B) (beta) of approximately 120.

Beta varies from various types of transistors and identical part number transistors. Transistors can also break down from thermal runaway. To compensate for this and to regulate our current gain, we use different forms of biasing.

For our example, we will illustrate one of the most stable forms of biasing known as **voltage divider biasing**.

It consists of R_B, R_E and for internal stability R_L. For our midpoint of operation, we will have 40 µA of I_B and 4.75mA of I_C to give us I_E = 4.79mA.

To determine the maximum load, use R_L = (V_{CE} cutoff) / (I_C saturation) → (12) / (8.25m) → 1.5kΩ. Deduct 100Ω to use as R_L since I_C will flow through it as well. This leaves us 1.4 kΩ for R_L.

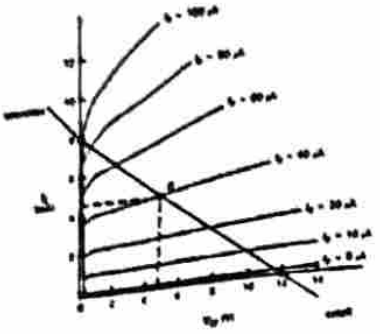


Figure 5

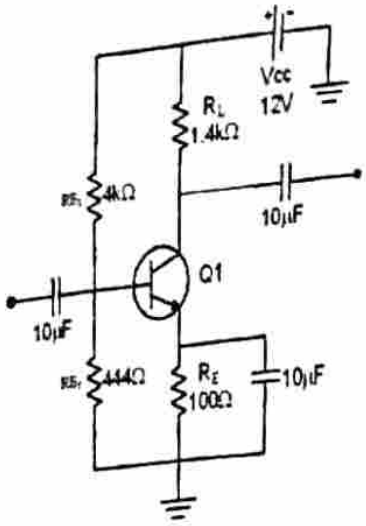


Figure 6

If R_E = 100Ω the $V_{RE} = I_E * R_E$ → 4.79 mA * 100 → .479V and the V_B must be .7 more positive to bias the device on, so V_B = .479 + .7 → 1.179V.

The V_{RB2} = 1.179V and V_{RB1} = 12 - 1.179 → 10.821V.

Next we will use a bias stability factor of 5 to calculate the size of R_{B2} by using the formula:

$$S = [(R_{B1} \times R_{B2}) / (R_E \times (R_{B1} + R_{B2}))] + 1.$$

We also know that R_{B1} is 9.18 times larger than R_{B2} because of the voltage drop ratios (10.821/1.179).

We can substitute (9.18 x R_{B2}) for R_{B1}. 5 = [(9.18
(100 x (9.18R_{B2} + R_{B2}))] + 1 → 444Ω for R_{B2} then w
R_{B1} = (9.18 x 444) = 4kΩ.

To verify, we use the voltage divider equation
[(R_{B2}) / (R_{B2} + R_{B1})] x V_{CC} →
[(444) / (444 + 4k)] x 12V → 1.2V.

One last component to calculate is the size of C_E, cap. For most applications, set its X_C value to 1/10 lowest frequency to be amplified.

Let's apply a signal to our circuit Figure 6 and var 60µA to 30µA causing the I_C to vary from 6.5mA to $\Delta I_C / \Delta I_B$ → (3.5m) / (40m) → 87.5.

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We can substitute (9.18 x R_{B2}) for R_{B1}. 5 = [(9.18
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[(444) / (444 + 4k)] x 12V → 1.2V

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Let's apply a signal to our circuit Figure 6 and var 60µA to 30µA causing the I_C to vary from 6.5mA to $\Delta I_C / \Delta I_B$ → (3.5m) / (40m) → 87.5.

A_V can be determined by first determining AC resi the emitter. r_E = 25mV / I_E → (25m) / (4.79m) → 5.2

To allow for variations we also determine the high using r_E = (50mV) / (I_E) → (50m) / (4.79m) = 10.4Ω

Next, A_V = (R_L) / (R_E + r_E) → (1.4K) / (100 + 5.2).

Since we added a bypass C_E, we can use

(R_L) / (r_E) → (1.4k) / (5.2) = 269 or (1.4k) / (10.4) 134.6.

Finally, let's determine power gain,

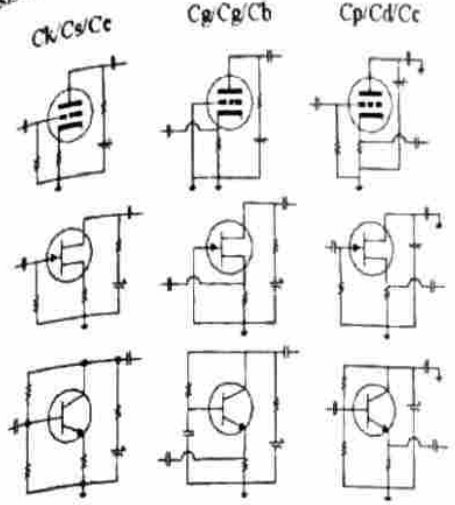
A_P = A_I x A_V → 87.5 x 134.6 → 11.8k.

This may be expressed in dB's:

A_P = 10 log 11.8k → 40.7 dB.

The values chosen to bias our various amplifier circuits may need to be adjusted slightly to compensate for amplifier configuration, input/output impedance matching, frequency losses and signal distortion.

Vacuum tube amplifier circuits compared with FET and transistor configurations:



A_v = Moderate	A_v = High	A_v = Low
A_i = Low	A_i = Low	A_i = High
A_p = High	A_p = High	A_p = Moderate
Z_{IN} = Moderate	Z_{IN} = Low	Z_{IN} = High
Z_{OUT} = Moderate	Z_{OUT} = High	Z_{OUT} = Low
180° Phase Shift	0° Phase Shift	0° Phase Shift

Figure 7

Figure 7 was created so you could visually compare the various amplifier configurations. The common-cathode (Ck), common-source (Cs), or common-emitter (Ce) amplifiers have high power. Voltage and current gain are most widely used.

The common-grid (Cg), common gate (Cg), or common-base (Cb) amplifiers are used primarily as RF amplifiers giving good power and voltage gain but no current gain. The common-plate (Cp), common-drain (Cd) and common-collector (Cc) amplifiers are used mainly as isolation amplifiers giving good power and current gain but no voltage gain.

IC Operational Amplifiers

As our technology has advanced, the IC (integrated circuit) has emerged. The operational amp is one such device. It incorporates a combination of the amplifier circuits internally in such a way as to give it a high input impedance so it won't load down the signal source and a low output impedance so it can drive a low impedance load. Op-amps may have open loop gains of 200,000. One added advantage of the Op-amp is its common-mode rejection, which allows it to reduce hum and noise.

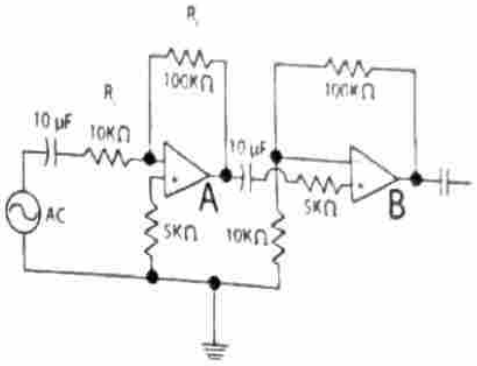


Figure 8 inverting (A) and non-inverting (B) Op-Amp

If a signal is applied to the non-inverted input as illustrated in Figure 8B, the $A_v = 1 + [(R_f) / (R_i)]$, and if the signal is fed to the inverting input as illustrated in Figure 8A, the $A_v = -[(Rf) / (R_i)]$.

Notice how gain is set by the ratio of R_f to R_i. Op-amp A_v tends to drop off rapidly at higher frequencies.

Types of Coupling

Three types of coupling are used in cascading amplifier circuits together:

1. direct
2. capacitor
3. transformer.

Figure 8 above uses capacitor coupling. Figure 9 below uses direct coupling and Figure 12 uses transformer coupling.

Negative Feedback

Negative Feedback in any amplifier reduces gain and increases bandwidth, where Positive Feedback causes oscillation.

Amplifier circuits

Darlington

A Darlington Pair is an example of a directly coupled pair of transistors. The collectors are tied in parallel while the base emitter junctions are in series. It has high input impedance to reduce circuit loading and low output impedance with good power gain. Often it is housed in a single package.

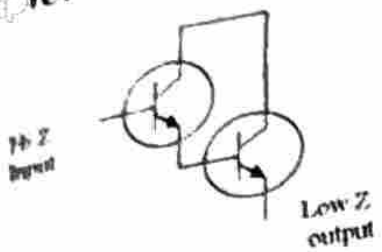


Figure 9 Darlington pair

Differential Amplifier

A differential amplifier is another circuit pair with matching transistors as shown in Figure 10. The object of a differential amp is to amplify the difference between the two inputs, and cancel any common signal between them. Out of phase signals are amplified because there is maximum instantaneous difference between any point, and in-phase signals are rejected. The output is taken off the collector of each transistor in a balanced output. Differential amps are used in Op-Amps as well as medical electronics devices where balanced sensor input need to eliminate noisy and unwanted interference.

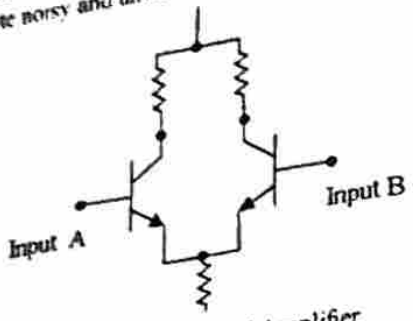


Figure 10 Differential Amplifier

Complementary-Symmetry MOSFET power amplifier

Complementary-Symmetry MOSFET is shown below. One N-channel and one P-channel FET is used in a series connected output circuit. High input impedance with low output impedance gives good power gain.

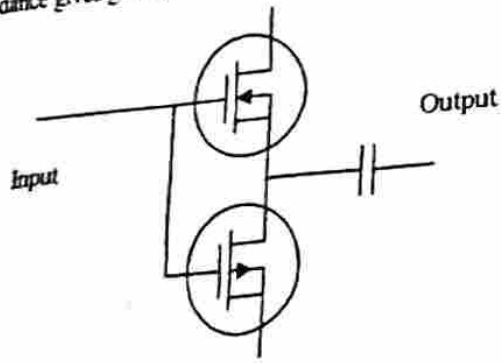


Figure 11 Complementary-Symmetry MOSFET

Push Pull RF Amplifier

Our focus so far has been on Class A or linear tion, but at times we may need to use 2 devices i amplifier configuration Push-Pull will allow one amplify one alternation of the input signal while vice amplifies the remaining alternation of the in realize any increase in efficiency we must bias h operate as Class AB or B and still amplify the si fashion. See Figure 12.

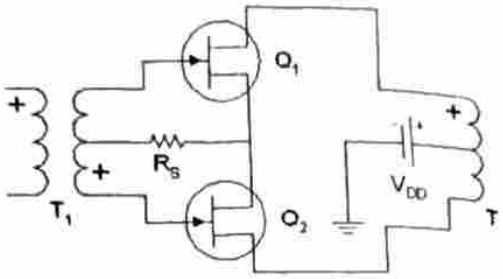


Figure 12 Push Pull FET RF ampl

At radio frequencies, a class C amplifier wit action of a tank circuit can be used to reproduc See Figure 13. Class C conducts for less than signal. The same frequency is applied to the in nant output tank circuit. The amplifier pulses t to make it ring at the resonant frequency to pr ous output.

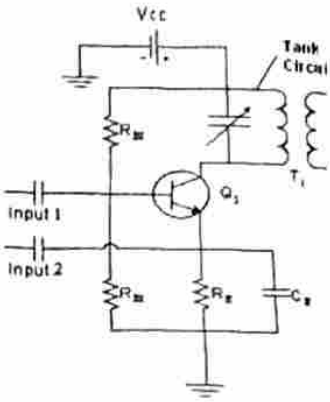


Figure 13

RF frequency conversion

An application of non-linear amplification (C is frequency conversion or heterodyning as use two radio frequencies are fed to an amplifier op linear fashion, they will combine to produce bo nal frequencies and the sum and difference of t frequencies. If the tank circuit in Figure 13 is t those four frequencies, for example the differe frequency will be amplified and passed on to t

The Q (quality) of the tank circuit will determine how wide of a bandwidth of frequencies may be passed along with the original tuned frequency of the tank. This allows conversion of a received radio frequency containing either amplitude or frequency modulated intelligence to an IF amplifier (intermediate frequency) in the receiver. When the received RF is mixed with a LO (local oscillator) frequency, the intelligence is transferred from the RF input to the IF stage of the radio receiver.

By gang tuning the first RF amp of the receiver with the LO, we reject any undesired image frequencies that may try to mix with our local oscillator and produce our given IF. This increases the receiver's selectivity (ability to reject image or unwanted frequencies). More RF discussion will follow in Chapter 13.

RF Detection

Another use of non-linear active devices is receiver detection. See Figure 14. Detection is the process of recovering the intelligence from the transmitted radio frequency. Once the intelligence has been transferred to our intermediate frequency, it is first amplified to a suitable level and then fed to the detection stage.

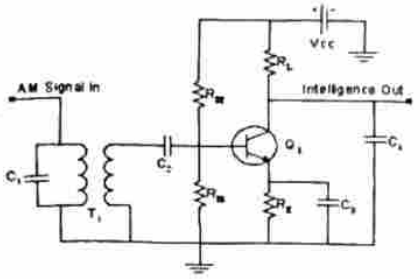


Figure 14 AM detector amplifier

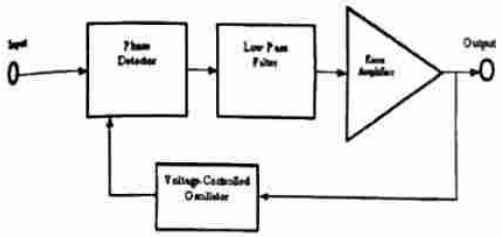


Figure 15 FM detector

For an FM (frequency modulated) signal, see Figure 15 where the IF is fed to a PLL (phase locked loop) circuit or discriminator. In a PLL, as the input frequency varies, the DC error voltage will vary in an attempt to keep the VCO on track. The variations in DC error correction voltage level and frequency of those variations of the DC error correction voltage equates to being equal to the intelligence that originally modulated the transmitted radio signal.

The Decibel (dB) *by Fitz Reid*

The Decibel (dB) is a sub-unit of the bel, named after the inventor of the telephone, Alexander Graham Bell. There are 10 decibels in one bel (1 dB = 1/10 Bel). Since the decibel is a ratio, it may represent volts, amperes, watts, acoustic power, etc. The decibel can be regarded as the smallest difference in sound intensity that a normal ear can distinguish.

An average conversation in a quiet room is about 55 dB above perfect silence or zero loudness.

Let us avoid confusion with *loudness* and *sound intensity*.

Intensity of sound is determined by the amount of energy carried by a sound wave.

Loudness is the actual sensation produced in the brain. Loudness depends on:

- The intensity of the sound waves which strike the diaphragm of the ear
- The functioning capability of the ear and its related hearing mechanisms

Notice that 0 dB indicates the presence of sound.

- It does not mean **NO SOUND**.
- It is the sound intensity to which other intensities are compared.

Some Facts To Remember:

1. dB is a ratio; therefore it needs a reference
2. dB is positive when it represents an increase above the reference level
3. dB is negative when it represents a decrease below the reference level
4. Gains are positive and losses are negative
5. dB may be added or subtracted directly
6. We hear sound in a logarithmic fashion

Power Levels

$$dB = 10 \log P_2 / P_1$$

Where: P_1 = input power
P_2 = output power

Chapter 12

Example: The output of an amplifier is 7 Watts and the input is 3.5 Watts. Calculate gain or loss in dB.

a) 3 dB
b) -3 dB
c) 0.3 dB
d) -0.3 dB

SOLUTION:

$$\begin{aligned} \text{Gain} &= 10 \log P_2 / P_1 \\ &= 10 \log 7 / 3.5 \\ &= 10 \log 2 \\ &= 10 \times (.3010) \\ &= 3 \text{ dB} \end{aligned}$$

Choice (a)

Example: A power of 10 milliwatts is required to drive an audio amplifier. The output of the amplifier is 120 milliwatts. What is the gain expressed in dB?

a) 10.8 dB
b) 12 dB
c) -12 dB
d) -10.8 dB

SOLUTION:

$$\text{Gain(dB)} = 10\log \frac{P_{out}}{P_{in}}$$

$$\text{Gain(dB)} = 10\log \frac{120}{10}$$

$$\begin{aligned} \text{Gain(dB)} &= 10\log 12 \\ \text{Gain(dB)} &= 10 \times 1.08 \\ \text{Gain(dB)} &= 10.8 dB \end{aligned}$$

Choice (a)

Example: A network has a loss of 16 dB. What approximate power ratio corresponds to this loss?

a) 40:1
b) -16:1
c) 1.6:1
d) 4.0:1

SOLUTION:

$$\begin{aligned} dB &= 10 \log P_{out}/P_{in} \\ 16 &= 10 \log P_{out}/P_{in} \\ \text{(Divide both sides by 10)} \quad 1.6 &= \log P_{out}/P_{in} \\ \text{(Taking antilog of 1.6)} \quad &= 39.8 \; P_{out}/P_{in} \\ &= 40:1 \end{aligned}$$

Choice (a)

Example: The output A.F. circuit of a pocket radio receiver is modified so that the power to the loudspeaker is changed from 450 mW to .65 W. Calculate this gain in decibels.

a) 1.6
b) -1.6
c) 16
d) -16

SOLUTION:

$$dB = 10 \log 650/450 = 1.6$$

Choice (a)

Voltage Ratio Levels

The true formula for finding voltage gain in decibels is

$$\text{Gain dB} = 20 \log (V_{out}/V_{in})$$

Example: The signal measured on a simple dipole antenna is 500µV. What dBmV does this signal represent?

a) -2
b) 2
c) 6
d) -6

SOLUTION:

The reference signal used in RF work is *1000 µV or 0 dBmV*. Therefore:

$$\text{Gain}_{dB} = 20 \log (V_{out}/V_{in})$$

$$\begin{aligned} dB &= 20 \log (500/1000) \\ &= 20 \times (-0.3) \\ &= -6 \end{aligned}$$

Choice (d)

dB Table

dB	RATIO	RATIO
1	1.1	1.2
3	1.4	2
5	1.7	3
6	2	4
7	2.24	5
10	3	10
12	4	16
16	6.3	40
18	8	64
20	10	100
24	16	256
30	32	1000
40	100	10000
50	320	100000
60	1000	1000000
70	3200	10000000
80	10000	100000000

Table 2

Let us look at ways of simplifying decibels by using a table. See Table 2.

(1) 12 dB = 6 dB + 6 dB
= 2 x 2 (Voltage Ratio)
= 4 times

(an increase, or amplification by the factor 4)

(2) 57 dB = 60 dB - 6 dB + 3 dB
= 1000 ÷ 2 x 1.4
= 700

(an increase, or amplification by the factor 700)

Expressing dBs

Three types of dBs

1. Acoustic sound pressure (SPL) levels referenced to .0002 dynes/cm² which is 0 dB for audio sounds
2. Absolute level (a signal, power or voltage) at a point with a reference level specified, as in dBm, (the "m" suffix means dB is referenced to 1 mW)
3. Relative difference between two points
   a. dB written just by itself
   b. Using TL as a relative dB measurement between two Transmission Level Points (TLPs)

Examples of each type

1) Acoustic sound pressure:
   - Also known as loudness of perceived sounds
   - Jet Engine@ 2000dyn/cm2 = 140 dB
   - 20 X log 2000/ 0002 = 140 dB sound level

2) Absolute:

• dBA	Acoustic threshold of hearing referenced to 0 dB
• dBm	Power level referenced to 1 mW (See dBm chart)
• dBV	Voltage level referenced to 1 volt
• dBmV	Voltage level referenced to 1 mV
• dBr n	Noise or signal level referenced to 1 pW
• dBr nC	Noise or signal as above with C weighting
• dBr nC0	As above, but also normalized to 0 TL
• dBw	Power level referenced to 1 Watt
• dBu	Voltage that corresponds to dBm in a 600-ohm circuit – about 78 volts

3) Relative:

dB = (+dB) Gain or (-dB) Loss of a circuit, NO Reference specified. (See "dB" chart)
TL = Net gain or loss of transmission path between two TLP points or dBm levels.

Tables on the following pages are provided for your use and reference.

NOTES

Chapter 12

Relative dB chart

Power ratio		dB	
		60 dB	
1,000,000		50 dB	
100,000		40 dB	
10,000		30 dB	Gain
1000		20 dB	
100		10 dB	
10		3 dB	
2		0 dB	
1	Unity Gain	3 dB	
.5		-10 dB	
.1		-20 dB	
.01		-30 dB	Loss
(10^{-3}) .001		-40 dB	
.0001		-50 dB	
.00001		-60 dB	
(10^{-6}) .000001			

Based on $dB = 10 \log P_2 / P_1$

Voltage ratio		dB	
		120 dB	
1,000,000		100 dB	
100,000		80 dB	
10,000		60 dB	Gain
1000		40 dB	
100		20 dB	
10		6 dB	
2		0 dB	
1	Unity Gain	-6 dB	
.5		-20 dB	
.1		-40 dB	
.01		-60 dB	Loss
(10^{-3}) .001		-80 dB	
.0001		-100 dB	
.00001		-120 dB	
(10^{-6}) .000001			

Based on $dB = 20 \log V_2 / V_1$

Absolute dBm Power Levels

Signal Power	dBm	dBW
1000 W (1KW)	60 dBm	30 dBW
100 W	50 dBm	20 dBW
10 W	40 dBm	10 dBW
1 W	30 dBm	0 dBW
.1 W (100 mW)	20 dBm	-10 dBW
10 mW	10 dBm	-20 dBW
1 mW	0 dBm	-30 dBW
.1 mW (100 mW)	-10 dBm	-40 dBW
10 mW	-20 dBm	-50 dBW
1 mW	-30 dBm	-60 dBW
.1 mW (100 nW)	-40 dBm	-70 dBW
10 nW	-50 dBm	-80 dBW
1 nW	-60 dBm	-90 dBW
.1 nW (100 pW)	-70 dBm	-100 dBW
10 pW	-80 dBm	
1 pW	-90 dBm	
1pW (100 fW)	-100 dBm	
10 fW	-110 dBm	
1 fW	-120 dBm	
.1 fW (100 aW)	-130 dBm	
10 aW	-140 dBm	
1 aW	-150 dBm	

Brownian Noise Floor
-173 dBm per Hz
-133 dBm / 10KHz
-113 dBm / 1 MHz
- 93 dBm / 100 MHz

Units

T	= Terra	10^{+12}
G	= Giga	10^{+9}
M	= Mega	10^{+6}
k	= kilo	10^{+3}
m	= milli	10^{-3}
m	= micro	10^{-6}
n	= nano	10^{-9}
p	= pico	10^{-12}
f	= femto	10^{-15}
a	= atto	10^{-18}

Charts courtesy of John Baldwin, CETsr

Chapter Quiz

1. **The formula [$(\Delta V_p) / (\Delta V_c)$ | I_p constant] is used to determine the _____.**

   a. transconductance of the vacuum tube
   b. AC plate resistance of the vacuum tube
   c. amplification factor of the vacuum tube
   d. forward current transfer ratio of the vacuum tube

2. **A class AB amplifier will allow output current to flow for approximately _____ of the input signal.**

   a. 100°
   b. 180°
   c. 280°
   d. 360°

3. **The formula [$(\mu * R_L) / (R_L + r_D)$] is used to calculate:**

   a. forward transfer current ratio of the JFET
   b. voltage gain of the JFET amplifier.
   c. amplification factor of the JFET.
   d. transconductance of the JFET.

4. **Choose three of the following amplifier configurations that yield the highest power gains.**

   a. common-grid
   b. common-cathode
   c. common-plate
   d. common-source
   e. common-gate
   f. common-drain
   g. common-emitter
   h. common-base
   i. common-collector

5. **Given the following information about a common-emitter amplifier, a .01Vp-p input signal causes transistor's base current to vary from 70μA to 50μA. This causes the collector current to vary from 3.5mA to 2.5mA through our 2.2kΩ load. Determine the dynamic current gain.**

   a. 10
   b. 20
   c. 50
   d. 100

6. **Which of the following amplifier configuration(s) would you likely find being used in a linear push-pull arrangement?**

   a. Class A
   b. Class B or AB
   c. Class C
   d. All of the above

7. **If a 1.02 MHz signal and a 1.475 MHz signal are fed to a common-emitter transistor amplifier biased to operate in a non-linear manner we would expect to see which of the following frequencies in the output?**

   a. 455 kHz
   b. 1.02 MHz
   c. 1.475 MHz
   d. 2.495 MHz
   e. All of the above

8. **Which of the following terms is associated with a commercial receiver being able to reject image frequencies?**

   a. Sensitivity
   b. Selectivity
   c. Discrimination
   d. Bandwidth

9. **Figure 14 on page 97 illustrates _____.**

   a. a non-active de-modulator
   b. an RF amplifier being used as a mixer
   c. a common-emitter amp used as a limiter
   d. an active detector

10. **Which of the following circuits can be used to recover audio from an IF in a FM receiver?**

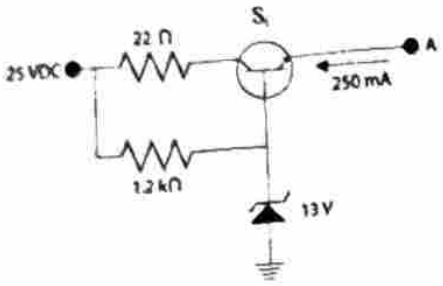
    a. Phase Locked Loop
    b. Class A, common-gate JFET amplifier
    c. Push-pull, class A JFET amplifier
    d. Mixer

ANSWERS on page 223

Unit 5 Review Questions - Analog Circuits, Power Supplies and Amplifiers Chapters 11 and 12

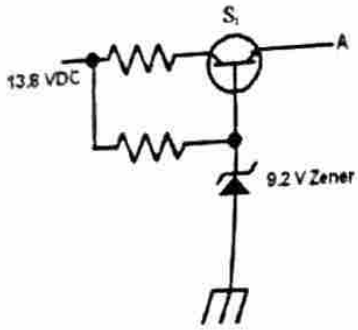
1. In the drawing below, what voltage is present at point A?

   a. 13 V
   b. 19.3 V
   c. 14.9 V
   d. 12.3V



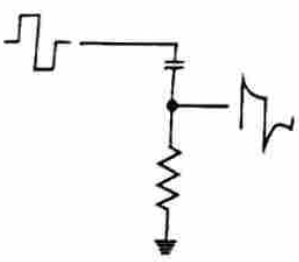
2. The output (A) of the regulator circuit below should be about:

   a. 13.8 V DC.
   b. 9.2 V DC.
   c. 10 V DC.
   d. 8.5 V DC.



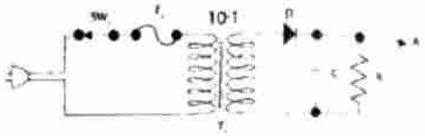
3. In the circuit below, to reproduce the square wave input more faithfully at the output:

   a. decrease the value of the resistor.
   b. lower the frequency of the square wave.
   c. decrease the value of the capacitor.
   d. increase the value of the resistor.



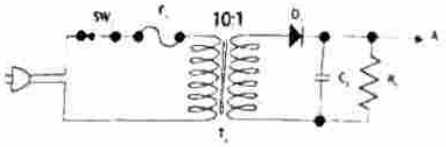
4. In the circuit below, what voltage would be expected at point A if 117 volt AC were applied?

   a. 11.7 V
   b. 15.5 V DC
   c. 24 V DC
   d. 117 V DC



5. In the circuit below, what voltage is expected across the secondary of the T1 transformer if the circuit is connected to 117 V AC?

   a. 11.7 V AC
   b. 17 V AC
   c. 117 V AC
   d. 1170 V DC



6. How do you calculate the gain of a non-inverting operational amplifier?

   a. $\frac{\text{input}}{\text{output}}$
   b. $\frac{\text{- output}}{\text{input}}$
   c. $1 + \left(\frac{Rf}{Ri}\right)$
   d. $1 - \left(\frac{Rf}{Ri}\right)$

7. If the frequency of the broadband input signal to an Op-Amp increases, then the output signal level _____.

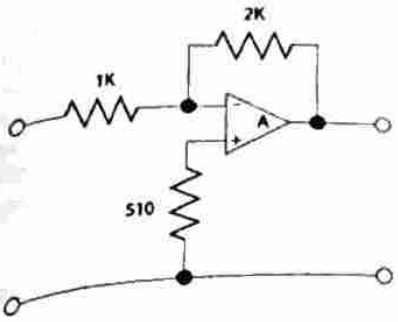
   a. increases
   b. decreases
   c. remains the same
   d. increases the frequency

9. When measuring power to a speaker, a doubling of the power represents a 6 dB increase.

a. True
b. False

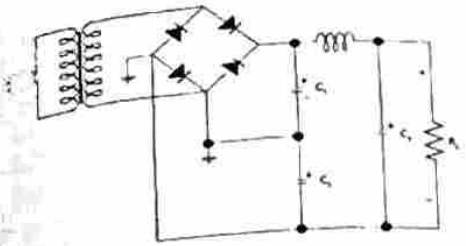
10. The input of the op-amp below is a .5 V sine wave (centered about +2 V). What will the output be?

a. A 1 V sine wave centered about +1 V
b. A 1 V sine wave centered about -1 V
c. A 1 V sine wave centered about -2 V
d. A 1 V sine wave centered about -4 V



11. What is wrong with this circuit below?

a. Bridge diodes are backwards.
b. C1 and C2 filters reversed.
c. Load polarity is backwards.
d. Diode grounds are shorting.



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Chapter 13

Basic Radio, RF Signal Propagation and Commonly Used Frequencies

by David Mark Somsell, CETsr

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Basic Radio

It has been my pleasure compiling the data within the next few pages. My sincerest hopes are that you receive a better understanding of radio, its principles, and the frequencies that are used. Later, we will discuss items such as modulation, antenna types, RF signal propagation and much more. So sit back, relax and enjoy the chapter that follows. I think you will.

History

Long before our fancy computers and GPS units, radio was a major part of our lives, since television was not even in existence yet. Technology has really come a long way and continues to advance daily. Even though we have all of the "latest and greatest" gadgets, our radios are still indispensable items.

Now, you might ask? Well, if you take a look at police, fire, ambulance and other rescue personnel, radios are used a lot. Even the space shuttle uses radio frequency communications. So whether we use radio communications as a monitor for a child's crib, or just listen to our car radios for the fun of it, we all can enjoy what they have to offer.

Lastly, it always amazes me that fads and fashions go away, games and gadgets disappear, but radio and communications continue to contribute so much for so many.

Basic Radio Receiver

What if I said that you would be able to receive (RX) radio communications using the following items or lack thereof, no batteries, but only a diode, capacitor, some wire and a pair of headphones? Many of us probably think that it cannot be done. On the contrary, this is what our first radios were like. Please note Figure 1.

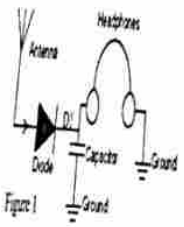


Figure 1

As rudimentary as this drawing is, it's how it all began. The radio signal hitting the antenna travels down to components D1. This component is a diode made out of galena, silicon or germanium. This diode rectifies the radio frequency signal and the capacitor removes the RF carrier. The resultant audio is applied to the headphones. You are then able to hear the broadcast. Truly, these were the "Golden Years of Radio."

Yes, we are able to receive RF communications, but not as good as the new radios that we have at home? There are many drawbacks with this simple, early design. For example, many channels would often intermingle with one another. Also, there was no tuning (selectivity) and minimal sensitivity.

So what is selectivity? Selectivity in a radio is the ability to separate one channel from another. Sensitivity, on the other hand, is the ability to receive weak signals reliably (some things that our basic radio receiver in Figure 1 doesn't have).

Can you think of any other drawbacks with this design? What about being able to tune in your exact favorite channel? What about being able to listen to your station without having it drift off frequency on you?

How about maintaining a constant volume rather than having it get louder or quieter as you listen? Since the tuning of this basic receiver is so rudimentary, it is almost impossible to separate one channel from another.

Also, there is no mixer or intermediate frequency (IF) found in this diagram. These are some key components in today's modern radios. They are those obviously lacking in this basic design. We will discuss these items in detail. Also, the volume would be varying constantly with the lack of an automatic gain control (AGC) or automatic volume control (AVC) (whichever acronym you prefer).

Having said that, it's still a thrill to receive radio signal communications with next to no components.

Superheterodyne Receiver

Today's Radio

The majority of today's radio receivers are what we call superheterodyne. This is just a fancy term meaning that two frequencies are mixed together to produce the sum and difference frequencies. As we go, try to follow from left to right on the block diagram below in the order of signal progression. Since most electronic schematics are written in this order, it is a good idea to get used to the logic now. Please note Figure 2 at the bottom of this page.

Antenna

On any good radio receiver or transmitter, we need the antenna to be electrically matched (AC resistance-wise), to the radio. The proper impedance and also the length of the antenna is important. This means that the antenna has to be physically cut for the particular frequency or band of frequencies that you need. In radio communications we use ¼ wavelength incremental units often. Please note Figure 3a.

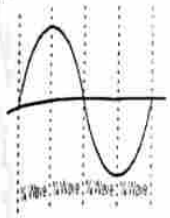


Figure 3a Basic Sinusoidal Waveform

The impedance that is generally used in radio communications is 50 ohms. So, what that means is that the antenna needs to have an impedance of 50 ohms, along with the internal circuitry of the radio. When the antenna and internal circuitry impedance are equal (AC resistance-wise), only then will we have maximum power transfer either in receive (RX) or transmit (TX) mode of the RF signal.

Let's apply what we have learned. The formula for finding the length of a ¼ wavelength antenna is as follows:

$$\frac{1}{4}WAVELENGTH\,(ft) = \frac{246}{freq.(MHz)}$$

Given a frequency of 7 MHz, the answer is 246 ÷ 7 = 35.14 feet. A long antenna, that is, you think? Not really, wrapping a wire around a ferrite core would electrically lengthen the antenna. This is why your AM broadcast receiver doesn't have a 246-foot antenna.

Another rule of thumb is that the higher in frequency you go, the smaller the antenna becomes. An easier way to remember it is just to think of microwave RF emissions. These tiny wavelengths are very small, yet they have very high frequencies, thus requiring small antennas. The interesting fact is that time and frequency are inversely proportional to each other. Therefore, the higher the frequency, the smaller the wavelength and vice versa. Here are some examples.

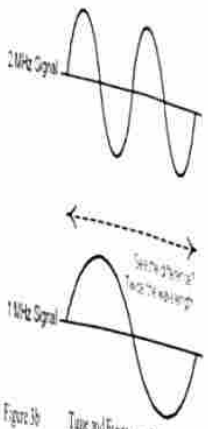


Figure 3b Time and Frequency Comparison

RF Section

The radio frequency section of Figure 4 is the first part of the receiver that receives the incoming RF electrical-magnetic signal. It is very broad banded (capable of receiving a wide range of frequencies). The signals entering the input are, for the most part, generally weak—sometimes only a few microvolts. A few millionths of a volt! The RF section must have enough bandwidth to accommodate the entire frequency span of the radio. For example, if we take the AM receiver shown in Figure 2, this radio is supposed to receive frequencies from 540 kHz through 1650 kHz (kilohertz). The RF input circuitry must be designed so those frequencies can pass through. This part "beefs up" the input signals since it acts as a RF amplifier. This section is where the most noise is produced.

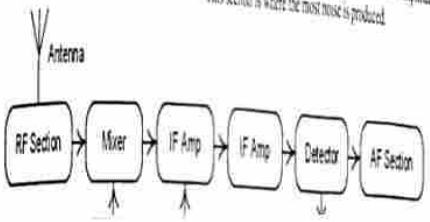


Figure 2 Superheterodyne Receiver

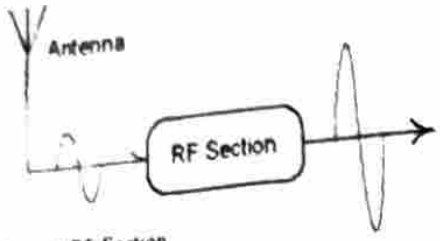


Figure 4 RF Section

Mixer/Local Oscillator

Once the signals are amplified, the receive signal is then heterodyned (or mixed) with a signal produced by the local oscillator circuit (Shown in Figure 5). The output of the mixer will contain the sum and difference frequencies. Now as a rule of thumb, the oscillator circuit is constructed to run above the received frequency by an equal amount to the IF (intermediate frequency).

For example: If you want to receive your favorite radio station at 1000 kHz and also to have an IF of 455 kHz (This is pretty typical for AM radios), what should our oscillator frequency be?

Formula:
Oscillator Frequency = 1000 kHz + 455 KHz = 1455 kHz

(desired freq) + (IF) = Osc. Freq.

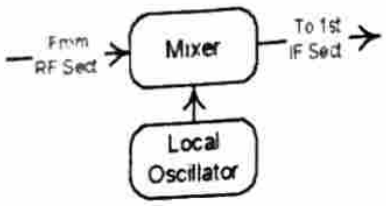


Figure 5 Mixer and Local Oscillator Circuit

All right, the oscillator signal is at 1455 kHz and the original station that we wanted to hear is 1000 kHz. Both will be mixed to produce the sum and difference frequencies. The difference of the two signals will be our IF band pass frequency.

1445 kHz – 1000KHz = 455KHz (IF)
We always subtract to find the IF.
Our old frequency of 1000KHz has been converted to 455Khz.

IF Section

Now that we have established an IF of 455 kHz, we can proceed and see how this section works. The IF sections use a set frequency throughout. Also it is really an amplification network comprised of transistors, resistors, and capacitors. This is one of the major differences with our superheterodyne receiver compared to the basic radio previously discussed. Like the basic receiver, we have heterodyned or mixed the incoming signal with a local oscillator. Then, we established an IF pass band frequency. Now you are amplifying the desired tuned frequency through the IF section. Hence frequencies above or below 455 kHz will not be amplified or passed. Thus it is imperative that the mixing process of both the incoming and local oscillator produces only this frequency (455KHz). It is much easier to amplify just one frequency – the IF. See Figure 6 below.

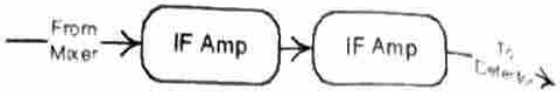


Figure 6 IF Section

In Figure 7 below, I have compiled a list of commonly used IF's and their uses.

Figure 7 Common IF Frequencies

Detector Circuit

Detection, as it is noted, is the rectification of the RF carrier. Note Figure 7a below:

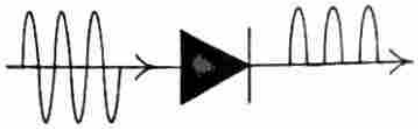


Figure 7a Simple Detector Diode

If you look at the diode, what is it doing? Is it amplifying, attenuating, filtering, or distorting the waveform? No, it's really rectifying the incoming RF signal. As the desired signal enters the anode and goes through the PN junction, the negative portion of the RF wave is "removed." Then the signal will proceed through other circuitry to have the RF 'carrier' removed, thus leaving only the modulation.

AGC/AVC Section

Automatic gain control (AGC) or automatic volume control (AVC), as it is called, actually samples the signal at the detector and develops a control voltage that is fed back to the first IF Amp. Ideally the operation of the AGC will amplify weak incoming signals and attenuate strong ones, allowing a constant, desirable output volume. Please note Figure 7 b.

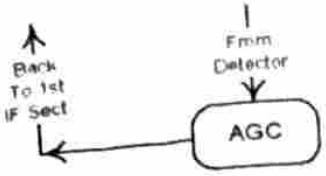


Figure 7b AGC Circuit

Modulation

Now we have the modulation to deal with. This is the precious cargo, if you will, that rides on the carrier. Modulation can range from static, electrical noise to voice, Morse code, tele-video information, and other analog, as well as digital information. If you took an oscilloscope and hooked up a microphone to one of its inputs and spoke, what would you see? You would see your voice on the display. Depending on the voice you have, it usually is different than anyone else's. As you spoke, you would see it rise and fall in amplitude, depending on how loud you spoke. Also, it would expand and contract, depending on your voice characteristics and frequency. See Figure 8a for some common modulation types.

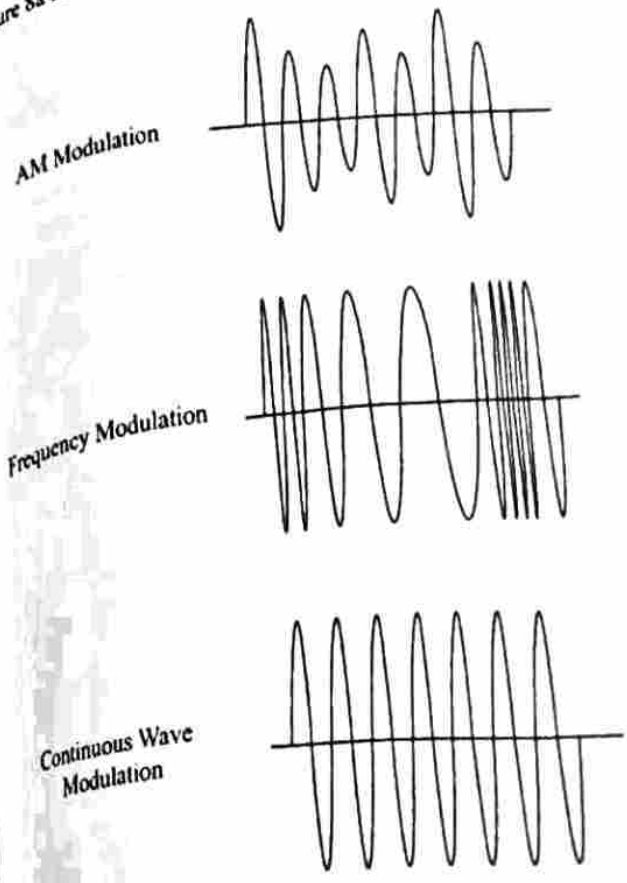


Figure 8a Modulation Types

AF Section

The last part of the superheterodyne receiver is the Audio Frequency (AF) Section. This is an audio amplification network that drives your speakers. Remember what we talked about before regarding antenna impedance matching? The load (in this case our speakers) must match the radio's circuitry (AC resistance wise). Proper impedance matching has to be observed not only to deliver maximum power to antennas, but must be observed with audio applications as well. This gives the speakers the maximum, matched output power possible. See Figure 8b.

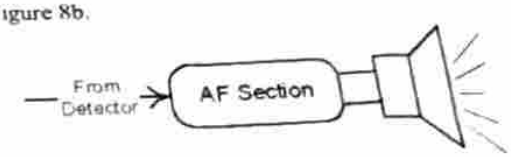
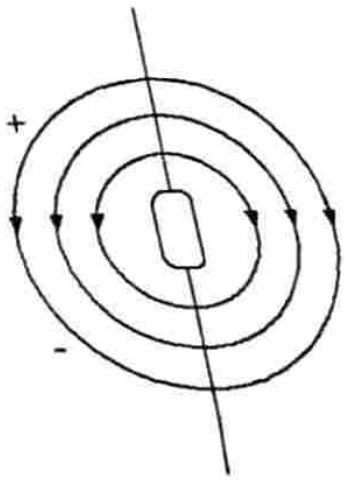


Figure 8b AF Section

RF Signal Propagation

Principles

Radio frequency emissions travel quickly. They move at a rate of over 186,000 miles per second, the speed of light! If you refer to Figure 9a and b below, they show a few simpl examples of signal propagation and antennas.



*(Electromagnetic field around dip

Figure 9a Signal Propagation

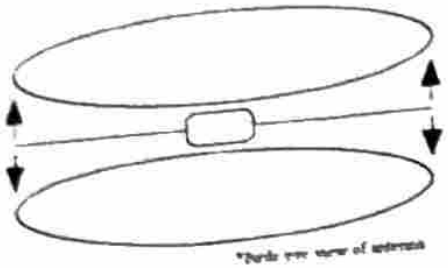


Figure 9b Simple 1/2 wave Dipole Antenna

When an electrostatic field is introduced in the above antenna, an electromagnetic signal is propagated outward from the antenna elements. Notice the radiation pattern of the antenna in Figure 9b above. It sort of looks like a figure eight pattern, doesn't it? Dipoles generally have a radiation and receive pattern of 180 degrees. Each antenna type will produce its own "polar pattern." The antenna shown in Figure 10 (which is a vertical "omni directional" antenna) has a polar pattern of a full 360 degrees.

A dipole is used as a reference and as a comparison in measuring "gain" of another antenna. Gain will be talked about shortly. When you think of a 1/2 wave dipole antenna, it is classified as having 0-dB gain, thereby having a reference for other antennas.

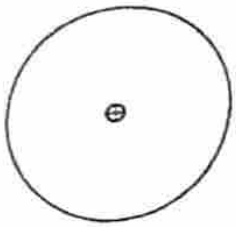


Figure 10 Omni directional Antenna (Vertical)

So what is antenna gain? Transmitting-wise, it is the amount of effective radiated power from the antenna as compared with that of a reference antenna. Given the same input power parameters, receiving also benefits, thus being able to receive weak and often very distant stations. The more gain the better.

Why is gain important? It is important not only for long distance reception but also for distant transmission of RF, since it actually concentrates the signals on transmit and receive.

Please note Figure 11. This type of antenna is called a yagi. *This type of "beam"* antenna focuses all of its energy in a narrow beamwidth as described below, for example, 20 to ... degrees. The more director elements that you have, the ... the gain rating and narrower beam width. The 2 elements ... est the left are the directors. Again, let me stress it once ... more directors that you have, the higher the gain and direct... ity. The second element from the right is the driven element ... and this is the part where the coax is attached. Lastly, the ... element is the reflector. It does just that – reflects received ... nals onto the driven element. Your beam has to be "pointed" ... at a specific station, unlike our omni directional antenna in ... Figure 10 which transmits and receives in a full 360 degrees. ... Without getting into antennas too deeply, hopefully this will ... give you a flavor of what some basic antenna types are like ... and their characteristics.

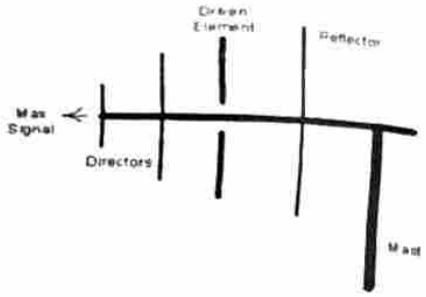


Figure 11 Yagi Style Antenna

Our Atmosphere

Ionosphere

What is the ionosphere? It is part of our atmosphere that is some 40-400 miles above our planet. The sun seems to be an endless supplier of the earth's radiant energy. As the sun warms up the ionosphere, bizarre things happen. If you refer to Figure 12, we will discuss what happens to radio signals when this layer of our atmosphere is heated.

First, let's look at the D Layer. It is approximately 30-50 miles above the earth. It is most prevalent during daylight hours, and practically disappears at nighttime. This is one layer that short wave radio listeners do not enjoy. This layer actually absorbs radio signals under 10 MHz, especially those from 1-8 MHz during daylight hours.

The E Layer is less discriminating in a sense. It is located 60-90 miles above our planet. This layer will, at times, pick and choose both High Frequency (HF) and Very High Frequency (VHF) signals. Some of these signals will be reflected back to earth, depending on the frequency used and time of day.

The F, F1, and F2-Layers of the Ionosphere are the furthest out of all the layers, from 250 miles and OUT! Now these

layers are the short wave listener's friend, since most of the HF signals will be reflected back here on earth. The reason for this is that the wavelength is so long at these frequencies, it is very difficult for them not to be reflected back to us, unlike the VHF and UHF frequencies in other layers, which pass right through.

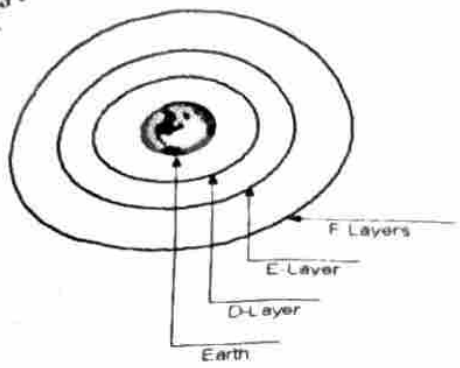


Figure 12 Ionosphere Chart

Coaxial Cable Chart

Any good radioman or radiowoman will be required to be fluent in coaxial cable terminology. In Figure 13 below, I have constructed a sample coax chart. These charts will generally list the type of coax, the ohms rating, (O.D.), outside diameter of the cable, the loss in dB for 100 feet of cable, at 100 and 400 MHz respectively, velocity factor, and the center conductor gauge. We will discuss each.

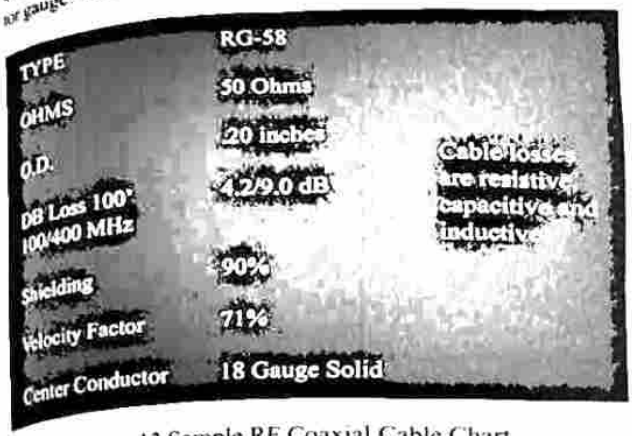


Figure 13 Sample RF Coaxial Cable Chart

Type

Our example shows RG-58. This is a very common 50-ohm cable used for communication work, test and repair. It particularly is used in mobile and short run installations. It is not well suited for long runs because the dB loss would be too great.

Ohms

As stated above, this cable is rated at 50 ohms. Every cable type has its own rating in ohms. Simply put, if you have ever seen the inside of your cable television line (which is 75 ohms), construction-wise it is very similar with a hard PVC outside jacket, a copper braided shielding, a dielectric (insulating material) and center conductor. Take a look at Figure 14.

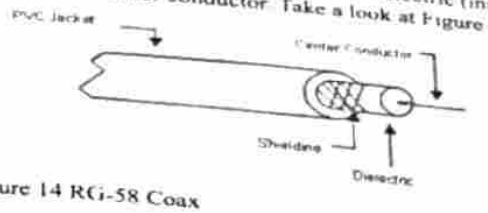


Figure 14 RG-58 Coax

The impedance of the coaxial cable is primarily determined by the spacing of the center conductor, through the dielectric, to the shielding.

OD (Outside Diameter)

This is the outside thickness of our cable. It is generally measured in inches. Since RG-58 is a relatively thin cable, only 2/10 of an inch can do the following: It can be somewhat bent around corners, coiled up, twisted into various shapes and able to fit in some pretty tight locations. As an electronics technician, you will be making decisions on which cable to use, how to install the proper connections and testing of the finished cable assembly with an ohmmeter.

dB Loss

No matter what materials manufacturers use, our RF cables will always have some signal loss, whether in receive or transmit mode. Even if we use gold and silver in our construction, these too have loss characteristics. But what should be done to keep these RF losses down to a minimum and what are the effects if we don't? In radio electronics, we use decibels (dB's) in measuring signal loss or gain—in this case, loss. If we refer back to Figure 13, the chart clearly indicates the loss at 100 and 400 MHz, both using 100 foot runs of cable. The loss is 4.2 dB and 9.0 dB, respectively. Notice the difference in loss as we go higher in frequency. As another rule of thumb, the higher the frequency, the greater the RF loss will be and vice

versa. But, if we don't keep our RF losses down to a minimum, we will have degraded both receive and transmit results. This can be reduced by using lower loss cable and making sure all connections are tight, also by grounding equipment and checking soldered connections, to name just a few.

Shielding

This part of the coax has multiple purposes. It not only sets up our impedance but also prevents RF from entering and escaping the center conductor. This copper braid encompasses the entire outside of the insulating dielectric housing. The shielding characteristics of a cable are extremely important, not only for transmission, but also for reception of RF. It is rated in percentages. We should always strive for maximum shielding and low loss, resulting in greater efficiency.

Velocity Factor

Don't get overly caught up with this item. For most applications the manufacturer has already taken this factor into consideration when designing the cable. RF energy slows down a bit when it travels through coax. The velocity factor of a cable is dealing with the speed of the transmission of RF through the coax, as compared with the same signal propagated in free air. Our example RG 58 cable has a velocity factor of 71%. In other words, the transmitted signal would be traveling at 71% of the same signal speed propagated in free air. As you already know, RF energy travels at the speed of light. Even if it slows down a bit, it is still traveling very quickly.

Center Conductor

This is the heart of the cable that carries the signals to and from the antenna. It is commonly rated in a standard gauge form. The gauge we used in our cable was 18 gauge solid copper. Some center conductors can be a stranded wire as well, or a combination of both! The thicker the center conductor of the cable, the lower the signal loss.

RF Safety—SAR

Specific Absorption Rate (SAR), as it is termed, is your body's ability to absorb RF energy. As a technician working with radio transmitters, antennas and cables, you should be aware of the potential dangers with RF. Generally, the higher the frequency you attain, the more it affects your body. It takes less power at higher frequencies to do the same damage as lower frequencies with higher power. Radio frequency emissions, at certain frequencies and power levels, resonate and produce a heating effect of human tissue. Be careful.

One of the main organs that we need to be concerned with is our eyes. There is evidence that strong, near field RF energy can affect your eyes over the long term in the formation of cataracts.

Now that I have scared you a bit, let's see how these hazards CAN be reduced. The first thing is using good sense. Secondly, never touch a transmitting antenna, serious RF burns can result. Never repair RF amplifiers with the shielded covers removed, with the power on and transmitting. Never look into a fiber optic cable or wave guide assembly. We should ground all equipment, make low power transmissions with our testing and terminate all connections that could leak RF energy. This is especially true with microwave emissions. If we just use a little good sense, observe the situation, analyze the possibilities and limit our problems, we should enjoy RF communications and pride ourselves in the fact that we ARE Certified Electronics Technicians.

The following chart will list some commonly used frequencies and their appropriate uses. It would be wise to commit them to memory. In closing, I truly hope that this information provided herein was informative, educational and fun. Good luck with your career in electronics, may it be a rewarding one not only for yourself, but also for others that YOU help along the way.

Commonly Used Frequencies

FCC Band Plans

The Federal Communications Commission (FCC) establishes all of our frequency rules, regulations, and frequency allocations. Here are some common frequencies that should be committed to memory.

SSB – Single SideBand

Single-sideband modulation (SSB) is a refinement of amplitude modulation that more efficiently uses electrical power and bandwidth. It is closely related to vestigial sideband modulation (similar to AM TV broadcasting)

Amplitude modulation produces a modulated output signal that has twice the bandwidth of the original baseband signal. Single-sideband modulation avoids this bandwidth doubling, and the power wasted on a carrier, at the cost of somewhat increased device complexity.

NOTES

Chapter Quiz:

1. What does the acronym FCC mean?

a. Federal Committee of Communications
b. Financial Communications Commissary
c. Federal Communications Commission
d. Federal Communications Command
e. Faraday Commission of Communications

2. What is not a part of a superheterodyne radio receiver?

a. Intermediate Frequency Circuit
b. Radio Frequency Circuit
c. Audio Frequency Circuit
d. Audio Radiation Resistance Circuit
e. Mixer Circuit

3. Match up the correct items:

a. RF	1. Amplitude Modulation
b. AF	2. Weather (no chapter reference)
c. IF	3. Audio Frequency
d. FM	4. Radio Frequency
e. AM	5. Intermediate Frequency
f. WX	6. Frequency Modulation

4. Which part of a superheterodyne receiver rectifies the RF carrier?

a. AF Circuit
b. Detector Circuit
c. Low Pass Filter
d. Modulation Circuit
e. RF Circuit

5. How long should a ¼ wavelength vertical antenna be that is designed for operation on 146 MHz?

a. 1.1 Feet
b. 1.6 Feet
c. 1.9 Feet
d. 2.3 Feet
e. 3.2 Feet

6. What is the common impedance of radio communication and television coaxial cable, respectively?

a. 35 and 55 ohms
b. 50 and 75 ohms
c. 60 and 93 ohms
d. 77 and 98 ohms
e. 80 and 100 ohms

7. Which layer of the ionosphere absorbs high frequency radio waves during daylight hours?

a. D Layer
b. E Layer
c. F Layer
d. F1 Layer
e. F2 Layer

8. Which style of antenna radiates in opposite directions, with a pattern resembling a figure 8?

a. Omni directional
b. Yagi
c. Dipole
d. Cubical Quad
e. Log-Periodic

9. What frequency band is an average AM commercial broadcast radio capable of receiving?

a. 535 to 1700 kHz
b. 630 to 1810 kHz
c. 810 to 2120 kHz
d. 2.5 to 3.10 MHz
e. 3.8 to 4.2 MHz

10. Why is RF safety important?

a. To insure maximum power output at all times
b. Always assuring variable tuning capability
c. Capability to receive distant radio stations
d. To limit human exposure to RF for health reasons
e. Maintain adequate FCC bandwidth spacing

ANSWERS on page 223

Chapter 14 Transmitters, Transmission Lines and Antennas

by Chief Petty Officer James Bruening, CET

James Bruening currently serves in the US Navy as an Aviation Electronics Technician, Chief Petty Officer. In his 20 years of Naval service, Jim has fixed radar, radios, sonar, data link, military computers, and electronic navigation systems for Navy aircraft. In addition to certification as an Industrial Electronics CET, he has a BS in Electronics Technology (Magna Cum Laude) from Regents College and an AS in Physics (Honors) from San Diego Mesa College.

Transmitters

Transmitters typically convert intelligence into a signal that can be radiated over some distance to a receiver. The receiver then converts the transmitted signal back into the original intelligence. The intelligence can be Morse code, voice, video, or digital information. The method of converting intelligence into a transmittable signal is most often by modulating a signal that then carries the intelligence to a receiver. Modulation is the process of imposing intelligence on a carrier signal.

The carrier frequency is usually in the Radio-Frequency (RF) region of the electromagnetic spectrum. RF signals lie in the frequency range of 3 kHz to 300 GHz. RF signals can travel great distances at the speed of light (3.0×10^8 meters per second).

The frequency of the radiated signal depends on the application. For example, communications transmitters typically operate in the kHz to MHz frequency range. Satellite communications occur in the GHz frequency range. RADAR also uses frequencies in the GHz. RADAR is used to determine the speed, distance, or the direction of an object.

Continuous Wave (CW) Transmitters - The output of a CW transmitter is a constant amplitude carrier wave that is either on or off. The CW signal is keyed on or off to transmit a code (usually Morse code) that can be interpreted at the receiving end. Figure 1 is an example of master-oscillator-power-amplifier CW transmitter with keying in the buffer stage.

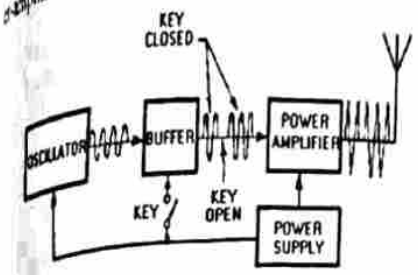


Figure 1 Basic CW Transmitter

Amplitude Modulated (AM) Transmitters – AM transmitters modulate the RF carrier wave with the intelligence to be transmitted. In amplitude modulation, the amplitude of the carrier is varied to create an envelope of RF.

The RF envelope will exactly resemble the modulating intelligence. Upon reception, the receiver demodulates the RF signal. In demodulation, the carrier is removed and the intelligence is again available.

Figure 2 is the block diagram of a typical AM transmitter. The intelligence is voice audio from a microphone. A current driver and a modulating amplifier strengthen the voice audio. A continuous RF wave is produced by an oscillator and applied to a power amplifier via a buffer stage. The buffer stage isolates the oscillator from external circuitry, keeping the oscillator frequency stable. The power amplifier amplifies the buffered oscillator signal. The amount of amplification is determined by the modulator output. This is accomplished by biasing the power amplifier with the modulator signal, creating an envelope of RF representative of the original voice input.

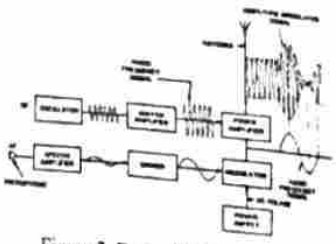


Figure 2 Basic AM Transmitter

Frequency Modulated (FM) Transmitters – In FM transmitters, the frequency of the carrier wave is shifted to represent the intelligence.

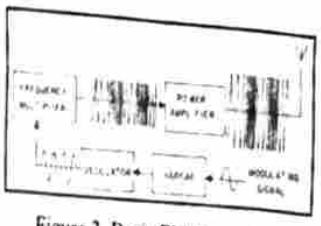


Figure 3 Basic FM Transmitter

Referring to Figure 3, a modulating signal is applied to a varicap. The changing amplitude of the modulating signal varies the reactance of the varicap. The varicap is part of the frequency determining circuitry of the oscillator. The oscillator frequency is thus varied about its center frequency as the reactance of the varicap is varied. The varying output of the oscillator is frequency multiplied to radio frequencies and

amplified for transmission by the power amplifier.

Television Transmitters - In a television transmitter, video intelligence is transmitted by AM modulation and audio by FM modulation. The band of frequencies for a particular channel of TV video and audio is 6 MHz wide. The primary difference between a TV video transmitter and a typical AM transmitter is the video sync signal. The TV transmitter sync generator provides camera sweep information for the receiver sync circuitry. This allows the receiver to display the picture in sync with the camera sweep.

Microwave Transmitters - The formula for calculating the wavelength of a signal is $l=v/f$, where 'l' is the wavelength in meters, 'v' is the speed of light (3.0×10^8 meters per second), and 'f' is the frequency of the signal in hertz. Frequencies between 1.0 GHz and 100.0 GHz, when applied in the preceding formula for wavelength, yield very small wavelengths. This is why frequencies between 1.0 GHz and 100.0 GHz are defined as microwave frequencies.

Microwave circuits require special considerations because of the high energy of the microwave signal. If not properly confined, much of the microwave signal would be lost as unwanted radiation.

Generating a microwave signal also requires special circuitry. Two electron tubes used to produce microwaves are the klystron and the magnetron. These tubes act as oscillator and amplifier for the microwave signal.

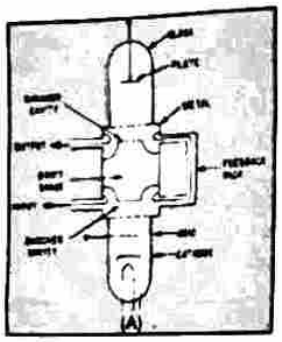


Figure 4 - A Klystron Tube

A klystron tube is constructed such that the electron stream emitted from the cathode transits the tube in bunches. The physical construction of the klystron determines the frequency at which the electron stream bunches up. Some of the energy of the pulsating electron stream is picked off by the tube itself to sustain oscillations. Most of the pulsating electron stream, however, is removed from the tube as a microwave signal (sine wave). Klystrons are used in low to medium power microwave applications.

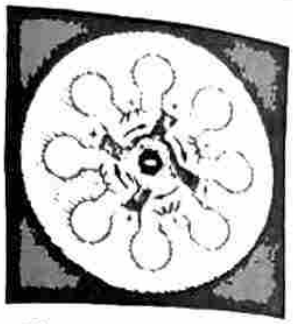


Figure 5 - Electron Behavior in a Magnetron

A magnetron tube produces a microwave signal for use in high power applications. The magnetron tube is constructed to resemble a ring of pockets, or cavities, surrounding the electron-emitting cathode as can be seen from Figure 5. A magnet mounted in contact with the tube causes the electron stream inside the tube to bunch up in a circular pattern around the cathode. Some of the electron energy is transferred into the cavities as the bunched electrons pass the cavity openings. The energy transferred into the cavities sets up current oscillations within the skin of each of the cavities. The frequency of oscillations is determined by the physical size of the cavities. A small probe mounted in one of the cavities picks off the microwave signal to be used.

RADAR Transmitters - RADAR transmitters are microwave devices. A typical RADAR transceiver will transmit a high frequency pulse of RF and wait for the pulse to return after having been reflected from some object. The time the pulse takes for the round trip will be used to calculate the distance to the object. The direction from which the pulse returns is used to determine the direction to the object.

A RADAR is designed to use the DOPPLER Effect if the speed of an object is desired. For Doppler RADAR, the amount of frequency shift of the returning pulse is used to determine the speed of an object.

Satellite Transmitters - High power microwave transmitters are required for satellite communications. The power rating for a satellite up-link transmitter is in kilowatts. Satellite up-link typically occurs with a 6 GHz carrier and down-link at about 4 GHz. The transmitter signal is beamed at the satellite via a highly directional parabolic antenna, which will be discussed later.

Transmission Lines

A transmitter is often separated from the transmitting antenna by some distance. When this is the case, it is necessary to route the signal to be transmitted to the antenna via a transmission line. Ideally, a transmission line will get the RF from the transmitter to the antenna with a minimum amount of power loss.

Some types of transmission lines you will likely encounter are the twin lead, flexible coaxial, and waveguides

Twin Lead Transmission Lines - The two leads of the twin lead transmission line are kept uniformly spaced by embedding them in a low loss dielectric. The simple construction of this type of line is an advantage. However, excessive losses due to unwanted radiation and excessive noise pickup are serious disadvantages of this type of line. Twin lead line is most often used to connect a television antenna to the television receiver.

Figure 6 Twin Lead Transmission Line

Flexible Coaxial Transmission Lines - A major advantage of coax is that losses due to unwanted radiation are minimized. This is because the electromagnetic fields of the line are confined to the space between the conductors. Higher frequency (GHz) signals, however, do experience unacceptable losses with flexible coax.

Figure 7 Flexible Coaxial Transmission Line

Two types of flexible coax in use are designated RG6 and RG59. Both RG6 and RG59 are 75-ohm line capable of carrying broadband signals. RG6 coax, with its 20-gauge center conductor and larger diameter, is capable of carrying a signal for a greater distance with less attenuation than RG59. RG59 has a smaller diameter center or inner conductor of 22-gauge wire.

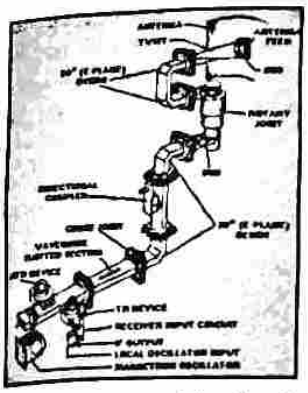


Figure 8 Example of Microwave Circuit using Waveguides

Waveguides - Waveguides are ideal for microwave transmission. Waveguides are hollow tubes with no center conductor. Waveguide circuits are often referred to as "plumbing" by technicians. Figure 8 illustrates a basic microwave circuit using waveguides.

The physical dimensions of the waveguide determine the wavelength and power of the signal it can carry. The width of the waveguide will determine the wavelength and the height determines the power handling ability. Since one dimension of a waveguide depends on wavelength, it would be impractical to use waveguides for lower frequencies than microwave. For example, a waveguide for a 1 MHz signal would be about 500 feet wide.

The microwave signal is carried through the waveguide in a series of electromagnetic wave fronts. The wave fronts bounce down the length of the waveguide from source to destination. Figure 9 illustrates wave fronts propagating down a waveguide.

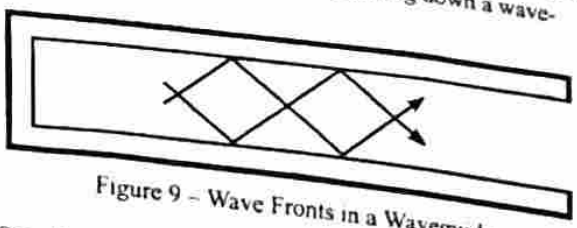


Figure 9 - Wave Fronts in a Waveguide

Standing Waves - Recall that impedance is the parameter of a circuit that takes resistance, capacitive reactance, and inductive reactance into account. $Z = \sqrt{R^2 + (X_L - X_C)^2}$. A transmission line will have a characteristic impedance. For maximum power to be carried to the antenna or other load, the load impedance must match the characteristic impedance of the transmission line. Maximum power transfer occurs when $Z_{Line} = Z_{Load}$. If there is an impedance mismatch between the transmission line and the load, standing waves will develop on the line. Standing waves are the result of energy reflected back into the line by the impedance mismatch at the load. If no standing waves exist on the line, maximum power transfer to the load can occur. A Standing Wave Ratio (SWR) Meter can be used to measure standing waves on a transmission line. Poor termination can cause standing waves.

Antennas

A transmitting antenna radiates RF as an electromagnetic field. On the receiving end, these electromagnetic fields induce RF voltages and currents in the receiving antenna.

The length of an antenna, for transmit and receive, depends upon the RF frequencies it will be transmitting or receiving. The same antenna may be used to both transmit and receive a range of frequencies, with the center frequency being the ideal frequency for which the antenna was designed.

Antennas are typically divided into two basic classes: Hertz antennas and Marconi antennas. A Hertz antenna is one half the wavelength long of the frequency it is to transmit. Hertz antennas are often used for frequencies above 2 MHz. A Marconi antenna is a quarter wavelength long. Marconi antennas are often used below 2 MHz.

Antenna Resonance - Antenna resonance is achieved when the length of the antenna is one half the wavelength of the signal to be transmitted or received. The Hertz antenna is constructed to be one half wavelength. The Marconi antenna, being physically a quarter wavelength, must be grounded at one end. The grounded end of the Marconi acts like an additional quarter wavelength to achieve resonance.

The importance of resonance is that maximum radiation of the transmitted signal occurs only at resonance. If the length of the antenna is wrong, and it is not at resonance, much of the signal to be transmitted will be dissipated as heat. Resonance is important for the receiving antenna as well, if maximum electromagnetic energy is to be induced in the antenna.

Atmospheric Wave Propagation – Electromagnetic energy travels from the transmitter to the receiver as either ground waves or sky waves. Refer to Figure 10.

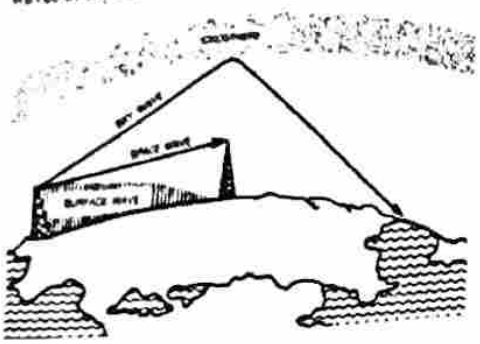


Figure 10

Ground waves are composed of a surface wave and a space wave. The surface wave hugs the surface of the earth. Frequencies below 2 MHz can be propagated via surface waves over great distances. Frequencies above 2 MHz are greatly attenuated because of absorption by the ground, making long-range ground wave communications impractical for higher frequencies.

Space waves travel directly or by reflection from the transmitting antenna to the receiving antenna. VHF frequencies or above, i.e., above 30 MHz, are often propagated as space waves. Space waves are known as line-of-sight when used for radio communications in the VHF and UHF bands. Tropospheric ducting often allows VHF and UHF signals to travels hundreds of miles.

Sky waves bounce off the ionosphere and return to the earth over the horizon from the transmitter. Ideal sky wave propagation is limited to a narrow range of frequencies. A signal with too high a frequency will simply shoot through the ionosphere and pass into space. This occurs because the energy level of the high frequency signal exceeds that of the atmospheric ionization. Low frequency propagation, as stated earlier, is best achieved with surface waves.

HF frequencies (3 to 30 MHz) are ideal for sky wave propagation. A HF signal does not have enough energy to simply pass through the ionosphere and radiate into space, but bounces back towards the earth as desired. A UHF signal, on the other hand, would simply pass through the ionosphere because of its high energy level. This is why AM radio stations can be received far from the transmitter and FM radio stations are local to the transmitter.

Antenna Directivity and Gain - The directivity of an antenna is the property that causes radiation or reception in a particular direction. Omni-directional antennas radiate and receive equally in all directions. Uni-directional antennas radiate strongest in one direction, and bi-directional in two. To say that an antenna is highly directional is to say that it is uni-directional or bi-directional.

Directivity is directly proportional to antenna gain. A uni-directional antenna has high gain in one direction because all of the radiated or received energy is focused to or from one direction. An omni-directional antenna, by comparison, has poor gain because the transmitter or receiver energy is radiated or received via all directions.

Antenna Polarization - Polarization is an important consideration in positioning an antenna. A horizontal antenna transmits a horizontally polarized wave and a vertical antenna transmits a vertical wave. A receiver's antenna must be polarized the same as the transmitter antenna for ideal reception.

Signal polarization is defined by the orientation of the electric lines of force of the electromagnetic wave passing through the antenna. An electromagnetic field is composed of both electric (E-Field) and magnetic (H-Field) lines of force at right angles to one another as can be seen in Figure 11. When the E-field is vertical to the horizon a wave is considered to be vertically polarized. A horizontally polarized wave has its E-Field horizontal to the horizon.

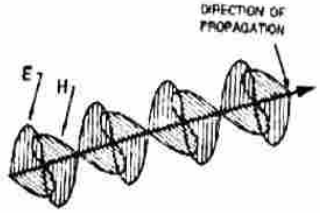


Figure 11 – E-Field and H-Field in an Electromagnetic Wave

Vertically polarized transmissions are usually omni-directional. This means that the strength of the transmitted signal is equal in all directions, which is ideal for radio communication. Also, vertical polarization is preferred for lower to medium frequencies because these signals propagate as ground waves. Vertically polarized ground waves are less prone to be affected by obstacles on the ground than horizontal ground waves.

A horizontally polarized antenna is more directional than a vertically polarized antenna. Recall that a vertically polarized antenna is omni-directional. Omni-directional vertical polarization could lead to reception of unwanted interference, man-made and natural. With a horizontal antenna, the direction of transmission and reception can be controlled to reduce interference.

Dipole Antennas – The most basic form of Hertz antenna is the dipole. Since the dipole is a half-wave antenna, it can be mounted at any height, as long as it is isolated from ground. The impedance of an ordinary dipole antenna is 70-75 ohms.

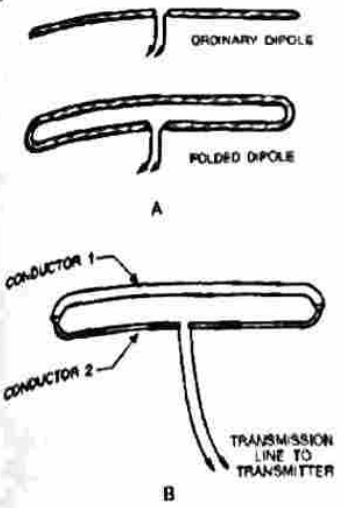


Figure 12 – Dipole Antenna

Folded Dipole Antennas – A simple folded dipole is a half-wave antenna that has a conducting element connected across its ends. The additional conductor is mounted parallel to the dipole elements at a distance of several inches. The advantage of adding the conducting element is that the antenna will resonate at a wider range of frequencies. This makes the folded dipole more desirable than the ordinary dipole if greater bandwidth is needed. The impedance of the folded dipole is 300 ohms. The gains of other types of antennas are normally compared with the half-wave dipole antenna.

Quarter-Wave Antennas – A grounded quarter-wave antenna resonates at the same frequency as an ungrounded half wave antenna. As illustrated in Figure 13, the grounded quarter wave antenna is frequently used in mobile receiver-transmitter applications. These are normally vertically polarized.

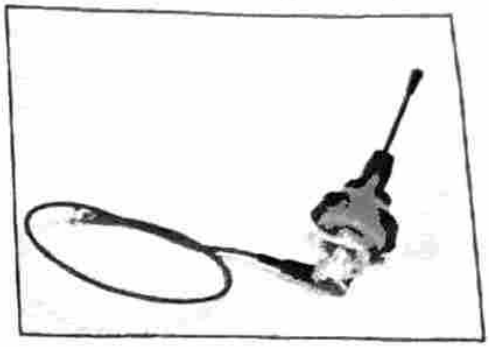


Figure 13 – Quarter-Wave Antenna used for Mobile Communications

Parasitic Arrays – Array antennas are constructed of more than one element. The parasitic array consists of a driven element, usually a dipole, and one or more parasitic elements. The driven element is the element that is driven by the transmitter or receives signals for the receiver. A parasitic element is mounted close enough for electromagnetic coupling with the driven element. Parasitic elements are not electrically connected to the driven element or to each other. Parasitic elements are classified as either reflectors or directors on Yagi type antennas as shown in figure 14.

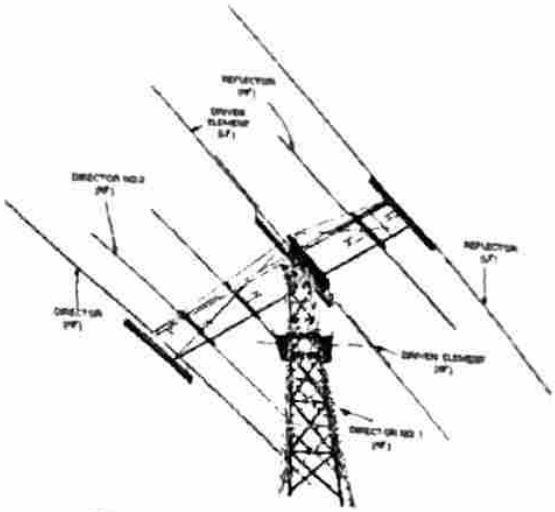


Figure 14 – A Parasitic Array Antenna

Parasitic elements concentrate the signal for transmission in the desired direction. For transmission, the parasitic elements get their energy from this electromagnetic coupling with the driven element. For reception, the parasitic elements transfer their received energy to the driven element. Parasitic arrays are highly directional.

Chapter 14

Microwave Antennas – Microwave antennas typically use a dish shaped reflector to concentrate the radiated energy into a beam. The dish is parabolic in shape and has a focal point located a specific distance out from the center-bottom of the dish. To help illustrate, a flashlight has a parabolic mirror that focuses the light of the bulb into a beam. In the case of the microwave antenna, all transmitter RF radiation is directed into the parabolic reflector and is concentrated into a beam. Likewise, all received energy beamed into the dish is concentrated at the focal point for reception.

Parabolic reflector antennas are uni-directional and have very high gain in that one direction. For this reason, parabolic reflector antennas are used in satellite communications.

Summary

- The process of imposing intelligence on an RF carrier is called modulation.
- For AM, the RF carrier is amplitude modulated to create an envelope of RF that exactly resembles the intelligence.
- For FM, the RF carrier is frequency shifted at a rate that represents the intelligence.
- The formula for calculating the wavelength of a signal is L=v/f, where 'L' is the wavelength in meters, 'v' is the speed of light (3.0×10^8 m/s), and 'f' is the frequency of the signal in hertz.
- The physical size of the klystron tube components determines the tube's frequency of oscillations.
- The physical size of the magnetron tube cavities determines the tube's frequency of oscillations.
- Transmission lines carry the RF signal from the transmitter to the antenna.
- Flexible coax lines are ideal for transmission of RF signals below the GHz range.
- Waveguides are used for transmission of microwaves.
- Standing waves on a transmission line are the result of an impedance mismatch at the load.
- Maximum power transfer to the load occurs when the impedance of the load matches that of the transmission line.
- A Hertz antenna is a half-wave antenna.
- A Marconi antenna is a quarter-wave antenna.
- Maximum radiation of a signal occurs when the signal to be transmitted is at the resonant frequency of the antenna.
- Ground waves are used for low frequency communications.
- Space waves are usually line-of-sight and are used for UHF frequencies and above.
- HF communications are ideally propagated via sky waves for maximum range.
- Antenna directivity is the antenna property that causes it radiation in a particular direction.
- Antenna directivity and gain are directly proportional.
- Electromagnetic field polarization is defined by the orientation of the electric lines of force of the field.
- The transmitter and receiver antennas must be similarly oriented with respect to field polarization for ideal reception.
- Dipole antennas are basic half-wave antennas.
- Folded dipole antennas are half-wave antennas with greater bandwidth than a basic dipole.
- Quarter-wave antennas must be grounded at one end for resonance at the desired frequency.
- Parasitic arrays are constructed with a driven element and one or more parasitic elements.
- Parabolic reflector microwave antennas are highly directional.

All illustrations courtesy of the U. S. Navy.

NOTES

Quiz

1. What is the process of imposing intelligence on a carrier wave?
   a. Extrapolation
   b. Modulation
   c. Transmission
   d. Oscillation

2. The basic formula for wavelength is: L=wavelength, v=velocity, f=freq.
   a. L=v/f.
   b. v=L/f.
   c. L=f/v.
   d. L=f/v.

3. Flexible coaxial transmission lines are ideal for what frequencies?
   a. MHZ only
   b. kHz only
   c. Above GHz
   d. Below GHz

4. Standing waves occur on a transmission line when there is a/an _____ between the transmission line and the load.
   a. impedance match
   b. impedance mismatch
   c. resistance match
   d. capacitive reactance match

5. Maximum power transfer to the load occurs when the load impedance matches that of the transmission line.
   a. True
   b. False

6. Antenna resonance is the characteristic that permits _____ radiation of the transmitted signal.
   a. minimum
   b. maximum
   c. zero
   d. omni-direction

7. Space waves are ideal for what frequency of communications?
   a. VLF
   b. HF
   c. Line-of-sight UHF
   d. Audible frequencies

8. Electromagnetic field polarization is determined by orientation of what part of the electromagnetic wave?
   a. Magnetic lines of force
   b. Electric lines of force

9. The orientation of the transmitter antenna and the receiver antenna must match for ideal reception.
   a. True
   b. False

10. Quarter-wave antennas must be _____ at one end for resonance at the desired frequency.
    a. shielded
    b. ungrounded
    c. in free space
    d. grounded

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ELECTRONICS TECHNICIANS ASSOCIATION ETA INTERNATIONAL

Chapter 15

Basics of Telephone

by John W. Baldwin, CETsr

John Baldwin has been involved in electronics for 37 years as a professional, 28 years teaching at the college level and 9 years in industry and the military. He taught courses and wireless communications at South Central College in Mankato, MN as well as at Riverland Technical College in Faribault, MN. He was an instructor for Spectrum for the Institute for Wireless Education, where he held seminars for working professionals. He holds a BS degree in Industrial Education from the University of Minnesota and teaching credentials in Minnesota and California. A charter member and Board member of the ETA, he is also an IEEE member, corporate affiliate and life member. He holds an FCC GROL license, Amateur Radio License and is a certification administrator for the ETA. John served in the USAF as a radio communications specialist and was formerly employed by EF Johnson Co. as a technical support supervisor and a technical instructor. He is the author of industry publications which highlight wireless and telecommunications. In 2006, John was the recipient of the GWEC Wireless Educator of the Year award for 2 year college programs.

Introduction

Telephones were first used in the late 1800's and developed by Alexander G. Bell and Elisha Gray who worked independently on early concepts during similar 19th century times. Talking machines, as they were called, enabled the speech of one person to be transported to other places where that person was not physically located. By converting acoustical speech sound waves to electrical signals, copper wire conductors could be used to extend the range of a person's audio communications.

Telephony, as it was later called, let a person's voice be transmitted or transported over great distances. The impersonal telegraph was no longer needed. The personality of a person's speech was carried along with the signal to a distant location to be heard by another person. Refinements allowed both persons to talk at the same time in either direction. This is called duplex.

Today we enjoy portability along with the convenience of having a private conversation with one or more persons. Cordless phones and wireless phones are replacing the last mile of connectivity, breaking the tethered connection to the telephone office. Various standards have risen to support different transport methods and techniques. In this chapter, we will look at the basics of telephone systems and how they are allowing us more connectivity than ever before. Wireless data is one of the fastest growing areas to allow access to the Internet any time and anywhere.

Basic Terms

A few of the basic telephony terms are defined below. These are necessary to understand the concepts that follow:

Duplex – Simultaneous transmission in both directions.

Half Duplex – Transmission in both directions, but one way at a time.

Simplex – Transmission in only one direction, such as a broadcast radio or TV station.

Deregulation – The US government allowing the manufacture of devices to an open standard and permitting the connection of these devices to the Bell Telephone network.

Divestiture – The 1984 breakup of the AT&T monopoly of local and long distance service into regional Bell operating companies (RBOCs).

Line – A connection from a local central office to customer premise equipment.

Trunk – A connection between central office switching centers, toll centers, regional tandem offices or long distance carriers that carries one conversation in both directions.

Central Office – A point of connection for all local subscribers to connect to the network for service.

Inter-lata – Telephone service connections between different area codes.

Intra-lata – Telephone service connections between the same area code. Sometimes called local long distance.

LATA – Local Area Transport Authority. Boundaries between area codes are called lata-lines.

Cordless Telephone – A telephone using a wireless unlicensed radio frequency service for low mobility connection.

Wireless Telephone – Wireless is a name for cellular and personal communications (PCS) service using licensed radio frequency spectrum.

Mobility – The degree of movement allowed in portable or mobile telephone service and still maintain a connection to the caller.

1G – Analog cellular service known as AMPS, advanced mobile phone service, in the US markets and NMT, Nordic mobile telephone, in European markets.

2G – Second generation digital cellular or PCS wireless telephone service.

3G – Third generation wireless service providing higher data rates and better speech quality

4G LTE – Fourth generation wireless service providing 4-10x higher data rates than 3G

Signaling – The transport of special commands needed by the telephone network to perform operations such as call setup, call delivery, call tear-down.

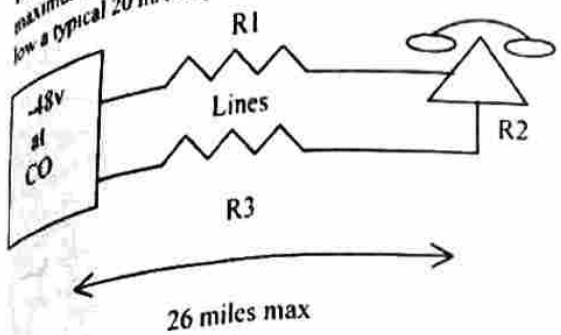
SS7 – The common channel signaling specification known as signaling system 7.

Basic Telephone Operation

Local Loop – DC analysis

Hello? Well, that's what we all say as a conditioned response to answer a call. Behind the scenes there's much more to it. It all starts with a DC circuit. The central office has a battery voltage of –48 v DC and the subscriber telephone set is the load.

The telephone lines in the circuit are represented as R1 and R3. Inside the telephone set, a hybrid coil represents R2. The maximum loop resistance can be as high as 2400 ohms to allow a typical 20 mA loop current.



$I = V/R$ –48v / 2400 ohms = 20 mA

Loop current is generally maintained between 16 and 23 mA by most telephone companies. If the hybrid coil in the phone set is 200 ohms, then the maximum resistance of the lines are 2200 ohms, and each resistance, R1 and R3, can be 1100 ohms as they are always the same length.

$RT = R1 + R2 + R3$, where 1100 + 200 + 1100 = 2400 ohms

Wire sizes vary from 19 AWG to about 26 AWG for telephone conductors in typical lines. This permits total length of a circuit of about 26 miles if the larger 19 gauge wire is used, or less distance for smaller gauge wire. From a wire table, 19 AWG has 8 ohms per 1000 ft. There are 5280 ft per mile, so each mile is 42.2 ohms and 1100 ohms / 42.2 ohms per mile = 26 miles maximum distance from the central office.

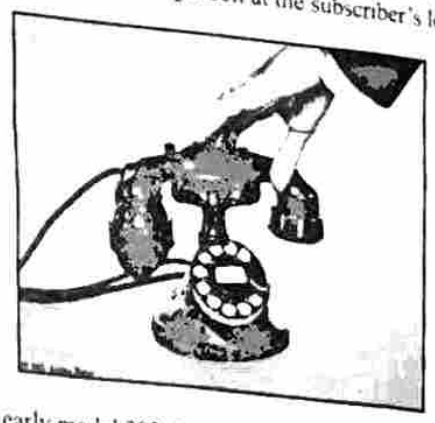
The two conductors of a telephone line are frequently called the tip (T) and ring (R) from the early switchboard patch cord designation. The tip is the green wire and the ring is red wire in RJ-11 style modular connectors. The red ring wire is –48 v.

The DC current is used to detect the "off-hook" condition of when the user or subscriber telephone is placed across the line. Two methods are used, called loop start and ground start. When a current flow is detected, the central office provides dial tone back to the subscriber set, so that the user knows that service is available.

Local Loop – AC analysis

All of the voice or speech signals, as well as the signaling tones, are AC signals sent over the local loop circuit. Yes, it is possible to send both DC and AC currents over the same circuit! I've had some electricians tell me that is impossible, but we electronic technicians know that it is an everyday occurrence! By using capacitors, coils and transformers, both DC and AC signals can co-exist in a circuit and have different functions as well.

The earliest signaling frequency used over a telephone line was the 20 Hz ring signal. It was originally generated at the telephone set to ring or "flag" the switchboard operator when the user desired to place a call. By turning a crank attached to a permanent magnet (PM) generator, the telephone user could send a low frequency 20 Hz signal to the switch-board operator, a flag would "drop" on the console next to the incoming line requesting service. Early directionality of the circuit was referred to as "line" and "drop." Line was toward the subscriber end and drop was toward the central office. The 20 Hz ring signal was also used to ring a bell at the subscriber's location.



In the early model 202 desk telephone shown in the picture, the base unit contained only the handset and rotary dial mechanism. The bell ringer was usually wall mounted and consisted of a single electromagnetic coil with a gong capable of moving between two bells. Some of the earliest telephones did not have a bell ringer inside them. The ringer was always connected to the line and was not disconnected from the circuit by the user in any way.

Chapter 15

It took a stronger signal to ring the early bells, and that was typically 90 volts RMS AC at 20 Hz, for 2 seconds duration, then off for 4 seconds and repeated until an off-hook condition is detected. Anyone working on a line when a ring signal was being sent could feel the tickle of this higher AC voltage. This voltage is still used on wireline telephones today.

Other signals sent on the line side of the connection are the dial tone, busy, fast busy and ring back. These are sometimes referred to as supervision tones and are in the audible range and are considered to be a form of "in-band" signaling. Out-of-band signaling uses a separate circuit or frequencies that we can't hear on the same circuit, such as the 20 Hz ring signal. In-band signals are what we can hear.

Two-tone frequency combinations are used for most of these supervisory alerting signals used today.

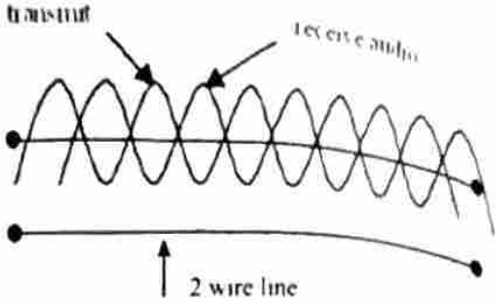
Dial tone	350 + 440 Hz
Busy	440 + 620 Hz, .5 second on and off
Fast Busy	440 + 620 Hz, .25 sec on and off
Ringback	440 + 480 Hz, 2 sec on, 4 sec off

In the early rotary dial telephones, the number of the person we were calling pulsed the DC voltage at 10 to 20 pulses per second for each digit dialed. Today dial-pad touch-tones are used to send numeric information from most telephones. This is called Dual Tone Multiple Frequency or DTMF. The frequencies used for this dialing system are:

697 Hz	1	2	3	
770 Hz	4	5	6	
852 Hz	7	8	9	
941 Hz	*	0	#	
	1209	1336	1477	1633 Hz

Note that the 1633 Hz is not used yet and is a spare set of tones for possible expansion.

One of the earliest problems to overcome in the telephone system was how to send voice signals in both directions over a single pair of lines, for one subscriber circuit. The easiest way to describe this is to say that the outgoing talk from the handset transmitter (mouth) and incoming receiver (ear) signals are out-of-phase with each other. Early systems referred to these as the mouth and ear signals. A hybrid circuit with coils and capacitors in the base unit of the subscriber telephone separates the talk and receive signals from the 4 wire handset, and places both signals on to a 2 wire line.



The bandwidth of the line circuit is also filtered for response between 300 and 3400 Hz for voice grade audio. Many different "weightings" for filters can change the slope of these filters, but a common one is called C message weighting. This helps reduce the unwanted noise, background noise, or wind noise from being added to and transmitted with the voice signal. The Bell system specifications ensured that a minimum amount of noise was permitted on the network.

To enable a user to hear a small amount of their own voice transmit signal in the earpiece of the same handset, some of the audio signal is fed to the earpiece when the phone is connected to a DC source, such as in a line from a local central office. This helps to provide the user with the sense that the phone is working and then the person feels that if I can hear myself, others can too! This is called sidetone in the telephone industry. When a subscriber picks up a handset, and blows or breathes into the mouthpiece, it gives the quick response that there is a connection and the system is working.

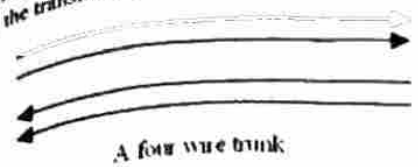
Dial tone is also provided as soon as the handset is lifted, the hook-switch connects the hybrid coil to start drawing DC current from the line. However, there may be times that the central office is too busy to furnish dial tone back to the subscriber right away, so the user may have to wait for dial tone during times of peak demand.

A line sometimes referred to as a twisted pair local-loop connection is also known as a plain old telephone service or POTS line. POTS lines are considered to be analog connections, where digital subscriber loop or DSL is able to transport high-speed digital bit rates. The length limitation of 18,000 ft. on DSL lines comes from the fact that it is designed as a line to a single subscriber without line repeaters. Repeaters to extend operating distance are available for interoffice high capacity trunk systems, but not individual subscriber lines.

Trunks

A line connects a subscriber to the central office. A trunk connects central offices and other switching facilities together.

Each trunk carries one phone call at a time. In long-haul and short-haul wireline systems, 4 wire trunks are normally used - 1 pair for the transmit signal, and 1 pair for the receive signal.



A four wire trunk

The number of trunks between telephone offices is calculated to serve a percentage of users, and not every subscriber or user needs a trunk at the same time. Trunks are time-shared for economies of scale. Traffic tables based on Erlang's or Poisson models are used to calculate trunking efficiencies. One Erlang is equivalent of one call lasting for one hour. Traffic tables are built on these models for a certain percentage of calls being blocked. Probability theory is used to predict levels of congestion of trunks, switching centers and access of service.

For example, if there are 50,000 subscribers in a city, the telephone company may have 672 long distance trunks available. Or a private branch exchange (PBX) in a company may have only 10 outside trunks to serve 150 employees each with office extensions.

In a short-haul system, multiple pair cables may be used for trunks for increasing the number of trunks bundled for short distances. However, it is more effective to use other techniques for longer distance to save on the amounts of copper wires needed. If one can remember the older telephone poles with individual wires strung on knob and tube wiring, each wire on its own insulator, you soon get the idea that we would run out of space for overhead wires. Old telephone wires were spaced on long wooden bars to reduce the capacitance and crosstalk between wires. Newer concepts allowed bundling many wires together in a small diameter cable. Twisting pairs of wires around each other reduced magnetic fields to help minimize crosstalk. Hence the name "twisted pair" is synonymous for one telephone circuit.

Instead of using multiple conductor cables, multiplexing concepts have been developed using frequency and time to separate user conversations on trunks.

Analog FDM Trunks

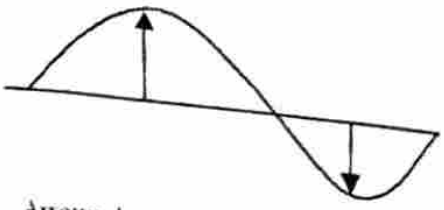
Analog frequency division multiple (FDM) systems used 4 kHz channels for each conversation, going up several hundred kHz in bandwidth to stack channels. 12 channels are called a group. 5 groups are called a Supergroup containing 60 channels. 10 Supergroups are called a Mastergroup and contain 600 channels. And, 6 Mastergroups are referred to as a Jumbogroup which has 3600 channels.

The Analog FDM trunk hierarchy

1 voice grade channel = 300 - 3400 Hz
1 Group (12 channels) = 60 - 108 kHz
1 Supergroup (5 Groups) = 312 - 552 kHz
1 Mastergroup (10 Supergroups) = 564 - 3,084 kHz
1 Jumbogroup (6 Mastergroups) = 564 - 17,548 kHz

Digital TDM Trunks

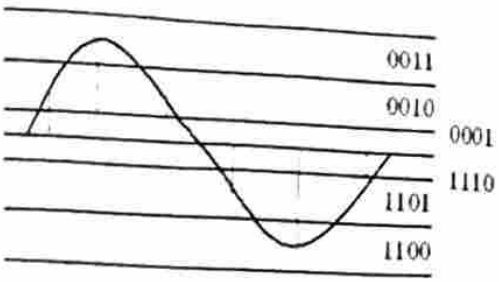
To overcome the problems of noise, distortion and cost of maintenance in FDM systems, digital time-division multiplex (TDM) systems were developed. In 1962, the first T-1 carrier system was installed in New York City. Instead of using noisy analog frequencies, the channels are first converted to digital bits using a high enough sampling rate to avoid distortion. 8 bits are used to describe the intensity of each sample, and 8,000 samples per second are used in accordance with Nyquist's theory. The basis of this theory is that at least 2 samples per waveform are needed to accurately reconstruct an analog signal.



Arrows showing 2 samples per waveform

This means that the sampling frequency must be twice as high as the highest frequency used in the system. If the highest audio is 4 kHz, then the sampling rate needs to be 8 kHz or 8,000 samples per second.

If 8 bits per sample are used to describe the amplitude of the sample and 8,000 samples per second are taken, then the bit stream that results is 64,000 bits per second needed to accurately reproduce speech and voice grade audio. This is called a DS-0 signal level, for digital signal level zero for one voice grade channel.



Quantizing shown with non-linear bands

To improve quality of sound with fewer bits per sample, narrower sample bands can be formed for the smaller values of signal voltage near the zero axis of a waveform, as shown

earlier. This reduces quantizing distortion, resulting from the approximation of the amplitudes during the signal reconstruction process.

Two types of non-uniform quantizing in digital transmission systems are widely used. The A-law is used in European systems and the mu-law is used in North American systems. These are sometimes referred to as companding standards by the ITU, the International Telecommunications Union. Companding is a form of compression and expansion used to reduce background noise in both analog and digital systems, such as in Dolby audiotape playback.

The bit stream is a linear progression of all of the bits from consecutive samples, such as 0010 0011 0010 1101 1100 1110 from the diagram shown earlier.

The resultant bit stream can be combined with digital samples from other conversations and multiplexed into a higher speed system. In TDM systems, 24 channels are frequently multiplexed together to form a DS-1 signal. This is referred to as a digital signal at the T-1 level. A T-1 results in 24 time slots each with 64,000 bits per second. Adding a few framing bits brings the bit rate up to equal 1,544,000 bits per second. Multiple T-1s can also be combined into higher-level multiplexing for more digital channel capacity.

All long distance voice traffic uses digital systems today that are nothing more than "bit haulers," which transport bits from point-to-point. The specifications for the North American digital hierarchy of TDM systems are shown below:

Electrical Spec	Bit Rate Mbps	# of Voice Chan	T carrier Designation
DS-0	64 k bps	1	
DS-1	1.544	24	T1
DS-2	3.152	96	T2
DS-3	44.736	672	* T3
DS-4	274.176	4032	T4M

** T3 - Name never existed until later*

European CEPT Digital TDM Standards:

CEPT Spec	Bit Rate Mbps	# of Voice Chan + signaling channels	
E1	2.048	30 + 2	
E2	8.448	120 + 8	(128)
E3	34.368	480 + 32	(512)
E4	139.264	1,920 + 128	(2048)
E5	565.148	7,680 + 512	(8192)

Separate signaling channels are needed in the CEPT (European Conference of Postal and Telecommunications Administrations) system as channels cannot be set up to have signaling bits taken from each voice channel as in the North American T1 systems. This process is called robbed bit signaling in the T1 carrier systems, and lowers the bit rate available for voice to 57.6 kb/s (56K) per channel.

In optical fiber trunk systems, the SONET or synchronous optical network standards for optical carriers are designated as OC levels of the STS (synchronous transport signal) specifications.

Electrical Equivalent	Bit Rate (Mbps)	# of Voice Chan	OC Designation
STS-1	51.64	672	OC-1
STS-3	155.52	2,016	OC-3
STS-9	466.56	6,048	OC-9
STS-12	622.08	8,064	OC-12
STS-18	933.12	12,096	OC-18
STS-24	1.2 Gb/s	16,128	OC-24
STS-36	1.9 Gb/s	24,192	OC-36
STS-48	2.5 Gb/s	32,256	OC-48
STS-96	4.6 Gb/s	64,512	OC-96
STS-192	9.9 Gb/s	129,024	OC-192
STS-256	13.3 Gb/s	172,032	OC-256

Higher optical rates are under development, as an OC-1024 specification now exists. Multiple wavelength sources of infrared light can also be used on the same fiber optic cable to further increase the capacity of optical systems.

Switching centers

Telephone lines were first connected together by manual patching of a call with the assistance of a switchboard operator. In 1905 New York City had over 6,000 telephone operators in the Bell system. An undertaker named Almon B. Strowger made the first automatic electromechanical switch in 1892, but it was slow to be adopted into the larger Bell system. He also was credited with the invention of the rotary dial telephone.

There are several types of switching systems. Electromechanical designs include relay driven step-by-step, also known as the Strowger switching system and the crossbar, a matrix form of connecting two lines. Electronic switching is generally of two types, space division and time division. Each line coming into a switch is assigned a number, and is associated with a certain location.

Electronic systems can be made smaller and faster than the older electromechanical types. They also have greater switching capacity due to faster microprocessors, and connect time is reduced.

As the telephone system grew, it became known as the public switched telephone network or PSTN. As part of the divestiture, area codes were added in 1976.

Numbering Plans

The North American Numbering Plan Association (NANPA), assigns area codes known as numbering plan area (NPA). Numbers are either 7 digits or 10 digits including an area code. The general arrangement is:

NPA (area code)	Central office prefix	Station # extension
3 digits	3 digits	4 digits

In the form of:

NXX-NXX-XXXX

where N = 2 thru 9

and X = 0 thru 9

The first digit of either the area code or the central office prefix cannot be a 0 or 1. These numbers are reserved for special functions, such as for calling the operator or for making a long distance call.

A smaller switching center, such as a private branch exchange (PBX), can perhaps connect 1,000 lines, or 1,000 different station extension numbers. A larger switch could connect all 10,000 numbers assigned to one central office prefix: 0000-9999. Larger switching systems yet can connect 100,000s of lines.

In doing numerical analysis of the NANPA telephone numbering plan, there are 7,999,999 numbers that could be assigned to one area code and 9,999 numbers assigned to one central office prefix.

In other words, each area code can support about 8 million telephone numbers and each office prefix can support ten thousand numbers. Every new device requiring a telephone number, such as another line for a PC modem, a fax machine, a wireless cell phone, or a pager uses additional numbers available in each area code. One of the rules for assigning numbers in an area code is that the central office prefix and the area code cannot be the same 3 digit number.

For example, you cannot have:

507-507-1234 as a telephone number

Area codes can be added by either of two methods: splitting or overlaying. Splitting involves dividing up an existing area code into smaller service areas. Overlaying assigns multiple area codes to the same service area. Each plan has its advantages and disadvantages. If overlaying is used, everyone must dial a ten-digit number in that service area. All new numbers may be assigned to the new area code, which means you could have two different area codes within your home or business. If splitting is used, only some of people and businesses will be assigned new numbers based on geographic location.

Special numbers are also assigned to each area. These are toll free 1-800 numbers and N11 numbers.

N11 numbers are:

211 Unassigned
311 Non-Emergency Police and Other Governmental Services
411 Local Directory Assistance
511 Reserved in Canada, unassigned in the U.S.
611 Wireless Customer Care Center or repair center
711 Deaf Telecommunications Relay Service (TRS)
811 Company Business Office
911 Emergency

800 numbers involve toll free and portability issues. Local number portability (LNP) is an attempt by federal mandate to provide:

1) Service provider portability - You can keep your old number if you change local telephone companies.
2) Location portability - You can keep your old number even if you move out of your area code.
3) Service portability - You keep your same features (caller ID, call forwarding, etc), no matter if you change telephone companies or service area.

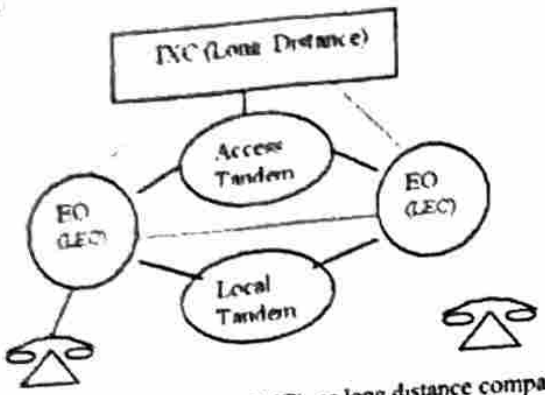
The implementation of LNP is part of the Telecommunications Act of 1996 to allow new competition into the local telephone markets. New telephone companies are called CLECs for competitive local exchange carrier. This Act allows that your next telephone company can be your cable TV provider, a long distance carrier from another state, or a wireless cellular or PCS provider.

Call Routing

In routing calls through the PSTN switching centers, a network hierarchy existed before the 1984 divestiture of the Bell System. This included 5 levels of call centers and classes of offices:

Regional Centers	Class 1
Sectional Centers	Class 2
Primary Centers	Class 3
Toll Centers (billing center)	Class 4
End office EO (local central office)	Class 5

After divesture and establishment of regional Bell operating companies (RBOC's) with the aid of digital and electronic switching centers, the PSTN became a non-hierarchical system.



The Interexchange Carrier (IXC), or long distance company, connects local calls to its long distance networks. A local exchange carrier (LEC) is the original established incumbent local telephone company servicing an area. From the changes in the Telecommunications Act of 1996, a LEC can provide long distance to its subscribers if it allows a long distance company to sell local access in its market. In the US, the original long distance company was AT&T. MCI became the second company after it installed microwave links between St. Louis and Chicago and challenged AT&T in court. Sprint is also one of the largest IXCs.

A LEC connects to an IXC at a point of presence (POP). Central offices became known as end offices (EO) after divesture and due to the dominance of the long distance companies controlling the access networks, they were the "end" points of their networks. In most areas, the original LEC is also the RBOC. After divestiture, there were seven RBOCs. These are:

- Ameritech
- Bell Atlantic
- Bell South
- NyNex
- Pacific Telesis
- Southwestern Bell
- US West

Presently, there are 4 RBOCs after recent mergers.

Trunks connecting an end office to access tandem or local tandem offices connect calls to either long distance carriers or other local offices within the area code. Also during divestiture, service areas called Local Access and Transport Areas (LATA's) were established. Boundaries between LATAs are called LATA lines and mark areas served within an established area code. Local long distance is Intra-LATA and within the same area code. A long distance call between area codes is an Inter-LATA call.

Switching centers are included in each of the office types: CO (or EO), access tandem, local tandem, and IXP to route calls to the called number.

Circuit switched vs. Packet switched

Calls can be routed by using either a circuit-switched network or a packet-switched network. Voice calls are typically circuit switched connections, and data has best performance over packet-switched networks.

Circuit-switched networks maintain only one path for the duration of the call, and then disconnect. Packet-switched networks use many routes and maintain an "always on" operation. If a circuit-switched telephone connection is used for a data call, it is called circuit switched data (CSD).

Telephone numbers route calls on a circuit-switched network, using 10 digit numbers such as 612.123.1234. IP addresses route data packets over the Internet routes in the telephone system using a 32-bit domain numbering system, in the form of 255.255.255.255. Internet routers maximize the bandwidth of connections by providing multiple routes and routers capable of dynamically assigning a trunk capacity. If a T-1 carrier is used for Internet traffic, it can be configured for higher bit rates than if channelized for voice traffic.

Signaling

Telephone connections involve both the switching centers and the network of transmission facilities that are under user control. Signaling accomplishes this by providing control to the individual as well as to the system operator. Two types of signaling involve signals on the subscriber local loop and those signals sent between central offices.

One common method to connect switching signaling systems to transmission signaling systems is called E&M signaling, or Ear and Mouth Signaling. It is commonly used for switch-to-switch and switch-to-network connections where either side may initiate actions. The E and M conductor leads connect trunk equipment to separate signaling equipment.

Four functions of signaling include: alerting, routing address information, supervision and the transmission of user information. Alerting includes requests for service and notifying the called party of an incoming call. Addressing is done either with telephone numbers or Internet Protocol (IP) addresses.

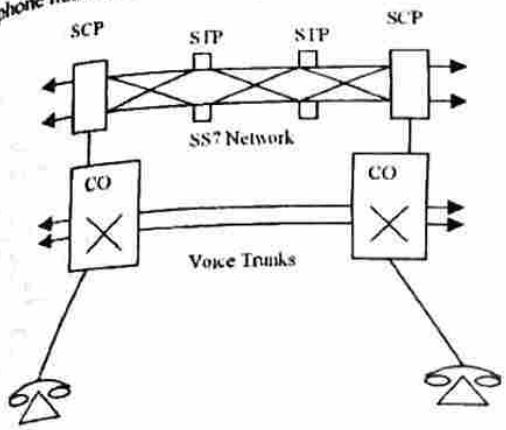
system status of calls is also sent to know whether the trunks are busy or not, can be seized for a new call, or if the call is being terminated.

Signaling between switching centers has been accomplished by two different methods. Associated Channel Signaling (ACS) or Common Channel Signaling (CCS). Before 1976, ACS sent signaling information over the same trunk that was used for the voice call. Tones such as 1600 Hz and 2600 Hz activated decoders to relay signals.

Today, common channel signaling is widely deployed. It is sometimes referred to as common channel interoffice signaling (CCIS) and a private channel separates signaling from the conversation. Protocols are enhanced in the Signaling System 7, known as an SS7 network in North America and CCITT 17 elsewhere. Integrated service digital network (ISDN) features such as call forwarding, call waiting, caller ID, billing information and other database information about special features a user may have are sent between offices by the SS7 private channel signaling network.

The commonly used "star" 2 digit feature codes are enabled with ISDN and the SS7 network. Feature codes can be used to enable or disable features such as call forwarding, last number call back, etc.

The speed of the SS7 network is derived from its separate signaling channel, usually a 56k connection using one DS-0 channel of a T-1 span. In the old days when systems used ACS, it took a long time for the dialed digits to reach the local central office on a long distance call. Now, using SS7 and digital switching, the dialed digits are sent on the private SS7 channel so quickly to the serving office that the ringback tone starts almost immediately after entering the last digit of the telephone number, even on a long distance call.



The SS7 network has single databases associated with each central office known as Signal Control Points (SCP). Redundant Signal Transfer Points (STP) are cross-connected for reliability.

Wireless Telephones

One of the fastest growing areas in telephony has been in the use of wireless telephones. The freedom that wireless gives users is its greatest asset. Two types of wireless connectivity are the unlicensed cordless phones and the high mobility cellular phones. Cellular and PCS mobile phones are approaching 90% market penetration in the US, and some countries are considered 100% with more than one connection per person.

Cordless Phones

Currently manufactured cordless phones use the unlicensed areas of the UHF frequency spectrum in the 902-928 MHz and the 2.4 GHz ISM band. Older cordless phones also used split frequencies in the 1.6 MHz and 49 MHz HF bands. Generally power output is less than 100 mW for unlicensed devices, however, the FCC regulations allow ERP rating to go to as high as 1 W in the newer UHF frequency bands. This ensures that cordless phones are low to medium mobility devices.

Several types of modulation and access technologies are employed in cordless phones. Narrowband FM is used for analog phones, and PM is used for digital phones. Most digital phones use QPSK or some form of it to increase the number of bits that can be sent over a narrow channel bandwidth.

Like the wired counterparts, cordless, as well as cellular phones, are duplex in operation. That means that 2 RF channel frequencies are needed for each direction of the call, to support simultaneous transmission in both directions. This is called frequency division duplex (FDD). Some cordless phones also use time division duplex (TDD), sharing one RF channel frequency in both directions, with the base transmitting at one time and the cordless phone transmitting at another.

Both analog and digital consumer cordless phone technologies share new frequencies, but the trend is toward digital spread spectrum for privacy. In business applications, the PACS (Personal Access Communications System) system is used in the US; DECT (Digital European Cordless Telephone) is a strong standard in Europe and the PHS (Personal Handy Phone) is prevalent in Japan.

Access technologies describe how wireless phones access base stations. In a cordless phone environment, generally only one–perhaps two cordless phones–share the same base station unit, so there is not as much contention for access in these unlicensed units as compared to cellular phones, where hundreds of users are vying for system access thru the same base station.

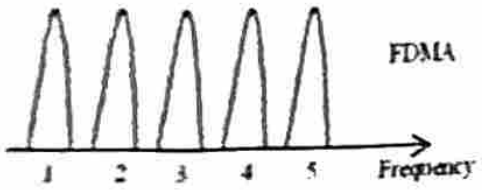
Cellular and PCS systems

Cellular and PCS (Personal Communications Service) allow high mobility services. Cellular service in the US is on 824-849 and 869-894 MHz, providing 25 MHz licenses for two carriers in each market area, often referred to as the A block and B block cellular. The licenses areas are split between metropolitan areas (MSAs) and rural service areas (RSAs). Analog cellular systems are being retrofitted with digital systems to increase channel capacity and to provide more features.

PCS licensing is more complex with 6 licenses (A thru F) originally auctioned in 1995 in the 1850-1990 MHz spectrum. However, due to the high prices bid, some of the smaller companies have defaulted, and there have been continual re-auctions of the C and F spectrum blocks. Total spectrum auctioned is 120 MHz, where A, B blocks in major trading areas (MTA) are 30 MHz each. D, E and F blocks are 10 MHz each and called BTAs for basic trading areas.

In the re-auction process, the C block has been split into smaller sections and has resulted in more spectrum available for larger carriers short of capacity in major metropolitan areas. Over 35 billion dollars has been spent to date for PCS licenses in the US. All PCS systems at 1900 MHz must use a digital access standard.

The three prevalent wireless access technologies are frequency division multiple access (FDMA), time division multiple access (TDMA) and code division multiple access (CDMA). TDMA and CDMA are referred to as 2G digital systems.



Multiple users using different RF channels

FDMA

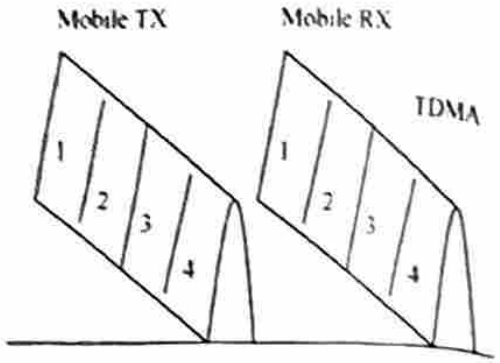
Frequency division multiple access or FDMA puts each user on its own RF channel. This is not very spectrally efficient and it is easy to intercept with an FM scanner receiver, so privacy becomes an issue.

The first generation (1G) cellular systems around the world use FDMA on different frequencies. The advanced mobile phone system (AMPS) in the Americas has 395 channels available on 824 to 894 MHz. AMPS uses a 30 kHz channel. The European Nordic mobile telephone systems (NMT) use 450 MHz and the ETACS (European Total Access Systems) uses 900 MHz frequencies with 25 kHz channels. In Japan, the JTACS also is an early analog FDMA system.

What makes cellular unique from other radio systems is that it allows handoffs of calls between cells and similar systems in different market areas. This is referred to as roaming. When a person travels to a new area from a home area, the home system must have an agreement with the visited system to permit roaming and a billing path. High capacity is also possible from the small cells and frequency reuse plans engineered into base station deployment.

TDMA

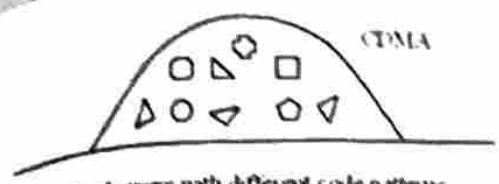
In TDMA systems, users share the same frequency by taking turns transmitting and receiving. In the diagram below, 4 users are shown on each channel. TDMA systems also use FDD to keep transmit (TX) and receive (RX) channels separate. Current systems use 3 to 8 users on one RF channel so it is more spectrally efficient and privacy is increased, as it is more difficult to intercept calls.



Four users sharing one RF channel

Time of transmit pulses vary from .5 mS to 6.6 mS in length and are used to send digitally encoded speech data. These are called "time slots" and contain 150 to 260 bits of information depending on the system used. Both the North American TDMA TIA-136 system and the European GSM (global system for mobile communication) system use TDMA technology. Battery life of mobile units is increased as well, because the transmitter is not on the air continually. TDMA is overlaid on analog 30 kHz channels, whereas GSM systems have a 200 kHz channel width for higher capacity.

The GPRS (General Packet Radio Service) and EDGE (Enhanced Digital Generation Evolution) technologies are based on GSM and are considered to be 2.7G. With higher data rates for "always-on" data, services will be delivered simultaneously with voice conversation. You can be on a call and still get your email in the background. These will be an intermediate step to faster 3G services.



Multiple users with different code patterns on a single RF channel

CDMA

CDMA is the newest of the access technologies. Originally the idea for code division came from Hedy Lamarr in World War II, where she used the tune of "Old Susanna" from a player piano to synchronize transmitters and receivers using an early version of spread spectrum technology. Starting in the early 1990's, Qualcomm is given credit for developing the TIA-95 wireless specification.

Today fast microprocessors generate random patterns to quickly code and decode "chips" that identify users in a broadband transmission channel. The 2G TIA-95 standard uses a 1.25 MHz channel bandwidth that has carried forward to 1xRTT EV-DO systems. A CDMA transmission appears to be just random noise to other receivers. It is very secure. One individual coded voice packet has very low power levels, and maybe less than the overall noise of the channel. Precise power control helps to reduce interference from other mobiles sharing the same frequency. Spreading gain also helps to detect weak signals by using the entire channel to send encoded packets.

Simultaneous transmission permits soft handoffs

With CDMA it is also possible to receive coded packets from more than one base station simultaneously. This is feasible because all of the base stations are on the same channel frequency. Mobile receivers can decode packets from up to 3 base stations or multipath signals. This helps reduce fading from signals canceling at the receive point as in other technologies, such as FDMA and TDMA, where multipath can be destructive. CDMA also makes "soft" handoffs possible, where there is no perceptible change in audio when moving from cell to cell.

Newer CDMA systems dominate the US market, with data, multimedia messaging, video media and TV services. 1xRTT (One 1.2288 MHz RF carrier Radio Transmission Technology) with EV-DO options for separate channels for voice and data services are in use today. EV-DV and 1xRTT using 1.68 MHz wide RF carriers will be backward compatible with 1xRTT systems, thereby not requiring new spectrum. GSM systems migrating to W-CDMA European standards require new spectrum. This is a more expensive solution since spectrum auctions have been very costly to wireless carriers.

Approximately 1 billion subscribers worldwide use GSM systems. 200 million users in the US use CDMA and GSM technology. Korea uses CDMA, while Europe is almost 100% GSM, and migrating towards W-CDMA.

One last thing about wireless networks is that in addition to having multiple radio "air interface" (TDMA, GSM and CDMA) standards around the world, we also have different network protocol standards that connect base stations and mobile switching centers (MSCs) together. In the US, the ANSI-41 (IS-41) standard sets the communications standards on the SS7 data networks for TDMA and CDMA wireless providers. In Europe and in GSM systems, providers that use the mobile application protocol (MAP) does not forward features to the North American IS-41 systems. Global roaming may get a boost from VoIP and IP networks forming the backbone for all continents worldwide.

There are many other wireless standards that we could discuss, such as Bluetooth, WAP and wireless LANs, but this chapter is on telephone systems, so I'll end it here.

References:

Noll, A. Michael, Introduction to Telephones and Telephone Systems, 3rd Edition, Boston, Artech House Publishers, 1998

Newton, Harry, Newton's Telecommunications Dictionary, 16th Edition, New York, CommonWeb, 2000

Hioki, Warren, Telecommunications, 4th Edition, New York, Prentice Hall, 2000

NOTES

Chapter Quiz

1. On hook DC voltage on a telephone local loop is_____.

   a. + 24 volts
   b. – 8 volts
   c. – 48 volts
   d. + 12 volts

2. Dial tone is composed of these two frequencies _____.

   a. 350 + 440 Hz
   b. 440 + 480 Hz
   c. 480 + 620 Hz
   d. 440 + 620 Hz

3. The DTMF tones for the number 5 on a telephone keypad is _____.

   a. 770 + 1336 Hz
   b. 697 + 1209 Hz
   c. 941 + 1477 Hz
   d. 852 + 1633 Hz

4. A trunk connects _____.

   a. two telephone lines
   b. two central offices
   c. two conductors
   d. two area codes

5. The form of signaling used on telephone systems today is _____.

   a. ACS
   b. ISDN
   c. POP
   d. CCS

6. A CLEC is_____.

   a. a cable TV company
   b. the incumbent telephone company
   c. a wireless provider
   d. either a or c

7. The maximum bit rate available on a T-1 transmission system is_____.

   a. 56 kbps
   b. 64 kbps
   c. 1.544 Mbps
   d. 2.4 Gbps

8. A call between area codes is_____.

   a. local long distance
   b. Inter-LATA
   c. Intra-LATA
   d. toll free

9. The device that causes the phase change between mouth and ear audio in a telephone subscriber unit is the_____.

   a. hybrid coil
   b. delay line
   c. hook switch
   d. transmitter

10. The 2nd generation digital wireless access technology that uses code patterns to separate users is_____.

    a. AMPS
    b. TDMA
    c. GSM
    d. CDMA

ANSWERS on page 223

Review Questions - RF, Cabling and Telecommunications
Chapters 13, 14 and 15

Unit 6

1. **Oscillators use quartz crystals that generate output by the:**

   a. Phase-shift effect.
   b. Piezoelectric effect.
   c. Doppler effect.
   d. Avalanche (or Zener) effect.

2. **Analog telephone lines have what DC voltage level across the tip and ring when off-hook?**

   a. 110 V DC
   b. 6 V DC
   c. -48 V DC
   d. Depends on length of line

3. **The velocity factor in a transmission line is:**

   a. equal to the termination impedance.
   b. determined by the line length.
   c. increased by the distributed capacitance and inductance of the line.
   d. compared to free space velocity.

4. **Cables of different impedances may be combined for short runs.**

   a. True
   b. False

5. **The bandwidth of a standard definition analog TV signal is :**

   a. 6 MHz.
   b. 16 MHz.
   c. 24 MHz.
   d. 88 MHz.

6. **What radio receiver section determines the radio's inherent noise level?**

   a. RF
   b. Mixer
   c. IF
   d. Detector

7. **Carbon microphones are much like variable:**

   a. capacitors.
   b. resistors.
   c. thermistors.
   d. inductors.

8. **In telecommunications work, _____ is replacing twisted pair cabling.**

   a. microwave
   b. wireless
   c. fiber optics
   d. both b and c

9. **A length of coaxial cable, used for computers, radio or television signals, is open at both ends. With an ohm meter, the resistance measured from the center conductor to the shield is 50 ohms. What does this indicate?**

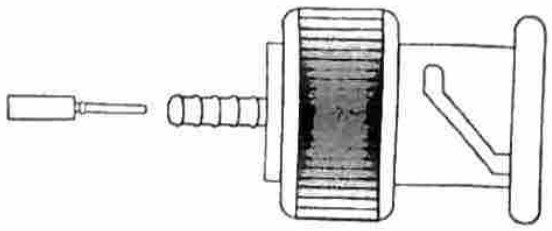
   a. The cable is partially shorted.
   b. The cable is completely open.
   c. That reading is correct for 50 ohm impedance RF cables.
   d. That reading is lower than 72 ohms for video cables.

10. **Waveguides are not utilized at HF or VHF frequencies because:**

    a. characteristic impedance is too high.
    b. resistance to high frequencies is too high.
    c. large waveguide dimensions required at lower frequencies are not practical.
    d. grounding problems.

11. **The item shown below is a (no chapter reference):**

    a. Mini UHF.
    b. PL-259.
    c. BNC.
    d. N.



ANSWERS on page 223

chapter 16

Digital Concepts, Gates, Logic Circuits and Computer Basics

by Richard B. Kirkland CETsr

Electronics II and Computer Technician teacher at the Area Technical Trade Center of North Las Vegas, Nevada. Richard has been teaching the Electronics program since 1983. He tells his students "I won't ask you to do anything I haven't done." He is a Vietnam veteran, U.S. Navy Electronics technician (10 Years). ETI1, A+ Certified, a CompTIA Professional, a Cisco Academy Instructor (CCAI) and a member of the National Society of Collegiate Scholars in addition to being a CETsr and Certification Administrator for the ETA.

Digital Concepts

Logic signals are called bits (short for binary digit). One bit is either on or off, high or low, or called logic one (1) or logic zero (0). Simple gates may change the state from 0 to 1, hold a bit, or combine a bit with other bits in certain combinations. The building blocks of all digital devices are logic gates. From the simple on-off switch to a complex microcontroller, all circuits include some form of logic gates. Some are used to change logic function and some are designed to route logic signals.

Analog circuits deal with variables and tolerances. In most analog circuits the term 'close' really means 'good enough.' In other words, the voltage doesn't have to be exactly 100-volts, or the resistor doesn't have to be exactly 100-ohms for the circuit to perform its function. Digital circuits, on the other hand, perform under more stringent circumstances. For example: an analog circuit can be compared to a dimmer-switch on the living room lamp. When the power is applied, the light comes on, a little. As the dimmer-switch is rotated the light gets brighter. The digital circuit, on the other hand, can be compared to the standard wall switch: the light is either OFF or ON. Another example is the variable resistor vs. the fixed resistor. Understand however, there is no perfect device, so yes, there are some variables and tolerances in the digital world too.

One area of digital is combinational logic circuits, which are made up of several gates types performing logic functions of AND, OR, NOT, NAND, NOR, XOR and XNOR.

A second area of digital electronics is called sequential logic, where the basic building block is a device called a flip-flop. Flip-flops are used to build binary counters and memory registers. Many different types of flip-flops have been developed, including RS, Clocked RS, D, JK and Master-Slave JK. To operate these circuits so that they work in sequence, a master control signal called a clock is required to keep all devices in a system in step with each other.

A third area of digital uses devices called tri-state logic, or three state logic. Since there are only two logic states, the third type of logic state is a controlled condition where the input and output of a gate can be disabled or un-terminated. This concept is the basis for all digital computers, where signals can be connected to a common bus, or set of paths, and the signals from various devices have to be stopped and started at different times, so that they can share the same digital highway without collisions.

Definition:

- Digital is on or off with nothing in between.
- Analog is on or off with everything in between.

In the case of most digital circuits, zero-volts would represent the number zero (0), and +5 volts would represent the number one (1). This assignment of voltage to number is how we incorporate the voltage measurement to the binary number system. A few word associations are now in order. +5-volts can also be expressed as follows: logic-1, high, on, yes, or just 1. 0-volts can also be expressed as follows: logic-0, low, off, no, or just 0.

For ease of understanding this chapter will focus on two types of digital circuit families: Transistor Transistor Logic (TTL) and Complementary Metal Oxide Semiconductor (CMOS). Logic levels are standard voltage levels and currents for specific logic families.

In the late 1960's, a positive true logic system began to emerge as a standard with the development of TTL, Transistor-Transistor-Logic. This was based on logic 0 = 0 volts and logic 1 = + 5 volts. It gets quite confusing when you look at these levels more closely, and what the minimum and maximum specifications are for each type of TTL family. Some of the TTL families are standard, low power (L), low power Schottky (LS), high speed (HS), and advanced low power Schottky (ALS), in the 5400 and 7400 series.

Assuming we use +5-volts as our binary digit one (1) and zero-volts as our binary digit zero (0), a TTL device is said to be one (1) when the voltage is between +2-volts and +5-volts.

A TTL device is said to be zero (0) when the voltage is between 0-volts and +.6-volts. For ease of mind we can say a high exists between 40% and 100% of Vcc, and a low exists between 0% and 10%. (+.6 is more than 10% of 5, but it's easier to remember that way.) So what happens if the voltage is between 10% and 40%? This is a condition known as 'undefined.' See Figure 1.

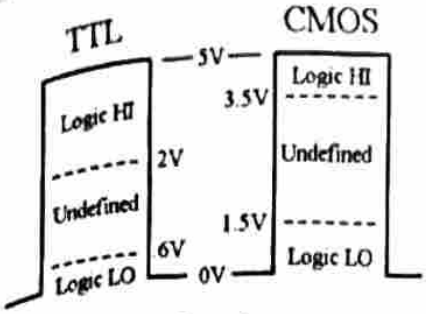


Figure 1

Another development which lowered the current required for gates was that of using FET devices in logic gates. These are called CMOS or Complementary-Metal-Oxide-Semiconductors. One N-channel and one P-channel MOSFET are used to create a complementary pair that operates on a voltage of 3 to 18 volts and uses microamps of current rather than the milliamps required for standard TTL devices. Many MOS technologies developed into families that are used in building complex IC designs. Newer microprocessor designs use .9 volt to 3.3 volt BiCMOS for lower current, less heat (power) dissipation and faster clock speeds.

Using the same assumption as for TTL (with power supply of +5-volts) a CMOS device is said to be one (1) when the voltage is between +3.5-volts and +5-volts, and zero (0) when the voltage is between 0-volts and 1.5-volts. Again, we say a high exists between 70% and 100% of Vdd, and a low exists between 0% and 30%. And yes, the voltage between 30% and 70% is termed 'undefined.'

Recap: TTL is undefined between 10% and 40% of Vcc. CMOS is undefined between 30% and 70% of Vdd.

A digital signal is either ON or OFF with nothing in between. But there is an 'in between.' The region between logic zero and logic one causes improper operation of digital circuits and can be visualized as a top hat shown in Figure 1.

How is a digital signal generated? Three basic circuits to generate digital signals are called multivibrators:

1. Monostable - generates a single pulse (sometimes referred to as a one-shot)
2. Bistable - gives a 'clean' high or low (sometimes referred to as a latch)
3. Astable - produces a series of square pulses (sometimes referred to as free running or clock)

There are many different ways to create clocks and multivibrators with discrete components or integrated packages. Frequency and pulse width characteristics are adjustable in some cases, while other clocks have fixed values designed during manufacturing. An example is a 555 I.C. which can be used to produce an astable multivibrator free running clock or a one-shot pulse generator.

Shown below are clock modules for system board applications where precise signal frequency and accuracy is required.

Many types of custom clock modules are available and are also known as VCXO, TCXO, OCXO, MCXO, DCTCXO which are abbreviations for voltage control crystal (xtal) oscillator (VCXO) or temperature control xtal oscillator (TCXO).

Summary:

1. Analog signals vary gradually, while digital signals are discrete levels commonly referred to as high and low.
2. Computers, calculators, and many other devices make use of digital circuits.
3. Most electronics equipment contains both digital and analog circuits.
4. Logic levels, referred to as high, low, and undefined, vary with different families of semiconductors such as TTL and CMOS.
5. Three basic circuits generate digital signals. They are monostable, bistable, and astable multivibrators. These circuits are sometimes referred to as 'one-shots,' 'latches,' and 'free running.'
6. Typically, +5-volts is high (red), or 1 and 0-volts is low (green), or 0, realizing that the voltage can be a little less than 5 for high, and a little more than 0 for low. 'No indication' on a logic-probe can mean an 'undefined level' or no voltage applied. (If you want voltage, then use a volt-meter.)

Digital Numbering Systems

A technician needs to be familiar with the number systems used by digital circuits and computers. The language of the digital circuit is binary, however, large numbers are usually expressed in the hexadecimal number system, and sometimes there is a need to express a value in the octal number system. All of these numbers may need to be converted to the decimal number system for clarity. Therefore, in addition to the decimal number system, the binary, octal and hexadecimal numbering systems are frequently used.

Counting in binary

Using voltage to express the number 0 and the number 1, binary results can be displayed as binary numbers to express this value using two LED's that 'light' for different binary values.

For example with two LEDs side by side

both LEDs OFF = binary 00 = decimal zero 0
left LEDs OFF, right LED ON = binary 01 = decimal one 1
left LED ON, right LED OFF = binary 10 = decimal two 2
with both LEDs ON = binary 11 = decimal three 3

This method (LED ON or OFF) gives four (4) combinations. See Figure 3.

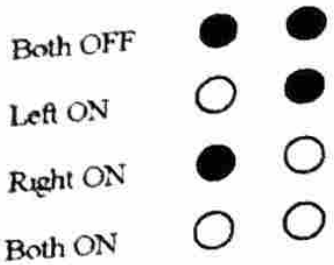


Figure 3 4 binary states with 2 indicators

Using two-character binary code: 00, 01, 10, 11, where LED=OFF=0 and LED=ON=1, 4 binary states result, with a maximum decimal count of 3.

To express values greater than 3, there are several methods of expression used in binary. ONE binary digit is called a BIT. A BIT can have a numeric value of 0 (zero) or 1 (one). A four-bit combination is called a NIBBLE, and an eight-bit combination is called a BYTE. Combinations of more than 8-bits are called WORDS. For example, 16-bit combinations are sometimes called 2-bytes or a 16-bit word.

3-bits express decimal values of 0 (zero) through 7 (seven), which is eight combinations. (Zero is a value too.)

4-bits express decimal values from 0 to 15, or 16 combinations.

Now, looking at the binary system: $2^1=2$, $2^2=4$, $2^3=8$, and $2^4=16$, the 'exponent' represents the number of combinations, and the 'largest value' is the number of combinations less one.

An 8-bit binary number (a byte) has 2^8 combinations (256) and a maximum value of 255.

A table helps to summarize number of bits with the maximum decimal number expressed.

	States	Largest Number
1	2	1
2	4	3
3	8	7
4	16	15
5	32	31
6	64	63
7	128	127
8	256	255
9	512	511
10	1024	1023
11	2048	2047
12	4096	4095
13	8192	8191
14	16384	16383
15	32768	32767
16	65536	65535

Binary is okay up to a point. Once we pass 512 (2^9), we get into some strange (hard to remember) numbers. For example, 2^{10}=1024, 2^{11}=2048, 2^{12}=4096, etc. A 16-bit binary number has 2^{16} combinations (65,536) and a maximum value of 65,535. So, to make life easier we change the 1024 to 1K, 2048 to 2K, 4096 to 4K, and 65,536 to 64K.

Using your fingers... ten fingers is 2 to the 10^{th} power, or 1K: twenty fingers is 2 to the 20^{th} power, or 1 megabit. 1024 x 1024 = 1,048,576 = mega in binary, and thirty fingers is 2 to the 30th power, or 1 gigibit.

If you associate your fingers with wires or bus paths so, 32-wires would be 2^{32} or 4-gigabits.

Putting this into perspective, if you have a 32-bit bus selecting 8-bit memory locations, you have address ability of 4-gigabytes (not bits). Remember that 8 bits make one byte.

Since writing a value of 65,535 in binary would be 1111111111111111 (sixteen 1's), convert this binary number into another form: either octal or hexadecimal to make it easier for people to handle. You need to be able to convert between any of the four (decimal, binary, octal and hexadecimal) number systems.

To convert from binary to **octal**, group the binary number in groups of three binary characters, starting from the right side.

Example

a. 1111 would be 1 111 = 17 octal
b. 11111111 would be 11 111 111 = 377 octal
c. 1111111111 would be 1 111 111 111 = 1777 octal

Each group can have a value between 0 and 7. Octal numbers are used in main frame computers

To convert binary numbers to **hexadecimal**, you group the binary numbers in groups of four binary characters, which gives you 16 combinations, and possible values of 0 through 15

Binary	Decimal	Hex
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

In addition you have to use 'single characters' for each group of four binary characters. So, we assign letters to values above nine. Ten is A, eleven is B, twelve is C, thirteen is D, fourteen is E, and fifteen is F. Using the examples above:

a. 1111 would be 1111
b. 11111111 would be 1111 1111
c. 1111111111 would be 11 1111 1111

Each group can have a value between 0 and F (0 and 15): a. would be F, b. would be FF, and c. would be 3FF. To convert back to decimal you add the place values. (This is a little tricky.) Since all number systems have a commonality of place value, a character in the first column (furthest to the right) has a value of the number-base raised to the zero power, and the next column (to the left) has a value of the number-base raised to the one power, etc., you simply multiply the value by its place, and add to the other 'place-values.'

Examples

In decimal 2374 is $4 \times 10^0 + 7 \times 10^1 + 3 \times 10^2 + 2 \times 10^3$ (From right to left)

Or $4 + 70 + 300 + 2000$, which is 2,374 in decimal

In octal, 2374 is $4 \times 8^0 + 7 \times 8^1 + 3 \times 8^2 + 2 \times 8^3$ (Again, from right to left)

Or $4 + 56 + 192 + 1024$, which is 1,276 in decimal

In hexadecimal, 2374 is $4 \times 16^0 + 7 \times 16^1 + 3 \times 16^2 + 2 \times 16^3$

Or $4 + 112 + 768 + 8192$, which is 9,076 in decimal.

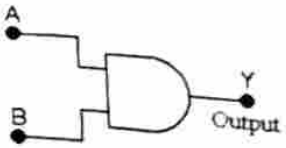
Logic Gates

There are seven (7) logic gates used in digital circuits. The first three (3) are called basic logic gates, and the other four (4) are referred to as simple logic gates. The reason for the distinction is we can construct the 'simple' gates from the 'basic' gates. Basic gates are just that, gates in the simplest form.

Basic Logic Gates

Logic gates have three characteristics. They are the logic symbol, the Boolean expression, and the truth-table. The basic gates consist of the AND, OR, and NOT (NOT is sometimes referred to as INVERTER), as follows:

The AND Gate Symbol



Boolean Expression

$$Y = A \cdot B$$

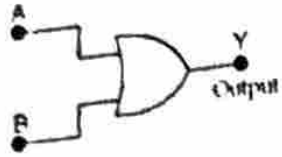
(Pronounced: Y equals A ***and*** B, not Y equals A ***times*** B)

Truth-Table

A	B	Y
0	0	0
0	1	0

The AND gate can be described as the 'all or nothing' gate. All inputs must be high for the output to be high. Another way to remember the characteristic is 'any low equals a low.'

The OR Gate Symbol



Boolean Expression

$$Y = A + B$$

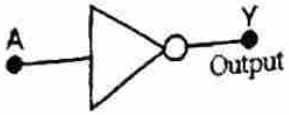
(Pronounced: Y equals A *or* B, not Y equals A *plus* B)

Truth-Table

A	B	Y out
0	0	0
0	1	1
1	0	1
1	1	1

The OR gate can be described as the 'any and all' gate. Any input must be high for the output to be high. Another way to remember the characteristic is 'any high equals a high.'

The NOT Gate or Inverter Symbol



Boolean Expression

$Y = \overline{A}$ (Pronounced: Y equals *not* A)

- or -

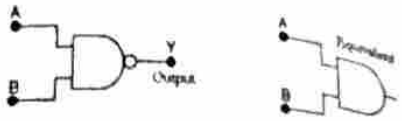
$A = \overline{A}$ (Pronounced: A equals *not* A)

Truth-Table

A	Y
0	1
1	0

Note: The truth-tables use 0 and 1 to indicate voltage values. A '0' indicates a low and a '1' indicates a high. Depending upon the family (TTL or CMOS), the high and low voltage levels are as discussed earlier. You can assume that a logic probe will be used to indicate both input and output. Also, in most circuits that have the input(s) disconnected (especially TTL circuits) will "float" high. Wh is, if you fail to connect the input of a logic ga high with respect to the reaction of the output troubleshooting tip: When you get an unexpec gate, you may have a disconnected input rathe tive gate. Many people insist that the NOT is a gate because it doesn't perform a Boolean calc matter of semantics; it depends on how you lo

The NAND Gate Symbol



Boolean Expression

$$Y = \overline{A \cdot B}$$

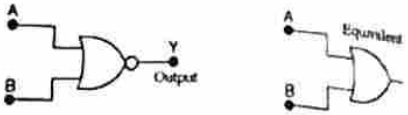
(Pronounced: Y equals *not* A and B, or Y equ

Truth-Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The NAND gate can be described as the "n versal gate. Any input must be low for the out Another way to remember the characteristic i a high.'

The NOR Gate Symbol



Boolean Expression

$$Y = \overline{A + B}$$

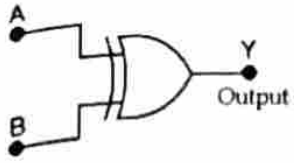
(Pronounced: Y equals *not* A *or* B, or Y *equa*

NOR Truth-Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

The NOR gate can be described as the 'not any' gate. Any input must be high for the output to be low. Another way to remember the characteristic is 'any high equals a low.'

The XOR Gate Symbol



Boolean Expression

$$Y = A \oplus B$$

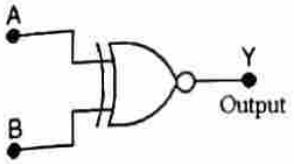
(Pronounced: Y *equals* A *xor* B) Note: The symbol between the A and the B is a plus-sign inside a circle.

XOR Truth-Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

The XOR gate can be described as the 'any but not all' gate. An odd number of inputs must be high for the output to be high. Another way to remember the characteristic is 'an odd number of 1's equals 1.'

The XNOR Gate Symbol



Boolean Expression

$$Y = \overline{A \oplus B}$$

(Pronounced: Y *equals* A *xnor* B) Note: The symbol between the A and the B is a plus-sign inside a circle.

Truth-Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

The XNOR gate can be described as the 'not any and all' gate. An odd number of inputs must be high for the output to be low. Another way to remember the characteristic is 'an odd number of 1's equals 0.'

Combinational Logic Circuits

Combinational circuits create multiple input, memory or arithmetic circuits from basic logic gates. Truth tables, Boolean expressions, Karnaugh mapping and gate circuits can be used to simplify logic design and to determine what basic gates are required.

These techniques provide a listing of gates and how they are connected together. Boolean expressions for example come in two forms--the minterm and the maxterm expression. Both expressions refer to the 'truth-table' output column (the Y column) and how it relates to the inputs.

The **"maxterm"** is called **"the product of the sums"** and the **"minterm"** is called the **"sum of the products."**

Looking at the OR gate truth table below:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

The OR gate has an output as shown in the truth-table consisting of one zero and three ones in the output column.

The zero output is used to create the maxterm expression.

The ones output are used to create the minterm expression.

The maxterm expression separates the terms (A and B) with an "OR" symbol, and the minterm expression separates the terms with an "AND" symbol. Parentheses, in the maxterm expression are separated by an "AND" symbol, and in the minterm expression are separated by an "OR" symbol.

The minterm expression from the truth table would be

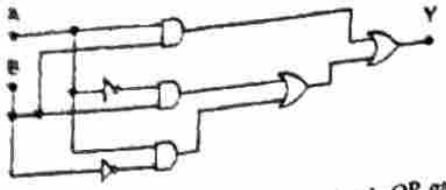
$Y = (A \cdot B)$ where Y = 0 in table

The minterm expression would be

$Y = (A \cdot B) + (A \cdot \overline{B}) + (\overline{A} \cdot B)$ (Or'ed OUTPUT)

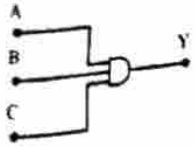
Note the Not A in the first term and Not B in the second term. The expression determines the 'output-gate,' or the 'parenthesis separator' is the output-gate

All of these terms are connected with an AND gates and an OR gate output, as shown



This would be an equivalent circuit to the simple OR gate.

The three-input AND gate



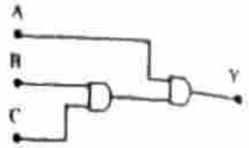
The Truth-Table is:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

In this Truth-Table we have seven instances of Y=0, and one instance of Y=1. The minterm expression would be:

$$Y = A \cdot B \cdot C$$

A possible combination circuit replacement could be



Flip-Flops: Sequential logic gates

Next, we look at sequential logic circuits that are driven by clock signals.

A flip-flop is a circuit whose output either flips (goes from 0 to 1) or flops (goes from 1 to 0) in response to data and clock input signals. The difference from a Boolean gate is that a flip-flop includes a latch function, once the output assumes a new state, it latches in that state until it's told to change again. As we'll see, the latch function is actually implemented by taking Boolean gates and cross-coupling them.

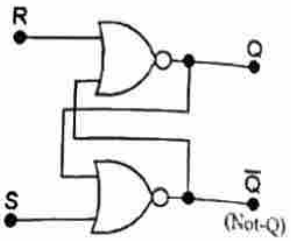
Three common types of flip-flops are:

1. RS and Clocked RS
2. D type
3. JK and Master JK

Inputs are labeled with letters depending on the type of flip-flop and the outputs have the letter Q and NOT-Q.

By definition, a flip flop is SET when the Q output is HIGH.

The first flip-flop we will look at is the **R-S flip-flop**



Inputs on the R-S flip-flop are (as you may have guessed) R and S. The 'R' stands for 'RESET' and the 'S' stands for 'SET.'

What input state on S will cause Q output to be high? The answer is: When R=0, when S=1, Q=1. You can prove this to yourself by looking back at the NOR gate truth table to see that if any input is 1, the output for that NOR is 0.

The Truth-Table for an R-S flip-flop

	R	S	Q	$\overline{Q}$
Hold	0	0	Previous State	
Reset	1	0	0	1
Set	0	1	1	0
Not Permitted	1	1	Prohibited	

When R is low and S is high, Q is high. This is called the "SET condition."

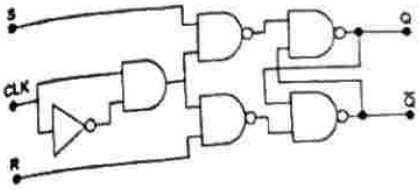
When R is high and S is low, Q is low. This is called the "RESET condition."

When R is low and S is low, Q is what it was when R (or S) changed from low to high.

If Q didn't change when one of the inputs moved from low to high–if it was already in the required state–the condition is called "HOLD."

When R is high and S is high, Q is the same as not Q. But, if Q is equal to not Q the condition is called "PROHIBITED." (Q can't equal not Q; it violates the rules of Boolean Algebra.)

Next the clocked R-S flip-flop is shown below.



Clocked RS FF's have three inputs The center input is called the "clock" input. If the CK (clock) input is low, the output of both NAND gates (connected to CK) would be high, and since the other two NAND gates are connected the same as the previous R-S flip-flop, this puts the output Q into the HOLD condition. Thus, the CK input must be high for the other two inputs to have an effect on the output.

Note that the left-hand ANDs go low when they become active, and the right hand ANDs (which were ORs in the RS flip flop diagram) latch when one of their inputs go low. Thus, the inner circuit in this flip-flop is "active low," where the normal concept of "active high" is reversed. There is nothing illegal or even rare about active-low circuits.

Clocked R-S flip-flop conditions of Q are:

Set input = 1 and when the clock goes high, then Q = 1

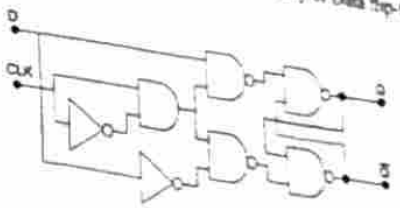
Reset input = 1 and clock goes high, then Q = 0

Hold when the clock is high, Q = previous state

Hold when the clock is not enabled, Q = previous state

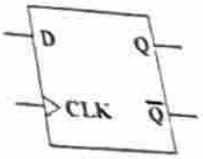
Prohibited state where Q = $\overline{Q}$ and not used

The **D flip-flop** is also known as the **Delay** or Data flip-flop and is shown below:

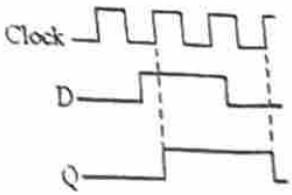


It looks similar to a clocked R-S with NAND gates except it has an inverter connected between what was formerly the S and R inputs. This limits the D type so that the bold and prohibited modes never occur.

The D FF has only two inputs: the D input and the CLK. The simple block diagram of a D type is shown below:



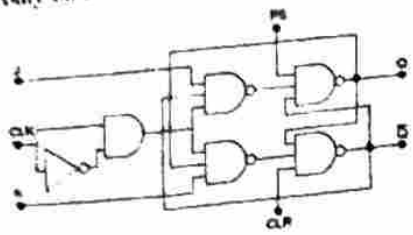
The small triangle indicates that data is transferred only during the clock transition: not during the time when the is high, but at the instant when the clock goes high. Eve clock stays high, further changes at the input will not c the Q output. In other words, the clock is "edge-enable



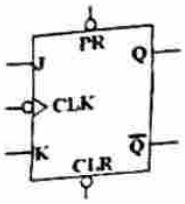
The D input data is delayed and the output is in step or in sync with the clock pulses, and it is used in memory and shift register circuits.

The J-K flip-flop is probably the most common flip-flop in use today, especially in counter circuits.

The J-K flip-flop is similar to a clocked R-S flip-flop except that both data inputs may be high. Note the extra links to the input NANDs, which have three inputs. When both J and K are high, the Q and Not-Q outputs toggle with every clock: 1-0-1-0-1-0. Also note the extra inputs. These are "preset" and "clear" inputs which instantly (asynchronously) flip or flop the gate, as opposed to the J and K inputs which flip or flop the gate, but only on clock transitions (synchronously).



So, instead of asking you to analyze the circuit, we put it in a box, and tell you to memorize its characteristics.



Note that, in the diagram, state change occurs when the clock input goes high. However, in the symbol, there's a circle or "bubble" at the clock input. This indicates that the clock function is inverted; state change occurs when the clock goes low. Many flip-flops are designed this way because experience has shown that it's useful. In this particular case, the clock, preset and clear are all low-enabled.

If the CK input is a "line" input, it is "high enabled."

If the CK input has a "bubble" input it is "low enabled."

If the CK input has an "arrow" (or greater than symbol), it is "leading-edge enabled." (The J-K diagram above is leading-edge enabled.)

If the CK input has a "bubble" and an "arrow," it is "trailing-edge enabled." (The J-K symbol above is trailing-edge enabled.)

Looking at a square-wave, it has four distinct characteristics, they are:

- Low: when the value is a steady zero volts
- High: when the value is a steady +5 volts
- Leading-edge: when value is traversing from low to high
- Trailing-edge: when value is traversing from high to low

We can use these characteristics to enable a flip-flop, when it is 'edge-triggered' or 'edge-enabled.' All J-K flip-flops are edge-enabled. Some are leading-edge, and some are trailing-edge.

The J-K flip-flop has eight (8) possible conditions of Q, as follows:

1. Asynchronous Set
2. Asynchronous Reset
3. Asynchronous Hold
4. Asynchronous Prohibited
5. Synchronous Set
6. Synchronous Reset
7. Synchronous Hold
8. Synchronous Toggle

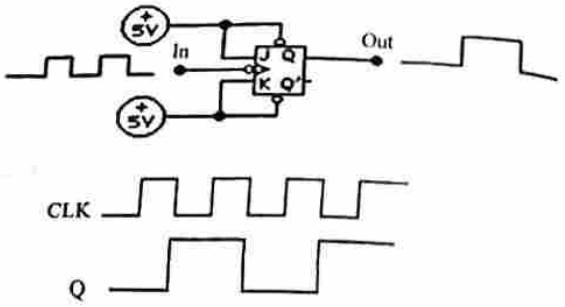
Asynchronous means not in step with the clock.

Toggle means change to opposite state. By setting both J and K inputs high, the J-K flip-flop is in the toggle mode after each CLK pulse, the output Q will change to opposite state. (If Q was 0, it will go to 1, and if Q was 1, it will go to 0.)

This characteristic then leads us to the next topic.

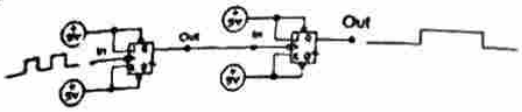
Binary Counters

Counters are digital circuits that count in binary states. Using the J-K flip-flop, with the inputs J and K enabled set high, the output Q will toggle at every instance of the clock being enabled. If, for example, we use a trailing-edge enabled J-K, setting J and K to high, upon each clock enabled the outputs will change to the opposite state. See the circuit below:



Normally, the PS and CLR inputs would be held low, but note the bubbles- these are low-enabled inputs, so they must be held high. No input should ever be left disconnected or "floating," especially with CMOS circuits.

Analyzing the circuit, we see that it takes one complete square-wave to cause the output Q to change state (because there is only one trailing-edge per square-wave). The result is the output is "half as many" square-waves. Or, in more technical terms, we have a "frequency divider"--in this case a "divide by two" frequency divider. Guess what? If we add another J-K, we get a "divide by four" frequency divider. Look at the figure below:

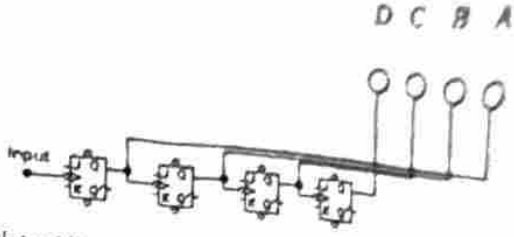


If a third circuit is added, the counter divides by eight, add a fourth JK circuit, it divides by sixteen, which results in 16 values for counter states as summarized in the truth table following:

D	C	B	A	Values
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

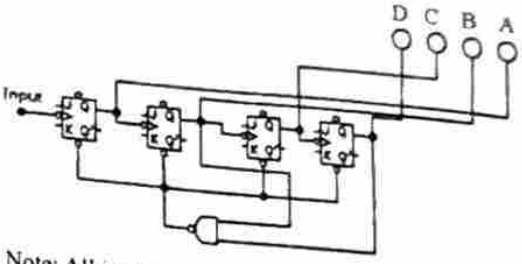
In the truth-table, the A-column changes every row, the B-column changes every other row, the C-column changes every 4th row, and the D-column changes every 8th row. Converting these row-values (binary) into decimal, the value column is the decimal count.

Look at the output of the first flip-flop in the circuit below. We'll call it FF-A (flip-flop A). FF-A changes output state (0 or 1) on every clock pulse. FF-B (the second flip-flop) changes output state every other clock pulse. FF-C every 4th, and FF-D every 8th. Looking at a truth table for four variables, and doing some adjusting to LED indicators, I can make a circuit that 'counts.'



Note: All inputs J, K, PS, and CLR are connected to +5 volts (not shown to reduce clutter). The outputs D, C, B, and A are connected to their respective flip-flops. FF-A is furthest left and FF-D is furthest right. Now I have a 'counter' that will count from zero in binary to fifteen in binary.

Now, let's do a little 'circuit modification,' to make this circuit count from zero to nine (a decade counter) as follows:



Note: All inputs shown not connected, are set to +5 volts.

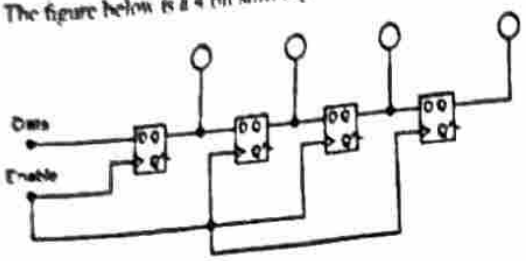
With the addition of the NAND-gate, when the 'count' gets to 1010 (decimal 10) the counter will 'reset' back to 0000. The 1010 will not be indicated because of the 'speed' of the flip-flop to respond to a 'clear.'

Counters come in various forms: there are up-counters and down-counters, synchronous counters and asynchronous counters. The circuit design can be modified to accommodate the application.

Shift Registers

Registers are sequential circuits created with flip-flops and other gates. They hold, or store, data as bits. Sometimes, in a microprocessor, a register is referred to as a "flag." A flag is a one-bit register. A register, then, is a storage circuit, or an indicator circuit. Some register circuits are also considered to be memory circuits.

The figure below is a 4 bit shift register



The shift register is made from four D flip-flops. This 4 bit shift register is called a serial-load, shift-right register.

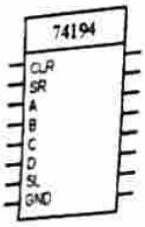
There are 4 types of shift registers: shift-right, shift-left, serial-load, and parallel-load

These may be called: serial in-serial out (SISO)
serial in-parallel out (SIPO)
parallel in-serial out (PISO)
parallel in-parallel out (PIPO)

The function of the register is to hold data for a period of time. To 'load' the above register with 1011 (A=1, B=0, C=1, and D=1) the input must be high for the first clock pulse, high for the second clock pulse, low for the third clock pulse, and high for the fourth clock pulse. Then the clock must be "disabled," otherwise the data (the output LED's) would change on the next clock pulse (the clock being the enable-input). As the clock pulses arrive, the data is "shifted" (in this case 0 to the right). So...

Data is shifted right when it moves from "A" to "D," it is shifted left when data moves from "D" to "A."

Registers also come packaged in I.C. form, as shown below:

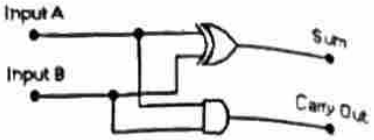


This (above) is the "universal" shift-register, the 74194. It can be parallel loaded. It can shift left. It can shift right, and it can hold its contents. Do you need to know what's inside? No...just memorize the way it works, as follows:

When S0 =0 and S1=0 it's HOLD
When S0 =0 and S1=1 it's Shift left
When S0 =1 and S1=0 it's Shift right
When S0 =1 and S1=1 it's Parallel Load

Arithmetic Circuits

Arithmetic circuits are "logic-circuits" that perform arithmetic functions made up from basic gates. The following circuit is a "half-adder."



The inputs represent the binary numbers to be added and are shown below in truth table form.

Input A	Input B	Sum	Carry out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

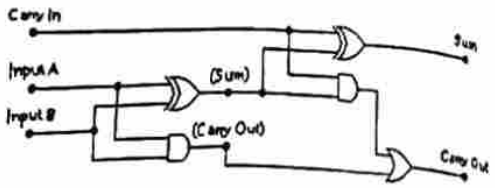
Note that the Sum output column is the same logic function as an exclusive OR gate (either A or B is high = logic HI out) and that the Carry Out column is the same logic function as an AND gate where both A and B inputs are high for a HI output.

The carry is the leftover after you add in binary.

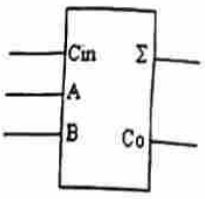
A binary addition table is shown below with all possible combinations of 0s and 1s.

0	1	0	1
+0	0	+1	+1
0	1	1	0 with a carry of 1

A full-adder is two half-adders connected together, as the following circuit shows:

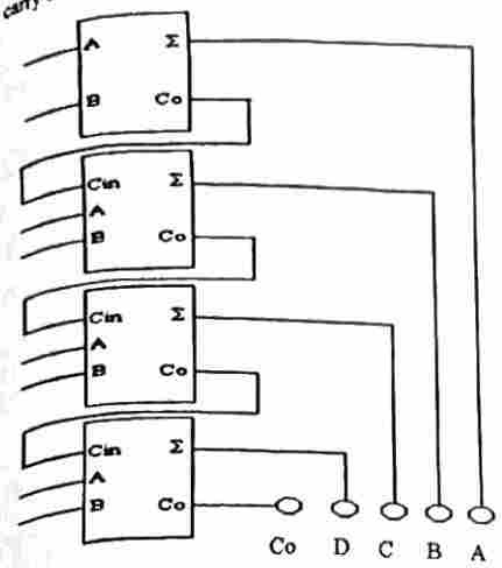


The full-adder block diagram symbol:



The sigma character (∑) represents the sum output.

Note that the full adder has a Carry Input where the half adder only has a Carry Out. This Carry Input allows the adder to be cascaded to connect multiple bit columns and account for the carry bits. Shown below is a 4 bit adder:



This circuit can "ADD" two 4-bit numbers. The output is 5 bits, including the (Co) Carry Output bit.

Decoders

One of the most common is the decoder-driver IC to decode BCD and binary bits to display on a 7 segment LED or LCD display. This circuit uses many gates to decode a binary number and convert it to a 7-segment display.

The IC has both a decoder for decoding bits, and driver to provide the current to drive a display. The inputs can be low or high enabled, and the outputs can be normal or complimented.

Definition:

- Normal output means the output is "high"
- Complement output means the output is "low"

Decoder Output is available in two configurations:

1. common-cathode display
2. common-anode display

Common TTL types are 7446 and 7447 ICs

Memory

Memory can be made from shift registers, or it can be made from a circuit specifically designed for storing data and/or instructions. Many types of semiconductor memory are available today with very large capacity and low current drain.

Computers use many types of memory, with two subsets, being semiconductor (fast) memory and magnetic or optical (storage) memory. Each subcategory has several other categories

Semiconductor memory types:

RAM	Random Access Memory (Read-Write memory) Two types: Static and Dynamic RAM Bipolar and MOS versions are available
ROM	Read Only Memory
PROM	Programmable Read Only Memory
EPROM	Electrical Programmable Read Only Memory
EEPROM	Electrically Erasable PROM

Flash Memory:

SD	Secure Digital Mini and Micro sizes available Memory Stick Compact Flash

Magnetic:

Hard Disk
Tape

Optical:

CD-ROM, CD-RWM,
DVD-RW, DVD+RW

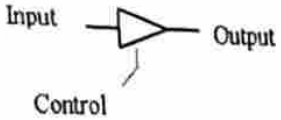
Tri-State Logic and Control Devices

Unidirectional devices

A one way tri-state logic buffer has one input, one output and a control line. The control line enables or disables the data transfer through the device. When the buffer is disabled, it is said to be in a high-impedance state and disconnects the input from the output.

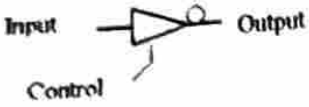
The input can only be an input and the output can only be a output. Signals can pass only in one direction, and only when the control line is activated. There are several types of tri-state buffers:

1. Non inverting, active high control

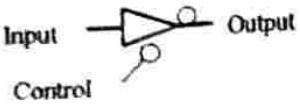


2. Non inverting, active low control

3. Inverting, active high control



4. Inverting, active low control



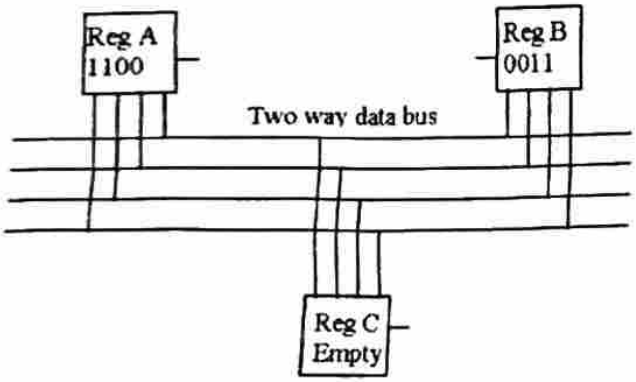
These devices are used where logic signals only pass one-way, for example, on an address bus output from a CPU. Some signal buffering and current drive is provided by these devices so that fan-out (number of gates driven) is improved.

Bi-Directional

Bi-directional tri-state devices allow signals to pass in either direction, so that an input can also be an output and vice versa. Common I/O pins are typical. The 4 types of devices as listed above also apply to the bi-directional tristate gates. These devices are used on two-way data bus lines to allow a bus to drive multiple devices, while minimizing bus loading.

Shown in the next diagram is how simple data transfer is accomplished on a 4 bit data bus. To transfer data between Register A and Register B, a third register is used as temporary storage during the transfer process.

- Data in Register A = 1100
- Data in Register B = 0011
- Data in Register C = EMPTY

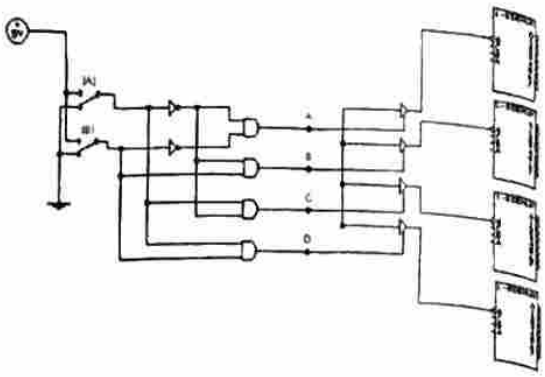


Tri-state control devices along with 4 bit Registers provide a basic computer model of how data can be moved or transferred from one memory register to another. Suppose that you want to swap the data in registers A and B without losing the original contents. The process is as follows:

1. Read the data from register A and write to register C
   Read 1100 from A >>>> Write 1100 to C
2. Read the data from register B and write to register A
   Read 0011 from B >>> > Write 0011 to A
3. Read the data from register C and write to register B
   Read 1100 from C >>>> Write 1100 to B

This is the simplified data transfer of a memory swap between two locations, by using a third memory register for temporary data. This process shows a common data bus similar to that in 32 bit or 64 bit computers used today.

Memory is designed to be a storage facility for data and instructions. Addressing memory, and retrieving data/instructions can also be accomplished in several different ways, such as 'sequential' for drives, and 'mask' or 'matrix' for random memories. For example, look at the figure below.



Point A would be high when Switch A and B are low, and the other points (B, C, and D) would be low. Since the Tri-State Buffer requires a high to be enabled, only the top circuit would receive a ground or low when Switch A and B are low. This can be interpreted as addressing the top circuit. This is just one method of addressing.

Schmidt-Trigger Devices

Schmidt-Triggers are used to clean up noisy switch contacts or signals that do not meet TTL specifications. They have a noise immunity hystersis characteristic that eliminates unwanted logic voltage levels.

Iso-Coupler/Opto-Couplers

Iso-Opto-couplers are input devices for sensing a paper moving in a printer or a flag sensor moving from a mechanical motion. They are usually made from an infrared LED and a phototransistor. When the light is blocked from the LED, the phototransistor is cutoff.

Computer Basics

What is a computer? Depending on whom you ask, you may get any description from "the circuit under the hood of your car," to some sophisticated "super computer." For this section I will deal with the common PC (personal computer).

In order to qualify as a PC, the unit consists of several distinct components: a processor, memory, input and output, storage, and power supply. That is a very basic description.

Today, we see two basic types of PCs–those built around the IBM Intel / Microsoft model, and those built around the Apple/Macintosh model. Both types use some interchangeable components, and each has unique or proprietary components. The primary difference is the language of the microprocessor. The IBM/ Microsoft unit uses an instruction set that is different than that of the Apple/Macintosh unit. This means the two types cannot use the same operating system or the same software without some intermediate communication method, or translation method. Binary code can be interpreted as a language and consists of bits, bytes, and words.

The computer (PC) has a main circuit-board. This "PCB" is usually referred to as the motherboard, main board, or the system board, which connects the microprocessor, ROM, RAM, and I/O interface connectors and ports.

Clock or processor speed can be a little confusing, such as 4.8 GHz or 2.4 GHz. This specification is typically reserved for the processing speed, or the speed at which the microprocessor manipulates data internally. The other speed, not typically advertised, is the "BUS" speed. Bus speed is typically characteristic of memory, or the speed at which information is stored and retrieved, usually at a much lower rate. These are 800 MHz, 1066 MHz, 1333 MHz, and 1600 MHz. Hard drive (or hard disk) can also have different characteristics, such as size and speed. The PC has an IDE (integrated device electronics) hard drive or SATA (serial ATA). Some have a SCSI (small computer system interface) hard drive. Both have advantages and disadvantages, but are described primarily by storage capacity, such as 1 terabyte. This specification is describing the number of bytes or 8-bit groups that can be stored on the drive track/sectors.

The other characteristics of the PC are peripheral devices, such as CD-ROM, DVD, CD-RW, monitor, printer, mouse, keyboard, etc. The last not typically advertised specification is the power supply. This box is usually described as 350 Watts or up to 850 Watts. This is an indicator as to how many peripheral devices can be added (internal devices such as another hard drive or other device that requires a power source). The larger the number, the more devices the computer (power supply) can handle.

Computer Literacy is being familiar with the operation of PCs, not necessarily the technical operation of components. How to start (boot-up) the computer is part of knowing the basics. A cold boot is a start from power-up. A warm boot on PC's is a restart from the keyboard by pushing Clt-Alt-Del keys together and using the task manager to start over.

Computer literacy generally means being able to use a keyboard, with mouse, and use special function keys, such as the Alt and Control characters. For example: the F1 key is typically used as the Help key in a Windows environment and the continue key from the start-up sequence (... press F1 to continue). A typical control character like Ctrl-C (holding down the Ctrl-Key and typing the letter C) would be 'copy' in a Windows environment. Being literate today also means being able to use the Internet to search, browse, email and use forms to fill in information.

Ports

As a technician, you should be familiar with the interfaces used to connect peripherals. Most connections are made at the back of the computer through some sort of connector. You should be familiar with at least the following:

Serial Ports: On most computers you will find a 9-pin male DB9 connector. This is the serial interface where a serial device can be attached, such as a serial mouse. On some older computers you may find a 25-pin DB25 male connector. Both of these connectors (9-pin male and 25-pin male) are called serial ports. A DB9 is a RS232 standard. Serial ports have two other characteristics: The address, and the interrupt, or IRQ. As a technician you need to be acquainted with these also... as follows:

- COM1 Address = 3F8 IRQ = 4
- COM2 Address = 2F8 IRQ = 3
- COM3 Address = 3E8 IRQ = 4
- COM4 Address = 2E8 IRQ = 3

S-R Flip-flops can be used as ripple counters

USB universal serial bus 9orts have generally replaced COM ports and transfer data much faster, up to 5 Gb/s with version 3.0.

Parallel Port: A PC may have a parallel port (sometimes called the Centronix printer port) 25-pin female connector. As with serial ports, parallel ports have an address and an IRQ,

as follows (parallel ports typically use the designator "LPT" which was used to indicate a line printer)

- LPT1 Address = 378 IRQ = 7
- LPT2 Address = 278 IRQ = 5
- LPT3 Address = 3BC IRQ = 7

Note: Sometimes the address is shown as 0378 or 378H as is with the serial ports, and one other "tidbit." Most PC's use 378 for LPT1, however IBM will typically use 3BC as their LPT1 address. In addition this connector is not always a parallel interface. On some older SCSI cards, a 25-pin female connector is used to connect to external SCSI devices.

Other ports on a PC sound card can be either a S/PDIF high-quality sound port or a MIDI (musical instrument digital interface) for example.

USB, DIN, and PS/2 ports often are used for input keyboard and mouse connectivity.

Network ports require connectors such as RJ-45 ethernet cabling for data transmission rates in ranges of 10/100/1000 Mb/S.

Ports used for monitor connectivity can include VGA, S-video, DVI, and HDMI.

Displays

Historically, in the 1970s, a computer was originally connected to a "dumb terminal" which was nothing more than a low resolution black and white TV monitor with an upper case only keyboard for input. 80 characters wide by 25 lines of text (or 40×25 larger text) were common. Since then, things have changed dramatically.

The evolution of display technology has paralleled the television industry for resolution and image quality.

The display resolution, and the associated setting such as the dot-pitch is the distance between 'pixels' on a monitor, normally measured in mm [millimeters]--(the smaller the number, the closer the pixels) and the resolution, which is a measurement of 'dots on the screen' horizontally vs. number of lines vertically.

Format	Horiz x Vert	Colors	Connector
VGA	640 x 480	256	15 pin
SVGA	800 x 600	256	15 pin
XGA	1024 x 768	16 mil	15 pin
	1366 x 768	HD	

Example: 1280 x 1024 means 1280 dots across and 1024 lines up and down.

To be able to display high resolution images clearly, more bandwidth is needed in the circuits that drive the output device. As the number of pixels increase in screen resolution format, each video frame scanned at 60 frames per second (Hz) dots get smaller, therefore more dots are required to make up the image in one still frame.

For example, at a horizontal scanning frequency of 15,750 Hz, 320 dots per line, if each line takes 63.5 uS sec
Time per line (t = 1 /f = 1 / 15,750 Hz = 63.5 uS),
then each dot (pixel) takes 63.5 uS / 320 = .2 uS of time

This corresponds to a video bandwidth of:

$$BW \text{ (in MHz)} = 1 / t = 1 / .2 \text{ uS} = 5 \text{ MHz}$$

This was typical bandwidth of early NTSC (National Television Standards Committee) analog TV with 4 x 3 aspect ratio (4 wide x 3 high).

Newer high definition displays double the horizontal scanning frequency 31,500 Hz with wide screen formats of 16 x 9 aspect ratio and 1920 dots per line yields:

Time per line of 1/ f = 1 / 31,500 = 31.75 uS
With 1920 pixels per line, 31.75 uS / 1920 = .016 uS / dot
which translates to a video bandwidth of:

$$BW = F = 1 / t = 1 / .016 \text{ uS} = 60 \text{ MHz}$$

Digital signal processing (DSP) creates compression techniques of video and audio signals which will require less bandwidth to transmit the same amount of information as in older analog systems. In a standard analog NTSC 6 MHz channel, 12 or more standard digital video channels can be broadcast, or a few less standard digital video channels if HDTV signals are broadcast. Data rates for HDTV are 15 Mb/s to decode properly, slower rates may cause pixilation or dropped pixels and image quality will be poor.

CRT (cathode ray tube) technology is quickly being replaced with flat panel technology such as plasma, LCD (liquid crystal display, and DLP (digital light processing).

Both digital television and computer technology use similar video formats for displays, and monitors can be used for either digital video system.

Laptop and notebook PCs are available with double layer DVD read-write drives, and BlueRay drives for storage and playback so that HDTV quality video can be recorded and displayed.

Graphic adapters have massive amounts of memory to store and quickly process rapidly changing video content. To store one video frame at a resolution of 1920 x 1080 takes 2,073,600 bits for each color displayed and with 256 colors, that is 530,841,600 bits or 66 Mbytes of data for each screen being refreshed every 1/30 of a second.

Small super bright LCD high resolution display panels in digital cameras and cell phones allow video and still pictures to be displayed under direct sunlight and high illumination.

With increased brightness comes more battery drain, therefore the light output efficiency becomes important. New super bright displays approach 150 lumens per Watt.

NOTES

Chapter Quiz

1. A Logic 1, for a TTL device is _____.

   a. more than 1 volt
   b. more than 1.5 volts
   c. more than 2 volts
   d. more than 3.5 volts

2. A Logic 1, for a CMOS device connected to a 5V supply is _____.

   a. more than 1 volt
   b. more than 1.5 volts
   c. more than 2 volts
   d. more than 3.5 volts

3. A green light usually indicates _____.

   a. a logic high
   b. a logic low
   c. a TTL logic level
   d. a CMOS device

4. Another name for an astable multivibrator is _____.

   a. Latch
   b. One shot
   c. Free running
   d. Mono

5. The number 12 in decimal is _____ in binary.

   a. 1200
   b. 1100
   c. 1010
   d. 1011

6. The number 25 in octal is ____ in binary.

   a. 010101
   b. 011001
   c. 110111
   d. 100101

7. The three Basic logic gates are _____.

   a. AND, OR, NAND
   b. AND, OR, NOT
   c. NAND, NOR, Inverter
   d. NAND, XOR, NOR

8. The NAND gate is sometimes called _____.

   a. all or nothing
   b. any but not all
   c. universal
   d. inverter

9. The Boolean expression $Y = \overline{A + B}$ represents _____.

   a. AND
   b. NAND
   c. OR
   d. NOR

10. The difference between minterm and maxterm is _____.

    a. minterm, Y = 1, Least # of gates is AND
    b. maxterm, Y = 1, Least # of gates is OR
    c. minterm, Y = 0, Least # of gates is OR
    d. maxterm, Y = 0, Least # of gates is AND

ANSWERS on page 223

Chapter 17

Microprocessors and Fundamental Control Systems

by Dan Siddall CETsr

Dan Siddall is a robotics programmer for SCP Global Technologies. He has taught electronics courses at ITT Technical Institute in Boise, ID. He also served in the US Navy as an electronics technician operating nuclear power plants on aircraft carriers.

Microprocessors

A microprocessor is a multipurpose device built on a single piece of silicon, called a wafer or chip. It can be programmed to perform a great number of information-handling tasks and performs all of the digital logic functions described in Chapter 16. It serves as the heart of a PC and Internet data communications, word-processing use, for controlling other machines, industrial processes, monitoring hospital patients, and for hand-held calculators.

Microprocessors are classified in terms of the number of "bits" of information that can be transferred in parallel and held in their registers. Intel® developed the first commercial microprocessor in November of 1971. It was the 4004, which was a 4-bit processor used in a calculator. It contained 2,300 transistors on a single chip, which was impressive in 1971, but now the Ivy Bridge Core i7® is a 64-bit processor and has over a billion transistors.

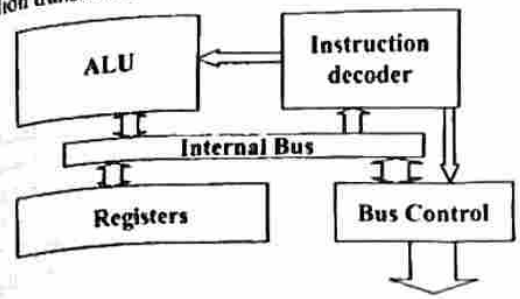


Figure 1

Figure 1 contains a simplified representation of a microprocessor. It will be used to describe how a microprocessor works. A microprocessor contains several sub-parts:

1. The Arithmetic Logic Unit (ALU) performs arithmetic and logic operations. Examples of the types of operations are addition, subtraction, logical AND, and logical OR.

2. Registers are temporary storage for the data used by the instruction decoder and the ALU. Letter identifiers (A, B, etc.) are usually used to designate the registers.

3. The Instruction Decoder determines how to perform the instruction and coordinates its execution.

4. The Bus Controller controls the flow of information into and out of the microprocessor.

5. The Internal Bus provides a path for the information to flow within the microprocessor.

Microprocessors are programmable using machine language. As the microprocessor is a digital logic device, machine language must be in binary. Assembly language is a more convenient language that has a direct correlation to machine language. For example, the assembly language instruction ADD could represent the machine language value 00000011. A device called an assembler is used to convert the word ADD to its machine language equivalent. Each microprocessor family has a finite set of instructions that it recognizes, called an instruction set. The manufacturer publishes the instruction set and information on how it is to be used. (Note: Examples can be found on the Internet at www.intel.com. Search for "instruction set").

We will use assembly language to illustrate the operation of a microprocessor. For example, to perform the addition of the numbers 4 and 3, two different assembly language instructions are necessary. The MOV instruction moves data from one location to another. It takes two arguments, destination of the data and the source of the data. The arguments are generally located in the memory locations immediately following the instruction. For example, MOV (register A, memory). This instruction takes the contents of the memory location specified and places them into register A.

The ADD instruction adds the contents of register A and register B, then places the sum into register A. Table 1 is a representation of a block of memory. The addresses and data are hexadecimal representations of the binary data.

Let's start the step-by-step analysis of adding 4 and 3. The Bus Control has a register that contains the address of the next instruction (in this example the address is 0001).

Step 1 – The first step is for the Bus Control to retrieve the instruction from that address. In this case the address is 0001 and the instruction is MOV. The instruction is sent to the Instruction Decoder for decoding. The MOV instruction takes two arguments. So the Instruction Decoder directs the Bus

Control to fetch the contents of the next two memory locations. The contents of memory location 0002 (Register A, the destination of the move) are accessed and sent to the Instruction Decoder. Memory location 0003 (0100, the source of the move) is accessed and its data is also sent to the instruction decoder. The Instruction Decoder directs the Bus Control to get the contents of memory location 0100 and put them in Register A. The first instruction is completed.

Step 2 – The Bus Control fetches the second instruction, which is located at address 0004. The instruction is MOV. The instruction is sent to the Instruction Decoder for decoding. The Instruction Decoder directs the Bus Control to fetch the contents of the next two memory locations. The contents of memory location 0005 (Register B, the destination of the move) are accessed and sent to the Instruction Decoder. Memory location 0006 (0101, the source of the move) is accessed and its data is also sent to the instruction decoder. The Instruction Decoder directs the Bus Control to get the contents of memory location 0101 and put them in Register B.

Step 3 – Bus Control then fetches the next instruction at memory location 0007, which is ADD. The Instruction Decoder instructs the ALU to add the contents of registers A (4) and B (3), then put the result back into register A (7). Finally, the instruction at location 0008 is fetched. The instruction is MOV. By using a process similar to the MOV instructions outlined above, the contents of Register A are placed in memory location 0102.

Address	Data		Address	Data
0001	MOV		0008	MOV
0002	A		0009	0102
0003	0100		000a	A
0004	MOV		0100	4
0005	B		0101	3
0006	0101		0102	7
0007	ADD			

Table 1

As seen by the example, it is a complex process just to add two numbers. For most applications, a high level programming language is used and the mechanics of the machine language are hidden. Although there are some applications that still use assembly language programming, they are mostly for smaller microprocessors and for specific applications.

For further information on microprocessors, visit the following web site:
http://www.howstuffworks.com/microprocessor.htm

Microcontroller

A microcontroller is a system that contains, at a minimum, the following four components: Memory, I/O, Bus, and MPU. The MPU (also called a CPU) is the MicroProcessor Unit or Central Processing Unit discussed in the previous section. MPU is the brain of the system where the instructions and data are processed. Memory stores the program(s) and the data necessary to accomplish the desired task. I/O connects to the outside world, through connections to such devices as a keyboard, monitor, printer, or modem. The Bus transfers information between all of the devices. Figure 2 shows a simplified microcontroller. A typical device is the 8051 which has self contained CPU, RAM and ROM.

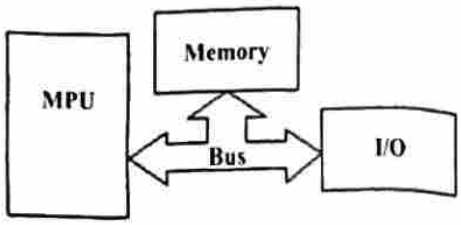


Figure 2

Common examples of microcontrollers are the PIC, Stamp and PLC, programmable logic controller. Units consist of a MPU, RAM / ROM, a display, keyboard interface and lots of input and output connections. As technology advances, manufacturers can fit more components on a chip and new devices are created. PIC and Stamps fit on small circuit boards and can control just about anything, from toy robots to home automation. Software programming is accomplished by easy commands and not in native assembly language. Advances in the memory sub-system include RAM and flash memory. The number of new I/O devices is too numerous to list here. For further information see: http://www.parallax.com

Basic Controls

A control system is a system of devices that are used to maintain a parameter at a specified level called a setpoint. The parameter can be anything that is measurable, such as voltage, current, temperature, or pressure.

In any control system there must be at least three components. The transducer is the device that converts the parameter being measured into an electrical signal that can be used by the controller. The controlled element is the device that the controller uses to change the value of the parameter.

In the case of monitoring temperature, the controlled element can be a heater that will warm the air and thus change the temperature. The controller monitors the parameter, compares it to the setpoint and manipulates the controlled element to affect a change in the parameter. The first control system we will discuss is also one of the simplest types, ON/OFF control.

ON/OFF Control

ON/OFF control operates just like it sounds. The controller either turns the controlled element 100% on or completely off. The example that illustrates this type of control is the typical home heating system as shown in Figure 3. The transducer in this case is a thermometer that converts heat into an electrical signal. The controller is the thermostat. The controlled element is the furnace.

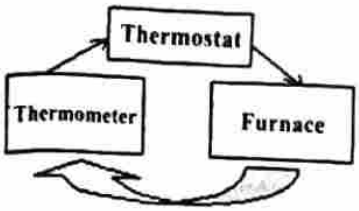


Figure 3

As temperature decreases due to losses to the outside environment, the signal sent to the controller by the thermometer also decreases. Once the temperature drops below the setpoint, the thermostat sends a signal to the furnace to turn on. This causes the heating elements and the fan to turn on. The fan will blow the heat from the heating elements throughout the house. As the temperature increases above the setpoint due to the added heat, the thermostat sends a signal to the furnace to turn off.

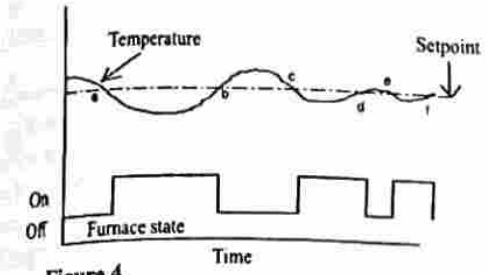


Figure 4

Figure 4 shows a graph of temperature over time and the Furnace State. While temperature remains above the setpoint the furnace remains off. Once the temperature drops below the setpoint (as shown at points a, c and e) the furnace turns on, adding heat to the house. The furnace remains on until the temperature rises above the setpoint (as shown at points b, d and f). This cycle continues the entire time the controller is enabled.

There is a problem with ON/OFF control. If the parameter is varying rapidly about the setpoint, it causes the controller to cycle the controlled element on and off excessively, which generally reduces the life of the controlled element. To alleviate this problem, hysteresis (also called a deadband), is added to the controller. The on and off points are different, with a deadband between them. In Figure 5, the setpoint is 80, the on setpoint is 78, and the off setpoint is 82. The temperature has to change by 4 degrees to cause the controlled element to change state.

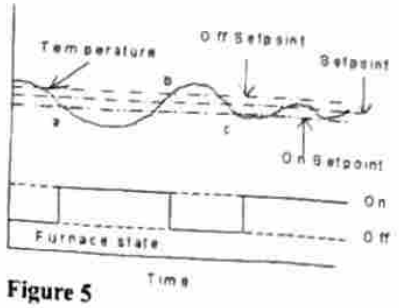


Figure 5

The furnace turns on at points a and c. The furnace turns off at point b. The same temperature curve is used in Figures 4 and 5 and the number of controlled element cycles is reduced in Figure 5 by adding the deadband.

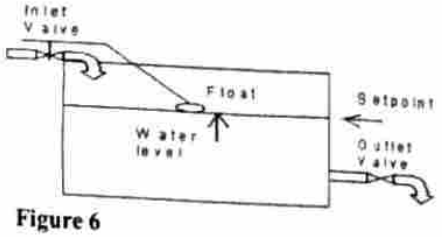


Figure 6

Proportional Control

Proportional control is a control system in which the controlled element can be at some partial position. The position of the controlled element is dependent on the difference between the setpoint and the parameter's actual value. The difference is known as the error. When the two values are the same, the controlled element is in a neutral position. The neutral position is dependent on the system design. The most common neutral position is at the 50%-of-full position.

The example used to illustrate this type of control is that of maintaining the water level in a tank as shown in Figure 6. To sense the water level a float is used. The controlled element is a valve. The controller is part of the valve. As the float's position lowers, it mechanically causes the valve to open, allowing more water to enter the tank.

First, let's discuss the case in which the outlet valve is partially opened from its initial position. This allows more water to leave the tank so that the water level in the tank drops. This, in turn, causes the float to lower and opens the inlet valve from its initial position to add more water to the tank. The float continues to lower, opening the inlet valve until the rate of out-flow is the same as the rate of in-flow.

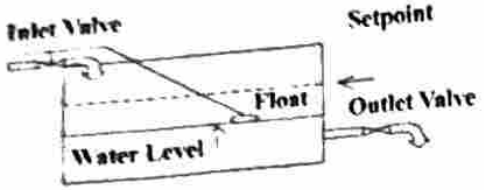


Figure 7

In Figure 7 the flows are equal and the float is lower, which is necessary to maintain the in-flow at the level to keep the system balanced. As you can see, the level is not at the setpoint. The reason that the level is not at the setpoint is that the difference is necessary to hold the valve at its new position to keep the system balanced. This is one of the drawbacks to proportional control.

Another case in this example is that the outlet valve is partially closed from its initial position. This causes the water level and the float to rise. This causes the inlet valve to partially close from its initial position and reduces the amount of water added to the tank. This process continues until the rate of in-flow is the same as the rate of out-flow. This results in a water level above the setpoint.

To correct these drawbacks of the proportional control system, something must be added. The system must be able to compensate for any error that persists which is not corrected by the proportional control. Integral control is used to correct these errors. Integral control uses the calculus operation of integration. Integration is the operation that takes the area under a curve, in this case, the error over time curve. The integral portion augments the control signal specified by the proportional control.

For an example of proportional-plus-integral control, let's modernize the tank example. Instead of a float connected to a valve, an ultrasonic level sensor is used. The ultrasonic level sensor sends sound waves down toward the water. The sound waves are reflected from the surface of the water back to the sensor. The time it takes for the sound waves to return is proportional to the distance the surface of the water is from the sensor. Thus it can be converted into an electrical signal that represents the water level in the tank and can be understood by the controller.

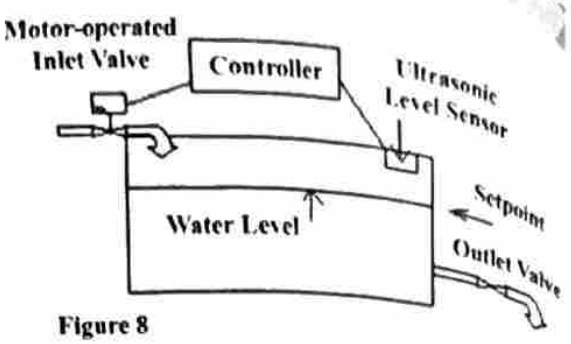


Figure 8

For the controller, a microprocessor is used, as it can handle the math necessary for integral control. The controlled element is a motor-operated valve. This allows the control system to open and close the valve. Figure 8 shows the system.

The first case to be considered is the one in which the outlet valve is partially opened from its initial position. The water level decreases and this lower level is reported to the controller by the ultrasonic sensor. The proportional control detects the error and commands the inlet valve to partially open from its neutral position, allowing more water to enter the tank. Since the error still persists, integral control commands the valve to open to restore the water level to the setpoint. With both types of controls working together, the inlet valve is open more than it was with just proportional control. This action continues until the water level returns to the setpoint and the in-flow rate equals the out-flow rate.

The second case is the one in which the outlet valve is partially closed from its initial position. This causes the water level to rise and the level sensor reports this to the controller. The proportional control commands the inlet valve to partially close from its initial position. Since the error still persists, integral control causes the valve to close partially, returning the water level to the setpoint. This action continues until the water level returns to the setpoint and the in-flow rate equals the out-flow rate.

Proportional control has another drawback. It does not respond to how fast the error is changing. The calculus operation of differentiation involves taking the rate of change of a parameter. Therefore, adding derivative control to the control system allows it to react to the rate of change of the parameter. The resulting control is proportional-plus-derivative control.

For this example, a system similar to the one in the previous example is used, except that the controller now has proportional and derivative elements. The first case that we will consider is the one in which the outlet valve is rapidly opened completely. The result of this is the water level rapidly decreases. It may decrease quickly enough to empty the tank before the proportional control opens the inlet valve far enough to prevent the tank from emptying. The derivative con-

and senses the large rate of decrease and causes the valve to open farther than the proportional control would have, adding enough flow to prevent the tank from being emptied. Figure 9 shows the graphs of water level and valve position.

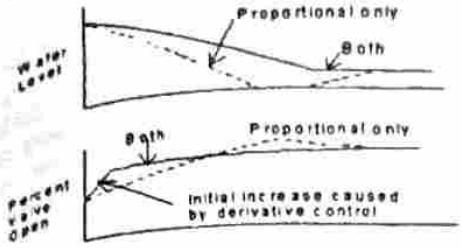


Figure 9

The most popular and most flexible control system is proportional-plus-integral-plus-derivative, also called PID. This system combines all three elements described earlier. A PID system handles changes in the parameter, long term errors not handled by proportional control, and compensates for rapid changes.

Control systems can also be classified by whether they have feedback or not. Open-Loop control does not have feedback. For example, the water level control can be made into an open loop system. The level sensor is removed and the controller is programmed to keep the valve open to 75%, which is the setting of the outlet valve. Under normal conditions this would maintain the water level. If the outlet pipe became clogged, the out-flow would be less and the level will start to rise. Since there is no level signal set to the controller, the controller cannot correct for this and the tank will overflow.

Closed-loop control uses feedback. The examples in the section on proportional control were all closed-loop control systems. In the tank example the water level signal was the feedback.

Web sites for proportional control:
http://www.controleng.com
and
http://www.fourmilab.ch/hackdiet/www/subsection1_2_3_0_5.html

Programming

There are numerous books on programming. This section will not try to cover the complex art of programming, but only cover some of the fundamentals of programming languages, writing programs and debugging programs.

Programming Languages

Machine language is the lowest level language and is used directly by the microprocessor. In this language the programmer must input every command and all data in binary form. Machine language programming is a very tedious and time-consuming task. Since the program is written in the language of the microprocessor it executes very quickly. However, the time saved in running the program rarely justifies the extra time needed to write the program.

Assembly language is a way of simplifying machine language by assigning a short (usually three-letter) mnemonic code to each machine language command. It is generally faster to write and debug assembly-language programs than their machine language counterparts. In assembly language, each mnemonic command and its symbolic operands equals one machine instruction. An "assembler" program translates the mnemonic "opcodes" (operation codes) and symbolic operands into binary language. One of the drawbacks to assembly language is that it can be used with only one type of CPU chip or microprocessor. Therefore programmers who spend much time and effort to learn how to program one computer have to learn a new set of commands each time they work on another machine.

Due to this shortcoming, a language is needed that uses one symbolic statement to represent a sequence of many machine language instructions, and allows the same program to run on several types of machines. These needs are met with high-level languages. Although most programmers use a high-level language, there are some successful programmers who use assembly language.

High-level languages often use abbreviated English words as commands that correspond to a sequence of tens or hundreds of machine language instructions. Some examples are list, print, and open. The commands can be entered from the keyboard or from a program in memory. A translator program is used to convert the commands into their machine language equivalents.

There are two kinds of translator programs: interpreters and compilers. An interpreter converts the program into machine language while it is running. Since the translation is occurring at execution time, the program runs slower than a compiled program. JAVA is an example of an interpreted language. A compiler translates the entire program into machine language prior to execution, so such programs run as rapidly as though they were written directly in machine language. C and C++ are examples of compiled languages.

Writing Programs

The first step in solving any problem or completing a task is to understand what must be accomplished. Programming is no different. In programming, the first step is to understand what inputs are available, the desired outcome, and the methods available for achieving the desired outcome. For example,

the problem of adding 4 and 3. The available inputs are the numbers 4 and 3, the desired output is their sum that is 7, and the methods are machine language, assembly language or any of the high-level languages.

If the task is complex, it will need to be broken down into steps. Flow-charting, state diagrams, and Venn diagrams can be handy tools in breaking down the task and organizing the parts. Figure 10 shows a high-level flow chart for a control system.

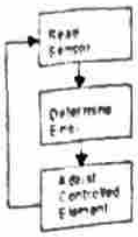


Figure 10

The next step is to determine which methods can be used and which will be the most efficient to use. Sometimes there are hardware limitations on which method can be used. If the task is programming a Programmable Logic Controller (PLC), the only programming option may be ladder logic programming. If the task is to be performed by an embedded microprocessor, the only choice may be assembly language. Even if the hardware allows higher level languages like C++, Perl, and Visual Basic, a choice as to which language to use must be made. Some languages are better suited to certain tasks than others. For example, Perl is very good at character string manipulation. For example, to add 4 and 3 in C++ only requires one line (int sum = 4 + 3;). The same operation in assembly language is more complicated. See Table 1 in this chapter which has the code for adding 4 and 3.

Now that the major decisions have been made, it is time to start working on the specifics. Take one of the sections from the previous flowchart and refine it. Create another flowchart with very detailed steps that must be done to accomplish the task. Keep in mind the language that is to be used and try to break the problem into steps that correspond to the language's instructions. One of the most common mistakes is to not put enough detail into the flowchart.

One exercise I do when teaching a programming class is to have the students write detailed instructions on how to make a peanut butter sandwich. In most cases, I do not get to the point of putting peanut butter on the bread due to the students making assumptions. The students often make assumptions about the initial conditions. For example, one assumption may be that I know how to open the jar of peanut butter. Another assumption may be that I know which end of the knife to use. In programming, assumptions can also have adverse effects on the program, e.g. assuming that the sensor input is in the range of 1 to 10 volts when the designer is actually using 4 to 20 milliamps may cause serious problems, assuming that the list to be sorted starts at 1 when in fact the list starts at 0 can cause erroneous results.

Once the detailed planning of what is to be accomplished is finished, give the plan to another programmer or coworker to review. If that option is not available, find someone to listen as you explain it. Many times in explaining it to someone else you will find the hidden errors. This review phase is very important as it can catch errors before the program is actually run on a machine, which could possibly prevent equipment damage. The review can also detect design flaws that can prevent a rewrite of the program.

With the design complete, it is now time to start writing the program. If the language being used supports the use of comments, use them. If the language does not support comments, you may want to consider using another language. Comments are helpful when you have to go back later and fix an error or make enhancements.

Since the actual writing of the program is largely dependent on the language, any specifics on writing a program would not be useful. However, anything that makes the code more readable is a good idea. Using white space to break up the program into functional sections is a common practice. Also, indenting the code inside a control structure, such as for, while, loop, can make the code more readable.

Debugging Programs

Finally, after the program is written, it is time to debug (fix the errors) the program. This task should be accomplished in an orderly fashion. The first check is done by the compiler, if the program is written in a compiled language. The compiler will catch syntax errors.

When you are preparing to write a program, you should also prepare a test plan. The test plan verifies that the program operates correctly under all operating conditions. The test plan should include instructions on how to put the system in the proper condition to start the test, the test to be performed and the expected result. This maintains consistency during the test. In addition to testing normal operating conditions, the test plan should also provide instructions on how to test the program under error/abnormal conditions.

An additional useful strategy is to write test code into the program. The program is then commanded to run the test and display the results. The drawback with this method is that the test code could have an error, resulting in an incomplete test or erroneous results.

After the bugs have been found and fixed, the tests that

fixed should be run again to verify that the bug has been fixed. Also, the tests that cover the unaffected portion of code should also be run. This verifies that the fix did not "break" something else in the program.

NOTES

Chapter Quiz

1. **Which portion of the microprocessor would perform the addition of two numbers?**

   a. ALU
   b. Bus Control
   c. Instruction Decoder
   d. Register

2. **Which of the following languages can run directly on the microprocessor?**

   a. Assembly language
   b. Interpreted language
   c. Machine language
   d. Compiled language

3. **Which component of a microcomputer connects all of the sub systems together?**

   a. MPU
   b. Memory
   c. Bus
   d. I/O

4. **What type of control handles the rate of change of the error?**

   a. ON/OFF
   b. Proportional
   c. Integral
   d. Derivative

5. **What type of control handles long term errors?**

   a. ON/OFF
   b. Proportional
   c. Integral
   d. Derivative

6. **What type of control has the controlled element in either the full open or full closed position?**

   a. ON/OFF
   b. Proportional
   c. Integral
   d. Derivative

7. **Which type of program runs the slowest?**

   a. Assembly language
   b. Machine language
   c. Interpreted language
   d. Compiled language

8. **A test plan should contain which of the following?**

   a. Expected results
   b. Initial conditions for the test
   c. How to perform the test
   d. All of the above

9. **What problem does adding hysteresis to an ON/OFF control system fix?**

   a. Excessive overshoot of the setpoint
   b. Excessive undershoot of the setpoint
   c. Excessive cycling at the setpoint
   d. Excessive setpoint

10. **Where are the results of math operations stored in a microprocessor?**

    a. ALU
    b. Bus Control
    c. Instruction Decoder
    d. Register

ANSWERS on page 223

1. **The +5V level of a TTL digital signal is called a logical:**

   a. 0 or low.
   b. 0 or high.
   c. 1 or low.
   d. 1 or high.

2. **165 (decimal) converted to binary would be:**

   a. 01110100.
   b. 10001111.
   c. 10010101.
   d. 10100101.

3. **The 1 in binary number 1000000 carries a decimal weight of:**

   a. 32.
   b. 16.
   c. 64.
   d. 128

4. **Which one of the following integrated circuit or semiconductor devices normally requires the most careful handling to avoid damage by static electricity?**

   a. MOS
   b. ECL
   c. MOV
   d. TTL

5. **What does RAM mean?**

   a. A forced memory data storage procedure
   b. Random Access Memory
   c. Real Address Mode
   d. Regulated Audio Methodology

6. **An LCD (liquid crystal display) is:**

   a. an inductive coupling device.
   b. a lithium/cadmium DC source.
   c. a low-current digital display.
   d. a charge-coupled device.

7. **Hexidecimal 3E is equivalent to a decimal number of:**

   a. 62.
   b. 46.
   c. 37.
   d. 48.

8. **What is the binary number for decimal number 9?**

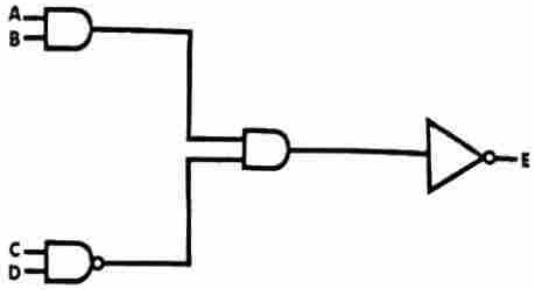
   a. 1001
   b. 0101
   c. 1100
   d. 0110

9. **Logic Pulser can be used as a:**

   a. logic probe.
   b. digital tracer.
   c. digital signal generator.
   d. filter.

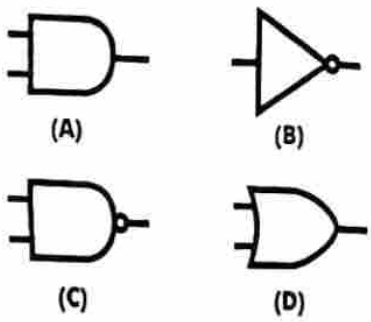
10. **In the figure below, if A is low and B is high and C is low and D is high, what is the output level at E?**

    a. Low
    b. High
    c. Floating
    d. Cannot be determined



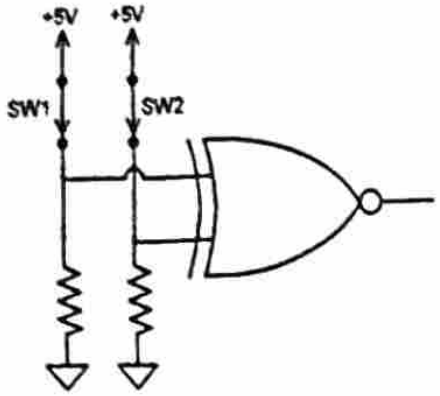
11. **In the figure below, which of these gates is an OR?**

    a. A
    b. B
    c. C
    d. D



12. **In the circuit below, with both SW1 and SW2 open, the output will be:**

a. high.
b. low.
c. floating.
d. approximately 2.5V.



13. **If SW2 were to open in the circuit above, the output of the gate circuit would be:**

a. floating.
b. low.
c. high.
d. approximately 2.5V.

14. **A three input NAND has two high inputs and a low input. The output will be:**

a. high.
b. low.
c. not enough information to determine.
d. can be neither high nor low.

15. **Adding the binary numbers 110111 and 001100 will give which one of the results below?**

a. 0000101
b. 0111101
c. 1000011
d. 1111011

ANSWERS on page 223

Chapter 18

Diagnostic Techniques

by Tom R. Janca CETsr

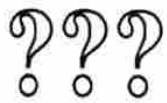
Tom Janca is actively working in the communications field for Power Engineers in Denver, CO. Previously, he worked for Bell Labs, Global Communications on the development team for developing a new Base Station Router. He has 20 years of experience with electronics, optical and radiometric systems engineering as a component level engineering technician. He has written an AFCEFT approved telecommunications curriculum for an adult education program and taught semiconductor electronics as an adjunct professor at Pueblo Community College. He has presented seminars at CDMA wireless technology and taught PC principles and techniques in a corporate environment and presently is a DOE instructor who recently officiated at the Skillful SA Telecommunications Cabling competition. He has been a member of the ETA since 1998, winning the Presidents Award and serving in various positions within ETA. Tom is an excellent teacher, writer and speaker on technical subjects, who brings the human element and understanding into his communication style and delivery.

Hardware-Software Technicians

Technicians and engineers continue to face a variety of equipment and systems that they are directly responsible for maintaining and restoring to operation if a failure occurs. Each profession faces its own complexities and challenges unique to its products and services. Compounding the daily difficulties, technical evolution has fused hardware and software within the fabric of high-tech problems. The days of a hardware technician and a software technician being two different specialties are disappearing. Modern equipment and systems require that the servicing individual be familiar both with its hardware construction and software operation in order to properly diagnose and restore operation.

How does one prepare then, for this evolving industry considering the fact that learning the common faults with a particular unit or system is only useful for a few years? To succeed, it is necessary to develop and practice a basic diagnostic technique that will serve you regardless of the technical environment you are immersed in.

It all begins with an alarm or a customer complaint. Something isn't operating the way it is expected to, and now the trouble ticket or customer is expecting you to correct the situation. Where do you begin?



Begin with the customer or system. Your most valuable starting point is the customer since they, not you, own and operate the equipment. Secondly, they know what the environment of the equipment was when it failed and what is no longer operating. This information can be extremely helpful when solving intermittent problems or preventing the overlooking of a special feature failure that does not affect the general operation of the equipment. Depending on your situation, customers can be questioned directly or a questionnaire with key questions can be filled out at the time a customer brings a product in for service.

A few good questions are:

- Please give a description of the problem or your difficulty with the equipment.
- What occurred at the time of failure?
- Is the problem consistent or intermittent?
- If intermittent, does it occur immediately or after a period of time?

Effective questions can help identify what mode of operation a piece of equipment was in when it failed. What external situations were taking place that might have affected the equipment, such as a lightning storm or an electrical brownout or surge? Every piece of equipment was designed for an ideal world and every piece of equipment is subject to a world that is less than ideal. Knowing what a piece of equipment has experienced can show you the trailhead to begin your troubleshooting journey. Consider your customer as your first contact in a foreign land and learn as much from him or her as possible before you begin your technical adventure. It could save you a great deal of time.

When working with systems, it is important to take advantage of the various alarm and records databases that maintain and store network and system health routinely. System history recording can help you discover what events were taking place when a given node or segment of the network became inoperative. Take advantage of these tools to inquire about the status of the system prior to and at the point of system failure. If it is a specific operation that causes failure, it may be a software glitch. Knowing what events took place prior to failure might help you locate or create a software patch to eliminate a software bug. It might even be a capacity issue that, during certain peak operations, failures occur and these failure records will help you present a convincing argument to management of necessary system upgrades or changes. Using the software interfaces provided to learn as much about the failure as possible before proceeding will reduce the amount of time to locate a system fault and assist in finding an effective solution.

Now What?

All right, Sherlock, you have done your investigation and questioned the witnesses. Now it is time to look at the gruesome body of the crime.

After inquiring about the failure and gathering the pertinent information, you need to verify that a crime actually has occurred.

If you have done a good job of investigating, you should know how the device failed and be able to reproduce or observe the failure. Your questioning should tell you if it will occur immediately or if you need to subject it to a burn-in phase. It is important that before you lift your first tool or fire up your gazillion dollar test equipment, you operate the customer equipment and verify its operation, or lack thereof, through observation. Sometimes it may be just an operator issue, and if you have taken out 100 screws and have panels scattered everywhere when you realize this, you might begin to feel a little silly.

If you are unfamiliar with the operation, most good service manuals contain a brief operations guide. Take the time to read this and become familiar with how it is supposed to operate and if there are any cautions you need to be aware of. Once the hound has seen the fox, it is only a matter of time before it is caught. You need to know what you are chasing before you begin the pursuit. Verification of the customer complaint is your second step.

Ah Ha!

Once you have completed your investigation and verification of the fault, it is time to begin to chase down the cause. Lets start the chase with the oldest set of troubleshooting tools available to you: your own senses. That's right! Your first step is to use your eyes, nose, ears and fingers to collect more clues. Visually check the equipment for signs of damage from a fall, moisture, and anything that might give physical evidence of a problem.

Many times power amplifiers, diodes and transistors will explode and leave a visible sign of failure in burn marks on heatsinks and printed circuit boards. Component failures can also be smelled with the nose. These aren't pleasant smells. Smells can point you in the right direction and sometimes reduce a good deal of troubleshooting time. Electrolytic capacitors and semiconductors have the most noticeable smell when they give up the ghost and join the great hereafter.

Another useful technique when dealing with equipment is to listen carefully to the sounds present. Televisions with high voltage circuits often whistle loudly when a failure occurs, and you can hear high voltage when it is generated by the static on the front of a CRT. Electromechanical devices often whine when a motor is bound up or an arm is jammed. A motor that is excessively noisy can indicate bad bearings that cause the motor drive circuit to heat up and an over-current fault to occur after a period of time.

When troubleshooting logic boards where a good number of integrated circuits are present, touch can help identify a chip that is operating at a higher temperature than its neighbor, thereby indicating the possibility of short. Remember that when a short is present, current rises. A rise in current always brings a rise in temperature. Some cautions are necessary, however, in interpreting the results of a touch test, and those are to know the function of the integrated circuit and to always wear ESD protection. Integrated circuits that are designed to power or control power often pull more current than purely logical circuits, and some familiarity and experience is required to develop an experienced sense of good and bad temperatures. Once this experience is refined, it can be quite helpful in getting you quickly to the stage or component level of troubleshooting. The one sense that isn't used often is taste, unless you're a person who prefers to test batteries with your tongue.

TOOL TIME!

What the senses have been unable to find, the brain must discover through logic and discipline using the technician's "toolbox."

- What common tools are in the box?
- Technical Education
- Experience
- Service Literature
- Test Equipment

Each of these is necessary to efficiently and accurately TEST a piece of equipment. Each one is dependent on the strength of the other three. If you have a good education but lack experience on a given piece of equipment, service literature becomes very important. If you have plenty of experience you may require very minimal test equipment or literature to solve a problem, but it should be common agreement that all four are necessary when encountering new products or unusual failures.

Service Literature

While service literature will never be found in a classical English lecture, it can be exciting and valuable reading for technicians and engineers. The service literature contains the following important documents necessary for repairing a given piece of equipment:

- Operating parameters or specifications
- Basic Operations Guide
- Interconnection Diagrams
- Block Diagrams
- Schematic Diagrams

If it is a detailed service manual, it will also contain circuit operational theories for new or innovative approaches to system operation.

Operating parameters can be useful when choosing a piece of test equipment to substitute a signal by informing you that it decodes NTSC video. It will prevent you from connecting a PAL or SECAM source for test. Parameters will technically describe what the equipment is designed to perform and the quality of performance that is necessary to accurately evaluate test equipment readings. Don't expect a Porsche when troubleshooting a VW. Both are German and both are cars, but they are completely different in performance.

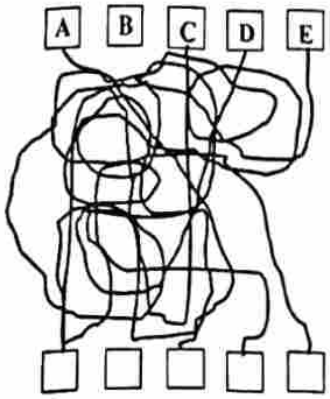
Basic Operations Guide

Where is the "on" button? How is it connected and operated?

This is helpful if you have never seen it before. However, if it is from an abandoned Air Force base, probably no manual is available and no one knows what it is anyway.

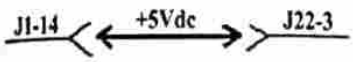
Interconnections or Cabling Diagram

When beginning as a technician, you might see a test question like this, or maybe you have seen a familiar puzzle on a fast food placemat.

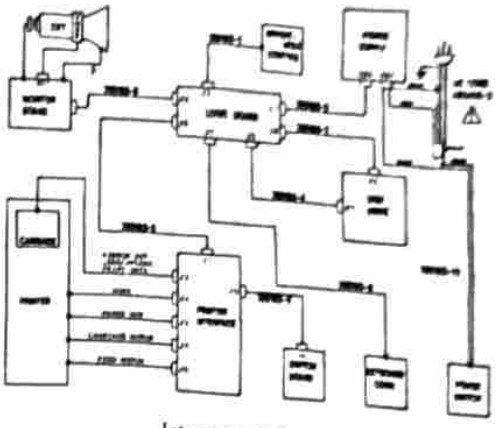


Which box is "A" connected to?

An interconnection diagram can answer this question. Interconnection diagrams identify the connection point and sometimes the cabling assembly that the power, clock, control or data signal travels through, identifying the connectors at each end by nomenclature similar to J2 to J16. Each connector is identified through PCB silk screening or a label immediately attached to the cable. The identification can also include the pin number of the connector similar to the following example:



In this way, a technician can trace the power source from one circuit board to the next.



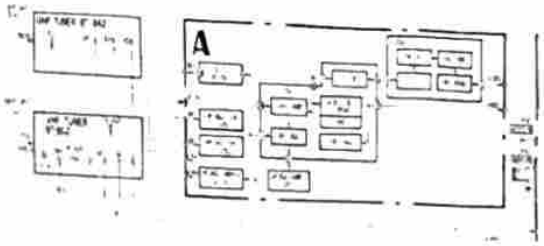
Interconnect Diagram

In this diagram the individual cable assemblies are labeled along with the connectors at each end. This diagram can also be called a 'block' diagram since it shows the primary blocks or sections of the piece of equipment.

Block Diagrams

The above interconnect diagram shows the different blocks or sections that work together to perform the equipment functions for the operator. Each block performs a certain task that is indicated by the section's name. An example is the Printer Interface section that has the task of operating and monitoring the printer. If the technician were experiencing a problem with the printer, this section would be a good place to start.

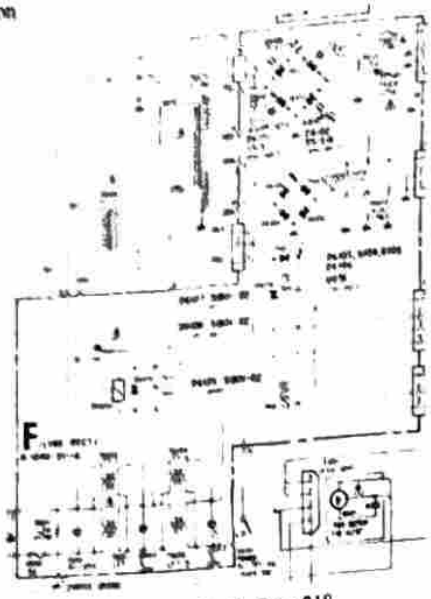
A more detailed block diagram might be available for the printer interface section that would give the technician an overview of the different sections involved in forming the printer interface function. The block diagrams allow the technician to divide a circuit quickly into operative and inoperative sections. The technician can then begin to evaluate each section's input and output for proper operation that allows him or her to locate the faulty area. This is what is classically known as 'divide and conquer.' Once the faulty section is identified, a schematic is necessary to follow the signal flow through the individual components that comprise the faulty section.



A Single Section Block Diagram See Appendix C—Page 217

Schematics

The schematic diagram details all the electronic magic that makes a given section operate. It is a topographical map compared to an interstate map of the nation. A schematic shows signal flow, biasing voltages and the electrical condition of an operational circuit. Being able to read and interpret a schematic is vital to the component level technician. Using various techniques like voltage/resistance analysis and signal substitution and tracing the schematic allows the technician to verify the proper operation of components in a given stage of a section.



See Appendix C—Page 218

Section, Block, and Component

Once the tools are in place, the fourth step is to begin to isolate the problem, applying TEST to the problem approach logically. Divide the operating sections from the inoperative sections. If multiple sections are faulty, look for something common to all of them, such as power or logic control, as the problem source. If a single section is bad, break it down into its defective blocks and then begin isolating it to component level.

TIPS

- Verify proper power levels
- Check for circuit clocks if present
- Always check IC signals at the chip body to ensure no PCB or other opens are present

Diagnostic Questions

Every electrical circuit consists of three basic sections - a source, a conductor and a load. When troubleshooting, it is good practice to first verify the source.

Source:

- Is the power applied correctly?
- Are the signal levels good?
- What is control doing?

Conductor:

- Is the power reaching the load?
- Is the signal reaching the load?
- Is control able to control the load?

Load:

- Is the load operational?

Oftentimes the load is the source of the next section, stage, and the troubleshooting repeats itself for the next section, block or component.

Methods

Voltage/Resistance Analysis

When it is necessary and possible to operate the equipment, live voltage analysis is used to determine the condition of a section, block or components. Since Ohm's law relates Voltage, Resistance and Current, the following rules of thumb may be helpful:

Short - Resistance decreases.

Parallel circuit Current increases and voltage decreases since a short in one branch shorts all branches.

Series circuit Zero, or a lower voltage is measured across the shorted component affected by the decrease in resistance. Other components show a higher V due to an increase in I.

Open - Resistance increases.

Parallel branch circuit Total circuit current decreases when one branch is inoperative.

Series circuit Entire circuit is inoperative. Nothing passes. An open has current, and the open component has total voltage applied minus the IR drop through the meter.

When a piece of equipment can't be operated as a result of the failure which is typical with power supply failures, resistance analysis can be useful in locating shorts or open circuits and components.

Remember that current seeks the path of least resistance and therefore use the Ohmmeter to measure and evaluate con-

power and circuit resistances. To be effective, one must know components and semiconductor biasing to determine good and bad values.

Signal Substitution and Signal Tracing

By substituting a good source signal from a calibrated signal generator, the technician removes the equipment source from the diagnostic equation. An example would be substituting a video signal source for a television to eliminate the possibility of a bad antenna, CATV or other source.

With a known good source applied, the technician can then use an oscilloscope or other device to trace the signal from input to output of the various sections, blocks and components to identify a faulty part. Being able to properly operate a variety of signal generators and oscilloscopes will be of great importance. Also, you must know how the circuit will affect the source signal. Often schematics will provide waveforms descriptions and key test points. If your waveform is missing or different than what is expected, you just passed your fault. Trace back until you isolate the first good signal. Now your fault should be between where it is good and where it is bad.

These are some introductory ideas to get you started. Each field will take and expand on this. It is important that you become very familiar with your basic test equipment and harvest what is presented in the chapter on test equipment. Knowing your test equipment's application will improve your TEST toolbox and help you diagnose the variety of problems encountered as a technician.

Repair or Replacement

Once you have isolated the fault it is time to replace the defective component(s) or repair the wiring or soldering responsible for the fault.

Two Facts:

1. "Your repair will only be as good as your soldering!"

2. "If you don't observe ESD handling precautions you will replace it twice."

The craft of a technician is in how well he or she completes the actual repair. A good soldering job is essential to a lasting repair and a satisfied customer. Secondly, apply new heat-sink compound when removing transistors and power devices that use a heat-sink to dissipate heat. Failing to reapply compound can cause equipment failure out of the warranty period and leave a bad word in the customer's mouth regarding your service. As a factory technician at one point, I followed behind another technician by one year. The individual replaced the faulty regulator IC but didn't apply heat-sink compound to the new device, which subsequently failed in a year's time.

The service literature will also contain a parts list or an exploded diagram to help identify different equipment parts. When possible, use the provided part numbers and OEM parts to properly repair the unit. If an OEM part is not available, use a replacement cross-reference book to locate a replacement.

Verify Operation

Once the parts have been installed, it is critical that you operate the unit for a period of time and verify that the fault has been corrected. Just replacing a component and sending it back is a poor troubleshooting technique. You should verify an equipment's operation for a given duration of time to ensure that all problems have been solved. It is possible that the immediate problem was the result of some other circuit failure that has not been identified. If the equipment must be immediately placed back in operation, make it a point to examine the unit again after it has been in service for at least 24 hours.

Extra Screws, Paperwork and Cleanup.... Next!

Proper operation has been verified and you are ready to move on to the next project. Pay close attention to your reassembly of the equipment to avoid little mishaps that leave a bad impression. Verify that all customer controls are accessible and operational with the unit fully assembled.

Complete your paperwork in order to be properly credited for the work. Paperwork should also be done as you work on the equipment to ensure that materials and parts are properly charged to the customer or correct account. This may be the most tedious part of your job, but it lets others know what work you have done and what materials and parts were used. If you don't do your paperwork immediately, you might forget to report a part used and this could cause problems with inventory. Worse still, the customer is notified at a later date than promised or not at all because the unit is sitting on the bench waiting for you to complete the paperwork.

With the unit fully assembled, the paperwork done and the unit returned the customer, you are now ready to begin your next adventure.

Chapter 18

Chapter Quiz

1. **Who knows the most about a piece of equipment?**

   a. The pharmacist
   b. The technician
   c. The operator
   d. The operator's friend

2. **Before beginning to troubleshoot, what should you do?**

   a. Verify the failure
   b. Call your friend
   c. Disassemble the equipment
   d. Divide and Conquer

3. **A short allows for an increase of current and _____.**

   a. voltage
   b. resistance
   c. heat
   d. trouble

4. **What is the oldest piece of troubleshooting equipment a technician has?**

   a. Oscilloscope
   b. Analog Volt Meter
   c. Galvanometer
   d. Five Senses

5. **Which of the following word(s) is in the acronym: TEST?**

   a. Time
   b. Ego
   c. Spectral Efficiency
   d. Technical Education

6. **What is the name of the diagram that shows all the general cable connections of a piece of equipment?**

   a. Schematic diagram
   b. Block diagram
   c. Interconnect diagram
   d. Parts list

7. **Diagnosing component failures requires the use of a/an _____.**

   a. schematic diagram
   b. block diagram
   c. interconnect diagram
   d. parts list

8. **A repair is only as good as _____.**

   a. the test equipment
   b. the equipment brand
   c. the part replacement
   d. the soldering

9. **Before returning the equipment to the customer it is important that you _____.**

   a. verify operation
   b. prepare the bill
   c. hide extra screws
   d. call the customer

10. **What is the most important service you provide?**

    a. Technical expertise
    b. Equipment repair
    c. Intelligence
    d. Customer service

ANSWERS on page 223

Chapter 19

Test Equipment

by Tom Schulte, CET

Tom Schulte is an application engineer at Sencore, Inc., located in Sioux Falls, SD. He is involved in test instrument design, plus training and support for electronics test equipment users. He has developed numerous training courses and seminars and has presented training/sales seminars across the U.S. and Canada for over ten years. Tom previously was an electronics instructor at Mitchell Technical Institute in Mitchell, SD, where he taught consumer electronics servicing, two-way radio servicing, and basic electronics.

Throughout your career as a service technician, you will be using test equipment during a high percentage of your servicing time, to provide information on a circuit's operating condition or to test the condition of individual circuit components. To be most efficient at servicing, you should have a good understanding of the proper use of all the test equipment at your disposal, and you should know just what each instrument can and cannot do for you.

To become a proficient service technician, you need to know how and when to use each of the different test instruments appropriate to your particular service field. Unless you have a good understanding of the full range of information these instruments are able to provide, you will often find your troubleshooting limited to memorized symptom/cure analysis (efficient only as an initial troubleshooting step) or blind parts shot-gunning (never an efficient troubleshooting process).

A good understanding of the proper operation of test equipment, and of the service information each instrument is capable of providing you, will tremendously increase your troubleshooting efficiency and will help place you among the top ranks of electronic service professionals.

Test equipment generally falls into one of three major categories; circuit measurement, circuit stimulus, or component test.

Circuit Measurement Equipment

This category of equipment includes voltmeters, ammeters, oscilloscopes, vectorscopes, logic probes, sound meters, logic analyzers, color photometers, AC leakage testers, RF level meters, audio, RF or light wattmeters, spectrum analyzers, and specialized circuit analyzers. Their output indicators may be analog or digital meters, CRT oscillograph displays, or LED status indicators.

These instruments are all used to determine the status of signals within or at the output of energized electrical or electronic circuits. You will use this type of instrument primarily to determine whether a particular circuit is operating within acceptable limits or whether an entire system is producing a desired output result. Circuit measurement equipment is often used during troubleshooting to trace DC supply paths and AC signal paths, to determine at what point the circuit operation becomes faulty.

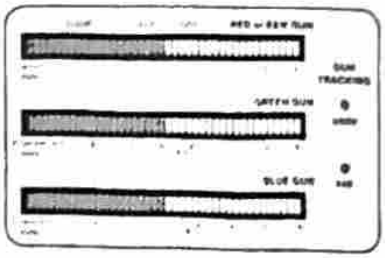


Fig. 1: An instrument's display may contain a number of types of output indicators.

Circuit Stimulus Equipment

Often called generators, this type of equipment produces single or multiple outputs that may be very simple or complex and may range from DC to high RF frequencies, or may be perceptible sound or light. Simple output signals include DC supply or bias voltages and single frequency AC voltages such as audio tone signals, and unmodulated IF or RF carrier signals. More complex generator signals include swept frequency voltages, random broadband signals (noise), video signals, logic bit streams, modulated digital data streams, modulated light signals, and modulated IF or RF carrier signals.

Generators are often used to provide an unchanging, known-good substitute signal to the input of a complete system, to closely duplicate the signal the system normally processes. This allows you to evaluate the operation of the system while troubleshooting or specification testing, without the confusion of a constantly changing, unknown-quality input signal. A signal generator can help you determine the full range of defect symptoms during an initial performance test of a defective system, and can deductively guide you to defective circuit blocks.

You may also use a generator to substitute known-good input signals to individual circuit blocks within a defective system, to help determine which stages are functioning properly and which aren't. A generator needs to have very low output impedance, with AC signal coupling, to effectively substitute signals at the inputs of individual circuit stages without affecting DC circuit voltages. Or, you may use a generator to

exercise an individual circuit or complete system over its full operating range (such as with a sweep generator, broadband noise generator, or multiburst video generator), to evaluate the circuit or system's performance in a way that wouldn't be possible with either the normal input signal or a simple generator signal.

Component Test Equipment

You will use component testers to test the characteristics of single or small groups of components, either out-of-circuit, or in an un-energized circuit. These testers include ohmmeters, transistor testers, semiconductor curve tracers, impedance bridges, capacitance meters, inductor ringers, capacitor/inductor analyzers, thyristor testers, IC comparators, CRT testers, cable testers, and custom component test jigs. These testers provide one or more stimulus signals to the component under test, measure the component output response, and drive an appropriate indicator display.

Fig. 2: A transistor tester can test bipolar and field-effect transistors in-circuit for good/bad, or out-of-circuit for gain and leakage.

You will usually use component testers in the final, isolation phase of servicing a defective circuit, after you have used functional analyzing to localize a problem to within one or two specific circuit blocks (usually using some combination of signal substitution and signal tracing). You may also use these testers to confirm the integrity of a replacement part before you connect it into a circuit or to perform an incoming quality test on components going into a manufactured product.

Commonly Used Test Equipment

Although a wide range of electronic test equipment is available, much of it is used only infrequently, for specialized testing. During most service or calibration, you will typically use only a couple of general purpose circuit measurement instruments (e.g. a voltmeter and oscilloscope), one or two signal generators appropriate to the systems you are servicing, and possibly a couple of component test instruments.

AC Isolation Transformer

An isolation transformer is one piece of test equipment that doesn't fit neatly into one of the three test equipment categories mentioned above, but should be on every service bench. This transformer has isolated primary and secondary windings with a 1:1 turns ratio. Whatever AC voltage is applied to the primary is also produced from the secondary. The difference is that the AC voltage produced from the secondary is "floating" with no fixed reference to earth ground. This prevents equipment that has internal AC voltage offsets from earth ground from causing safety problems during servicing.

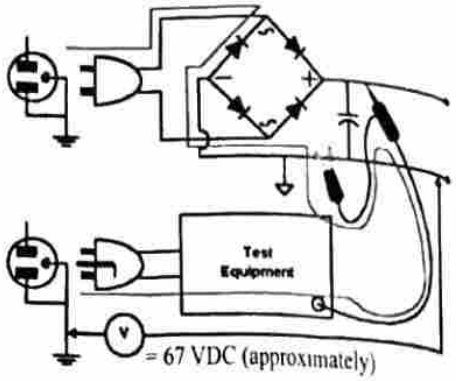


Fig. 3: A non-isolated chassis can cause damaging current to flow from hot chassis ground to grounded test equipment or personnel.

The isolation transformer should be used to provide power to any equipment you are servicing. While servicing, it is especially important to isolate any equipment that uses a full-wave bridge AC rectifier power supply.

A full-wave bridge produces a common reference point at its output which is always at a potential of one-half the AC line voltage, compared to the earth ground reference used by the AC power line and most test equipment. Note that a non-true RMS AC voltmeter measures this voltage as approximately 67 volts, due to the rectified, non-sinusoidal wave shape. We say that equipment of this type has a "hot chassis."

If you touch any internal circuit point on non-isolated equipment, and an external earth ground connection, you will likely receive an AC shock. If you connect test equipment to non-isolated equipment under test, either the test equipment or the equipment under test will be damaged, or a shock hazard will be created for yourself or others.

Fig. 4: All equipment under test should be connected to AC power from an isolated supply during service.

For personal safety and for the protection of your test equipment and the equipment you are servicing, always connect the equipment you are servicing (but not your test equipment) to the output of an AC isolation transformer, rather than directly to an AC outlet. For maximum safety and performance, test equipment should always be powered directly from a properly grounded AC outlet, not through an isolation transformer.

Realize that variable AC transformers are autotransformers and do not provide AC line isolation. If you are going to use a variable AC transformer, it should be in combination with an AC isolation transformer for safety.

DC Voltmeters

There are two general types of DC voltmeters: non-amplified and amplified.

Non-Amplified DC Voltmeters

The non-amplified DC voltmeter is generally an inexpensive or older meter that consists of only a d'Arsonval analog meter movement and a range multiplier resistor for each voltage range (Figure 6). This type meter is often called a Volt Ohm-Meter (VOM). Although this type of meter is quite suitable for electrical work (in low resistance circuits), it has at least two disadvantages for use in electronic circuits. The meter resistance between the two input leads is relatively low, and the meter resistance is different for each of the voltage ranges.

The low meter resistance means that when the voltmeter is used to measure the voltage difference between two points in a circuit, the meter will draw a small, but in some cases significant, current from the circuit. This additional current drawn from the circuit under test may cause the voltage between the two points to drop significantly when you connect this type voltmeter, resulting in a misleading voltage measurement. This effect is known as voltmeter loading, and is more pronounced when the internal meter resistance is lower, or when the external circuit resistances are higher.

A non-amplified voltmeter's internal resistance is specified in ohms/volt (often called its sensitivity), and is figured by taking the inverse of the full-scale meter movement current. Thus, a typical meter movement with a full-scale current of 50 microamps has a 20,000 ohms/volt sensitivity (1/.50 uA). A voltmeter with this sensitivity has 20 kilohms of internal resistance on its 1 Volt range and 2 megohms of internal resistance on its 100 Volt range.

If you use this type of voltmeter on its 1 Volt range (20 kilohms of internal resistance) to measure the voltage across a 20 kilohm circuit resistor with a 20 kilohm series resistance, the measured voltage will be one-half what the voltage was before the voltmeter was connected.

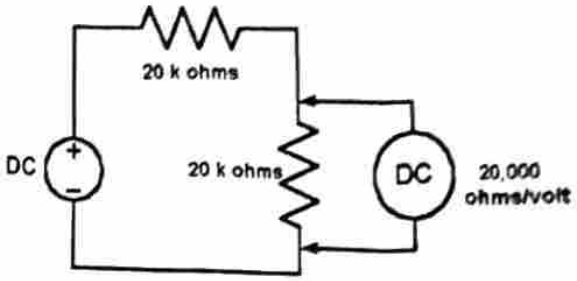


Fig. 5: If the voltmeter internal resistance isn't significantly higher than the circuit resistance, the voltage measurement will be very inaccurate.

To create a simple voltmeter, just add a series multiplier resistor to the basic meter movement. The multiplier resistor for any voltage range will be the resistance value (less the meter movement internal resistance) that limits the meter current to the meter movement's full-scale current when the maximum voltage for that range is applied. For a 2 V range with a 200 ua meter movement, for example, the total meter resistance needs to be 10,000 ohms (2 V / 200 uA). If the meter resistance is 300 ohms, the multiplier resistor will be 9,700 ohms.

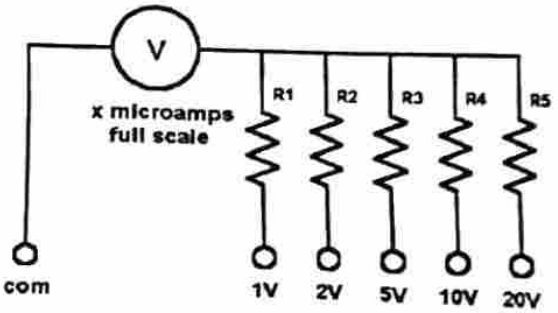


Fig. 6: The multiplier resistor for each range limits the meter current to its full-scale sensitivity with the full-scale range voltage applied.

Amplified DC Voltmeters

An amplified (sometimes called electronic) voltmeter uses an amplifier with high input impedance at its input. The input amplifier senses the difference in potential at the voltmeter's

input leads, while drawing very little current from the circuit under test. The input amplifier drives control circuits with switched voltage ranges, which then drive either an analog or digital meter display. The effective input resistance of this type meter is very high (typically 10 or 15 megohms), and is the same for every voltage range.

This high input resistance ensures that, even in moderately high resistance circuits, the small amount of current drawn from the circuit under test by the voltmeter is not enough to significantly change the circuit voltage when the voltmeter leads are connected to the circuit.

Another advantage of a meter of this type is that, since the meter's input resistance is constant for all ranges, it is quite easy to design a voltage extender probe for the meter. A probe can be designed to extend the voltage measuring range by a factor of ten, for example, by designing a probe that increases a meter's input resistance to be ten times its normal value.

For a voltmeter with a normal input resistance of 15 megohms, the extender probe would contain a series resistance of 135 megohms, nine times the meter resistance (Figure 7). This increases the total resistance of the meter plus extender probe to 150 megohms, ten times the resistance of the meter alone.

When you connect the meter and extender probe across a circuit voltage potential, one-tenth of the circuit voltage is dropped across the meter, and the other nine-tenths is dropped across the extender probe. Note that the extender resistor must be capable of withstanding this high voltage. Whatever voltage is measured by the voltmeter, you then simply multiply by ten to determine the actual circuit voltage.

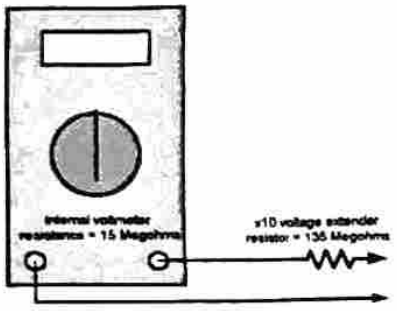


Fig. 7: The extender probe is nine times the meter resistance, so nine-tenths of the probe-tip voltage is dropped across the extender probe and one-tenth is measured by the meter.

The same method is used to create a 100x high voltage probe, to allow a standard voltmeter to measure the 20 kV to 35 kV high voltage developed in CRT display devices. The extender resistor will be 99 times the meter resistance–990 megohms for a 10 megohmmeter and 1485 megohms for a 15 megohmmeter.

A 10x or 100x extender probe also reduces the voltmeter loading effect by a factor of 10 or 100. This is an advantage since higher voltage-producing circuits are usually also higher resistance and more subject to loading, even with a 10 or 15 megohm voltmeter.

AC Voltmeters

Average Reading, Calibrated to RMS

Most AC voltmeters display the voltage of an AC signal in a way that is precisely correct for sine waves, but is not as correct for other AC wave shapes. These voltmeters generally use a precision half-wave rectifier to detect the average voltage of the AC signal, then multiply the average value by 1.111 to derive the displayed RMS value.

This is a generally accepted practice in electronics, although we should understand that the 1.111 conversion factor from average to RMS is correct only for pure sine wave signals (the average value of one-half sine wave cycle is 0.636 of the peak value; the RMS value is 0.707 of the peak value). For AC signals that aren't pure sine waves, the AC voltage measured by this type meter will be somewhat in error. Some techs like to use an analog meter to locate intermittent circuit problems.

As long as everybody uses an AC voltmeter of this same design, even if the signal measured is not a pure sine wave, we all agree on what AC voltage should be measured at a particular point in a properly functioning circuit.

True RMS

There are times, though, that you will need to know the true RMS AC voltage for AC signals which aren't sine waves. In this case, you will need a true RMS AC voltmeter. This is a more expensive voltmeter that includes a more sophisticated true RMS voltage detector. This type detector determines the true "heating effect" equivalence of an AC signal to a DC voltage.

Crest Factor

One of the limitations of a true RMS voltage detector is its inability to accurately measure narrow voltage peaks that are much greater amplitude than the RMS level. The crest factor of a signal is its ratio of peak value to its RMS value. More capable RMS detectors can accurately detect signals with greater crest factors. Typical general-purpose meters maintain acceptable accuracy with AC voltage crest factors from two to five.

Bandwidth

All AC voltmeters are limited to a particular range of signal frequencies they can measure. Be sure an AC voltmeter's bandwidth matches or exceeds the highest frequency content of the AC signal you are measuring. AC voltmeter bandwidth ranges from 1 kHz to 200 or 300 kHz.

Ohmmeters

The basic action of an ohmmeter is to apply a known voltage to a component or circuit and measure the resulting current draw. The meter scale is calibrated in ohms for the V / I equivalent.

Low Voltage Ohms Function

A normal ohmmeter circuit usually applies enough voltage to a circuit to turn on forward-biased semiconductor junctions. These forward-biased semiconductor junctions present a low resistance to the ohmmeter, which can make it difficult to obtain meaningful resistance measurements in-circuit.

The alternative is to design an ohmmeter circuit that applies only a couple tenths of a volt to the circuit, to ensure that no semiconductor junctions are turned on. This allows you to measure a resistor in the circuit, without any effect from semiconductor junctions in the circuit. One disadvantage of these low ohms functions is that they are usually limited to lower resistance ranges than standard ohms functions.

Ammeters

A conductive circuit that has a difference of potential across it will have electrons flowing through the circuit. An ammeter measures the rate of electron flow (current) through an energized circuit. An ammeter can be as simple as a mechanical meter movement with an appropriate full scale sensitivity which is at least slightly more than the current expected in the circuit to be measured.

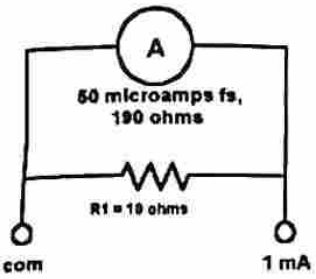


Fig 8: An ammeter current shunt allows a fixed percentage of the input current to bypass the meter movement.

For higher current ranges, a precision low-resistance shunt is placed in parallel with the meter movement. The shunt is designed to allow a fixed percentage of the circuit current to bypass the meter movement. The current passing through the meter movement is then always that percentage of the full circuit current.

For the simple ammeter shown in Figure 8, it can be seen that 950 microamps needs to flow through the shunt resistor when the meter is measuring 1 mA (1000 uA - 50 uA). For the shunt current to be 19 times the meter movement current (950 ua / 50 uA), the shunt resistor needs to be 1/19th the resistance of the meter movement internal resistance (190 / 19 = 10 ohms).

DMMs

Technicians typically use a DMM (Digital MultiMeter) for most of their circuit measurement tasks. A DMM usually includes, at a minimum, a voltmeter capable of measuring DC and AC voltages, an ohmmeter, and an ammeter capable of measuring both DC and AC current up to one amp or more.

Fig. 9: A DMM contains a number of auto-ranged measurement functions in one convenient instrument

A significant advantage with most digital multimeters is that they are able to automatically switch ranges, allowing you to keep your hands on the probe and other tasks, rather than having to manually switch ranges. In addition, some DMMs may include the following functions:

Diode Test

This test passes a constant current through a PN diode junction under test and measures the voltage drop across the junction. This works well for detecting open or shorted junctions, as well as for testing the front to back ratio of PN junctions.

Continuity Tone

This function places an ohmmeter into its lowest range and sounds a tone when the resistance between the test leads is less

than some low value, such as 20 or 50 ohms. This provides a convenient even-off continuity test for checking components or circuit paths.

Transistor Test

Some multi-function meters may include a transistor test function, designed primarily to test the gain of small signal transistors (higher power transistors have very low gain and fail primarily by shorting). You can insert a small-signal transistor into the built-in lead socket for a quick go/no-go test indication.

Capacitance Meters

Meters used to measure a capacitor's value are of two general types. One type determines a capacitor's value indirectly, by measuring its AC impedance at a particular frequency, then calculating the effective capacitance at that frequency. The other type measures the capacitor's RC charging curve to determine the value of the capacitor.

Impedance Bridge

An impedance bridge applies a single frequency AC voltage to a capacitor under test, usually at 60 Hz or 1 kHz, across one leg of a balanced bridge circuit. A variable, calibrated capacitor in the opposite leg of the bridge is then adjusted to balance the voltage across the bridge circuit, as indicated by a center zero reading on the balance-indicating meter.

The AC impedance of the unknown capacitor is then known to be equal to (or some multiple of) the calibrated value of the variable capacitor, at the test frequency. Once the impedance of the capacitor is known, the effective value of the capacitor can be computed according to the capacitive reactance formula:

$$X_c = 1/(2\pi FC), \quad \text{or} \quad C = 1/(2\pi FX_c)$$

A capacitor's effective value usually decreases at higher frequencies, both because of the physical properties of the materials used to form the capacitor (especially the dielectric material), and because of the inductive properties of parts of the capacitor (especially in a leaded capacitor). A capacitor's frequency dependence will be greater with some materials than with others. An aluminum electrolytic capacitor's value varies greatly with frequency. The capacitor's effective value is often specified by its manufacturer at a test frequency of 60 Hz, as many electrolytic capacitors are used to filter low frequencies. If you are using an adjustable frequency impedance bridge, it would be best to test the capacitor as closely as possible to the frequency at which the capacitor will operate. If the meter doesn't provide frequency adjustment, you should at least know what test frequency is used.

Capacitance RC Charging Curve

The second general type of capacitor value tester charges the capacitor under test with a constant DC current and measures the length of time it takes to charge the capacitor from a starting voltage to a higher voltage. The instrument then determines the value of the capacitor according to the basic formula:

$$C = dt / dV * I$$

that is, capacitance value equals change in time (charge time) divided by change in voltage across the capacitor, multiplied by the charge current.

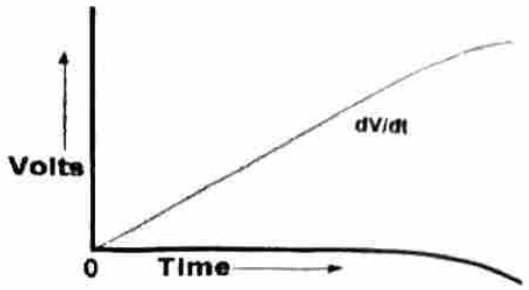


Fig. 10: When a capacitor is charged with a constant current, it will have a specific dV/dt charge slope, determined by the capacitance value.

Capacitor Analyzer

Since capacitors usually fail in ways other than just having a wrong capacitance value, a complete capacitor analyzer measures all the normal failure modes of capacitors typically used in electronic circuits. Four capacitor tests cover the failure modes most often encountered in servicing: equivalent series resistance (ESR), leakage, dielectric absorption (D/A), and capacitance value.

Capacitor Equivalent Series Resistance (ESR)

All capacitors have equivalent series resistance, which is the resistance in series with the plates of a capacitor due to resistive mechanical connections and lossy dielectric material. Increased ESR limits a capacitor's filtering or signal coupling capability, especially in high current or high frequency circuits.

High ESR often occurs as a failure mode in aluminum electrolytic capacitors, due to the drying of the electrolyte solution. It is important to be able to test the ESR of electrolytic capacitors, to isolate defective capacitors in circuits being serviced. Capacitor ESR testers are available as stand-alone

...or as part of complete capacitor analyzer instru-
ments.

Fig. 11. A capacitor and inductor analyzer measures all failure parameters for those components.

Capacitor Leakage

As a capacitor is charged and discharged, extra electrons are accumulated on one plate of the capacitor from the charging circuit and depleted from the other plate. However, there should be no appreciable flow of electrons from one plate to the other, through the capacitor dielectric (except in electrolytic capacitors). When a DC test voltage is applied to a capacitor, some current is drawn from the test supply to charge the capacitor. Once the capacitor is charged, the current should drop to zero in some types of capacitors (such as film), or almost zero in others (such as ceramic).

You should perform a capacitor leakage test at or near the capacitor's rated voltage, to be sure to exceed the voltage threshold that some leakage defects exhibit. For most capacitors the acceptable leakage current is very low, or zero, after the capacitor is charged.

Only electrolytic capacitors are designed to continually draw a small leakage current, in order to keep the electrolytic dielectric material formed as an insulator. The maximum acceptable leakage current for electrolytic capacitors depends on the electrolytic type (aluminum or tantalum), as well as the capacitor's rated voltage and value.

A capacitor leakage tester should apply the capacitor's rated voltage, wait for the capacitor to charge, then check whether the continuing leakage current exceeds acceptable limits for the type of capacitor being tested.

Capacitor Dielectric Absorption (D/A)

Dielectric absorption (often called battery effect or soakage) is caused by the high resistance isolation of portions of a capacitor's value. When a charged capacitor is discharged, the resistively isolated capacitance sections aren't completely discharged, due to their isolating resistance. When the discharge circuit is opened, the remaining charge on the isolated sections slowly recharges the main capacitor.

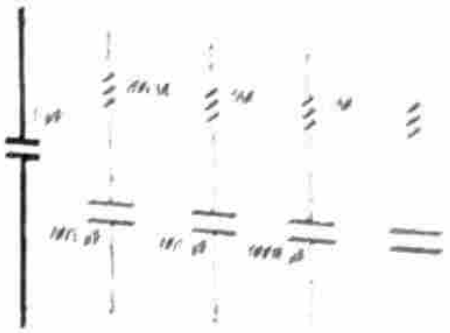


Fig. 12. A representation of D/A with many different resistor/capacitor combinations to show time constants ranging from seconds to days, in some cases.

This effect causes problems in precise detector filters and high frequency circuits. Increased dielectric absorption is also often an early indication of electrolytic aging. The standard test for dielectric absorption takes 12 hours to perform, but techniques have been developed to perform a capacitor D/A test in less than 10 seconds. This capacitor test is available on complete capacitor analyzer instruments.

Capacitor Value

A capacitor's value is important in a circuit to provide the proper resonating action at the circuit frequency for proper timing or filtering. The value may change due to a failure in the capacitor's plate structure or dielectric. A test of a capacitor's value can sometimes detect a capacitor defect, although tests for the other three failure modes are often earlier indicators of progressing capacitor failure, especially in electrolytic capacitors.

Inductor Analyzer

Inductors exhibit only a couple of failure modes: a winding opens, leakage develops between windings of a multiwinding inductor (transformer), the inductor core is damaged, or a short develops between the turns of a winding. An open winding or low resistance between isolated windings can be detected with an ohmmeter, but the most common defect in inductors, especially power inductors, is a short between adjacent turns in a winding.

Inductor Value

Inductance value may be used to identify a component or to verify that a proper component is in place in a circuit. A value test may indicate a value change due to a damaged core, but a value test alone is usually not sufficient to determine whether an inductor is defective.

Shorted Turns

A shorted turn in an inductor forms a very low resistance single turn load around the inductor core, absorbing almost all the energy from the inductor magnetic field. The shorted turn is a very low Q load and prevents the inductor from producing a normal expanding and contracting magnetic field around the inductor core.

Fig. 13: A specialized test is required to detect a shorted turn in an inductor.

This type of defect prevents normal inductor action, but the defect is very difficult to detect without specialized test equipment. An ohmmeter won't detect the small resistance change caused by a single shorted turn, and the inductance change usually isn't significant.

The specialized test equipment built to detect inductor shorted turns is called an inductor ringer tester. It applies a resonating capacitor in parallel with the inductor under test and pulses the inductor/capacitor tuned circuit. It then counts the number of oscillating pulses produced before the pulses damp below a particular test threshold. The test threshold is set such that ten or more ringing pulses occurring above the threshold indicates that the inductor does not have a shorted winding.

Oscilloscopes

An oscilloscope provides a visual representation of a changing voltage wave shape over time (a line graph). It also measures the amplitude of the voltage changes. A scope waveform shows us much more, though, than just the peak-to-peak amplitude of voltage changes in a circuit. The complete signal information that a scope displays allows you to make informed judgments about the performance of the circuit. This makes an oscilloscope a versatile and effective test instrument in the hands of a capable technician.

An oscilloscope's multiple controls and infinitely varied display waveforms can be confusing and intimidating, though, for a technician not fully trained and experienced in its use.

An oscilloscope has four main circuit functions; the CRT, horizontal sweep, vertical amplifier, and the trigger circuit.

A dual-trace scope has one focus control; alternating horizontal sweeps and dual inputs, position and trigger controls.

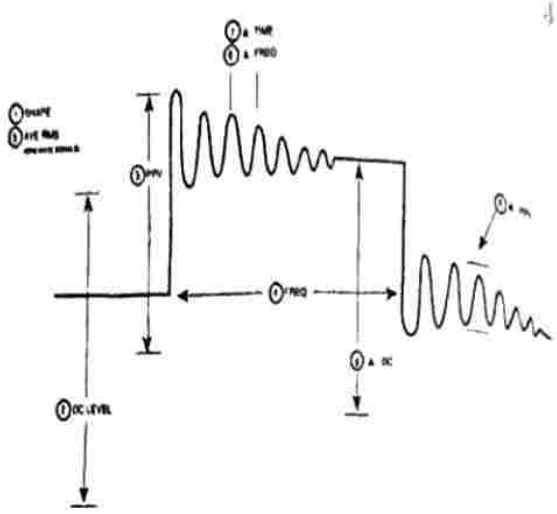


Fig. 14: An oscilloscope can be used to determine at least nine different parameters of an AC signal.

CRT Circuit

The CRT produces a stream (beam) of electrons that are directed, at any one instant, to a single point on the phosphor-coated display screen, producing a point of light. This stream of electrons travels along what is known as the CRT's Z-axis. Anything that changes the beam current, such as blanking, is said to be Z-axis modulation.

Controls associated with the CRT circuit are the intensity and focus controls.

Horizontal Sweep Circuit

The horizontal sweep circuit pushes the CRT beam from left to right across the screen (along the X-axis), forming a straight line of light. It then blanks the beam while the deflection is reset to the left edge of the screen to begin another visible trace. Controls associated with the horizontal sweep circuit are the time/division and horizontal position controls.

Vertical Deflection Circuit

The vertical amplifier circuit pushes the CRT beam up and down the screen (along the Y-axis), in step with changes in the input signal voltage. This changes the straight line of light into a "wiggly" line, which duplicates the shape of the input voltage changes. Controls associated with the vertical deflection circuit are the input coupling, volts/division, and vertical position controls.

As indicated by the simplified oscilloscope block diagram (Figure 15), any voltage at the signal input is applied, through the switched attenuator stage, to the vertical amplifier stages. The vertical output amplifier applies an amplified replica of

the input voltage to the vertical deflection plates in the oscilloscope CRT.

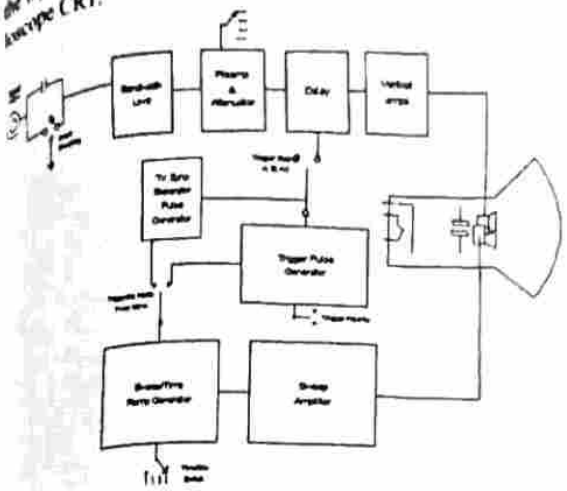


Fig. 15: The four main circuit functions of a scope are the CRT, horizontal sweep, vertical amplifier, and the trigger circuit.

The input impedance of an oscilloscope is actually quite low, at 1 Megohm and approximately 30 pf of capacitance to ground. If you connect the scope vertical input directly to a circuit point (using a direct probe), the low 1 MΩ input resistance will tend to load down any DC or AC signal in the circuit, regardless of its frequency (see voltmeter loading, above). In addition, the 30 pf of capacitance to ground will shunt higher frequencies to ground, increasing the loading effect, especially above about 5 kHz, where its capacitive reactance becomes less than 1 MΩ.

The solution to the scope input loading is to use an x10 extender probe; similar in design to the one we use with a DC voltmeter. Since we will use the scope to measure AC as well as DC voltages, we need a capacitive extender element as well as a resistive one.

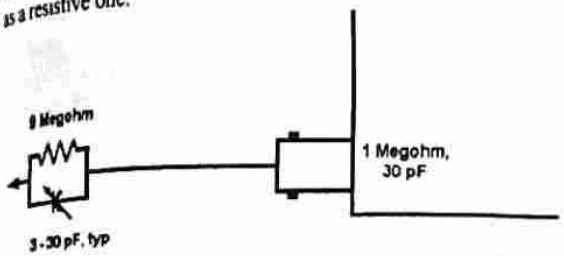


Fig. 16: Compared to a scope direct probe, a scope x10 extender probe (low cap probe) presents one-tenth the loading resistance and capacitance to a circuit point.

The extender resistor will be nine megohms; nine times the scope input resistance. The extender capacitor needs a capacitive reactance of nine times the capacitive reactance of the scope input capacitance (plus some parallel capacitance of the probe itself). That means the extender capacitor needs to be approximately one-ninth the 30 pF input capacitance. In practice, the capacitor is made variable, to be able to compensate for the slightly different capacitances of different scope models. The x10 extender probe allows only one-tenth the DC or AC probe tip voltage to reach the scope.

Trigger Circuit

Finally, the trigger circuit locks onto regularly recurring changes in the input signal and resets the horizontal deflection to start at the left edge of the screen at the beginning of each detected waveform trigger point. This causes the same portion of a recurring waveform to be displayed repeatedly, on top of one another. This allows us to examine a stationary portion of the waveform. Controls associated with the trigger circuit are the trigger mode, trigger source, trigger polarity, and trigger level controls.

Oscilloscope Operation

There are a couple of steps to follow when you connect an oscilloscope probe to a circuit test point. This will allow you to obtain a stable, useable waveform on the scope.

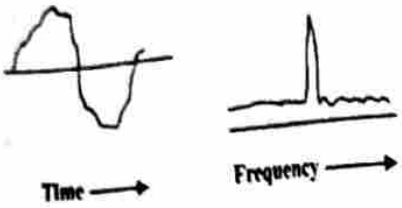
First, set the trigger circuit for auto trigger, set the trigger source to the channel you will connect to the circuit, set the input coupling switch to AC-coupled, and center the trace on the scope display with the vertical and horizontal position controls.

Then, preset the volts/division control to a voltage value between one-fifth and one-tenth of the expected circuit peak-to-peak voltage (divide the expected peak-to-peak voltage by ten, then set the volts/division control to the next larger voltage step above that). Now, when you connect the scope probe to the circuit, you should see some vertical trace deflection. Adjust the volts/division control so that the waveform is at least two divisions tall on the display (don't worry yet if the trace is still a scrambled waveform).

Next, adjust the trigger level control until the waveform stabilizes. If necessary, reverse the trigger polarity and readjust the trigger level. You should now have a stable waveform display, although it may be too compressed or stretched out in the horizontal direction.

Finally, adjust the time/division control to stretch out or compress the signal to view the desired signal detail.

Note: An oscilloscope displays a waveform in time domain view. A spectrum analyzer displays a signal in frequency domain view.



NOTES

Chapter Quiz

1. **An average-reading AC voltmeter _____.**

   a. displays the average AC voltage value
   b. converts to a peak reading before displaying the voltage
   c. multiplies the average value by 1.111 before displaying the voltage
   d. multiplies the average value by 0.636 before displaying the voltage

2. **Use of a x10 scope probe instead of a direct probe:**

   a. reduces circuit loading.
   b. increases the effective input impedance.
   c. decreases the signal to the scope.
   d. all of the above.

3. **Retrace blanking in an oscilloscope is _____.**

   a. not used
   b. applied to the Z-axis
   c. used to blank the vertical retrace
   d. applied to the X-axis

4. **To accurately measure a resistance across a transistor's emitter-base junction, use a _____.**

   a. higher ohmmeter range than you normally would
   b. lower ohmmeter range than you normally would
   c. high ohms function
   d. low ohms function

5. **An inductor shorted turn can be detected reliably using a _____.**

   a. leakage test
   b. ringer test
   c. AC impedance bridge
   d. all of the above

6. **An AC voltmeter will measure a non-sine wave signal accurately only if _____.**

   a. an extender resistor is used
   b. it is an average reading meter
   c. it is a true RMS meter
   d. the signal crest factor is very high

7. **A type of transformer which has a 1:1 turns ratio_____.**

   a. provides a ten-times voltage step-up ratio
   b. provides a floating secondary reference point
   c. is an autotransformer
   d. transforms the ground reference at its input to be the same at its output

8. **Shock hazards exist especially in a power supply which has _____.**

   a. a full-wave bridge rectifier connected directly to the AC power source
   b. an earth ground reference at its voltage output
   c. an isolation transformer at its input
   d. an aluminum electrolytic filter capacitor

9. **A DC voltmeter multiplier resistor _____.**

   a. limits the meter movement current to its full scale value
   b. multiplies the voltage across the meter movement
   c. provides an alternate path for current flow
   d. reduces the meter resistance for less loading effect

10. **A simple DC voltmeter with a 200 ua meter movement needs to have how much internal resistance on its 50 volt range?**

    a. 4 k ohms
    b. 10 k ohms
    c. 20 k ohms
    d. 250 k ohms

ANSWERS on page 223

Chapter 20 Soldering, Desoldering and PCB Repair Tools

by Dick Glass, CETsr

Dick Glass is an author of previous FTA publications as well as more than a dozen books dealing with electronics technology and business management, published by Howard W. Sams Co, TAB Books, and McGraw-Hill. He has also written hundreds of articles in trade journals and association publications. Dick is a former Navy Avionics Technician, Industrial Electronics Computer Technician, Technical Writer, FCC Commercial License examination modernization committee Chairman in 1995 and 2005, inductee into the Electronics Hall of Fame and currently CEO Emeritus of FTA. He has a broad background in all phases of work performed by electronics technicians.

Soldering

Technicians have always needed skills in replacing electronics components such as capacitors, resistors, coils, transformers and so forth. In most instances bad parts can't be removed and new ones inserted without using a soldering iron to heat and melt the connection for the old defective part. Also solder must be used to make perfect electrical connections for the new component.

Soldering is not difficult but there are some definite rules. Rules might be put in categories:

1. Care of soldering irons and equipment
2. Safety considerations
3. Soldering preparation
4. Soldering electronics components
5. Desoldering

Care of Soldering Irons

It is easy to keep your soldering iron in good shape. Most technicians use the small pencil-type irons which are rated at 25 watts to as much as 50 watts. Higher wattage irons than that might be used to do large jobs such as soldering a UHF fitting on a RG11 type transmission coax cable, or for securing jacks or brackets to the metal chassis.

The small pencil irons have a heating element with a copper or plated tip. The tip may be about 3/16" in diameter and an inch long -screwed into the larger portion of the heating element. The tip of either type is the problem. After using it for less than an hour, the shiny copper begins to tarnish. Then it builds up a crust of carbon-like material. The plated tip does the same.

Either tip works best if it is shiny and has no discoloration buildup of carbon. But the hot copper is "cooking" the air. After all, the irons run at 360°F+ (for 60% tin 40% lead solder).

With copper tips, cleaning can be done by simply filing the point (which may be chisel, conical or pyramid shaped) to get rid of the carbon deposits which prevent full transfer of heat to the solder joint.

Plated iron tips should be cleaned with an abrasive cloth, never a file, as the file will remove the plating.

Whether copper or plated, there are two things that can be done to reduce the buildup of carbon. One is to frequently rub the tip against a wet sponge. This cleans the thin layer of buildup off down to the copper or plating. Secondly, try to keep the tip covered with a layer of melted solder. Melting a coating of resin flux solder onto the soldering area of the tip reduces contaminant buildup. Because the solder coating is in its flowing or plastic state, a layer of the solder can be wiped off with the wet sponge thus keeping the tip in perfect condition to solder or unsolder a connection.

Lead-Free Solder

For decades, 60-40 solder, 60% tin and 40% lead, was the standard mixture in solder alloy used in electronic applications. Resin flux core is used–never acid core or pastes. Acid used for cleaning or flux will corrode or arc or cause damage to electrical components. Lead-free solder has universally replaced lead solder for worldwide ISO certification standards.

There are other combinations of lead and tin, 63/37 being one that is used in special cases, called 'Eutectic' solder. The object is to get the solder to melt at the lowest temperature, solidify immediately upon heat removal and not have any effect on surrounding circuitry or components. 63/37 melts between 355–365°F, whereas 60/40 melts between 361–376°F; Silver solder (2% silver, 59% Tin, 39% Lead) is used for surface mount components (SMT) for stronger bonding.

Danger of Lead and Fumes

About the turn of the millennium, concern was voiced for the potential health hazards leaded solder presents. The Restriction of Hazardous Substance Directive for banning lead type solder took effect July 1, 2006. Lead can enter the blood stream through a cut, or lead can be ingested which is poisonous and toxic to humans. The fumes from melting lead

solder and from resin flux cores of solder are toxic to the lungs and may cause eye irritation. The entire industrial world has banned lead-tin solder alloys and has replaced the lead with another metal. Tin-silver-copper and tin-copper are the choice for wave flow and rework service jobs. Be aware of the dangers from lead. Work in a ventilated area. Reduce contact with lead by using lead-free solder.

Solder Guns

Years ago, all service shops used soldering guns. Shaped like a real pistol, the "barrel" was a copper or plated arrow-shaped piece that was attached to two prongs by set screws (for easy replacement). Cleaning requirements are the same as for pencil irons. A big difference is the size. While few pencil irons are rated at over 40 watts, solder guns may be 100 to 250 watts and some even 500 watts.

In the early days of radio and TV, components were most often connected or mounted to terminal lugs or strips. You wound a resistor or capacitor lead around the lug stud tightly, then melted solder over the mechanical connection to provide a sure electrical contact.

Few parts today are mounted on terminal strips. Rather, printed circuit boards contain most of their components mounted through the PC fiber board - often soldered to both sides of the board and often with the leads poked through a hollow rivet. Surface mount technology (SMT) and tiny circuits with close placement of PC board traces and components requires delicate and precise soldering, thus solder guns are used primarily for connections to large metal ground lugs or for securing jacks and other parts mechanically to the chassis.

Solders guns have two problems which pencil irons do not. One is that the set-screw attachments which hold the gun tip into the two connecting cylinders will come loose. It is the set-screw physical connection which builds up an insulating carbon film where the tip is screwed onto the barrel shafts. It may seem the copper under the set-screw is eaten away by electrolysis. Whatever the cause, the screws need retightening frequently. If they aren't tightened to make a sure connection, the resistance at the set screws will result in the resistive heat being concentrated at the set-screws, not in the solder tip.

The solder gun "barrels" and tip are actually the stepped-down secondary of a simple AC transformer, built into the "handle" or bulk of the gun housing. Consider the 115V, 1 AMP power for a 100 Watt soldering gun: If the step-down from primary to secondary is 100 to 1, then the step-up in current in the secondary (thru the gun's tip) is now not 1 amp, but 100 amps! 100 amps through the copper tip loop will heat the copper tip to red hot. Guns use a trigger switch to control the heating of the tip. Electromagnetic damage can occur to nearby sensitive components.

Soldering of Components

It is natural for a person to try soldering, but to not quite "get it right." You will find trained technicians who still leave "cold solder joints." These later can require extremely difficult troubleshooting tasks to ferret them out. It is the technician's' job to recognize the reasons for "cold joints" and to assure that none are the result of his or her work.

A proper solder joint is one in which the component lead(s) and the PC feed-through or terminal, stud, etc have each been simultaneously heated to where the solder flows over and between them and embeds itself in the metal surfaces of each of the leads, traces or lugs at the joint. A good joint is shiny, not dull and gray.

It is natural to cover the joint with solder and assume the connection is OK. It may be that one or more of the leads, traces, lugs or a ground terminal has not been heated sufficiently to embed the solder. Thus you have rosin or air interface between the parts or leads. This creates a gap. In some circuits that gap could even result in an arcing and thus "static generator" within the circuit. A nightmare problem. T joint may "look" secure. Looks are deceiving.

Make sure you heat all the connecting leads and any test point or leg connection so that each and every one can melt the solder and so that solder flows onto each component an thus embeds itself in the surface of all leads. You must "fee that this has been done. Even when danger to components i a problem, you cannot let that excuse you from "sweating t joint," so to speak.

Tinning

Be sure to have "tinned" your soldering iron tip. This is simply melting solder onto the tip to help prevent oxidatic of the copper, and to give the tip a liquid (or plastic) inter with the metal your iron is touching (and attempting to he the melting point of the solder).

You will also have wires or component leads which m soldered better by tinning prior to physically being mour at the lug or joint or PC trace. To try to solder a stranded without first tinning the last 1/4" to 1/2" (intended to be dered to a connection) is to take the chance that one or n strands will not be soldered well. Some of the strands m look soldered, but really they may be "cold." The mech security of the wire will be less if all strands aren't secu

Tinning the iron or gun, or the wire or component leads takes a few seconds. It may save a few hours of troubleshooting time or downed equipment.

For HV connections, always leave a smooth and well-rounded covering of solder to reduce corona.

How much is too much?

You must make sure the solder connection "sweats" or flows onto all of the leads and wires at the connection. Sometimes you will find the heat buildup in a "stubborn" connection heats up the component leads to the point where you wonder now about the integrity of the component(s). How do you handle this?

As a professional technician, you will acquire a "sixth sense" that warns you something is not right - or that you are about to overheat the connection. Sometimes the plastic insulation on a wire starts melting or stripping back. Maybe a PC solder connection has gotten so hot that the connections nearby also appear to be affected by the heat.

Obviously, if a resistor starts "sweating" and discoloring; if a capacitor (lytic) starts "steaming" or the PC board starts crumbling and the copper traces are pulling off - you have applied too much soldering iron heat. One reason for this may be that the parts to be joined have lacquer on them or have oxidized to where solder resists bonding. PC traces always have a protective insulating coating. You can heat these impediments enough to blast through to the metal, but in the process you may destroy nearby components. Transistors and diodes, inside their protective case, have a very delicate and precise junction. Too much heat in the way of current, as well as too much heat physically, will ruin the junction.

Heat Shunts/Heat Sinks

Heat sinks are merely alligator clips with copper or aluminum welded onto the teeth so that you can use more heat at the joint, yet not have the heat travel up the transistor leg to overheat the junction. Once you have several heat sinks at your work station, you will get in the habit of using them regularly on delicate component leads.

In today's circuitry, with CMOS semiconductors that can be destroyed by even small static discharges, you should assure that your soldering iron and your body is at ground potential. Use a static safe bench and a wrist strap ground and frequently make sure everything in the area is grounded. This can help reduce soldering-caused troubleshooting problems.

In a pinch, you can tighten your long nose pliers on a lead and keep the pressure on by using rubber bands on the handles. Another very professional heat sink is built like a doctor's scissor-like clamp which tightens up on the lead - then clicks in a locked position until you decide to relieve the pressure.

Desoldering

Sometimes it seems smart to blast all of the solder off of a connection. Let's say you are having trouble with a large junction of wires and component leads, and you need to get rid of a gob of solder in order to unravel the wires and remove the single component you are replacing. It is tempting to heat up the joint and before it has time to cool - blast it with high pressure air or even freon! To do so is to violate a lot of rules including eye safety and co-worker safety. That is a No-No. Also it is tempting to heat a connection in a small board then "flick" the board to use centrifugal force to detach the solder on to the shop floor! Both the air blast and the "flick" will cause other troubles or the melted solder will not all detach, or it will lodge between nearby parts or circuits. Don't flick - Don't blast! At best - desoldering is an imperfect science. Even an experienced technician will find some jobs are nearly impossible to do neatly.

Wick

One tool universally used by techs is "solder wick." It started out as merely the copper braid from coaxial audio and video and RF cables. This was removed in convenient lengths - then when solder was to be removed from a joint, the braid was flattened - dipped in resin flux, then placed against the joint or component lead to be unsoldered, then heated with the soldering iron. Since the braid gets hot first, as the solder from the joint melts into its plastic state, it is "wicked" up the braid. The bulk of the solder in a joint or in a feed-through rivet on a PC board can be wicked away - often leaving such a thin layer of solder that leads can easily be wiggled free or unwound or untangled. Solder wick now can be purchased in rolls and is a must for working techs.

Suction

Bench techs have as a major tool of the trade - solder suckers. Several varieties are made. Their action is provided by a small hole in a Teflon tip - then a barrel which contains an air plunger. You "cock" it by pressuring the plunger to the tip end. A release button lets the spring inside retract the plunger - thus creating a vacuum which sucks air - and melted solder up the plunger tube and away from the heated joint. These work well. They need to have the solder particles (captured inside) removed frequently.

Chip Desolderers

Multi-pin IC chips require patience and delicate handling. Since most modern circuitry contains LSI (large scale integra-

(now) circuits, to replace one requires heating up 8 or more pins simultaneously to over one hundred in modern microprocessors. It's tough enough to unsolder one component from a PC board-what can be done with one which uses dozens of chip legs. Fortunately, several companies make special shaped desoldering templates which can fit flat against a 14 pin chip socket and melt all leg joints at once allowing the chip to be extracted. Once extracted, the chip's 14 leg holes can be cleaned out with wick or the even nicer air vacuum desoldering unit (PACE makes these).

Because it is tricky to remove all the solder from each leg socket in the PC board, it is not easy to pull a number of the legs loose with a IC remover tool. It is easy to rip one or more copper traces from the PC board. They seem to like staying attached to the chip leg and pulling up from the PC fiber surface.

Should you do a lot of work with (serviceable) PC boards, you may find yourself replacing feed-through solder rivets, replacing portions of copper circuit traces, jumpering broken traces and even gold-plating edge connectors that have become worn.

Real experts carry tiny drill bits (for cleaning out feed-through rivets), scribes, tweezers, kits of replacements PC board hardware, several sizes of wick, alcohol cleaner, glyptol shellac or poly coating and various types of chip desoldering heads.

Surface Mount Technology (SMT)

Some SMT PC boards are repairable. Fortunately, as PC boards have become so densely populated and PC traces so close, and magnifiers become necessary, fewer and fewer PC boards are repaired - especially in consumer electronics and computer throw-away products.

If you repair surface-mount electronics PC boards, chances are you repair a steady stream of them. You should have the replacement SMT components for the brands of products serviced. Where historically a solder connection is supposed to be mechanically connected, as well as electrically bonded, SM parts are small and are held onto the traces by solder alone. Magnifying bench lights, tweezers or vacuum component pick up tool and possibly even a hot air rework system may be your tools. SMT is a specialized segment of electronics repair. Silver solder paste is frequently used for new and rework. A product maker may set up a surface-mount station dedicated to repair of that company's specific PC boards. Field service techs normally would replace the entire unit, or a major PC board, sending the SM board back to the factory and then to the SM station for repair.

Hand Tools

Most technicians have schooling or experience with common hand tools. We won't spend any time on hammers, chisels, screwdrivers, hacksaws and so forth. Power tools, as well as hand tools, are a daily part of technician life, just as soldering equipment is. Safety includes not wearing loose clothes, chains, bracelets or long hair. Safety includes wearing goggles every time you use a grinder or power drill. It includes assuring there is adequate ventilation while soldering or using toxic cleaning materials. It includes keeping all tools in perfect condition.

NOTES

Chapter 20 Quiz

1. **The solder traditionally used is a metal alloy consisting of zinc and copper.**

   a. True
   b. False

2. **The most common solder used by electronics technicians consists of:**

   a. 50% aluminum, 50% tin.
   b. 50% zinc, 50% silver
   c. 63% copper, 37% tin.
   d. 60% tin, 40% lead.

3. **The proper process for soldering two objects together is:**

   a. heat both surfaces at the same time - apply solder.
   b. heat one surface at a time, assuring full coverage of the melted solder.
   c. heat the solder until it melts - flow it over the joint surfaces.
   d. clean any resin or flux off the joint - then apply heat.

4. **Overheating the component lead will likely damage _____ first.**

   a. transistors
   b. diodes
   c. capacitors
   d. transformers
   e. transistors and diodes

5. **Tinning of _____ makes the best connections.**

   a. soldering iron tip
   b. stranded wire
   c. component leads
   d. all of the above

6. **Movement of any component of the solder connection before the solder has hardened:**

   a. results in a poor electrical connection.
   b. results in a poor mechanical connection.
   c. results in a poor mechanical and electrical connection.
   d. is OK so long as the solder doesn't migrate away from the joint.

7. **Which device is capable of causing magnetic or inductive damage to electronics devices?**

   a. Pencil soldering iron
   b. High wattage resistance soldering iron
   c. Solder suckers & rubber vacuum bulbs
   d. Soldering gun

8. **To remove solder from a joint using copper braid:**

   a. heat the joint then quickly dip the cold braid into the melted solder and remove.
   b. heat the component leads at the joint first, then the braid.
   c. wrap the braid around the soldering iron tip and while hot, wipe off solder from the joint.
   d. touch the joint with the braid - heat it and let the solder wick up the braid.

9. **The best way to remove an IC chip which is soldered into a PC board is to:**

   a. use a desoldering adapter tip which heats all pins simultaneously.
   b. use wick and remove all solder from the pins - pry loose.
   c. heat each pin sequentially while prying up on the IC.
   d. use a solder-sucker to remove all pin solder.

10. **Surface mount components do not require mechanical bonding because:**

    a. the components are so small the solder connection holds them adequately.
    b. their sockets provide adequate bonding.
    c. their leads always "hook" onto the opposite side of the PC Board.
    d. special strength solder is used on SM boards.

ANSWERS on page 223

Review Questions - Troubleshooting, Test Equipment and Repair Chapters 18, 19 and 20

Unit 8

1. **A circle is obtained on the scope when:**

   a. two sine waves that are 90 degrees out-of-phase fed to the vertical and horizontal inputs.
   b. two sawtooth waves, in phase, are fed to the vertical and horizontal inputs.
   c. a sine wave and a sawtooth wave are compared on the scope screen.
   d. two in-phase sine waves are fed to the horizontal and the vertical inputs.

2. **Which frequency is most commonly used for audio troubleshooting?**

   a. 1000 KHz
   b. 1 KHz
   c. 1 Hz
   d. 1 KHz

3. **Why is a coupling capacitor used when troubleshooting with signal injection?**

   a. Improve frequency response
   b. Block DC level present
   c. Prevent loading of the circuit
   d. Provide a correct impedance match

4. **Which would be the likely cause of hum in an amplifier, but only present when the CD position is selected?**

   a. Bad filter capacitor in the power supply
   b. Open ground in the CD power amplifier
   c. Bad shielded cable from the CD player
   d. Defective speaker

5. **An stereo amplifier has severe hum whether CD, AUX, FM or AM RADIO are selected. What is the most likely cause?**

   a. Volume control shorted
   b. Defective output transistor grounding
   c. Power cord polarity wrong
   d. Power supply filtering

6. **Which of the below will cause distortion at high audio levels in an amplifier system?**

   a. Scratchy tone control
   b. Rubbing speaker voice coil
   c. Bias on a transistor
   d. Bass-boost control On

7. **The Y axis of an oscilloscope is:**

   a. brightness.
   b. intensity.
   c. focus.
   d. vertical.

8. **A power supply is found to have several 250 uF capacitors connected in parallel. To avoid shock from retained charge after disconnecting the power, the capacitors should be:**

   a. ignored.
   b. disconnected.
   c. discharged to chassis ground, with a suitable resistor.
   d. isolated with a non-conductive cover.

9. **A dual-trace scope has:**

   a. two focus controls.
   b. two sets of position controls.
   c. two vertical inputs.
   d. both b and c.

10. **A x10 probe has what effect on circuit testing?**

    a. Increases circuit loading.
    b. Increases the signal level by 10 times.
    c. Lowers the effective input impedance.
    d. Decreases circuit loading that a direct probe presents.

11. **A voltmeter that converts AC to DC by rectifying the voltage, would be least accurate measuring what types of alternating voltages?**

    a. Pulses
    b. Sine waves
    c. Peak voltages
    d. UHF signals

12. **When connecting an analog voltmeter into a DC circuit, what should be observed?**

    a. RMS rating of meter
    b. Loading of meter
    c. Polarity
    d. Peak to Peak voltage of circuit

ANSWERS on page 223

Chapter 21

Record Keeping, Productivity Calculation & Personal Behavior

by Robert (Bob) Ing, DSc, FAPSc, CESma

Robert (Bob) Ing has been with FTA-I since 1992, served as Chairman in 1998 and developed much of the material in the first edition of the CES Study Guide. He is a forensic scientist and security advisor specializing in technology crime. He has given workshops and lectured extensively on technology issues in the United States and Canada. Some specific areas of practice and expertise are Internet security, technology crime, Internet crime, computer security, electronic privacy, electronic surveillance and electronic evidence recovery. Bob has had diverse electronics industry experience throughout his professional life and has worked as a customer service representative, technician, engineer and industry representative.

Record Keeping

"Until input is linked to a goal there can be no intelligent accomplishment."

- Paul G. Thomas

Although the author of this quote was referring to input as thought and to the goal as a purpose, when it comes to record keeping this quote really hits home. For our discussion, the input is the written records we make, the goal is the reason we make them and the intelligent accomplishment is having a useful tool that will make our job easier.

Most of us, or according to statistics, 6 out of 10 people are considered paperwork minimalists. These are people that just try to do as little paperwork as possible, which may not necessarily meet even the most liberal record keeping standards. The reason people avoid or dislike paperwork is that they really don't see the value, purpose or goal of doing it. For most technicians, their employer will have a recordkeeping system in place with different forms to be completed with very little explanation as to why, but a lot of emphasis on simply "that's the way we do things."

It is important for you as a technician to understand the recordkeeping process and how it can affect your work, your client and company. Once you understand this relationship and the goal of keeping records you will be able to be more focused in your paperwork, reduce the time you spend on it and see it as less of a chore.

Why Keep A Record?

There are many reasons for keeping records just as there are different types of records. For our purposes, let's take a look at the more common reasons for keeping records in a technical service environment.

Customer Service

...tomer service purposes enables the ...clients

Preventive/Scheduled Maintenance

Keeping a record for preventive and scheduled maintenance purposes enables the technician to ensure that client equipment is kept in working order; view equipment service histories, and monitor and order consumable inventory items in readiness for scheduled service.

Service Contracts

Keeping a record for service contract purposes enables the technician to know in advance the resources required to service a client on a service contract. Resources such as spare parts, special test equipment or tools, and even if a second technician or technical team will be needed on a contract service call, can all be determined by looking at the service contract record.

Warranty Service Validation

Keeping a record for warranty service validation purposes enables the technician to determine if the unit for service is in or out of warranty and is crucial in obtaining a manufacturer's or vendor's warranty reimbursement on behalf of the client.

Wage/Salary Verification

Keeping a record for wage/salary verification purposes enables the technician to verify the billable hours worked, not only for the purpose of invoicing the client, but also to verify his or her own remuneration against hours worked.

Profitability

Keeping a record for profitability purposes enables the technician supervisor/manager to ensure that client invoices are properly processed and that parts and other consumable supplies are maintained at acceptable inventory levels.

Business Planning

Keeping a record for business planning purposes enables the technician manager to see how well the department or business ...service statistics and actual invoiced revenues.

...based on this information, decisions may be made regarding ...remodeling of some service areas and the downsizing of others, ...as well as the streamlining of services.

Tax Filing

Although last in our list of reasons for keeping a record, tax filing whether in the form of business, sales or employer tax is usually listed as the most obvious reason for record keeping. For the technician this means ensuring that sales tax is maintained on client invoices are accurate and are clearly calculated. It also means that sufficient service call information is provided to ensure proper and timely invoicing of the client. Thus, any taxes charged or taxes on business revenue can be properly remitted to the applicable agency.

As you can see, there are many reasons why a technician must ensure that accurate records are kept. In knowing the why or goal of record keeping, your task in creating a record every time you fill out a form or make a note is much easier once you know its purpose.

Creating Records

In the previous section we covered the most common reasons for keeping a record. It should come as no surprise that many of the reasons listed tend to overlap or complement one or more of the others listed.

All businesses depend on the use of forms to create permanent records. These forms have predefined fields that must be filled in. Once filled in, the form may be posted to a file or input into a computer database for processing and storage.

As a technician, you must ensure that the forms you fill out are as accurate and as legible as possible. Generally, when filling out a form you will want to print the information in the fields using a black ink medium ballpoint pen. A black ink medium ballpoint pen will provide a cleaner photocopy or computer scanned image than a standard blue ink ballpoint pen. A red ink medium ballpoint pen will also do the same, but it is general business practice not to fill forms out in red ink. Unless specifically indicated on the form, a form should never be filled out using a pencil or a non-indelible (water based) type pen.

Today much of the record keeping that you do will be completed electronically. While you may still carry a paper notebook, it is much more likely that your notebook will be in the form of a smartphone or tablet. You are probably relying on today to locate your customers' records from the company database, check out the service calls you have been assigned for the day and pull directions for reaching the customers' location. After your service call you will probably also use the software designated by your employer to record the activities of your service call, additional customer comments or requested projects, parts orders or inventory updates. Your time and productivity may also be charted electronically. You may be using an app to charge your customer for the service call and the calendar to reschedule any unfinished or additional work needed.

The use of these types of technology has helped to streamline work completed in the electronics service and other industries to increase productivity and add value for the technician, customer and company. Being able to access schematics or have immediate access to technical information in solving a complicated service problem is all available immediately to the technology-equipped tech today.

Now let's take a look at some of the specific forms or records that you may encounter as a technician.

Service / Repair Ticket

The service or repair ticket is used as a work order to document the details of who, what, when, where, why and how regarding the service call. The who is the client, the what is the item to be serviced, the when is the date/time of the request for service, the where is the location of the service call (home, office, in-shop), the why is the reason for the service call (signs and symptoms of malfunction, preventive maintenance or recall) and the how is a detailed account of how the technician performed the service (including parts used/replaced). When a service ticket is created and awaiting completion it is said to be "open." When the service or repair has been completed to both the satisfaction of the technician and client, it is said to be "closed."

Invoice

The invoice includes a summary of the information on the service / repair ticket with a detailed breakdown of labor and parts charges. Also the invoice will include important warranty information and payment terms for the client regarding the service performed. As a technician it is very important that you ensure that any information you provide relating to the invoice is accurate and concise.

Customer Service Record

The customer service record provides details regarding the relationship history between the client and company. It lists all the equipment purchased, the service and repair calls made and other relevant information. Selected information from the service/repair tickets and invoices issued to the client are listed in chronological order with brief notes cross referenced to the original documents.

Service Inventory Form

The service inventory form is used to indicate the parts and consumable supplies you have used on the service call so they can be replaced (in shop, service vehicle or department inventory) and re-ordered. This avoids lost productivity, down time, frustration and general delays in closing the service ticket caused by back ordered or out of stock parts.

Employee Time Ticket

The employee time ticket identifies the billable hours to client accounts made by the technician. It helps the technician keep track of hours billed out and most of all, helps in the processing of the technician's pay check.

Personal Field Notebook

Although not a form, every technician should keep a small pocket sized, lined notebook to help keep track of special requests, to do lists, reference material and as a memory aid. The best pocket notebook for the job is one that fits easily into the pocket of your service uniform, and has lined pages. The most efficient way to set up your notebook is to use the first 2-3 pages at the front of the book for important telephone numbers. The last 3-6 pages at the back of the book should be used for reference information such as formulas, tables, etc. This reference material may be written in, or photocopied from technical service manuals using a copier that can reduce the image to a size that can be pasted in the notebook.

Having set up the notebook, entries are made from the front of the book starting on page 4. To easily find the next blank page, an elastic band is placed around the part of the book that has the most pages just before or after the next blank page.

Today, technicians also have access to smartphones as well as tablets to organize their workday. They are small, portable and offer many different 'apps' to choose from.

Productivity Calculation

"Today is the beginning of a new day. You have been given this day to use as you will. You can waste it or use it for good. What you do today is important because you are exchanging a day of your life for it. When tomorrow comes, this day will be gone forever, in its place is something you have left behind . . . let it be something good."

All businesses and employees perform work to earn money. When most of the efforts of the business and employee are directly related to generating income, it can be said that the productivity level is high. However, in the real world obtaining maximum productivity is not always that easy.

The true productivity of a technician is based on the amount of time he or she spends on activities that are directly billable to the client. Time spent on non-billable or indirect client activities, also known as overhead activities, however necessary, reduces the productivity of the technician.

The 30-30-30 Rule

Generally, when a service business establishes a labor rate, they use what is known as the 30-30-30 rule. This is a general rule of thumb which states that 30% of the labor rate should be set aside for wages, 30% for overhead expenses, 30% for re-investment in the business and the remaining 10% to be used as a contingency fund.

For example, if a company charged $100 for a service call, this means that in theory up to $30 would be paid to the technician in gross wages, $30 to cover overhead expenses such as rent and administration, etc., $30 would be profit for re-investment in the business and $10 would be available to cover any unforeseen expenses that might occur in servicing the client.

The Productivity Formula

When a technician is billing a client for his or her time, it is quite clear that the technician is a highly productive profit center for his/her employer. However, as most technicians are scheduled for a specified number of hours to be worked per week, not all of their scheduled hours may be billable directly to a client. As a result of this, they must receive payment out of the overhead expense allotment in addition to any residual funds in the wage allotment.

The productivity, or utilization as it is sometimes referred, of a technician is the ratio between actual client billable hours to total scheduled hours worked in a given period of time (i.e. week, month, year). For an employer to know the productivity or utilization level of technical staff helps determine profitability, if more staff is needed or if efforts should be made to increase the client base.

For a technician, the ability to calculate personal productivity helps identify time management issues, personal profitability and can help in performance reviews when it comes to wage increase requests or clearly identifying instances when the technician is "maxed out" and needs help.

To calculate productivity, all that is required is to divide the billable hours to the client into the total hours of scheduled work for any given time period (week, month, year). This calculation will give productivity as a percentage.

The formula is: $B / T = P(100) = P\%$

where B is the Billable hours to a client, T is the Total scheduled hours worked, P is the Productivity indicated as a decimal number and P% is the Productivity or utilization multiplied as a percentage.

For example, if a technician has been scheduled to work 40 hours in a given week and has billed to various clients 24 hours in the same week, his or her productivity would be 0.6 or 60%.

This is expressed by our formula:

24 / 40 = 0.6(100) = 60%

The higher the productivity percentage the better. However, if you recall the 30-30-30 rule we discussed earlier in this section, if we use this rule of thumb model and relate it to our productivity formula model, it becomes apparent that a 60% productivity score may be considered a break-even point. Therefore, to sustain a business over the long term, productivity must be over this percentile.

Ways To Increase Productivity

There are several ways to increase productivity. In some cases it may involve increasing the clientele of the business to increase billable hours and fully utilize staff. In other instances better scheduling, personal time management, training staff to handle clients in a much more efficient manner and streamlining any activities not billable to the client can greatly improve productivity.

Personal Behavior

"There is no accomplishment so easy to acquire as politeness, and none more profitable."

George Bernard Shaw

As a technician, you have worked hard to become a professional and the last thing you need is to have your employer, peers and clients think otherwise.

Here are a few things to consider to maintain your professional standing.

Dress & Grooming

The first rule is to look and see how your peers dress and groom themselves. About 20 years ago, all technicians wore service uniforms. However, today technicians wear everything from traditional uniforms and business suits to casual wear. Where you work, your clientele and nature of your work will determine how you dress. One thing is for sure, you and your clothing must be clean. This means if you are working on something that is greasy or dusty, care must be taken to wear a smock, apron and or gloves to keep yourself clean when applicable.

Either with or without a uniform to wear some rules apply:

Tie – Unless the tie is a clip-on or bow-tie, it should be adjusted so that the bottom tip touches between a half to a quarter inch of the top of your belt buckle. For a technician, a tie bar should be used to secure the tie as opposed to a tie tack. Whether a clip-on or regular tie, a tie bar will allow you to quickly remove the tie or tuck the tie inside your shirt when working around machinery when it may pose a safety threat.

There are too many stories where a clip-on tie, that would normally pull off if caught in something, was tie-tacked to the technician's shirt with a fatal result. A tie bar will go with the tie if it is pulled hard enough, a definite safety benefit. If in doubt – remove your tie around this equipment.

Jewelry – Generally the only acceptable jewelry items are a wedding band, a wrist watch with a non-metal band and medic alert bracelet. Ladies may also wear small earrings, one to an ear. That's the general rule of thumb, safety conditions permitting.

Belts & Shoes – The color of your belt and shoes should match. If you wear black shoes, then you should wear a black belt. As a rule, black is the best color choice, but if your uniform requires you to wear brown shoes, then a brown belt is in order. Shoes should be polished and kept in good condition, this means new heels and soles as required.

Socks – Socks should only be black or dark blue.

On the subject of personal grooming, everyone should know that daily baths or showers and other personal hygiene activities are a must for anyone living in a civilized society. Here are some personal grooming tips:

Bath/Shower – If you suffer from dry skin, use a liquid soap instead of a bar soap for your bath / shower.

Deodorant vs. Anti-Perspirant – A deodorant will reduce body odor, an anti-perspirant will reduce underarm wetness and odor, however your body will increase perspiration elsewhere to compensate.

Fingernails – Technicians should keep their fingernails short and clean.

Hair – Long hair should be tied or put up so as not to get caught in machinery or contaminate an area or equipment you are working in and on. Remember, a static discharge can come from the end of a single human hair and that a loose human

hair that has found its way into a micro-electromechanical device has the potential to do damage over a period of time.

Personal Communication

How you communicate with others, both verbally and through body language, sets the type of relationship you will have with them. Even if you know that you may never see the person again, it is always best to communicate with them as if you will. How you communicate will have an impact on the level of respect, trust, intelligence and sincerity people will associate with you.

Here are some rules to make you a better communicator:

Introduce Yourself – The first words that you should say when you see a client on a service call are your name, your company and the reason for the call.

Be Personal – Always try to use the name of the person you are talking to at least twice in the conversation.

Face-To-Face Contact – When talking to someone, always look at the area of their face. Some books suggest eye contact but this may make some people feel uncomfortable, particularly if you are talking with someone of the opposite sex in a closed area.

Avoid Slang or Lazy Words and Phrases – While we could fill an entire book with these, here are some common ones and what you should really say.

- Yeah – Yes
- You Guys – You
- O.K. or Okey Dokey – All Right
- Fix – Repair or Service
- I don't know – I will have to look into that
- Cheap – Inexpensive
- What can I do for you – How can I help
- Crapped Out – Stopped Working

Hiding Your Hands – Research has shown that people who talk to others with their hands in their pockets or behind their backs give the impression that they are not being totally honest or are withholding information. As a technician this is not the message you want to send to clients, so keep your hands out of your pockets and where your client can see them during a conversation.

Chewing Gum & Mints – Chewing gum or having a mint in your mouth while in the presence of a client is a definite way to look unprofessional. Don't do it.

Smoking – It is never acceptable to smoke in front of a client or on client premises. As a technician you are there to perform a service and a client may take exception to the la charge if you were not completely focused on the service

NOTES

Quiz

1. **Record keeping is needed for which activities?**
   a. customer service
   b. warranty service validation
   c. business planning
   d. All of the above

2. **A technician should _____.**
   a. understand the record keeping process
   b. let administrative staff do all record keeping
   c. make multiple copies of all records they file
   d. allow the client to fill in their own forms

3. **Keeping a record for profitability purposes ensures that _____.**
   a. the appropriate taxes may be remitted
   b. client invoices are properly processed
   c. preventive maintenance schedules are correct
   d. none of the above

4. **Keeping a record for customer service purposes ensures that _____.**
   a. parts inventory levels are maintained
   b. the technician can verify billable hours
   c. follow-up service can be provided
   d. none of the above

5. **A service or repair ticket provides _____.**
   a. information about the who, what, when, where, why and how regarding the service call
   b. information to the client regarding warranty and payment terms
   c. information on out of inventory parts
   d. information on productivity

6. **Forms should be filled in using _____.**
   a. pencil
   b. erasable ink
   c. a black ink ballpoint pen
   d. a blue ink ballpoint pen

7. **A personal field notebook _____.**
   a. can help the technician keep track of special requests and reference material
   b. can be used to issue an on the spot invoice or receipt
   c. can be shared by other technicians on a single shift
   d. is a book of forms provided by the company

8. **The formula for calculating productivity is _____.**
   a. B(T) = P(100)
   b. B / T = P(100) = P%
   c. P / T (B) = P%
   d. P (T) = P(100)

9. **The 30-30-30 rule refers to _____.**
   a. billable hours, scheduled hours and productivity
   b. wages, overhead expenses, productivity
   c. wages, scheduled hours, overhead expenses
   d. wages, overhead expenses, re-investment

10. **The safest accessory item for a tie is _____.**
    a. a tie tack
    b. a safety pin
    c. a tie bar
    d. none of the above

ANSWERS on page 223

Chapter 22

Technical Writing

by David E. Caldwell, CETsr

David E. Caldwell is a senior certified electronics technician in avionics and communications. He retired from the USAF after working on advanced fighter aircraft and state-of-the-art weapons systems for twenty years. He then taught avionics technicians at Pittsburgh Institute of Aeronautics before developing and teaching his current two-year electronics technician course for the PA Department of Corrections. His course enables graduates to become certified as associate level electronics technicians. He has written several articles for TAB, chaired the Tech review of element 3 of the General Radio Operators License (GROL), and is actively involved with his church and Habitat for Humanity.

TECHNICAL WRITING

For his chosen field it was fit
To match his learning and his wit

DEFINITION

Writing is a means of making known the thoughts and ideas of the writer to a reader who is interested in that information. Technical writing involves the communication of specific knowledge and ideas, precisely and in such a manner that a specific audience will find it understandable. Technical writing is essential in all fields of engineering, applied science and technology.

The technical writer must have proficient knowledge of the technical field in which he works. He or she must be capable of skillfully and precisely illustrating and describing something unfamiliar to the reader. Unless the writer can adequately communicate his or her ideas to the reader, the information will be useless.

Technical material will be understood much more readily if it is presented in an accepted and familiar form. I will attempt to describe some standard techniques and arrangements for you in this chapter. However, the proper use of grammar, punctuation and spelling will still play a key role in your technical writing success.

LEVEL

Information can be technical and yet be written on various levels. If you follow all of the guidelines for proper writing, you might still encounter problems. As a writer, what is perfectly clear to you may seem confusing to the reader. You cannot use out-of-the-way terms and take for granted that the reader understands everything you write. Since there is no official technical dictionary for both you and the reader to use, it is important that your writing be properly directed to the particular level of the reader.

You must take into account the amount of training and experience the reader has. Readers will not read material that is written too high above their vocabulary level. Very few people will do additional research just to increase their comprehension of a written article. Your writing effort should be to express ideas rather than impress the reader with your vocabulary skills. You must also avoid trying to oversimplify topics by using words beneath the reader's vocabulary level. The ideal level should provide the reader with mostly new facts or at least stimulate the reader into thinking about the topic.

You might consider dividing your readers into one of three general levels. The first level can be described as elementary. This level consists of amateurs or beginners. Anyone who has received only minimal training might qualify for this category. A newspaper or magazine article should be written at this level. The second level can be designated as the intermediate level. This level is comprised of technicians and/or students. Any person who has specialized training and experience would qualify for this level. A memo about changes in equipment or maintenance procedures should be written at this level. The third level would be characterized as the advanced level. This level consists of scientists, senior engineers and technicians. Anyone holding a degree and having extensive experience in the subject field could be considered for this category. A report on research concerning a new theory may be authored at this level.

Since these various reader levels do exist, the writer must adjust his style to the specific purpose at hand. When writing at lower levels, remember to keep the language simple, use more illustrations, present less complicated theory, give more definitions, keep instructions short and offer more step-by-step explanations.

STYLE

When writing about technical topics, keep your composition simple, logical and objective. Technical writing should be accurate rather than convincing, specific rather than theoretical, reasonable rather than impulsive. Technical writing should be crystal-clear, to the point and precise rather than poetic and imaginary. A good technical writer will be sincere, direct and utilize clear composition. A clever technical composition will be critical and complete, leaving no unanswered questions.

...competent writer will use discipline and tact, omitting ... and individual opinion.

...writers use poetic and colorful explanations to ... nearly any item or circumstance they encounter. ... writers must refrain from this temptation. But this ... mean that good technical writing has to be drab or ... Technical writing is not meant to be so inflexible that ... writer's style cannot be expressed.

...things that may help you ... focused on your technical ... are called the ABCs of ... writing. 'A' stands for ac... your credibility as a writer ... your ability to present material as factually as pos... on your ability to present material as factually as pos... brevity. In today's hectic world, no one ... represents brevity. In today's hectic world, no one ... the time to wade through an unnecessarily lengthy article. ... composition can be kept brief by first letting the reader ... your reasons for writing the article, then give the facts, ... finally explain your expectations. 'C' symbolizes clarity. ... accuracy and brevity will contribute towards clar... short paragraphs, sentences and words will further ... clarity.

TYPES

When you compose **instructions** or **directions**, be sure to ... a design with which your readers are already familiar. ... every step in sequence from beginning to end of the task ... only one step to be presented per sentence. Number ... and make them obligatory by insuring the first word ... the reader an order to do something. Use exact quanti... measurements to promote understanding. Utilize key ... such as 'WARNING' to inform your reader of potential ... and other points of importance. Finally, you will need ... plenty of illustrations to insure your instructions are not ... different interpretations.

Manuals are usually written according to specific guide... and you must heed these specifications if you intend to ... a manual. You must also understand the purpose for writ... the manual and become familiar with all of the equipment, ... and tools you will be writing about. Gather all of ... information such as drawings, illustrations and manufac... data sheets prior to writing. While taking into account ... specifications, plan a framework as to how to present the ... information.

Whether **descriptions** are intended for a manual, report, or ... used on their own, be sure to describe accurately. One of ... important questions to ask yourself is whether your ... understands what you are trying to describe. You must ... utilizing occupational terminology and assuming that ... reader knows everything you do.

Describe what the item looks like, its parts, how it works and what it does. If a process or operation is involved, describe what happens, and how and why it happens. *Remember*, describing is different from writing out directions or instructions.

In a formal **proposal**, submit objective data that leads to a conclusion. Your plan that something be done must specify who, when, where, what, why and how, or it is of no value. It should include a description of current methods of doing things, available facilities and equipment, a discussion of how much time and expense would be involved, anticipated problems and the expected results.

Laboratory reports are generally written on a company's standardized fill-in-the-blank report forms. Occasionally you might have to present a final report that will tie together the details obtained from a series of blanket-form reports. Your best bet is to follow the same form, insuring you include all of the following information: report title, number and date, purpose, results, conclusions and appendices.

Articles composed for trade, company or popular magazines will vary in technical content. You must write a trade paper at a professional level because experienced and well-informed professionals will read it. It must be thoughtfully organized, direct and quite specific. If you are writing an article for a mass media magazine, you can be much more general and focus on making it interesting to the layperson who knows little about your subject. The layperson will want the general picture rather than the minute details.

The **technical** or **formal** report is a precise description of important data. It is based on complete and accurate information. It may deal with research discoveries, experimental data or an answer to a problem. The report's purpose should always be to inform in clear, concise language. A report may be short and informal, such as a memo or letter. It can also be a long and formal communication in a structured format. It may be complete in itself or be one of a series of progress reports. It is always aimed at a particular audience who wants and needs the information it contains. Let's take a look at the generally accepted standard format for the formal report:

FORMAT OF THE FORMAL REPORT

A. Cover and Title Page
B. Letter of Transmittal
C. Acknowledgements
D. Distribution List*
E. Purpose of the Report
F. Summary of Conclusions and Recommendations
G. Table of Contents
H. List of Illustrations

I Introductions
J Text or Body of Report
K Conclusions
L Recommendations
M Appendices
N Glossary*
O Bibliography or Reference
P Index

** Optional, depending on specifications of report.*

WRITING THE REPORT

A. THINK AND PLAN

Choose the purpose of your report and then ask yourself some of the following questions: "Who am I writing the report for?" "What do I want it to achieve?" "What do I want to get across?" "How do I present *just enough* information, but not too much?" "How long should the report be?" "What is the level of the reader?" "How and where will I get the data needed?" "Do I have a deadline to meet?" "Have I completed a time schedule to estimate the dates when I will finish each step or section of my work?" "Have I allowed enough time for the approvals needed from my supervisor?"

B. GATHER YOUR INFORMATION

You must decide what you will need for the report. Assemble your notes, records, published papers, journals, field work, test results, surveys, interviews, photographs, diagrams, etc. You may also find yourself researching textbooks, encyclopedias or even the Internet for information. If you do any researching, be sure you take notes so you may reference quotations or mention your sources of information. The use of 3" x 5" cards for note-taking will allow you to easily arrange your data. Write down the following for each source you consult: title, author, publisher and facts of publication. You will need this information for your bibliography or list of references at the end of the report. Read your data over and reflect upon it for a while before completing the next step.

C. MAKE AN OUTLINE

Choose the order in which you will present your data. When your information is arranged in the best manner, it will be easier for you to write and for the reader to read. Proper arrangement will help you to put just enough emphasis on each topic. Your outline should show what items are to be included and each item's relative importance in the overall report. You may use any of the types of outlines that exist. You may feel comfortable with the topical phrase outline that simply names the topics to be dis a word or simple phrase. Some people find this type to be too brief and prefer the sentence or paragraph o which the topics are written as complete sentences or paragraphs.

D. WRITE THE ROUGH DRAFT

This is a rough draft, so plunge right in and write d of your thoughts on paper without worrying about spe errors or grammar. The focus at this point is to get yo on paper. Later you will be able to make cross-outs ar notes to yourself in the margin, since probably no one this draft but you. Mark where your illustrations will Follow your outline and try to rough draft the entire te sequence. Remember, you are going to write the text not the full report in its final order. Your text will have finished before you can write your purpose of the repo summary of conclusions and recommendations. Your should contain the following parts:

Introduction - a brief, separate section of the report in which you tell the reader exactly what subject is covered in the report. It is important that the introduction be as interesting as possible because the reader will decide at this point whether to continue reading your report.

Body - describes all of the main points in your report interesting fashion, begin to tell your reader about the su Use headings and subheadings in the body of the report sect long passages and to keep the reader informed as to is being discussed next. Headings of major divisions sho in capital letters and may begin a new page if you wish subheadings, the first letter of all significant words is usu capitalized. Both may be placed on the left margin, inde or in the middle of the page, as you choose – just be con with the pattern you use.

Conclusion - explain the significance of what you have just described in the text and, for the first time, present yo own opinions. State your conclusions in order of importan Conclusions must be logical, reasoned and soundly suppo by evidence presented in the text. Summarize in your con ing paragraph(s), end all discussion and give the reader a of completeness. Emphasize and focus on your most impo information so that the reader is left with the desired persp tive.

You may use illustrations to help to make your message more clear and effective. They should have a useful and specific function. Choose the type of illustration (photo, gr line drawing, etc.) that conveys the information most clear

...graphically. Keep them simple, uncluttered and neither ... should look more simplified than the text. Link illustrations ... the text by numbering them consecutively, captioning them ... placing them near the text they support. Keep in mind ... the report will be reproduced, and remember they must be ...

...An abstract is a very compressed paragraph that tells the ... something about the report. If you are following the ... format for a long formal report, you may have a descriptive ... purpose of the report) and an informative abstract ... of conclusions and recommendations). Your intro... has discussed what is covered in the report. If your re... short and more informal, you might want to provide all ... information in just one section and call it something like ... Abstract," "Summary." It should be brief, useful and arranged ... a separate page, before the body, so the reader can decide if ... she wants to read your entire report.

REVISE AND/OR REWRITE IT

...Set your first draft aside for a couple of days. When you ... pick it up again you will be able to be more objective as ... you begin to review it. Use a dictionary to check your spell... (e.g. a thesaurus to find better words, and perhaps a grammar ... workbook if you are unsure of your usage. Modern day word ... processors often include these features to save you a lot of ... time and work. It may be a good idea to read over your first ... draft several times, each time concentrating on a particular ... purpose. Use the following checklist as a guide during your ... review. When you are satisfied with your report, ask a skilled ... co-worker to read it through and advise you of any improve... ments needed.

CHECKLIST FOR REVIEW

___**PURPOSE** - Does your report accomplish its purpose? Is it complete and comprehensive?

___**SUBSTANCE** - Did you really say something and use only relevant, needed information?

___**UNITY** - Is every part related to your central purpose?

___**COHERENCE** - Are your ideas and words logically connected?

___**CLARITY** - Do your sentences say only what you want them to say, in the best possible way?

___**CORRECTNESS** - Are all your facts and figures completely accurate?

___**PROPORTION** - Have you placed the proper emphasis on each part of your work?

___**DIRECTNESS AND SIMPLICITY** - Have you kept your sentences brief and simple?

___**USE OF THE RIGHT WORDS** - Do you use specific, concrete words rather than vague ones?

___**STYLE** - Is your personal literary style suitable for the report?

PREPARE THE FINAL COPY

Your final draft, after it is corrected and polished, must be typed in standard manuscript style. Use one side of white 8 1/2" x 11" paper with ample margins on sides, top and bottom, text double-spaced, and triple-spacing between paragraphs. Indent the first line of each paragraph and begin a new chapter on a new page. If you place footnotes at the bottom of the page, use a short horizontal line on the left to separate them from the text.

After completing the review process, you will need to add the fine touches to your report. For formal reports, you will need to include the following sections in your work. If your report is informal, you may omit some of them.

Footnotes are used with direct quotations or with specific facts taken from an author and expressed in your own words. The reader may assume that anything not footnoted is your original work. The use of footnotes also offers you legal protection. An accepted method for footnoting is listing the author's full name, first name first; the book or magazine title underlined or in italics; the title of the article, in quotes; the publisher and date, and the page number. It is best to place footnotes at the bottom of the page on which the quotation or statement appears. Use a reference number in the text after the passage and the same number in front of the footnote. Remember to number your footnotes consecutively.

A *bibliography* is an alphabetical list of the books and writings you considered in writing your report. It can also consist of books and articles you want to list as other sources of information or reference books. The books and writings are indexed much the same as in footnotes, except that the author's last name appears first and the number of pages in the book is listed. A list of references is used when you do not footnote. In the text, at the end of the sentence you want to acknowledge, place the number in parentheses. The list of references will follow the report, listing each reference by consecutive number in the order cited, not alphabetically.

An *appendix* contains supplemental data that is appropriate to, but not directly pertaining to the main discussion. It can also contain graphs, tables, mathematical calculations, etc. An appendix can also contain data that is too lengthy, detailed or distracting to include in the text.

Pages should be numbered with Arabic numerals in the upper-right corner of right-hand pages or at the bottom center of the page. Take into consideration any full-page illustrations to be inserted. Numbering should begin with page one of the text and continue through appendices, bibliography, etc., or number these end pages as A-1, A-2, etc. For those pages before the text (table of contents, abstracts, etc.) use small Roman numerals at the bottom center.

A *table of contents* can be made once the pages are numbered. It lists in sequence all sections down to the level of major subheads, exactly as in the text, and page number of each. If you have formulated a good, organized outline, you may have this job almost finished.

The *transmittal*, or *cover letter*, can be placed within the report or sent separately to the designated reader of the report. This letter identifies the report and formally presents it to the designated reader. It should contain the title and number of the report, a brief description of the report and its findings, and the signature of the person responsible for the report.

An *index* should be included if the report is detailed and long so that readers can find information about a specific topic quickly. The best way to compile one is to write out a 3" x 5" card for each topic, making sure to cross-index everything. When you have cards covering every possible topic, arrange them alphabetically.

A *glossary* is an alphabetically arranged short dictionary of special terms in the report. Commonly used or known terms should not be included in this list.

Illustrations should be listed by number, title and page number and placed on a separate page unless the list is a short one that can be included at the end of the table of contents.

A *distribution list* lets the reader know who else has read the report, in case the reader wants or needs to discuss the report with other readers. Include each recipient's name, title, company and the copy number of the report.

An *acknowledgment page* contains a sentence or paragraph on this page to briefly offer credit where credit is due to persons or companies for information used in the creation of the report. Any permission granted for reprinting material is also acknowledged here.

The *cover* and *title page* give information about the report's subject and purpose. The title should not be too brief or too long. The cover should be made of protective and attractive paper.

It should contain the title, author's name and affiliation, the date the report was issued, the number, the contract number and, if appropriate, the security classification notice and control number.

The title page, just inside the cover, should include all of the above, plus perhaps the authorization for the preparation of the report or a routing schedule. This page has no page number.

ENDING COMMENTS

Technical writers put much research and effort into their work, and sometimes that work needs to be protected from persons who have no right to read or use such information. If your material is considered restricted information, it should be marked as such on the top and bottom of the covers and each page. If you are handling U.S. Government classified information you will have to mark the classification conspicuously on the top and bottom of both covers and every single page in capital letters. Control numbers are maintained for classified documents and appear on the title page.

The information presented in this short article is only meant to be a guideline for your technical writing. You can find additional information in your local library or bookstore. If you decide to start writing on a full time basis, you may even benefit from a college course on technical communications or technical writing.

I wish you the best of luck.

NOTES

Quiz

1. Which of these would be written in the most detailed, concise manner?

   a. Manual
   b. Article
   c. Proposal
   d. Technical report

2. An 'abstract' _____.

   a. is a short descriptive paragraph preceding a report
   b. describes the information sources
   c. includes appendices
   d. lists copyrighted sources

3. Information on an advanced level would be directed toward which readers?

   a. Senior engineers
   b. Entry level technical students
   c. Public readers
   d. Graduate technical students

4. One of these would include page numbering the same as the text. Which one?

   a. Table of contents
   b. Abstract
   c. Index
   d. Illustration list

5. Notice of government security classification will be shown _____.

   a. on the title page
   b. on every page
   c. in the acknowledgements
   d. just on the front cover

6. Limited recipients would be identified _____.

   a. on the first text page
   b. in the Glossary
   c. in the Distribution list
   d. on the title page

7. The results of a DVD performance test and evaluation would probably be described in a _____.

   a. report
   b. manual
   c. description
   d. transmittal

8. After printing your first draft of a publication, which of these should occur next?

   a. Distribute to intended recipients.
   b. Submit to management for approval.
   c. Ask a co-worker to proofread it.
   d. You should proofread it.

9. One of these should be a separate list. Which one?

   a. Subjects
   b. Illustrations
   c. Chronography
   d. Transmittals

10. Which of these would be the first effort toward writing a report?

    a. Outline report
    b. Compile list of illustrations
    c. Consider and note objectives
    d. Make table of contents

ANSWERS on page 223

ELECTRONICS TECHNICIANS ASSOCIATION ETA INTERNATIONAL

Appendix A

Practice Exa

Associate CET Practice Exam #1

Included here are two practice tests of 100 questions each. These are typical of what a technician would find if they took the FTA Associate CET examination cove the general field of electronics, as well as the material contained in this book.

1. **8.33 meters is the full wavelength in air for what RF frequency?**

   a. 72 MHz
   b. 83.3 MHz
   c. 36 MHz
   d. 16.66 MHz

2. **Decimal 164 would appear as _____ in binary?**

   a. 10010101
   b. 10100100
   c. 10001111
   d. 01110100

3. **A resistor has color stripes yellow, violet, violet, silver. It is:**

   a. 46 MegOhm +/- 10%.
   b. 470 MegOhm +/- 10%.
   c. 37 MegOhm +/- 10%.
   d. 48 kOhm +/- 5%.

4. **What is the total capacitance of three capacitors valued at 100pf, 100pf and 50pf when connected in a series?**

   a. 25 µf
   b. 250 µf
   c. 25 pf
   d. 250 pf

5. **What is the ripple frequency of a full-wave single phase bridge rectifier operated at 60Hz?**

   a. 180 Hz
   b. 90 Hz
   c. 360 Hz
   d. 120 Hz

6. **When heated or cooled, a _____ emits a voltage.**

   a. thermocouple
   b. thermometer
   c. thermostat
   d. thermocline

7. **Which wire size is physically largest?**

   a. #8
   b. #12
   c. #16
   d. #24

8. **Adding the binary numbers 101011 and 001110 will giv you what result?**

   a. 111001
   b. 111111
   c. 001010
   d. 111011

9. **A component that reduces in resistance as it is heated:**

   a. is a negative temperature coefficient thermistor.
   b. is a positive temperature coefficient thermistor.
   c. is a varactor.
   d. is a thermocouple.

10. **An autotransformer:**

    a. has the secondary unattached to the primary.
    b. has a single winding.
    c. is used only in automobiles.
    d. uses a buck-boost coil.

11. **A dual trace oscilloscope differs from a single trace in that only the dual trace scope:**

    a. has two cathode ray guns.
    b. has a Z input.
    c. alternates sweeps.
    d. has intensified traces.

12. **If an RF signal voltage is reduced by 6 dB, it is then:**

    a. 25% less.
    b. half the original size.
    c. 75% less.
    d. not changed in voltage level.

16. Applying positive voltage to the P end and negative to the N end of a germanium diode:

a. will forward bias it.
b. will reverse bias it.
c. probably will damage it.
d. should have no effect on it.

17. You would probably prefer a digital VOM over an ordinary analog meter:

a. for most accuracy.
b. to insure operator safety.
c. for convenience.
d. when the signals are intermittent.

18. You should expect the video amplifier in a radar or TV receiver to have a frequency response of:

a. 20 Hz -20 kHz.
b. 20 kHz -180 kHz.
c. 0-6 MHz.
d. 88 MHz -108 MHz.

19. The radio communication mode that utilizes the basic frequency and one of the side bands:

a. is called "SSB."
b. is FM.
c. must be line-of-sight.
d. utilizes CB technology.

20. When using fiber optics to transfer a signal, the light must:

a. be visible.
b. be LED generated.
c. be between 3 GHz and 30 GHz.
d. enter the "cone of acceptance."

21. The energy stored in a coil is measured in:

a. Henrys.
b. Joules.
c. Farads.
d. Volts.

22. The telephone communication term "Cellular" refers to:

a. undefined telephone radio channels.
b. a provider trade name.
c. Caller Environment Lone Line service.
d. small geographic areas covered by each base station.

23. Integrated Circuit failures often exhibit:

a. open or short conditions.
b. solder leakage.
c. audible noise.
d. radiation hazards.

24. Which of these would you normally choose to cause an LED to light?

a. 3 mV AC
b. 3 mV DC
c. 30 mV DC
d. 3 V DC

25. A capacitor (3/8" x 1/8" ceramic or mica) marked 47k is:

a. 47pf+/-10%.
b. 4.7 µf +/- 5%.
c. 47 µf +/- 20%.
d. 47 mf+/-10%.

26. A "Digital Storage Oscilloscope":

a. uses micro processing and memory to allow close analysis of signals.
b. is specially designed for analyzing digital signals.
c. converts analog input signals to a digital display.
d. alternatives between digital and analog display traces.

27. Carbon microphones are much like variable:

a. Capacitors.
b. Resistors.
c. Thermistors.
d. Inductors.

28. An antenna that is longer than another:

a. will have a higher resonant frequency.
b. will have a lower resonant frequency.
c. has a higher Q.
d. delivers higher power.

29. An 8-bit binary code allows _____ logical combinations.

a. 16
b. 64
c. 256
d. 512

27. **An inductor in an electronic circuit:**

a. resists AC current flow
b. appears capacitive to AC current
c. reduces DC power dissipation
d. is a near short to AC current.

28. **A printer directly connected to a personal computer would use a _____ connection.**

a. VGA
b. Ethernet 100 Base-T
c. USB or parallel Centronics
d. DB15

29. **The digital method of representing characters by using combinations of zeroes and ones is called:**

a. decimal
b. alpha-beta.
c. duplex.
d. binary.

30. **FM utilizes _____ where AM utilizes signal level variations.**

a. frequency variations
b. power variations
c. sideband modulations
d. period modulation

31. **What decimal number is represented by the binary number 01101010 ?**

a. 6A
b. 53
c. 106
d. 152

32. **Decimal 50 would appear as ____ in Hex.**

a. 32
b. 62
c. 101000
d. 110010

33. **Of these choices, the display of a DMM would probably be:**

a. LCD.
b. Pointer.
c. CRT.
d. Floating ball.

34. **A disadvantage of LCD (liquid crystal display) is:**

a. high current demands.
b. environmental temperature sensitivity
c. slow reactance to input
d. vibration sensitivity limitations.

35. **A series DC motor, with no load:**

a. tends to stop or not start
b. may be damaged by excessive speed
c. reverses direction
d. becomes an AC generator.

36. **Which motor maintains a constant speed with changing loads?**

a. Series
b. Shunt
c. Servo
d. Buck-boost

37. **A small power drill or electric mixer would probably be powered by a/an _____ motor.**

a. three-Phase
b. induction
c. series
d. parallel

38. **The rotor in an induction motor is not connected to AC or DC power.**

a. True
b. False
c. It depends on speed limitations.
d. Induction motors have no rotor.

39. **0 dBm, when referring to audio signals means:**

a. decibels referenced to 1 milliwatt.
b. decibels referenced to 1 millivolt pk-pk.
c. decibels referenced to 1 volt pk-pk.
d. zero signal output.

40. **If a class A differential amplifier is using only one of its two possible inputs, what will the output signal shape be?**

a. Zero
b. Exactly the same as the one input
c. Equal in amplitude and frequency to the input
d. Smaller than if two in-phase input signals were applied

41. **Voltage is:**

a. potential difference between two points
b. amount of current flow.
c. quantity of electronics moving in a conductor
d. opposition to resistance

42. **"_____" is a term describing many op-amp circuits.**

a. Inverting
b. Bi-directional
c. Low-Pass
d. Open Quadrature

43. **The gain of an op-amp is inversely related to band width as input frequency increases.**

a. Always
b. Never
c. Either can be changed without affecting the other
d. It can affect bandwidth but higher gain affects bandwidth so little that it is effectively insignificant

44. **The audio detector circuit in an AM radio most commonly is a:**

a. discriminator circuit.
b. bridge circuit.
c. transformer.
d. diode.

45. **A differential amplifier with two identical in-phase signals on the inputs will have an output _____.**

a. of zero
b. twice the input, but same shape
c. that cannot be determined with this information
d. one half the input signal

46. **A free-running oscillator is also called a monostable multivibrator.**

a. True
b. False

47. **A fuse which has its fusible wire wrapped around an insulator is:**

a. a Slo-Blo type fuse.
b. an Inductive fuse.
c. a capacitive fuse.
d. intended for high voltage use.

48. **The power transformer in a power supply:**

a. always steps voltages up
b. always steps voltages down
c. might step voltages up or down
d. always steps current up

49. **Which of the below would be considered "basic" components of a non switching power supply?**

a. Zener, regulator, transformer, SCR
b. Regulator, diodes, AC power, load
c. Transformer, bridge, capacitor, load
d. Filter, rectifier, transformer

50. **The average of a full sine wave rectified voltage is:**

a. 637 x Peak.
b. .707 x Peak.
c. .707 x Peak to Peak.
d. 1.414 x Average.

51. **The ripple frequency for a half wave rectifier operating from 60 Hz electrical power is:**

a. 30 Hz.
b. 60 Hz.
c. 120 Hz.
d. 240 Hz.

52. **The effective voltage of a 100V Peak-to-Peak sine wave is:**

a. 35.35 VAC.
b. 70.70 VAC.
c. 141.4 VAC.
d. 31.8 VAC.

53. **A pi filter is an inductor with:**

a. a capacitor at each end to common.
b. a capacitor at each end in series.
c. a resistor at each end to common.
d. a resistor at each end in series.

54. **An inductor opposes any change in:**

a. voltage.
b. current.
c. frequency.
d. reactance.

55. A capacitor opposes any change in:

a. voltage
b. current
c. frequency
d. reactance

56. A capacitor will charge to what percent of full charge in one RC time?

a. 70
b. 33.3
c. 63.2
d. 67.6

57. Moving one plate of a capacitor further away from the other will:

a. decrease capacitance.
b. increase capacitance.
c. decrease voltage rating.
d. increase mutual transconductance.

58. The fact than an inductor opposes any change in current is true of:

a. AC signals only.
b. DC signals only.
c. AC and DC signals.
d. VHF signals more than UHF.

59. If you increase the value of a capacitor, the capacitive reactance will:

a. increase.
b. decrease.
c. depend on connected components.
d. not be affected by the value.

60. In checking a capacitor to see if it is shorted, open or leaky, you would use the ___ ohm meter range.

a. lowest
b. highest
c. You can't check a capacitor this way.
d. You will ruin the ohmmeter unless the capacitor is charged.

61. The capacitor in a power supply capacitive filter:

a. is in series with the load.
b. is in parallel with the load.
c. requires a serial choke.
d. must be in parallel with a choke coil.

62. The time required to discharge a capacitor dep the _____ across its terminals.

a. the total capacitance
b. the total resistance
c. the total inductance
d. the combination of LRC

63. With a given filter type, pure filtering is easies

a. low frequencies.
b. high frequencies.
c. mid frequencies.
d. any frequency

64. Using a resistor instead of an inductor in a pow supply pi filter:

a. is less expensive.
b. is less efficient.
c. does not filter the AC as well.
d. all of the above.

65. A power supply whose output voltage does not whether the load is light or heavy, would have percent of regulation?

a. 100
b. 50
c. 20
d. 0

66. A power supply which measures 50 volts when no load and 40 volts under a full load has what regulation using the % of regulation formula: (% = (EnL - EfL) / EfL x 100)?

a. 100
b. 50
c. 25
d. 0

67. Correctly connecting a properly rated zener di across a power supply load will:

a. protect the power transformer.
b. increase the output DC.
c. limit the output DC.
d. eliminate ripple.

68. **A Triac in a power control circuit:**

a. provides regulation.
b. limits surges.
c. acts as a fuse.
d. acts as an on/off switch.

69. **Which wire gauge has the largest diameter?**

a. 20
b. 24
c. 28
d. 32

70. **What specification should NOT be of concern regarding wire conductors?**

a. Resistance
b. Capacitance
c. Length
d. Cross sectional area

71. **The normal open circuit voltage for each nickelcadmium cell is approximately _____.**

a. 1.2
b. 2.1
c. 3.4
d. 6.3

72. **Lead-free type solder for electronics work is made from:**

a. silver and bismuth.
b. lead and tin.
c. silver and tin.
d. acid core.

73. **Soldering stations used for electronics circuit-board work have historically been (approximately):**

a. 5 watts.
b. 30 watts.
c. 100 watts.
d. 250 watts.

74. **Solder flux mainly:**

a. removes foreign material on the soldering iron.
b. removes oxides on metals to be soldered.
c. controls the temperature of the solder.
d. causes conductor-to-solder-to-conductor chemical adhesion.

75. **Laminations used for the core of a transformer:**

a. must physically make metal to metal contact with the adjacent laminations
b. must be insulated from each other
c. increase eddy currents
d. reduce counter emf in the windings.

76. **I=1 amp, R=10,000 ohms. What is the value of the voltage?**

a. 1 volt
b. 10 volts
c. 100 volts
d. 10,000 volts

77. **If R=200 ohms and I=.02 amps. What is the value of P (power)?**

a. 08 watts
b. 8 watts
c. 4 watts
d. 8 watts

78. **Which frequency will have a higher loss when using coaxial cable transmission line?**

a. 550 kHz
b. 27 MHz
c. 470 MHz
d. 1 GHz

79. **A _____ is a reference antenna that all other antennas with gain are compared to.**

a. Yagi
b. omni
c. mast
d. dipole

80. **A high frequency antenna _____ than a low frequency one.**

a. is thicker
b. is shorter
c. has more gain
d. is longer

81. **Standing waves on a transmission cable:**

a. indicate maximum power out.
b. indicate less than optimum signal transmission, but should cause no concern.
c. can be caused by poor termination.
d. can be located with a voltmeter.

Appendix A

82. Sideband frequencies are produced when a radio carrier frequency is:

a. modulated.
b. transmitted at minimum power level.
c. fed to a splitter.
d. radiated into space.

83. Doubling the power in a transmitter output signal is a ____ dB gain.

a. 1
b. 3
c. 6
d. 10

84. 0 dBmV is what level of received TV signal?

a. Zero
b. .0001 Volt
c. 1000 microvolts
d. 1 Volt

85. A byte is:

a. 4 bits.
b. 8 bits.
c. 16 bits.
d. 32 bits.

86. In a 3-input AND gate:

a. 1 input must be high in order to have a high output.
b. 2 inputs must be high for a high output.
c. 3 input must be high for a high output.
d. all three inputs must be high to have a low output.

87. An LCD operates by polarizing the light through filters. The fluid rotates the light _____ degrees.

a. 30
b. 45
c. 90
d. 180

88. To test an SCR with an ohmmeter:

a. this cannot be done.
b. check resistance the same as transistors.
c. with + to anode and - to cathode, short gate and anode together. Should go from near "open" to near "short."
d. with - to anode and + to cathode, short gate and cathode together. Should go from near "open" to near "short."

89. Using the R x 100 range and testing between the source and drain of an N channel JFET:

a. readings should be equal in both directions.
b. with positive on anode, reading should be 1/100th that of the opposite polarity.
c. with negative on anode, reading should be 1/100th that of the opposite polarity.
d. this check provides no useful information.

90. With + voltage to drain or source and a - voltage to the gate on an N channel JFET, internal resistance of the JFET will be:

a. low.
b. high.
c. high to drain, low to source.
d. low to source, high to drain.

91. Which is LEAST likely to cause ESD damage to a MOSFET?

a. Flux
b. Solder sucker
c. Human finger
d. Wick

92. A Logic Probe:

a. injects signals into logic circuits.
b. indicates high and low voltage conditions.
c. measures dB levels.
d. displays voltage levels on an analog scale.

93. In a typical electrical generator, a small AC voltage applied to the field windings is able to control the output level of the generator.

a. True
b. False

94. When working with and testing lead-acid batteries:

a. wear eye protection goggles if battery caps are off.
b. short between terminals only if in a well ventilated area.
c. when charging, keep room closed off.
d. always wear eye protection, never cause a spark and charge in a ventilated room.

94. Audio signals, such as the output from DVD players, satellite receivers and preamplifiers, usually are of what amplitude for input to a TV, modulator or other equipment?

a. 100 microvolts peak-to-peak
b. 1000 microvolts peak-to-peak
c. 1 millivolt peak-to-peak
d. 1 volt peak-to-peak

95. Which of these signal parameters does a spectrum analyzer show?

a. Current and phase
b. Leakage capacitance
c. Video color signal separation
d. Magnitude and frequency

96. Permanent, non-volatile, non-removable data storage in a computer is:

a. RAM.
b. ROM.
c. Flash.
d. Optical disk.

97. What should be used first to clean a corroded plated soldering iron tip?

a. Coarse emery cloth while cold
b. Wet sponge with acid core solder after heating
c. File
d. Flux, cloth after heating

98. When soldering high voltage connections:

a. it is important that a thermal shunt be in place.
b. use a minimum of solder.
c. leave the terminal with a smooth and rounded covering of solder.
d. leave the connection with rough solder tips.

99. Class AB operation of an audio amplifier means that current flows through the active device (tube, transistor, FET, IC) for _____ degrees of the input signal wave.

a. 360
b. more than 180 but less than 360
c. 45 but less than 180
d. 90

ANSWERS on page 223

Associate CET Practice Exam #2

Included here are two practice tests of 100 questions each. These are typical of what a technician would find if they took the ETA Associate CET examination covering the general field of electronics, as well as the material contained in this book.

1. A track on a computer removable disk is:

a. one record.
b. divided into sectors.
c. created as data is written.
d. one continuous path.

2. A PC has Conventional RAM and ROM memory area (in a PC it is 0 to 640 kilobytes); Upper memory area (640 Kb to 1 meg) and Extended memory area (above 1 megabyte). What other names for internal memory types do PCs have?

a. Cache
b. Tape drive
c. Resistor array
d. Expended

3. Which video system has the highest resolution?

a. EGA
b. VGA
c. SVGA
d. XVGA

4. You would want to use a dual trace oscilloscope to measure:

a. signal differences between two points.
b. suspected resistance problems in one circuit.
c. one faulty circuit in voltage level while another is faulty in a current level.
d. one fault in video and the other in audio.

5. Common Intermediate Frequencies (IF's) for AM and FM commercial broadcast receivers are:

a. 455 kHz and 1.2 GHz.
b. 288 kHz and 455 kHz.
c. 455 kHz and 10.7 MHz.
d. 288 kHz and 10.7 MHz.

6. [illegible] radio receiver signal compression:

a. uses D/A converters.
b. reduces the high frequency received signal amplitude.

7. Coaxial cable line losses at RF frequencies are due to:

a. wire resistance, capacitive and inductive reactance
b. wire resistance, dielectric fading and transconductance
c. capacitive and inductive reactance and braid leakage
d. magnetic effects of nearby materials

8. You are troubleshooting audio reproduction equipment and measure the signal to be input into the audio amplifier. What approximate RMS level would you expect that signal to be?

a. Audio signals vary too much to be measured
b. 1 microvolt
c. 1 millivolt
d. 1,000 millivolts

9. The noise level in a radio receiver is generated mostly in the:

a. IF stages.
b. Local Oscillator.
c. Audio section.
d. RF section.

10. The fast charging rate for a NiCad or Ni-MH battery should not exceed_____% of its AHr, or ampere-hour rating.

a. 5
b. 100
c. 25
d. 10

11. Which one of the following in a PC would be part of disk operations?

a. Serial port board
b. DMA ports
c. CGA controller
d. SATA or IDE controller

12. An LED:

a. only emits visible light when forward biased.
b. can be used as a fiber optic signal source.
c. does not need high current flow protection.
d. may send or receive light signals.

13. **A dual-trace oscilloscope may have:**

a. two electron guns.
b. one frequency control.
c. one set of position controls.
d. a Z input for intensity modulation.

14. **The gain of an op-amp connected as an inverting amplifier is determined by the external _____.**

a. supply voltage
b. + input DC voltage
c. series input resistor
d. feedback and input resistors

15. **Diodes are made from one type of material.**

a. True
b. False

16. **Which propagation condition would be significant for a UHF signal to be received at a distance of 500 miles?**

a. Line-of-sight
b. Faraday rotation
c. Groundwave
d. Tropospheric ducting

17. **A three input NAND has two low inputs and one high input. The output is:**

a. low.
b. high.
c. either high or low.
d. neither high nor low.

18. **Which of these would be considered a component of a "basic" power supply?**

a. Feedback circuit
b. IC regulator with short circuit protection
c. Diode rectifier
d. Circuit breaker box

19. **A "Video signal" is defined as:**

a. modulated RF.
b. baseband including sound information.
c. baseband including visual information.
d. broadband RF.

20. **You are servicing a power supply and have disconnected power. It has several 250 uF capacitors connected in parallel. To avoid shock, these capacitors:**

a. should be disconnected.
b. may be ignored
c. should be discharged to chassis ground, through a suitable resistor
d. should be isolated with a non-conductive cover and avoided.

21. **A schematic used in electronic servicing:**

a. shows the size and shape of all the components in the unit.
b. is sometimes called a block diagram.
c. shows the location and arrangement of components in the unit.
d. is much like a road map for following signals.

22. **High voltage should be discharged before servicing a product. After disabling AC input power, discharge an HV PS by:**

a. shorting output to the plastic cabinet.
b. disconnecting the HV lead with rubber gloves.
c. draining through a large value resistor to chassis ground.
d. shorting output of the collector heat sinks.

23. **Which of these would be an important electrical consideration in the use of a transformer?**

a. The operating frequency
b. Space requirements
c. Environmental temperatures
d. Number of laminations

24. **If a 12v product is dissipating 50w, how much load resistance does the device represent?**

a. .24 ohms
b. 24 ohms
c. 4.16 ohms
d. 2.9 ohms

25. **An "Audio signal" is often defined as:**

a. frequencies in the 20 Hz to 15,000 Hz range.
b. baseband including visual information.
c. frequencies above ultrasonic.
d. broadband RF.

Appendix A

26. _____ will cause a generator to produce DC voltage rather than AC.

a. Slip rings
b. Load
c. Armature loops
d. The commutator

27. You suspect the unit has an internal short. To prevent further damage when you apply power, you should:

a. short around the main fuse or breaker and proceed, watching for heat or arcing
b. replace the unit without attempting repairs, as it probably can't be repaired safely or within reasonable time or cost limits
c. measure the resistance at the dc connectors and compare to factory specifications.
d. insert a low resistance high wattage (100 ohm, 10 w) resistor in series with the power input.

28. A ripple counter:

a. is a meter that indicates the power supply ripple frequency.
b. is a network of several XNAND gates that regulates the computer oscillator.
c. is a synchronous counter that stores the counts of other counters in a computer.
d. is two or more logical flip-flops connected sequentially with the input pulses toggling the first one.

29. When a voltage is applied to a capacitor:

a. current is maximum at first, then reduces as the capacitor becomes charged.
b. current is constant until full charge is reached.
c. the voltage difference between the plates reduces as full charge is reached.
d. there is no difference between the plates after charging begins.

30. An electrolytic capacitor in series between the power transformer and the rectifiers in a power supply:

a. protects the diodes from power surges.
b. eliminates the need for filter capacitors.
c. is part of a voltage doubler circuit.
d. provides that earth ground be isolated from DC ground.

31. Information travels through optic fiber via the flow of:

a. protons.
b. lumens.
c. photons.
d. electrons.

32. You would most likely prefer an analog VOM over a DMM for:

a. intermittent fluctuations.
b. accuracy.
c. convenience.
d. safety.

33. Which of these circuits would most likely use a Zener diode?

a. A receiver I F section
b. A television tuner
c. A power regulator
d. A bridge rectifier

34. A block in a digital diagram is called a NOR. Which is true?

a. It may also be called a NAND.
b. It is a NOT OR or Inverted OR.
c. It will pass any single negative signal, but not more than one.
d. It must be reset after any positive signals are applied.

35. Which of these statements is most true relating to schematic diagrams of electronic devices?

a. They provide pictorial representations of components.
b. They supply manufacturing specifications of components.
c. They represent signal paths.
d. They provide user operational procedures.

36. In Personal Computer technology, ROM indicates:

a. memory that may not be addressed randomly.
b. memory that may only be read from.
c. real address mode memory.
d. resident operating system memory.

37. A light emitting diode will light when:

a. the gate is turned on.
b. at avalanche.
c. it is forward biased.
d. it is reverse biased.

38. **A Zener diode is rated 6.3 v. This tells you:**

a. it conducts to limit the voltage across it to 6.3 v
b. requires 6.3 v gate voltage in order to turn on
c. generates 6.3 v if activated by light
d. it does not allow PS voltage to go below 6.3 volts.

39. **A Triac is frequently used:**

a. for high voltage generation.
b. as a power supply rectifier.
c. for power surge protection.
d. to electronically turn equipment on and off.

40. **A technician must be knowledgeable about "ESD":**

a. to be able to select the proper soldering device.
b. because voltage sensitive devices come in numerous sizes.
c. in order that he will have the proper test equipment for them.
d. because static discharge can destroy many components.

41. **In a motor, the counter-emf voltage induced in the armature_____ the source current.**

a. adds to
b. causes a decrease in
c. does not affect
d. is unaffected by

42. **One significant difference between a TV antenna and a CB antenna would be:**

a. TV is a much lower frequency, thus longer.
b. TV is a narrow band, thus the antenna length is not critical.
c. CB is FM while TV is AM.
d. TV is polarized horizontal while CB is vertical.

43. **"Switching Power supplies":**

a. are sometimes called DC to DC converters.
b. are normally controlled from remote locations.
c. do not convert incoming 117v AC power to a higher frequency.
d. must operate off AC sources only.

44. **Which of these would you likely find in a TV motor speed control circuit?**

a. A klystron
b. An op-amp
c. An MOV
d. LCD's

45. **A transformer on a schematic has two leads connected to the primary and three to the secondary, one on each end and one in the middle. This is a/an:**

a. audio transformer
b. I F transformer
c. center tapped secondary transformer
d. high voltage transformer

46. **The 9 pin DB9 connector serial port on a PC is commonly called the _____ standard.**

a. Centronics
b. RS232
c. DIN
d. USB

47. **Three capacitors, of 470 pf, 560 pf and .47 pf are in parallel with each other. The total is:**

a. .47 pf.
b. 2.13 pf.
c. 1030.47 pf.
d. 1.03047 µf.

48. **Ohm's law provides equations relating to:**

a. volts, amps, and resistance.
b. amps, joules, and resistance.
c. decibels, resistance, and volts.
d. power, emf, and current.

49. **A power source such as a generator or battery might be compared to what component in a water supply system?**

a. A valve
b. A pump
c. A filter
d. A pipeline

50. Which of these is a good indication of the quality of an analog meter?

a. The lowest current scale
b. The highest voltage scale
c. The ohms-per-volt rating
d. The number of resistance scales

51. The vertical axis of an oscilloscope display is also called the:

a. X axis.
b. time base.
c. Y axis.
d. sawtooth output.

52. If you increase the value of an inductor, the inductive reactance will:

a. increase.
b. decrease.
c. depend on connected components.
d. not be affected by the value.

53. Which would least likely be a high-voltage (above 1,000V) capacitor?

a. Mica
b. Ceramic
c. Paper
d. Electrolytic

54. The rechargeable cells currently used in many consumer portable products would be:

a. Lead-acid.
b. Carbon-zinc.
c. Lithium.
d. Lithium-Ion.

55. A solid-state regulator:

a. is made up of SCRs.
b. varies conduction to maintain a level output voltage.
c. is simply a Zener diode.
d. uses Diacs to sense the output voltage level and vary the input voltage.

56. A current-carrying conductor is placed in a magnetic field perpendicular to the lines of flux. It tends to move in _____ direction to the force of magnetic lines of flux, as determined by Flemings left-hand rule.

a. a perpendicular
b. a parallel
c. an opposite
d. the same

57. The antenna used to compare the gain of other antennas is called a _____.

a. dipole
b. rhombic
c. conical disk cone
d. co-linear stacked

58. The component that effectively becomes a short when turned on, and is turned off by disconnecting the path from the anode to cathode is the:

a. silicon controlled rectifier.
b. Schottky diode.
c. unijunction transistor.
d. field effect transistor.

59. Which of these terms would you most expect to see when referring to types of resistors?

a. Electrolytic
b. FET
c. Metal film
d. Henry

60. "Microwave," as used in radio electronics, refers to

a. a kitchen appliance.
b. 1 GHz - 100 GHz radio frequencies.
c. a style of television.
d. an oscilloscope display interpretation.

61. A differential amplifier whose two input signals ar equal, but 180 degrees out of phase, will have a /a _____ output signal.

a. maximum
b. zero
c. undeterminable
d. one half

62. **Which of these can be used to control the output level of a generator?**

a. A rheostat in the armature winding
b. A small DC voltage applied to the field winding
c. A negative feedback circuit connected to the commutator
d. A centrifugal switch

63. **Impedance (Z, in ohms) includes:**

a. Inductance and Capacitance.
b. Henries, Farads and Ohms.
c. Resistance, Capacitive Reactance and Inductive Reactance.
d. Joules, Farads and Henries.

64. **The velocity factor in an RF transmission line is determined by:**

a. line length.
b. center conductor resistivity.
c. the dielectric material between the center conductor and shield.
d. how the line is terminated.

65. **A fuse, marked 250v, 650ma can be used in circuits with an "open-fuse" supply voltage of:**

a. 125 volts.
b. 650 volts.
c. 5000 volts.
d. 250 kvolts.

66. **A thermistor:**

a. increases resistance with higher voltage.
b. decreases voltage output with heat.
c. changes resistance with temperature.
d. is unaffected by heat or cold.

67. **When a transformer is designed, what most determines the output voltage from the secondary?**

a. The ratio of primary to secondary turns
b. The type of core material
c. The power rating
d. The secondary load

68. **Adding armature windings in parallel to a generator's existing windings:**

a. increases its voltage output
b. increases its current output
c. improves its voltage stability
d. increases the output frequency

69. **A 10W signal is received at 5 W. This is a ____ dB loss.**

a. 3
b. 6
c. 1.5
d. 4.5

70. **It is desired to raise the power to a speaker from 15 W to 30 W. This is a:**

a. 2 dB increase.
b. 3 dB increase.
c. 6 dB increase.
d. 15 dB increase.

71. **Which of these IC or semiconductor devices norm requires the most careful handling to avoid dama static electricity?**

a. TTL
b. ECL
c. MOV
d. CMOS

72. **Which of these components most closely con the output of a solar cell?**

a. A NiCad battery
b. A light sensitive diode and capacitor
c. An alternator
d. A FET

73. **Three series inductors have values of 2 and 40 mH respectively. Total inducta**

a. 90 mH.
b. 13.33 mH.
c. .075 mH.
d. .011 mH.

Appendix A

74. **Phototransistors:**

a. require visible light
b. requires LED's in order to operate.
c. emit light when activated.
d. can be sensitive to invisible light.

75. **To simplify the following of an electrical signal path through circuitry, you could:**

a. draw a detailed schematic
b. remove the complicated components.
c. restrict troubleshooting to taking voltage measurements.
d. use a block diagram.

76. **AC power outlet strips with surge suppressors will include at least one:**

a. MOV.
b. MOS.
c. Circuit breaker.
d. Zener diode.

77. **Connecting 3 capacitors of : .05 µF, 20 µF and 40 µF in series results in a total capacitance of :**

a. 60.05 µF.
b. .0498 µF.
c. 19.95 µF.
d. 65 µF.

78. **You would use the formula (Lt = L1 + L2 + L3):**

a. in a capacitive series circuit.
b. to find total inductance in a series circuit.
c. for series-parallel inductive impedance.
d. in a resistive parallel circuit.

79. **The fifth color band on a 5% resistor is yellow. This tells you:**

a. it has an extremely high degree of accuracy.
b. the manufacturer.
c. tolerance.
d. the recommended frequency range.

80. ***VHF or UHF antennas don't use waveguides as feed lines because:***

a. they create grounding problems.
b. their resistance to high frequency is too high.
c. the characteristic impedance is too high.
d. the large dimensions at these wavelengths makes other transmissions lines more practical.

81. **To trace a logic path, you might use:**

a. a Logic probe.
b. a DMM.
c. a Logic Pulser.
d. a Pulse Tracer.

82. **Ordinary alpha/numeric information transmitted by a computer is converted to binary using:**

a. a Decoder.
b. a Crystal.
c. a Multiplexor.
d. an Encoder.

83. **In electronic devices, what term refers to residual magnetism?**

a. Hysteresis
b. Flux
c. Induction
d. Kinetic energy

84. **Some variable capacitors are made so that the plates interleave. The capacitance varies as the plates open (separate) and close. As they open:**

a. the capacitance increases.
b. the capacitance decreases.
c. the voltage rating increases.
d. the dielectric becomes more dense.

85. **In VOM's, voltage multiplier resistors are important:**

a. in the voltage scales.
b. in the current scales.
c. in the resistance scales.
d. in protecting the movement.

86. **In a series-tuned circuit, to increase the resonant frequency, you need to:**

a. decrease the inductance.
b. add resistance.
c. increase the inductance.
d. increase the input signal level.

87. **A MOSFET:**

a. may be called a Zener.
b. typically has a low input impedance.
c. retains data even when input power is lost.
d. amplifies signal voltage variations.

Toroid cores are used in transformers for:

a. concentrating the magnetic flux.
b. stepping voltage up or down.
c. shielding from external influences.
d. reforming flux line of force.

A PLL or phase-locked-loop circuit, contains a VCO, or voltage-controlled- oscillator. The output of the VCO is a/an _____.

a. DC constant level
b. in phase clock signal
c. AC variable frequency signal
d. 60 Hertz pulse

A variable capacitor depends on:

a. applied voltages being variable.
b. the plate separation being adjustable.
c. pressure on the dielectric being adjustable.
d. provisions for feedback.

While anode and cathode voltages are maintained, the gate level on an SCR drops below the trigger point. What happens?

a. The SCR turns off.
b. The SCR turns on.
c. The SCR stays on.
d. A pulse will be seen on the collector.

The combination of resistance and reactance which restricts the flow of alternating current is called:

a. impedance.
b. resistance.
c. volt-amperes.
d. power factor.

The output of a thermocouple would most likely be measured in:

a. milliwatts.
b. millivolts.
c. volts.
d. amps.

The Z axis on an oscilloscope is:

a. vertical.
b. horizontal.
c. depth.
d. intensity.

95. When soldering transistor or IC chip legs, a _____ should be used to prevent overheating of the circuit discrete components.

a. ground lead
b. flux shield
c. heat sink clip
d. length solder wick

96. Which of the below would normally be used for the core of a transformer?

a. Paper
b. Ferrite
c. Rosin
d. Silver

97. The "1" in binary number 100000 carries a weight of

a. 32.
b. 10000.
c. 10.
d. 16.

98. You would choose an LCD if you needed:

a. a lithium/cadmium DC source.
b. a low current digital display.
c. an inductive coupling device.
d. a charge-coupled device.

99. Germanium diodes:

a. are also called Tunnel Diodes.
b. normally drop about .3 v with forward current.
c. may be used as Zener diodes.
d. more commonly are called Varactors.

100. RF signal propagation that is normally consid "line-of-sight" would be:

a. HF.
b. MF.
c. UHF.
d. VLF.

ANSWERS on page 223

Appendix B

ETA Certification Administrator Listing

Certification Administrators (CAs)

The following listed examination administrators are the heart of the CET and FCC testing program. Without them there would be no professional certification for electronics technicians. Finding a test site to take FCC Commercial License exams would require traveling long distances and being there on specific dates. We all owe a lot to the CAs.

When you are ready, ready to attempt either type of exam, contact the closest CA. Understand that some CAs are electronics instructors in schools who cannot just drop what they are doing and administer an examination for you. Ask the CA when it might be convenient for he/she and the institution to monitor your exam. Be sure the CA knows which exams you want to take. If you are in the military service, your education office and the DANTES TCO will obtain exams and set a time for you to take the exams, either CET or FCC. Bear in mind that military education offices are usually less aware of the details of the CET and FCC programs than civilian electronics technical schools, thus you should be aware of the rules and options of the program and help to make sure you are taking the correct exam and abiding by the other criteria of the FCC and CET program.

If you are unable to schedule your exam with a CA, please call the ETA office. You do not have to register in advance for ETA exams, and you pay the exam fee at the time of sitting for that exam. A money order, or your charge card numbers are the preferred method of exam payment. Some CAs may charge a sitting fee such as $6.75.

You can help your CA and the ETA Testing Staff by making sure you include the examination number on your information sheet as well as on each answer sheet (online as well as paper). Be sure every word you write on the information form is legible. If your school teaches electronics and is not an established CA site, have the school contact ETA by phone (800) 288-3824 or e-mail eta@eta-i.org.

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Texas

Appendix B — Certification Administrators Listing

Texas cont.

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Utah

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Virginia

[illegible]

Washington

[illegible]

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Ladder of Career Progression:

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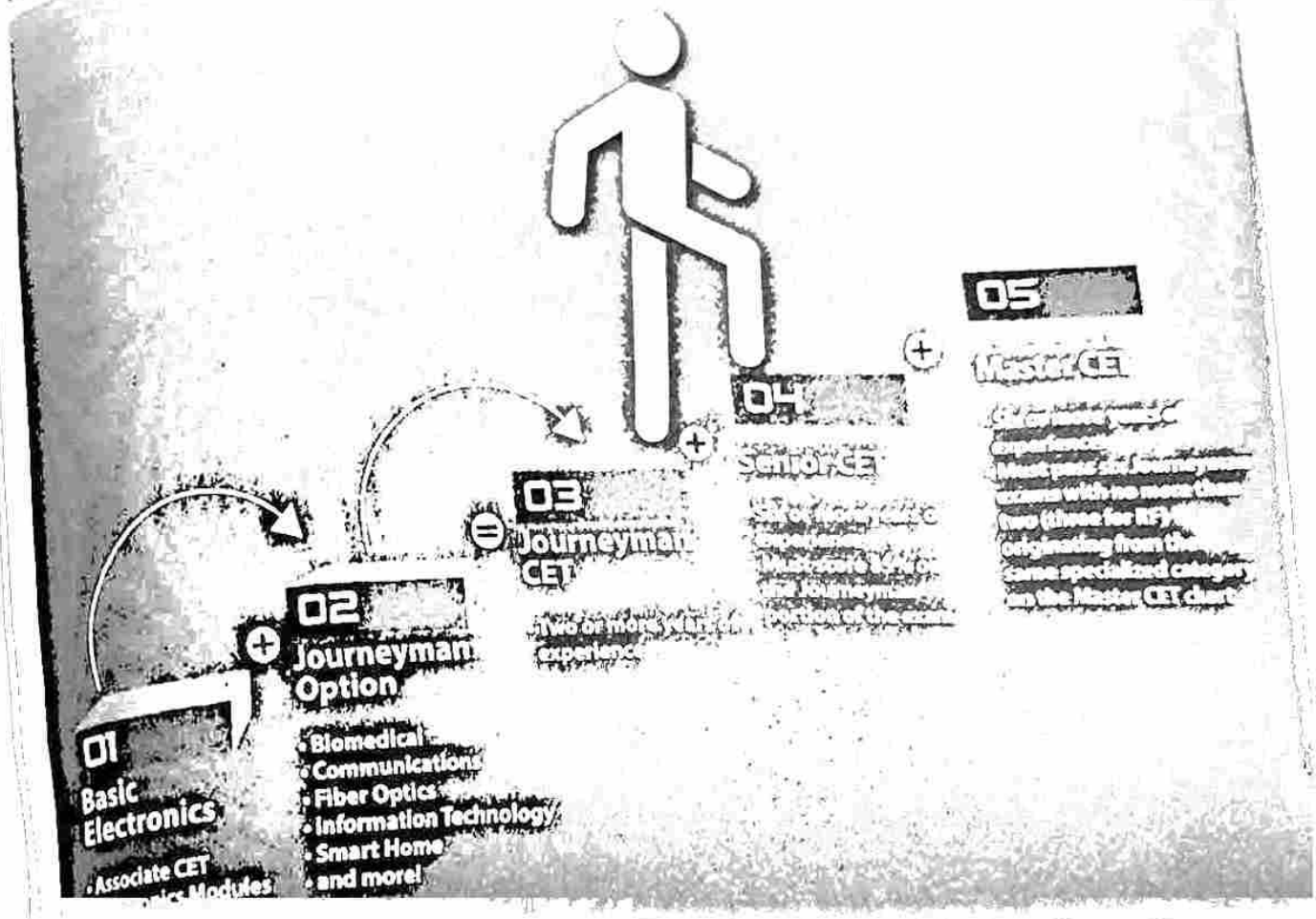

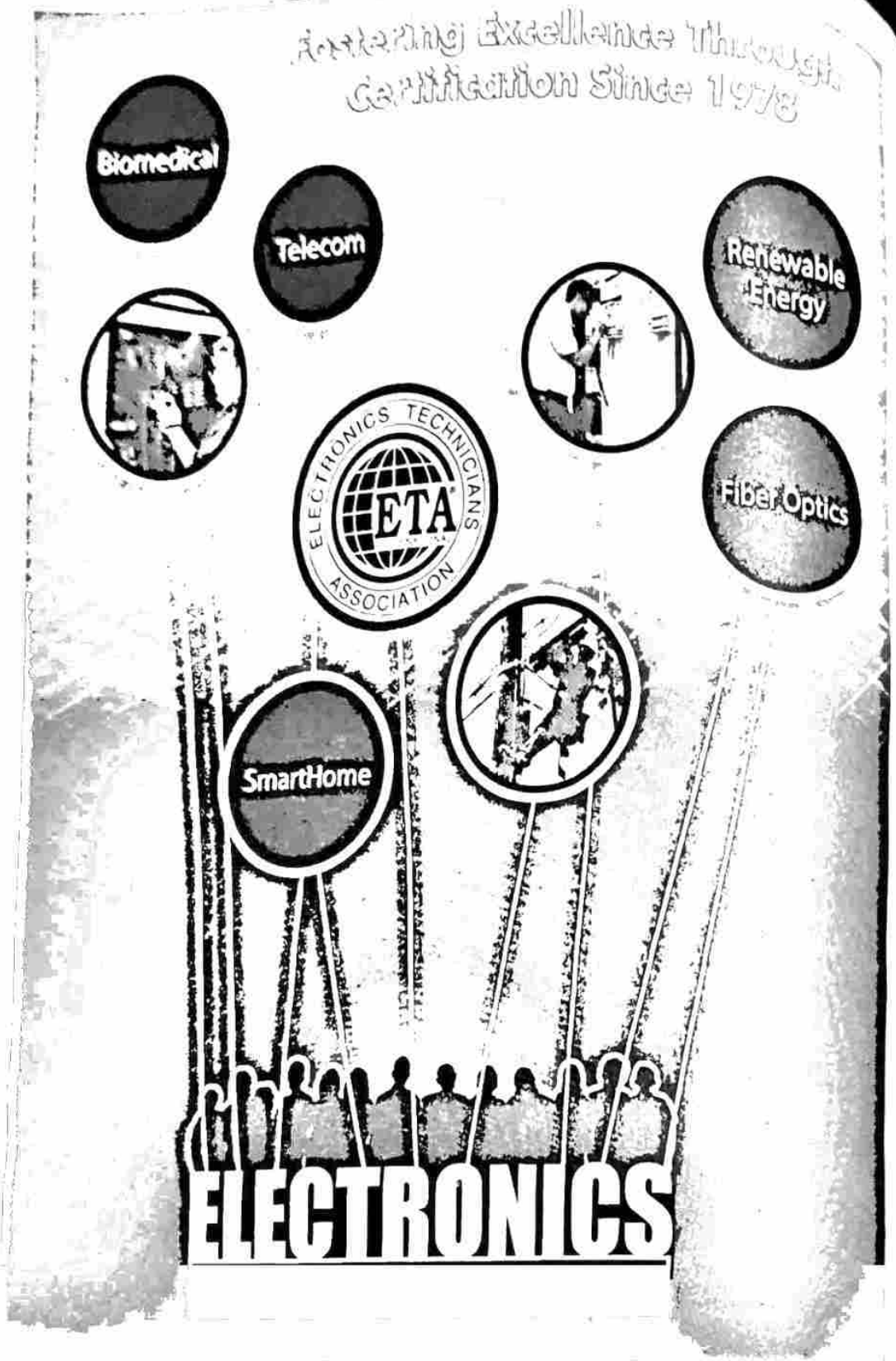
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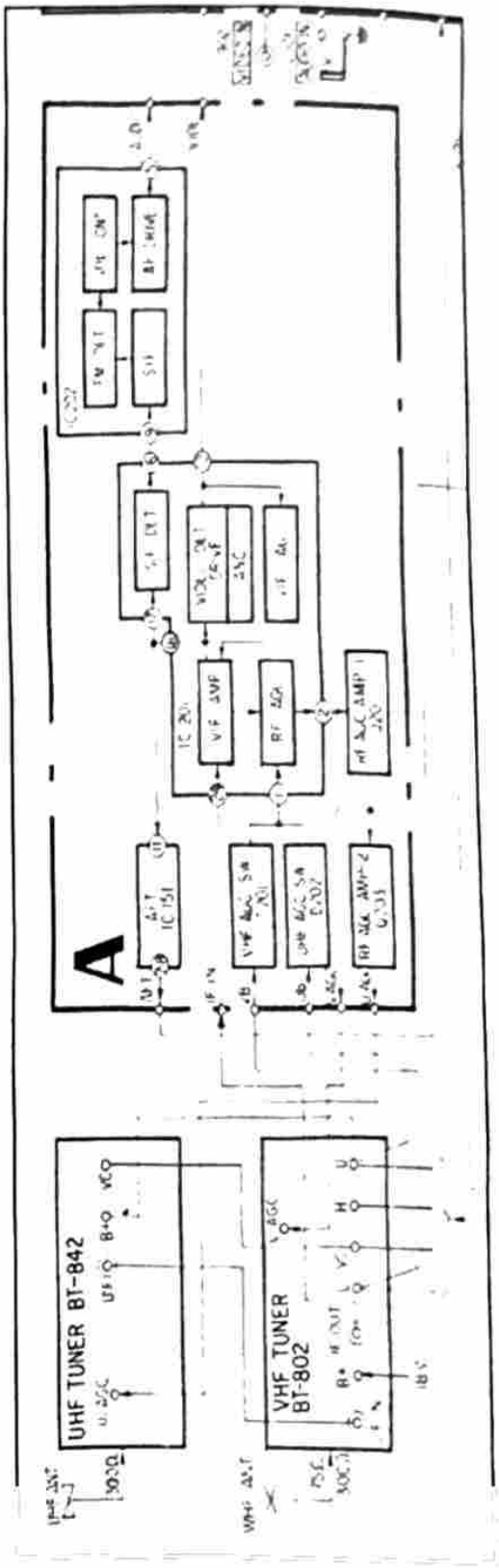
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Appendix C

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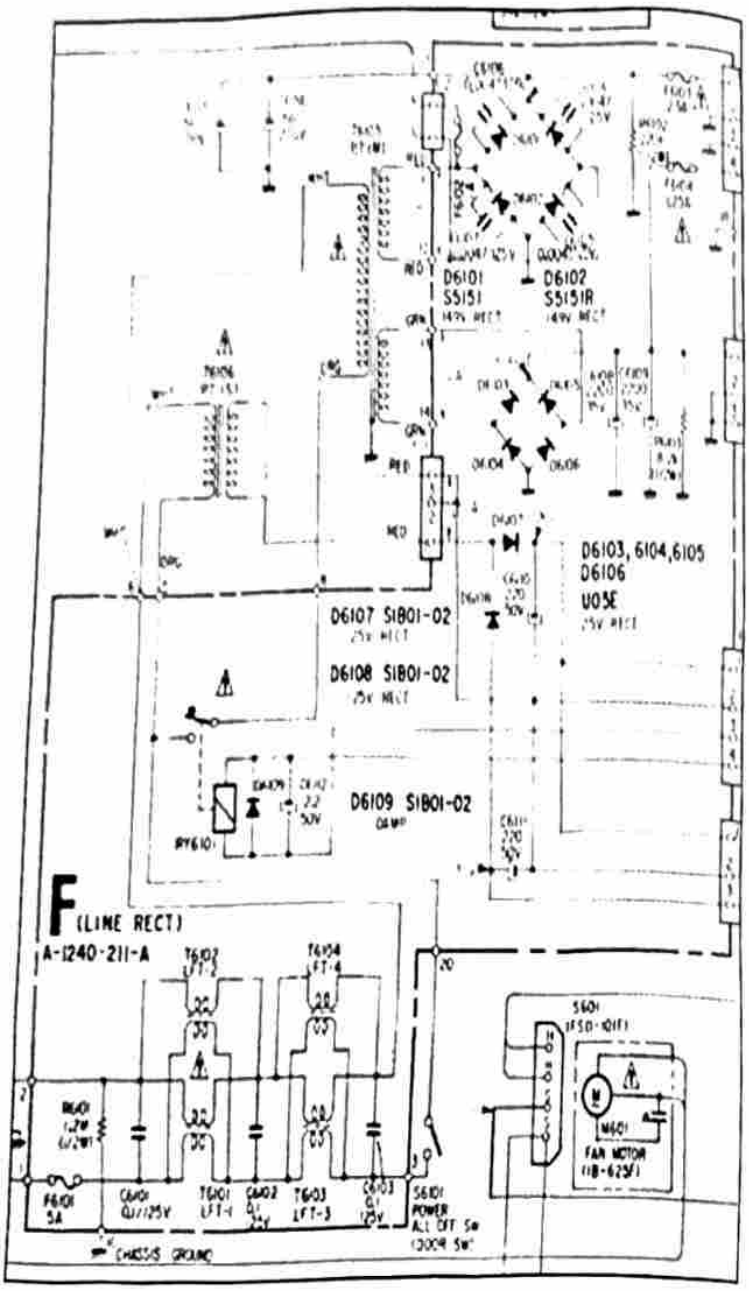

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Chapter 22: 1-d, 2-a, 3-a, 4-c, 5-b, 6-c, 7-a, 8-d, 9-b, 10-c

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Practice Exams

Practice Exam 1:

1-c, 2-b, 3-b, 4-c, 5-d, 6-a, 7-a, 8-a, 9-a, 10-b, 11-c, 12-b, 13-a, 14-a, 15-c, 16-a, 17-d, 18-b, 19-d, 20-a, 21-d, 22-a, 23-a, b, or c, 24-b, 25-b, 26-c, 27-a, 28-c, 29-d, 30-a, 31-c, 32-a, 33-a, 34-b, 35-b, 36-b, 37-c, 38-a, 39-a, 40-b, 41-a, 42-a, 43-a, 44-d, 45-a, 46-b, 47-a, 48-c, 49-d, 50-a, 51-b, 52-a, 53-a, 54-b, 55-a, 56-c, 57-a, 58-c, 59-b, 60-b, 61-b, 62-d, 63-b, 64-d, 65-d, 66-c, 67-c, 68-d, 69-a, 70-c, 71-a, 72-c, 73-b, 74-b, 75-b, 76-d, 77-a, 78-d, 79-d, 80-b, 81-c, 82-a, 83-b, 84-c, 85-b, 86-c, 87-c, 88-c, 89-a, 90-b, 91-a, 92-b, 93-b, 94-d, 95-d, 96-d, 97-b, 98-d, 99-c, 100-b

Practice Exam 2:

1-b, 2-a, 3-d, 4-a, 5-c, 6-b, 7-a, 8-d, 9-d, 10-d, 11-d, 12-b, 13-d, 14-d, 15-b, 16-d, 17-b, 18-c, 19-c, 20-c, 21-d, 22-c, 23-a, 24-d, 25-a, 26-d, 27-d, 28-d, 29-a, 30-c, 31-c, 32-a, 33-c, 34-b, 35-c, 36-b, 37-c, 38-a, 39-d, 40-d, 41-b, 42-d, 43-a, 44-b, 45-c, 46-b, 47-c, 48-a, 49-b, 50-c, 51-c, 52-a, 53-d, 54-d, 55-b, 56-a, 57-a, 58-a, 59-c, 60-b, 61-a, 62-b, 63-c, 64-c, 65-a, 66-c, 67-a, 68-c, 69-a, 70-b, 71-d, 72-a, 73-a, 74-d, 75-d, 76-a, 77-b, 78-b, 79-c, 80-d, 81-c, 82-d, 83-a, 84-b, 85-a, 86-a, 87-d, 88-a, 89-c, 90-b, 91-c, 92-a, 93-b, 94-d, 95-c, 96-b, 97-a, 98-b, 99-b, 100-c

Formulas

Common Formulas

Conversion Factors

π (Pi) = 3.14 $2\pi = 6.28$

$\pi^2 = 9.87$ $\log\pi = 0.497$

1 meter = 3.28 feet

1 inch = 2.54 centimeters

1 radian = 57.3°

Resonant frequency formulas

Where f is in kHz, L is in microhenries, C is in microfarads

$$f_{kHz} = 159.2 \div \sqrt{LC}$$

$$f_{resonant} = \frac{1}{2\pi\sqrt{LC}}$$

Frequency & Wavelength formulas *f = frequency, λ = wavelength*

0.5λ = 180° = half wave 0.25λ = 90° = quarter wave

$$f_{kHz} = (3\times10^5) \div \lambda meters \text{ or } f_{MHz} = 984 \div \lambda feet$$

$$\lambda meters = (3\times10^5) \div f_{kHz} \text{ or } \lambda feet = 984 \div f_{MHz}$$

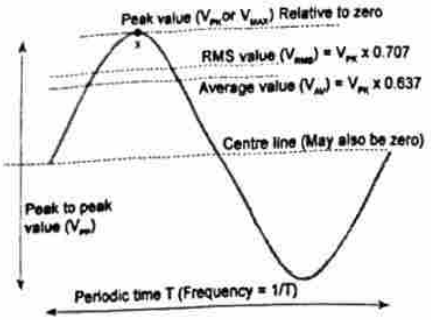
Sine wave conversion (RMS = root mean square).

Effective value (RMS) = 0.707 x Peak Value = 1.11 x Average Value

Peak Value = 1.414 x Effective Value (RMS) = 1.57 x Average Value

Average Value = 0.637 x Peak Value = 0.9 x Effective Value (RMS)

Identify: Waveform, Peak (amplitude), RMS, 1 cycle over time period (frequency), Peak to peak, and practical average.



Voltage gain in decibels

$$Gain\ dB = 20 \log (Vout / Vin)$$

Ratio of 2 power levels in decibels

$$Gain\ dB = 10 \log (P_1 + P_2)$$

Resistors in series

$$R = R_1 + R_2 + R_3...$$

Resistors in parallel

$$1 / R = (1 / R_1) + (1 / R_2) + (1 / R_3)...$$

Inductors connected in series

$$L = L_1 + L_2 + L_3 + ...$$

Inductors connected in parallel

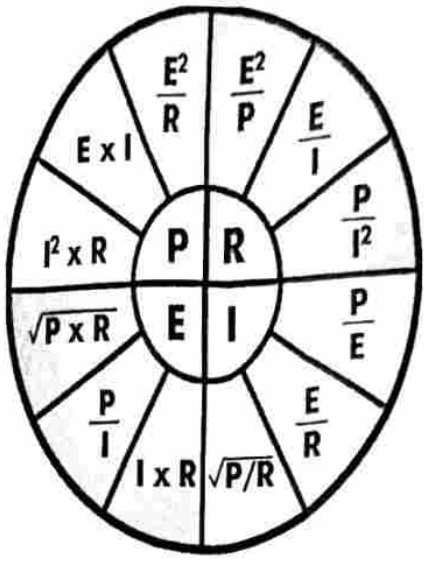
$$1 \div L = (1 \div L_1) + (1 \div L_2)...$$

Reactance of inductors

where X_L is reactance, f is frequency, and L is inductance

$$X_L = 2 \times \pi \times f \times L$$

Ohm's Law



E=Voltage I=Current
P=Power R=Resistance

P = I * E, the power being dissipated by the resistor is a product of the current and the applied voltage.

Time constants

T (Greek Tau), R (ohms), C (microfarads), L (microhenries)

RL circuit: **1 T (sec) = L (μH) ÷ R (Ω)**

RC circuit: **1 T (sec) = R (Ω) × C (μf)**

How to Compute Charge or Quantity of Electricity

where **Q** is the charge (in coulombs), **C** is the capacitance (in farads), and **V** is the potential difference (in volts).

$$Q = C \times V$$

Energy Storage in a Capacitor

where **W** is the energy (in Joules), **C** is the capacitance (in farads), and **V** is the potential difference (in volts).

$$W = \tfrac{1}{2} C \times V^2$$

Capacitors connected in parallel

$$C = C_1 + C_2 + C_3 + ...$$

Capacitors connected in series

$$1 \div C = (1 \div C_1) + (1 \div C_2) + (1 \div C_3)...$$

Reactance of capacitors

$$X_C = 1 \div (2 \times \pi \times f \times C)$$

Impedance Formulas for a Series Circuit

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ where Z is impedance

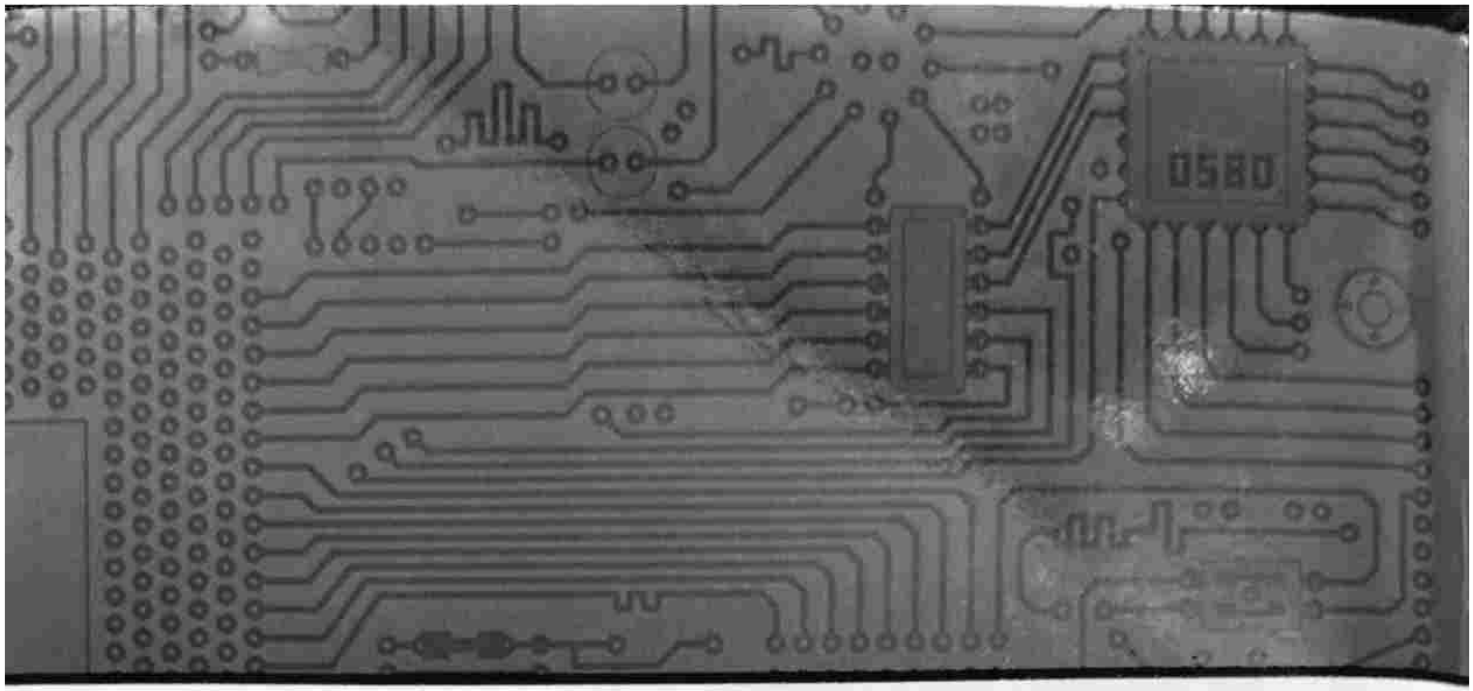
Impedance Formulas for R and X in Parallel

$$Z = \sqrt{\frac{RX}{R^2 + X^2}}$$

Battery internal resistance

$$V_{out} = EMF - V_{terminal}$$

Sine wave figure © Eric Coates MA BSc (Hons) Used with permission, 2013



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